
GX-S200/S400

The GX series is a new-concept PAC (Programmable Automation Controller) system integrating motion control, PLC, HMI, network interface, and data handling.

The modular PAC system GX-S200/S400 enables flexible system configuration tailored to user needs by supporting high-speed sequence control, high-precision/high-performance motion control functions, and standard industrial communication protocols.

1.1 GX-S200/S400 Overview

1.2 GX-S200/S400 Features

1.1.1 GX-S200/S400 Applications

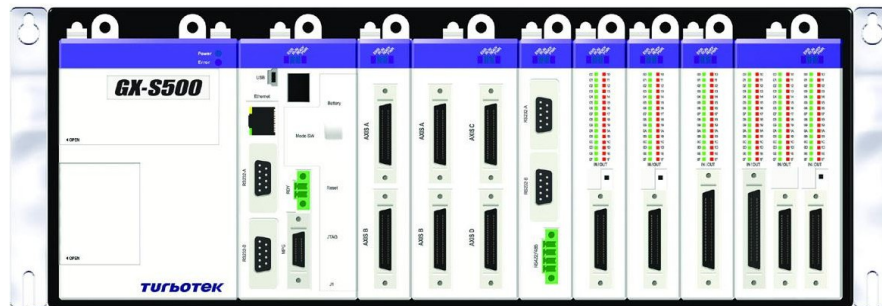
1.2.2 Motion Control

1.2.3 Sequence Control

1.2.4 User-Centric Programming Tools

1.1 GX-S200/S400 System Overview

The GX-S200/S400 is a high-performance PAC (Programmable Automation Controller) in a modular design, combining CNC's high-precision motion control technology with high-performance sequence control technology.



■ High-Precision / High-Function Motion Control Technology

- ◇ Simultaneous control of up to 4 channels
- ◇ Simultaneous support for motion commands and G Code
- ◇ Position determination, linear interpolation, circular interpolation, helical interpolation functions
- ◇ Synchronous control, position control, speed control

■ High-performance sequence control technology

- ◇ User function addition capability
- ◇ Support for Start PLC and Normal PLC
- ◇ Provides over 150 basic/functional commands
- ◇ Motion program call
- ◇ Ladder-style programming

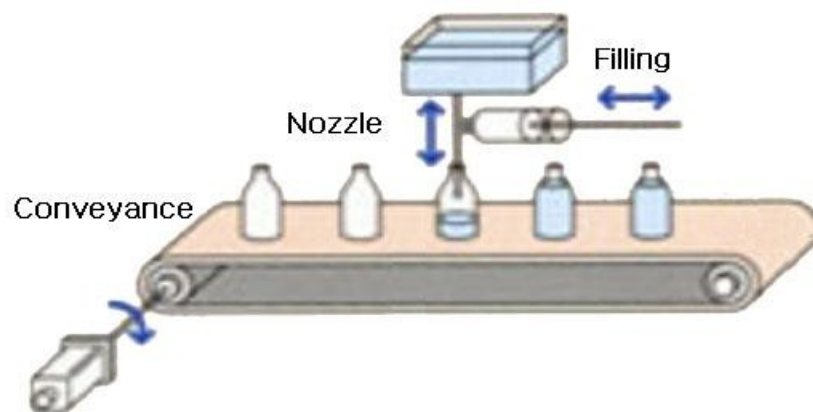
■ User-centric interface

- ◇ Provides a Windows-based programming tool
- ◇ System environment configuration and monitoring
- ◇ Motion / PLC program editing
- ◇ User-centric HMI screen editing

1.2 GX-S200/S400 Features

1.2.1 GX-S200/S400 Applications

The GX-S200/S400 utilizes a 4-channel motion program with motion commands, high-speed sequence control, and high-precision/high-functionality motion control to enable diverse motion control system configurations for automation applications.



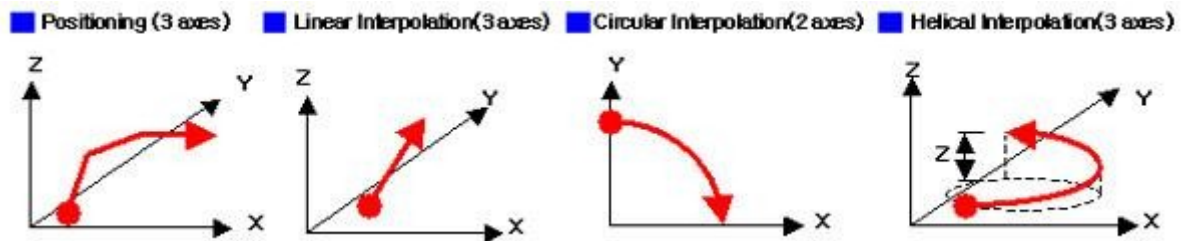
■ Application Fields

- ◇ Semiconductor / FPD-related equipment
- ◇ PCB / Electronic Product Assembly Equipment
- ◇ Machine tools
- ◇ Medical device and optical component processing equipment
- ◇ Cartesian coordinate robots, 3D palletizing robots
- ◇ Filling Machine, Drawing Machine, Punch, Rolling Machine
- ◇ Packaging and Logistics Equipment
- ◇ Inspection and measurement equipment

1.2.2 Motion Control

■ Motion Control

- ◇ Analog/Pulse/Digital Servo Interface Capable
- ◇ Provides various safety functions including Interlock, SoftLimit, and transfer restriction features
- ◇ Synchronized Phase Control
- ◇ Position Control
- ◇ Position Compare and Position Latch Functions
- ◇ Positioning, Linear, Circular, Helical Interpolation
- ◇ P I and Feed-Forward Control functions



■ Motion Program

- ◇ Motion Programming Support (Simultaneous Support for Motion Language and G Code)
- ◇ Simultaneous processing of 4 Motion Programs possible
- ◇ Master channel can issue axis commands to other channels
- ◇ Up to 12 axes can be commanded within a single block
- ◇ Simultaneous support for diverse Motion commands and G Code
- ◇ Programming using system registers and internal variables
- ◇ Support for various program control statements
- ◇ Supports calling up to 4 levels of subprograms
- ◇ Modal processing capability for Motion commands and axis-specific commands

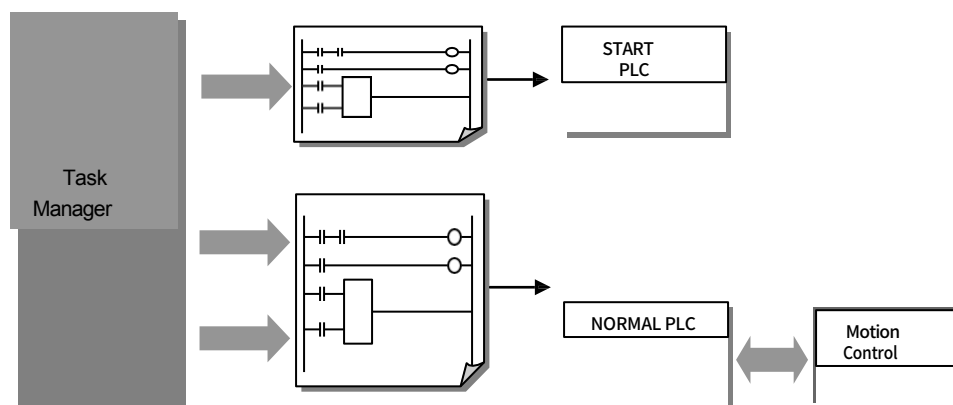
1.2.3 Sequence control

The GX-S200/S400 supports sequence control using both Start PLC and Normal Speed PLC methods, enabling high-speed scanning processing.

It supports a programming method using ladder-style language.

■ Sequence Control

- ◇ High-Speed Processing
- ◇ Direct Register Access enabled
- ◇ Large Programming Memory
- ◇ Support for a wide variety of rich instructions
- ◇ Motion program call function
- ◇ Subprogram call function
- ◇ Supports PLC programming using the ladder diagram method
- ◇ Convenient PLC program editing

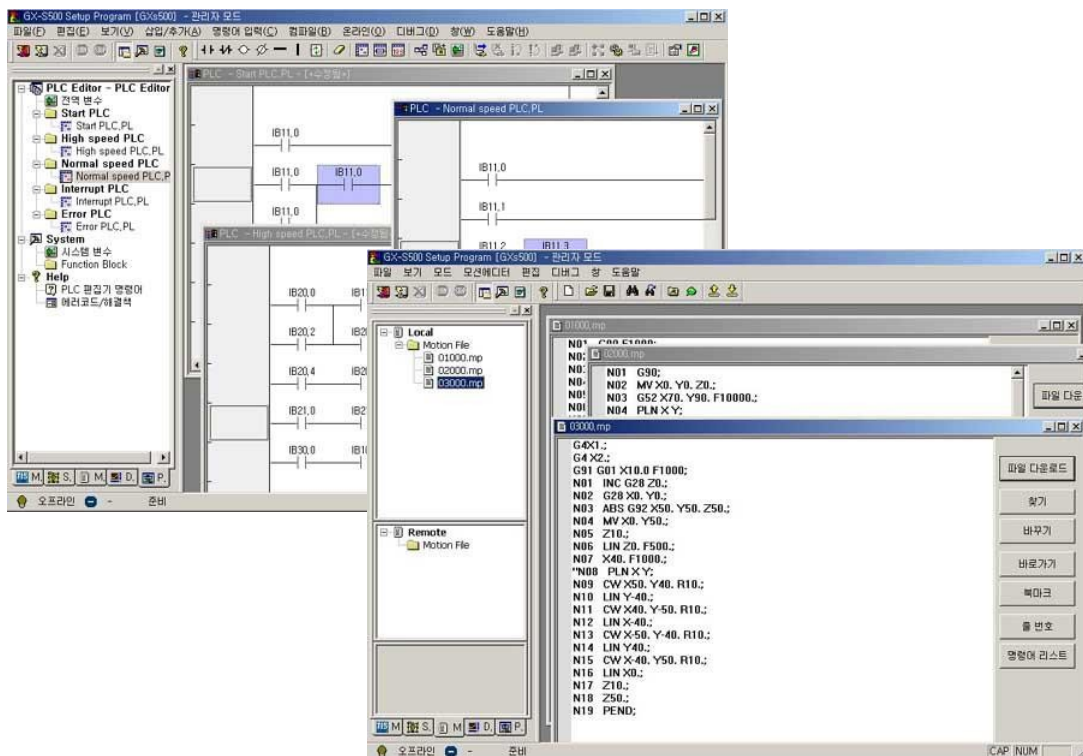


1.2.4 User-Centered Programming Tools

The GX-S200/S400 provides an integrated programming environment that allows you to manage a Windows-based PLC editor, Motion program editor, HMI editor, servo monitoring, and system environment settings all within a single program.

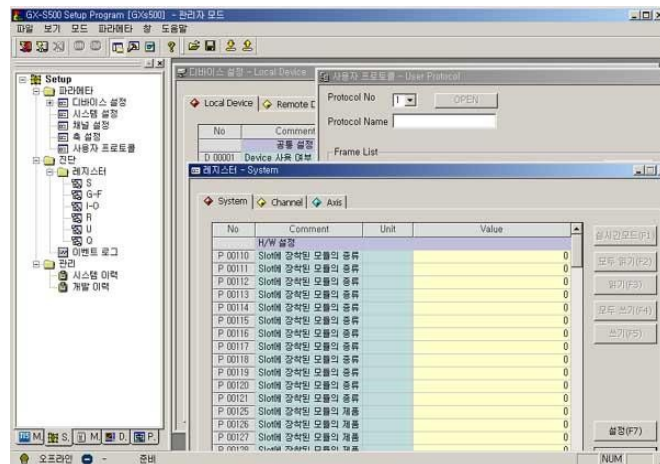
■ Ladder/Motion Program Editing

- ◇ Project-specific folder structure (Project-based management)
- ◇ Over 150 basic/functional commands editable
- ◇ PLC program editing using Multi Window
- ◇ Supports Ladder-style programming
- ◇ Symbol Table editing and display
- ◇ Various editing support functions
- ◇ Program user management function
- ◇ Program debugging and compilation
- ◇ Program monitoring function
- ◇ Contact Comment Function



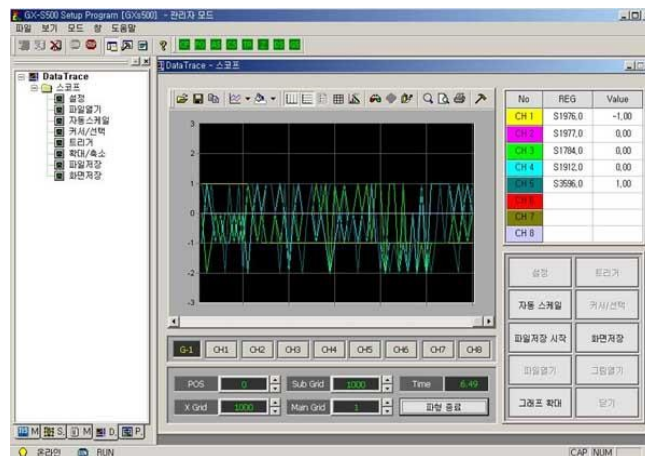
■ System environment and parameter settings

- ◇ Project Creation and Management
- ◇ Status variable and parameter setting functions
- ◇ H/W Device Configuration Function Using Graphics
- ◇ System Monitoring Function
- ◇ Motion program, PLC program editor built-in
- ◇ Program and data downloading via RS232C, USB, Ethernet, etc.



■ Real-time graphic display

- ◇ Real-time trend analysis of data fluctuations
- ◇ Servo waveform monitoring function
- ◇ Sequence monitoring via PLC timing chart analysis
- ◇ Support for 8-channel data trace environment
- ◇ Provides functions such as Trigger, Rising Time, Zoom, and Scale



GX-S200/S400 Specifications

This section describes the overall configuration of the GX-S200/S400, its basic hardware/software specifications, key features, command list, and characteristics.

2.1 Overall System Configuration

2.2 Hardware Configuration

2.2.1 System Configuration

2.2.2 General Hardware Specifications

2.2.3 Hardware Module Configuration and Specifications

2.3 Key Function Specifications

2.3.1 Motion Control Function Specifications

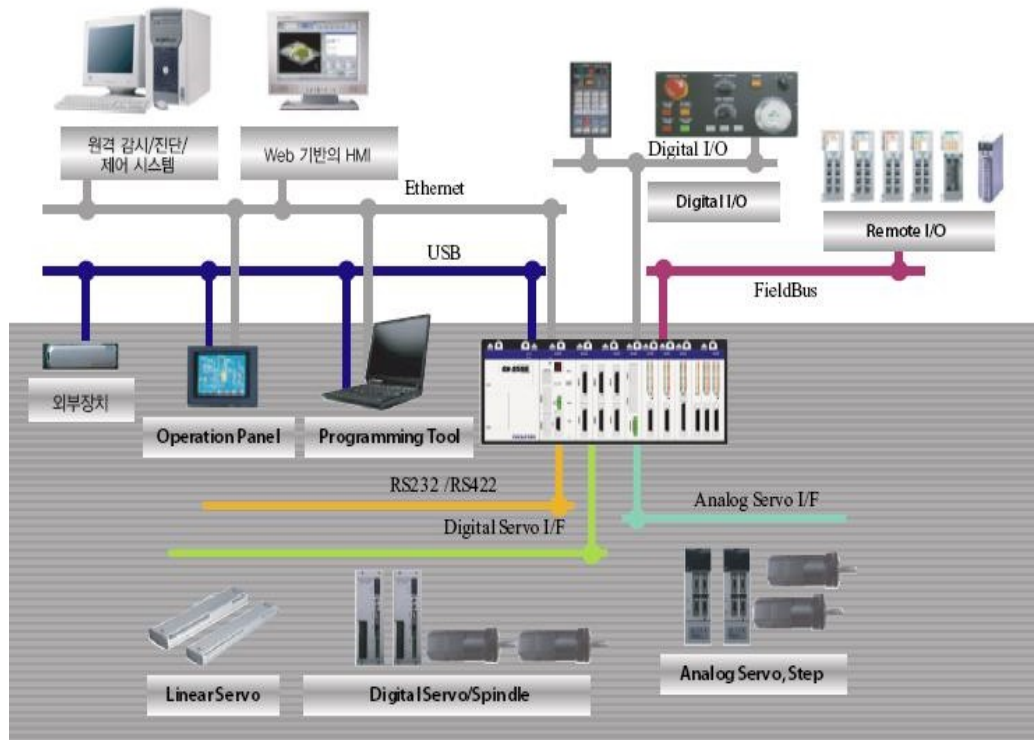
2.3.2 PLC Control Function Specifications

2.4 Command List

2.4.1 Motion Control Commands

2.4.2 PLC Control Commands

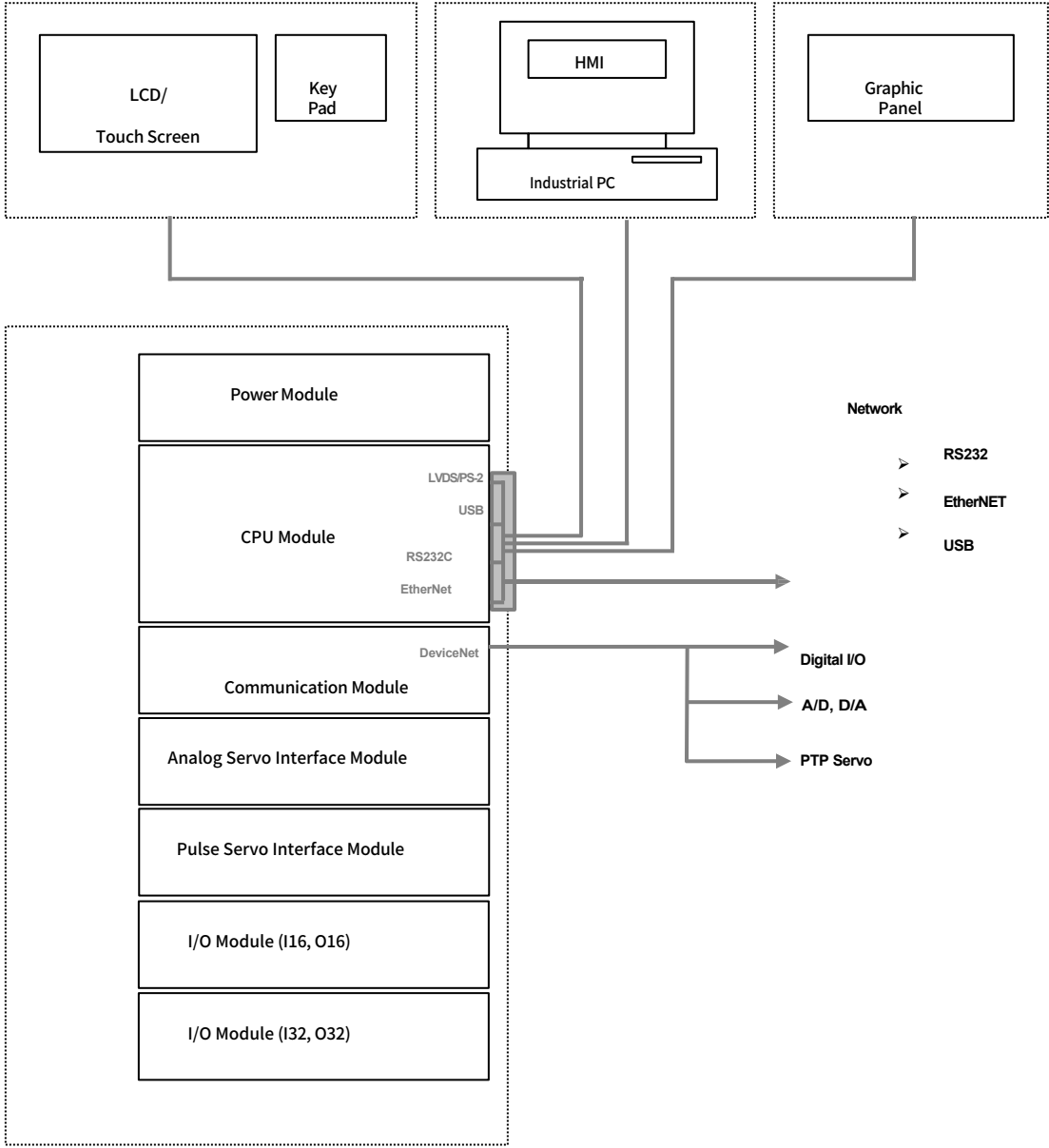
2.1 Overall System Configuration



- The GX-S200/S400 is a Stand Alone Type General Motion Controller operating in an Embedded environment , featuring a modular structure based on the architecture .

2.2 Hardware Configuration

2.2.1 System Configuration



■ The GX-S200/S400 consists of an integrated H/W module

2.2.2 General Hardware Specifications

The general hardware specifications are as follows.

Item	Item
Control Specifications	Max 4 Axes
	Max Input 48, Output 48 Points I/O
Servo Interface	Analog Interface
	Pulse Interface

2.2.3 Hardware Module Configuration and Specifications

The hardware configuration is as follows.

Type	Name	Designation	Description
Power	Power Module	GPWT	DC18V ~ 32V Input Voltage
CPU	DSP Module	G5CPU	Main Mode Module
Axis Module	Analog 2-Axis Module	GAXA2	Axis Module (Base)
	Analog 4-Axis Module	GAXA4	Axis Module (Extension)
	Pulse 2-Axis Module	GAXP2	Axis Module (Base)
	Pulse 4-Axis Module	GAXP4	Axis Module (Extension)
I/O Module (S400)	Input Module	GDI32	Input: 32 points/module, 8 points/common
	Output Module	GDO32	Output: 32 points/module, 8 points/common
I/O Module (S200)	Input Module	GDI16	Input: 16 points/module, 8 points/common
	Output Module	GDO16	Output: 16 points/module, 8 points/common
Communication Module	Serial Communication Module	GSC	RS-232/422/485
	EtherNet Module	GEN	Ethernet Communication Module

■ Power Module

The specifications and standards for the power supply module are as follows.

Item	Specifications
Name	Power Module

Name	GPWT	
Input Conditions	Input Voltage: DC18V ~ 32V	
		Input Current: 1.3A at 20W
		Inrush Current: 1A or less
Output Characteristics		Output Voltage: +5V, $\pm 12V$
		Output Current Range: +5V (4A), $\pm 12V$ (150mA)
		Rated Rise Time: 5ms or less
		Efficiency: 80% or higher
Safety		Safety Standards: Compliant with CE Standards (Class A)
External Wiring Terminals		External Power: DC24V, 24VGND, F.G
		Protective ground: FG
Operating Ambient Temperature		-10°C to +60°C (or 45°C)
Operating Ambient Humidity		90% or higher
Cooling Method		Natural air cooling
Storage Ambient Temperature		-20°C to +70°C
Storage ambient humidity		90% or higher

■ Analog Axis Control Module

The specifications and standards for the Analog Axis Control Module are as follows.

Item	Specifications
Name	Analog Axis Module
Name	GAXA2: Speed-type 2-axis Analog Axis Module
	GAXA4: Speed-type 4-axis Analog Axis Module
Servo Interface	Pulse Input Method: A/B/C Phase Input (Quadruple)
Analog Output	D/A Speed Command: 16-bit 4-point
	Output Range: -10V to +10V
Digital Input	Servo DI: 4 points
	SV_RDY (Servo Ready), SV_ALM (Servo Alarm)
	PSO (Absolute Encoder Signal), PL (Position Latch)
	ZERO, SV_SEN (Absolute Encoder Signal)

Digital Output	Servo DO: 4 points
	SV_ON, SV_RST, TQ_MD
Servo Control	Speed Control Mode
Indicator Lights	LED

■ Pulse Axis Control Module

The specifications and standards for the Pulse Axis Control Module are as follows.

Item	Specifications
Name	Pulse Axis Module
Name	GAXP2: 2-axis Pulse Axis Module
	GAXP4: 4-Axis Pulse Axis Module
Servo Interface	Pulse input mode: A/B/C phase input (quadruple)
Pulse Output	Positioning Pulse Rate Setting Range: 32-bit
	Reference Clock: 33MHz
	Supports both Open Collector and Line Driver
Digital Input	Servo DI: 4 points
	SV_RDY, SV_ALM, PL, ORG, INP
Digital Output	Servo DO: 4 points
	SV_ON, SV_RST, PC, P_CLR, ABSM
Servo Control	Position Control Mode
Frequency	130s ~ 0.3us
Pulse Width	50% Duty Cycle
Indicator Light	LED

■ DI Module

The specifications and standards for the Digital Input Module are as follows.

Item	Specifications
Name	Digital Input Module
Name	GDI16: Input Module with 16 Contacts
	GDI32: Input module with 32 contacts
Input signal	Input Points: 32 points/module, 8 points/COMMON

Allowable Current	200mA per pin, maximum 1A per contact
Fuse Rating	1A
Indicator	LED

■ DO Module

The specifications and standards for the Digital Output Module are as follows.

Item	Specifications
Name	DO (Digital Output) Module
Module Name	GDO16: Output module with 16 contacts
	GDO32: Output module with 32 contacts
Output Signal	Output Points: 16 points/32 points, 8 points/Common
Allowable Current	Max 500mA per pin, max 1A per contact
Fuse Rating	1A
Indicator	LED

■ Serial Communication Module

The specifications and standards for the serial communication module are as follows.

Item	Specification
Name	Serial Communication Module
Name	GSC
Signal	2x RS-232, 1x RS-422/485 (8-bit input/output each)
Speed	38,400 bps

■ EtherNet Communication Module

The specifications and standards for the EtherNet communication module are as follows.

Item	Specifications
Name	Ethernet Communication Module
Name	GEN
Speed	100 Mbps

Key Function Specifications

2.2.1 Motion Control Feature Specifications

The motion control-related feature specifications for the GX-S200/S400 are as follows.

Item	Specification
Number of Controlled Axes	S200 (up to 2 axes), S400 (up to 4 axes)
PTP Control	Linear, rotary, infinite length, independent axes
Position control	Position determination, external position determination, origin return, interpolation position detection function Interpolation with these features, constant speed transfer
Speed command output	-10V to 10V
Interpolation	Maximum 4 axes for linear, 2 axes for circular, 3 axes for helical
Command Unit	mm, inch, deg,
Minimum Command Setting Unit	1, 0.1, 0.01, 0.001, 0.0001, 0.00001
Maximum command value	-2147483648 ~ +2147483648
Speed command unit	mm/min, inch/min, deg/min
• Reduction Type	Linear, Asymmetric, S-curve
Coordinate System	Cartesian Coordinate
Programming Language	Motion commands, Ladder, G Code
Number of Program Tasks	Up to 4 programs can be operated in parallel simultaneously. Up to 4 subprograms can be called.
Number of Programs	Up to 128
Program capacity	4 MB
Axis movement commands	MV, LIN, CW, CCW, ZRN, ZRN2, SKP, SKP1, SKP2
Basic Control Commands	ABS, INC, PLN
Control Commands	IOW, PEND, IF~ELSE~END, WHILE~WEND, FOR~FEND, GOTO, SCALL, LOOP, RET

PLC Control Function Specifications

The functional specifications related to PLC control for the GX-S200/S400 are as follows.

Item	Specification	
Program Capacity	Memory Usage: 516 Kbyte	
I/O Area	Maximum 2048 Points I/O	
Control Method	Level 2 Normal Speed Scan (Level 1, Level 2)	
Programming Language	Ladder Diagram	
Number of Supported Commands	Supports over 150 commands	
PLC Channels	◇ Start PLC ◇ Normal Speed PLC	
PLC executable file	Startup Operation Processing Ladder File	
	Normal Speed PLC Ladder File	
	Subprogram Ladder File	
	◆ Subprograms of different types (S/H/N) cannot be called ◆ Sub PLC Program can be called in two stages from the Main PLC Program in two steps	
Data Memory	G/F	PLC & System Interface Data
	X/Y	Input/Output
	R	PLC Program Global Variable
	#	PLC Program Local Variable

2.3 Command List

2.3.1 Motion Control Commands

The following is a list of supported command codes for motion control.

■ Transfer Commands

Motion Commands	G-code	Description
MOV	G00	Rapidtraverse
LIN	G01	Linear Interpolation
ACW	G02	Clockwise Interpolation
ACC	G03	Circular Interpolation (Counter Clockwise)
DWL	G04	Predefined Dwell Time
INP	G09	In-position Check, Exact Stop
XYP	G17	Plane Selection 1-2 Geometry Axis
ZXP	G18	Plane Selection 3-1 Geometry Axis
YZP	G19	Plane Selection 2-3 Geometry Axis
PLN	G17.1	Arbitrary Plane Selection (PLN A1 A2)
INCH	G20	Input system Inches
METR	G21	Input system Metric
REF	G28	Machine Zero Position Return
REF2	G30	2 nd Machine Zero position return
REF3	G30	3 rd Machine Zero Position Return
REF4	G30	4 th Machine Zero Position Return
SKP	G31	Skip 1 function
SKP2	G31.3	Skip 2 function
MCALL	G65	Macro Call (Custom Macro Command)
ABS	G90	Absolute Dimension
INC	G91	Incremental Dimension
WCS	G92	Workpiece Coordinate System Setup
MCS	G53	Machine Coordinate System Setting

■ Control Command

Motion Command	M Code	Content
SCALL	M98	Subprogram Call
LOOP	M99	Subprogram Loop
RET	M99	Subprogram Return
PEND	M02, M30	Program End
	M01	Optional Stop
	M00	Program Stop

■ Conditional/Repeat/Jump Commands

Command	Code	Description
M	Parameter Setting	Synchronized Control / Synchronized Start with Other Channels Synchronized Control / Synchronized Start
Conditional Statement	IF [Condition] GOTO N_	IF [#W1 LE 10] GOTO N1
	IF [condition]	IF [#W1 ≤ 10]
	ELSE	ELSE
	IEND	END
Loop	WHILE [condition] DO N_	WHILE [#W1 LE 10] DO 200
	WEND N_	WEND 200
	WHILE [condition]	WHILE [#W1 LE 10]
	WEND	WEND
Jump Statement	FOR [INIT, END, STEP]	FOR [#W1 = 1, #W1 LE 10, 1]
	END	FEND
Jump Statement	GOTO N_	GOTO N100

■ Math Function

Command	Code	Description
SIN	SIN[30]	SINE
COS	COS[30]	COSINE
TAN	TAN[30]	TANGENT
ATAN	ATAN[30]	ARC TANGENT
SQRT	SQRT[4]	SQUARE ROOT
ABSV	ABSV[-2]	Absolute Value
ROUND	ROUND[3.5]	Round
AND(&)	IF [#W10 AND 1] GOTO N1	Logical AND (AND = %)
OR()	IF [#W10 OR 1] GOTO N1	Logical OR (OR =)
XOR(^)	IF [#W10 XOR 1] GOTO N1	Exclusive OR (XOR = ^)
FIX	FIX(3.5)	Discard
FUP	FUP(3.5)	Round Up
=	A = B + C	Substitution OR Value Assignment
->	B + C → A	Substitution OR value assignment
GT(>)	Both options supported	Large
LT(<)	Both supported	Small
GE(>=)	Both supported	Greater than or equal to
LE(<=)	Both supported	Below
EQ(==)	Both supported	Equal
NE(<>)	Both are supported	Not the same
+	#W10 = 10 + 3	ADDITION
-	#W10 = 10 - 3	SUBTRACTION
*	#W10 = 10 * 3	MULTIPLICATION
/	#W10 = 10 / 3	DIVIDE
MOD(%)	#W10 = 10 % 3	Remainder operation
BIN	#W1 = BIN(0x1234)	BCD TO BINARY
BCD	#W1 = BCD(1234)	BINARY TO BCD
SFTR	SFR #B1.1 N5 W10	BIT RIGHT SHIFT
SFTL	SFL #B1.1 N5 W10	BIT LEFT SHIFT

2.3.2 PLC Control Command

The following is a list of supported command codes for motion control.

■ Program Control Commands

Name	Symbol	Description
Subroutine Call	SCALL	Call Subprogram
Call motion program	MCALL	Call motion program
IF statement	IFON	Execute IF statement if OP1 value is 1
	IFOFF	Execute IF statement if OP1 value is 0
	ELSE	ELSE
	IEND	IF statement END
FOR statement	FOR	FOR statement
	END	FOR statement END
WHILE statement	WHILEON	If OP1 value is 1, execute While loop
	WHILEOFF	Execute the While loop if OP1 value is 0
	WEND	WHILE statement END
BREAK	BREAK	Exit the loop
Jump processing	GOTO	Jump to specified LABEL number
LABEL display	LABEL	LABEL Display
Program End	PEND	Indicates program end
Execution Halt	STOP	Stop all system operations

■ Input/Output Command

Name	Symbol	Description
READ	RD	Stores the value of a bit-type register on the stack
READ NOT	RDN	Stores the bit register value inverted onto the stack
READ Rising Edge	RDUP	contact specified, 1 cycle when rising edge (OFF->ON) HIGH for 1 cycle
READ falling edge	RDDN	When the specified contact is rising (ON->OFF) on , 1 cycle LOW
WRITE	WR	Store the result of the logical operation in the specified register.
WRITE NOT	WRN	Stores the inverted result of the logical operation in the register
Rising edge (output mode)	WRUP	When the input signal rises (, OFF->ON), the specified contact for 1 cycle
Falling Edge Differential (Output Mode)	WRDN	When the input signal goes from ON to OFF (), the specified contact for one cycle.
Set Coil (Output Meter)	SET	Self-holding operation
Reset Coil (Output)	RESET	When the Set signal is On, the output signal remains HIGH
KEEP (Output Circuit)	KEEP	Reset When the signal is On Output The signal is Continues LOW

■ Relay circuit stack operation command

Name	Symbol	Description
AND	AND	Performs a logical AND operation between the value of the specified register and the value on the stack and stores the result on the stack
AND NOT	ANDN	Invert the value of the specified register and perform a logical AND operation with the value on the stack and store the result in the stack
AND Ascending Differentiation	ANDUP	Logically ANDs the value of the specified register's rising edge derivative with the value on the stack and stores it on the stack
AND descending derivative	ANDDN	Performs a logical AND operation between the descending derivative value of the specified register and the value on the stack and stores the result in the stack
OR	OR	Performs a logical OR operation between the value of the specified register and the value of the first stack in ` ` and store the result in the stack
OR NOT	ORN	Invert the value of the specified register and store the result of the logical OR operation with the value of the first stack and store the result in the stack
OR Ascending Differentiation	ORUP	Logically XOR the rising derivative value of the specified register with the value in the stack and store it in the stack
OR Falling Edge	ORDN	Logically OR the falling derivative value of the specified register with the value on the stack and store the result in the stack
Read Stack	RDS	Shift the stack value 1 bit to the left and perform a logical OR operation with the specified register Store the value of the cell on the stack
Read Not Stack	RDNS	Shift the stack value 1 bit to the left and store the specified register value inverted on the stack and store it on the stack
Read Rising Edge Stack	RDUPS	Shift the stack value 1 bit to the left and store the inverted value of the specified register on the stack and store the rising derivative value of the specified register on the stack
Read Decreasing Differential Stack	RDDNS	Shift the stack value 1 bit to the left and store the rising derivative value of the specified register on the stack

		and store the descending derivative value of the specified register in the stack
And Stack	ANDS	Perform a logical AND operation on the first and second stack values and store the result in the stack and shift the stack value one bit to the right
Or Stack	ORS	Perform a logical OR on the first and second stack values and store the result on the stack and shift the stack value one bit to the right.

■ Logic Operation Instructions

Name	Symbol	Description
Logical AND	ANDMOV	Stores the result of the logical AND operation in the specified register
Logical OR	ORMOV	Stores the result of logical OR in the specified register
XOR	XORMOV	Stores the exclusive OR result in the specified register

■ Numeric operation instructions

Name	Symbol	Description
+	ADD	Store the value of OP1 + OP2 in OP3
-	SUB	Store the value of OP1 - OP2 in OP3
*	MUL	Store the value of OP1 * OP2 in OP3
/	DIV	Store the value of OP1 / OP2 in OP3
Remainder	MOD	Remainder
INCREMENT	INC	Adds 1 to the specified register
DECREMENT	DEC	Subtracts 1 from the specified register
TIME Add	TIMEADD	Hour/Minute/Second Addition
TIME Subtract	TIMESUB	Subtract hours/minutes/seconds

■ Numeric stack operation commands

Name	Symbol	Description
Numeric Read	NMRD	Stores the value of the specified register onto the stack
Numeric Read Stack	NMRDS	Shift numeric stack value one position left and store specified register value in numeric stack to the numeric stack
Addition	NMADD	Adds the value of the specified register to the value in the stack at address ` ` and stores the result in the specified register. and store the result in the stack
Subtraction	NMSUB	Subtracts the value of the specified register from the value on the stack and stores the result on the stack and store the result on the stack
Multiply	NMMUL	Multiply the value of the specified register by the value on the stack and store the result on the stack and store the result in the stack
Divide	NMDIV	Divides the value in the specified register by the value on the stack at address ` ` and stores the result on the stack
Addition Stack	NMADDS	First, add the first and second numerical stack values and store them on the stack. and then shift the stack value one position to the right.
Subtraction Stack	NMSUBS	Subtract the first and second numeric stack values and store the result on the stack and shift the stack value one place to the right
Multiply Stack	NMMULS	Multiply the first and second numeric stack values and store the result on the stack and shift the numeric stack value one place to the right.
Divide Stack	NMDIVS	Divide the first and second numeric stack values and store the result on the stack and shift the numeric stack value one place to the right.
Numeric Write	NMWR	Store the result of the logical operation in the specified register

■ Numeric Comparison Instructions

Name	Symbol	Description
<	LT	Size Comparison
≤	LE	Size Comparison
>	GT	Size Comparison
≥	GE	Size Comparison

=	EQ	Equivalent comparison. Stores the comparison result in ST0.
=	EQ2	Equivalence comparison. Stores the comparison result in ST0.
≠	NE	Equality comparison. Stores the comparison result in ST0.
Range check	RNG	Checks if the register value is within the range. Stores the comparison result in ST0.

■ Data conversion instructions

Name	Symbol	Description
Sign inversion	INV	Inverts the sign
Bit Invert	INVB	Inverts the bit values in a register (0->1, 1->0)
Two's complement	COM	Converts to 2's complement
Absolute value conversion	ABSD	Absolute Value Conversion
BCD Conversion	BCD	BIN ⇒ BCD Conversion
Binary Conversion	BIN	BCD ⇒ BIN Conversion
Data Type Conversion	WTOF	Convert WORD-type register to FLOAT-type
Data Type Conversion	FTOW	Convert FLOAT type register to WORD type
Bitstream-> Bit sequence conversion	BITRTOC	Set the specified bits in each of the 32 specified registers to the specified bits of the 32 specified registers
Bit String-> Bit row conversion	BITCTOR	Set the data of the specified bit () in the specified register () to the specified bit in the specified register () from the specified register (32-bit register) to the specified bit () to the separately specified register
Set data to a separately specified register.	PARITY	Calculate the number of On bits in the binary representation
ASCII Conversion	ASCTO BIN	Convert ASCII to 32-bit binary data
ASCII Conversion	BINTOASC	Convert 32-bit binary data to ASCII

■ Data Transfer Command

Name	Symbol	Description
Bit Copy	MOVB	Copies a bit value to the desired location
Register Copy	MOVD	Copy register value to desired location
Bit Swap	CHGB	Swaps the bit values between each other
Register exchange	CHGD	Swap the values of each other's registers
Byte Swap	SWAP	Swaps the upper and lower bytes of a word variable
Data Initialization	SETD	Initializes specified range of registers
Data Initialization	SETD2	Initialize specified range of registers
Set Bit Data	SETB	Writes a value to a specific location in a register
Set Bit Data	SETB2	Write a value to a specific location in a register
Data Search	SRCH	Searches for the location in the register that matches the data
SORT	SORT	Sorts registers within the specified range
=	INL	Assigns the right value to the left operand
->	INR	Assign the left value to the right argument

■ Timer/Counter Command

Name	Symbol	Description
On Delay Timer (1 ms)	TMRON	On Delay Timer (0 to 2147483 seconds adjustable)
Off Delay Timer (1 ms)	TMROFF	Off Delay Timer (0 to 2147483 seconds adjustable)
Integrating On Delay Timer (1 ms)	TMRONT	Accumulative ON Delay Timer (0 to 2147483 seconds configurable)
Integrating Off Delay Timer (1 ms)	TMROFFT	Accumulative OFF Delay Timer (0 to 2147483 seconds adjustable)
UP Counter	CNTUP	UP Counter
DOWN Counter	CNTDN	DOWN counter
UP/DOWN Counter	CNTUPDN	UP/DOWN Counter

■ Data Manipulation Commands

Name	Symbol	Description
Bit Left Rotation	ROTL	Rotates the bit value of the specified register to the left
Advanced Bit Left Rotation	ROTL2	Rotates the bit value to the left within the specified range
Bit Right Rotation	ROTR	Rotates the bit values in the specified register to the right
Advanced Bit Rotation Right	ROTR2	Rotates the bit value to the right within the specified range
Bit Left Shift	SFTL	Shifts the bit values of the specified register to the left
Advanced Bit Left Shift	SFTL2	Shift the bit value to the left within the specified range
Bit Right Shift	SFTR	Shift the bit values of the specified register to the right
Advanced Bit Shift	SFTR2	Shift the bit values to the right within the specified range

■ Math Function Command

Name	Symbol	Description
Square Root	SQRT	Square root
Degree Conversion	RADTODEG	Convert Radian to Degree
Radian Conversion	DEGTORAD	Convert Degrees to Radians
SIN	SIN	SIN
COSIN	COS	COS
TANGENT	TAN	TAN
ARC SINE	ASIN	ASIN
ARC COSINE	ACOS	ACOS
ARC TANGENT	ATAN	ATAN
Exponential	EXP	Calculate the exponential function with base e
Exponential	POW	Exponential calculation
Natural Logarithm	LN	LN
Common Logarithm	LOG	LOG
Rounding	ROUND	Round
Up	FUP	Up
Discard	FIX	Discard

■ Table Data Processing Commands

Name	Symbol	Description
Stack Area Setting	STSET	Set the stack area and initialize the values in the set area to 0 (only possible with the R register)
Store stack data	STPUSH	Store data in the stack area (Push)
First-in, first-out	STFIFO	Retrieve (pop) the data stored first from the specified stack area Command
Last-in, First-out	STLIFO	Pop data from the stack area () that was last stored () from the specified stack area (STLIFO)
Read table value	TBRD	Reads the value from the specified position in the table
Write table value	TBWR	Writes a value to the specified position in the table
Retrieve table data	TBSRCH	Search for the location of data in a table
Search for the maximum value in a table	TBMAX	Searches for the maximum value in the specified table
Search for the minimum value in a table	TBMIN	Find the minimum value in the specified table
Calculate the table sum	TBSUM	Calculate the sum of the specified table

■ Data control commands

Name	Symbol	Description
Dead band control	CTRLDB	Controls output data based on whether input data falls within the dead band range Controls output data based on whether input data falls within the dead band range
Dead Zone Control	CTRLDZ	Adds the input value when the input is positive Subtract the input value when the input is negative and output
Upper/Lower Limit Control	CTRLMT	Controls output data based on whether input data falls within the upper/lower limit range Controls the output data based on whether the input data falls within the upper/lower limit range
PI control	CTRLPI	PI Control
Primary delay	DELAY	Primary Delay
Linear Accelerator	LINEACC	Linear Accelerator
Rotary Control	Rotary Control	Control of rotating components such as tool changers, ATCs, and rotary tables Commands used for controlling rotating components such as tool changers, ATCs, and rotary tables

■ Other commands

Name	Symbol	Description
User Event	SETEVT	User Event Setting Command
Register Write	SETREG	Write specified register value command
Register Read	GETREG	Command to read the value of a specified register

System Basic Operation

This section explains the fundamental system operations required to operate the GX-S200/S400. It covers the system's operating modes, dip switch operation methods, registers, and provides information on the progress of the motion

/P LC program execution process.

3.1 System Operation

- 3.1.1 RUN Mode
- 3.1.2 STOP Mode
- 3.1.3 Dip Switch Operation
- 3.1.4 Memory Initialization

3.2 Register

- 3.2.1 Register Types
- 3.2.2 Register Data Types

3.3 Program Structure

- 3.3.1 Motion Program Structure
- 3.3.2 PLC Program Structure

3.1 System Operation Modes

System operation modes include RUN mode and STOP mode, each with the following characteristics. The system operation mode status can be confirmed by the LED off state or via the User S/W.

3.1.1 RUN Mode

■ RUN Mode Characteristics

- ◇ When power is applied to the GX-S200/S400, the [RDY] or [RUN] LED illuminates.
- ◇ If communication with GX-Builder is established, the Status Bar displays the [RUN] status.
- ◇ The RUN mode status indicates that the system and programs are operating normally without any errors.
- ◇ The RUN/STOP mode can be changed using the GX-Builder software.

3.1.2 STOP Mode

■ STOP Mode Characteristics

- ◇ The [RUN] LED on the GX-S200/S400 turns off.
- ◇ PLC/Motion programs operated by the system are stopped

■ When switching to STOP mode

- ◇ If the system scan time is not set
- ◇ When program memory is not initialized
- ◇ When a critical error occurs within the system
- ◇ When a command to switch to STOP mode is sent from GX-Builder S/W
- ◇ When the RUN/STOP switch is set to OFF and power is applied

3.1.3 Dip Switch operation

When controlling system operation, manipulate the Dip Switch on the CPU module.

The DIP Switch consists of 8 switches, and the setting method for each switch is as follows.

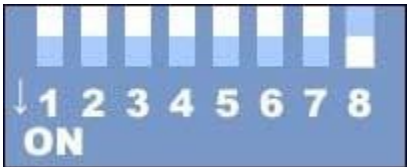
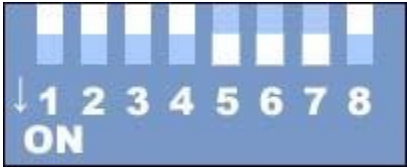
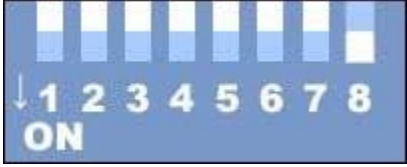
■ Dip Switch Function



Number	Purpose	Setting	Operation
①	Reset	ON	RESET
		OFF	ON LINE
②	Reserved	-	-
③	Default Mode	ON	Emergency Communication Default Mode
		OFF	Normal Operation
④	Reserved	-	-
⑤	Clear	ON	Program Memory Initialization
		OFF	Normal Operation
⑥	Clear	ON	Battery Backup Memory Initialization
		OFF	Normal Operation
⑦	Init	ON	Memory Initialization
		OFF	Normal Operation
⑧	Run/Stop	ON	Program RUN Mode
		OFF	Program STOP Mode

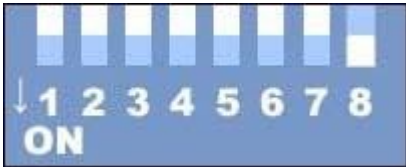
3.1.4 Memory Initialization

Memory can be initialized by performing Dip Switch settings and system operations in the following sequence.

Number	Operation	Dip Switch Status
①	Turn off the power to the GX system. Select and set one of the following conditions based on the range you wish to initialize.	
②	Initialize All Memory Dip Switch All switches OFF, and turn ON only Switches 5, 6, and 7.	
	Program Memory Initialization Turn on only Switches 5 and 7.	
	Register Storage Memory Initialization Turn on only Switches 6 and 7.	
③	When the system power is turned on, the memory initialization operation is performed.	
④	Verify that the [r] and [] symbols flash on the CPU's 7-segment display.	
	The [r] symbol flashing indicates that the system is currently initializing. [] symbol flashing indicates initialization complete	
⑤	Turn the system power OFF, then reset the Dip Switch to its original state.	
⑥	Turn the GX system power ON.	

3.1.5 Default Parameter Settings

Performing Dip Switch settings and system operations as described below will apply default values to all parameters. Refer to [Chapter 8: Registers] in this manual for the default value of each parameter.

Operation	Dip Switch Status
Turn on only switches 3 and 7. R e b o o t the system.	
Set the DIP switches to normal mode.	

3.1.6 Default Communication

If the system fails to operate normally due to an unexpected error while in RUN or STOP mode, switching the system to Default mode allows you to download the program from a PC via Default communication to restore the system.

Operation	Dip Switch Status
Turn Switch 3 ON. Rebooting the system switches it to Default mode.	

3.2 Register

In the GX-S200/S400, all data processing for operations is performed through designated registers, and modules are interconnected via interface processing registers.

3.2.1 Register Types

Classification	Name	Explanation
System	P	System Parameters
	D	Device Parameters
	S	Status Information Parameters
	E	System Event Information (Alarm/Warning)
	U	Motion Program Global Variables
	#	Motion Program Local Variables
	Q	System Control Data
PLC	G/F	PLC & System Interface Data
	X/Y	Input/Output Data
	R	PLC Program Global Variables
	#	PLC Program Local Variables

3.2.2 Register Data Type

■ Data Type

The register data used in the GX-S200/S400 is categorized into the following types.

Internal Internal	
Bit Type (32 Bit)	32 Bit
Word Type (32 Bit)	-2,147,483,648 to 2,147,483,647
Float Type (32 Bit)	3.4E ± 38 (7 digits)

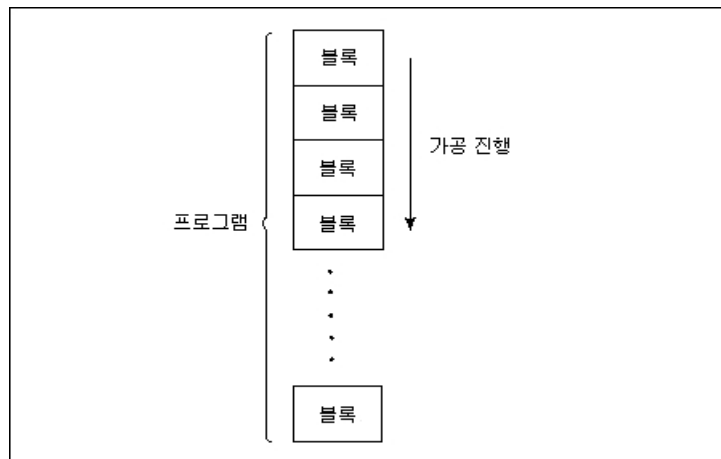
3.3 Program Structure

3.3.1 Motion Program Structure

■ Program Execution

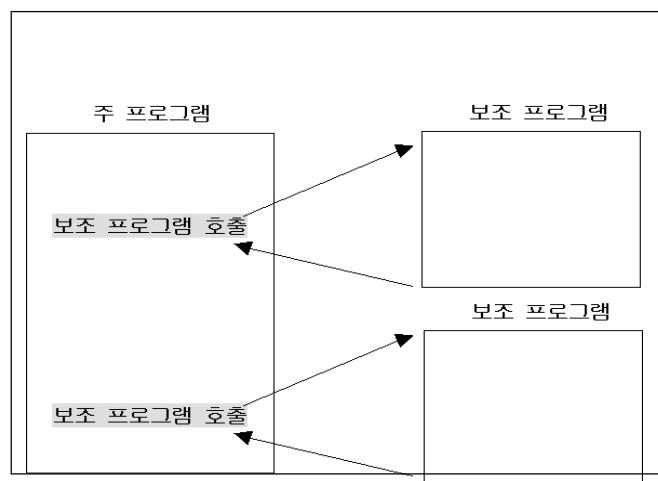
The program is composed of a combination of blocks, and the corresponding functions are executed based on the contents of each block. Primarily, the first block indicates the program file name, and the comment function is used to write program descriptions.

The very last block must contain the program termination auxiliary command M02 or M30.



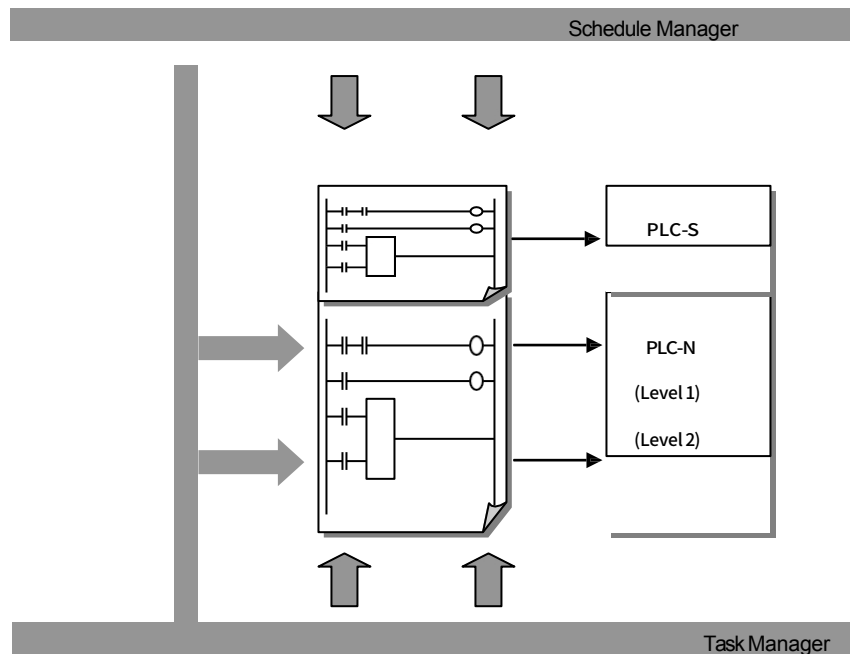
■ Main Program and Auxiliary Program

Programs are divided into main programs and subprograms. The main program is the program that first initiates machining from a stopped state. Subprograms are executed upon being called by the main program and return to the main program upon completion.



3.3.2 PLC Program Structure

PLC programs are controlled through the Schedule Manager and Task Manager of the Kernel system, which manage the execution conditions and priorities to determine the overall operation sequence of the entire PLC program.



■ PLC programs consist of Cyclic PLC and Non-Cyclic PLC.

■ The Normal Speed PLC, a Cyclic PLC, is organized into two levels and executes with the priority N-PLC(Level 1) > N-PLC(Level 2).

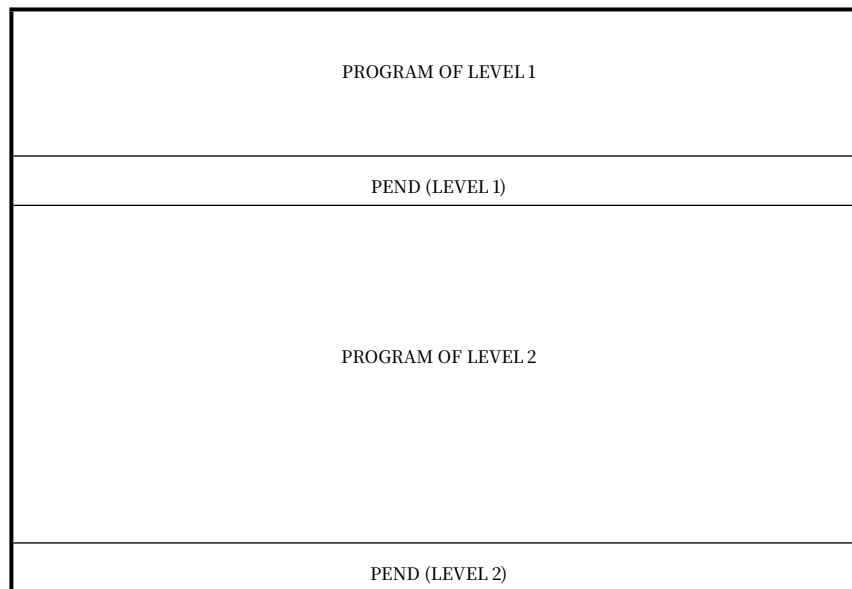
■ PLC-S Channel

◇ The Start PLC program executes only once during booting.

◇ You can insert initialization code to be executed before the actual program runs.

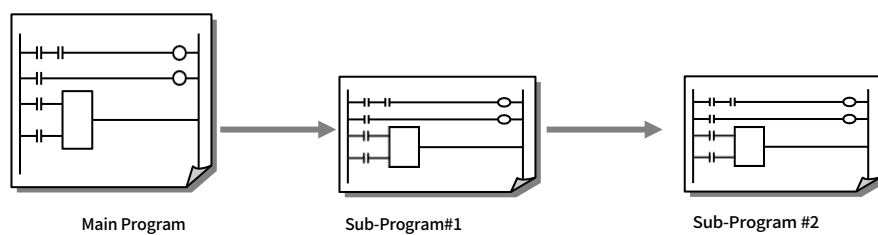
■ PLC-N Channel

- ◇ This is the Normal Speed PLC program, which is the primary PLC for writing most programs.
- ◇ Programs can be written in two levels to assign execution priority.



- ◇ The PEND command must be present in the program structure.
- ◇ LEVEL1 is programmed to handle functions that require priority processing during Normal Sequence execution. LEVEL1 is written to complete in a short time.
- ◇ LEVEL2 is a routine that handles sequences other than LEVEL1.

■ The Main PLC program for a channel can call up to two levels of subprograms.



System Operation

This document systematically outlines the installation and operation procedures for the GX-S200/S400, providing detailed explanations for each step.

It explains the types of system operation modes and the characteristics of each mode, and describes how to initialize memory to operate the system.

4.1 System Operation Sequence

4.2 System Installation

4.2.1 Installation Preparation

4.2.2 Module Installation

4.2.3 Wiring

4.2.4 Initial Setup

4.2.5 Project Creation

4.2.6 System Configuration

4.2.7 Programming

4.2.8 Program Transfer

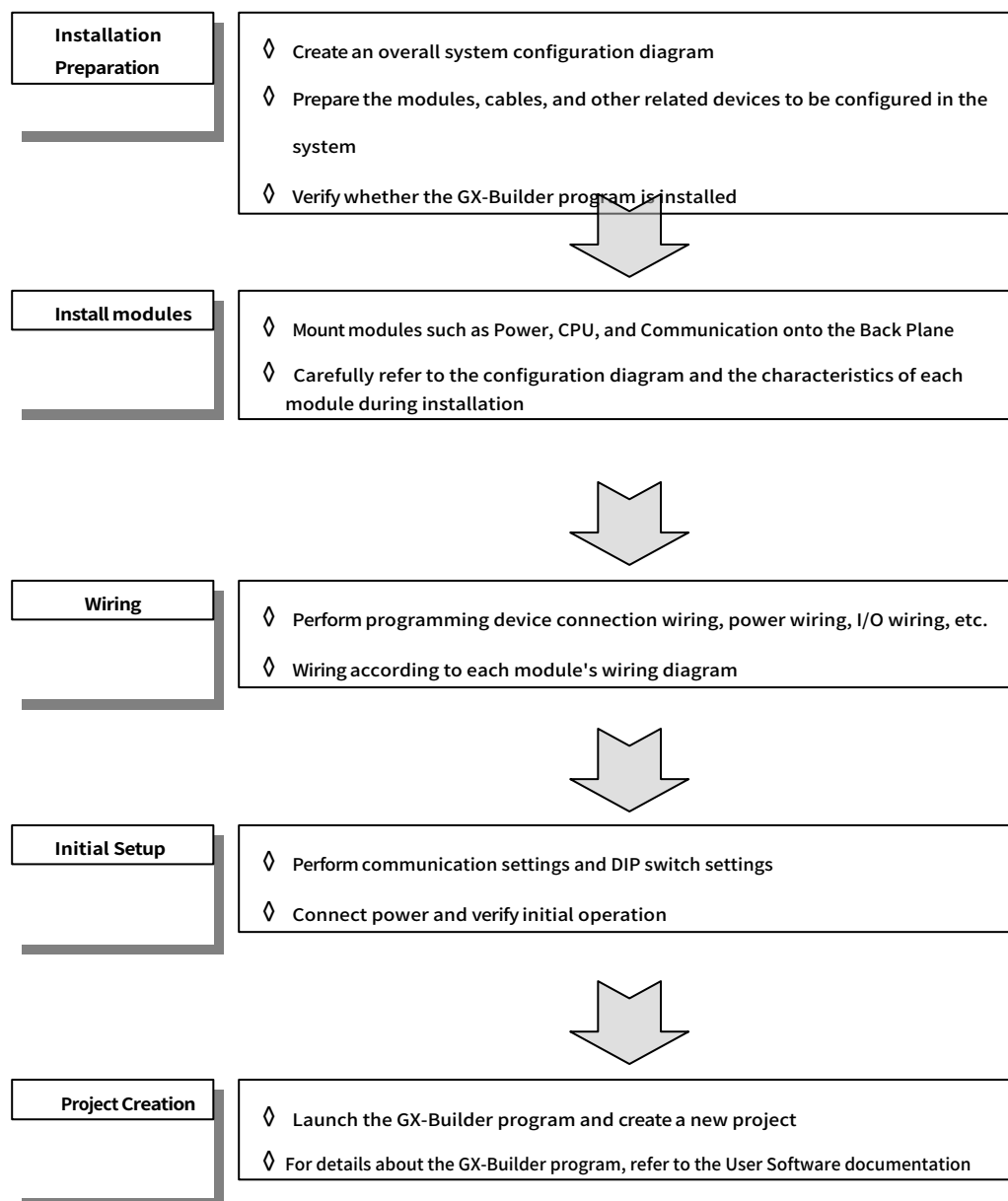
4.2.9 Operation Verification

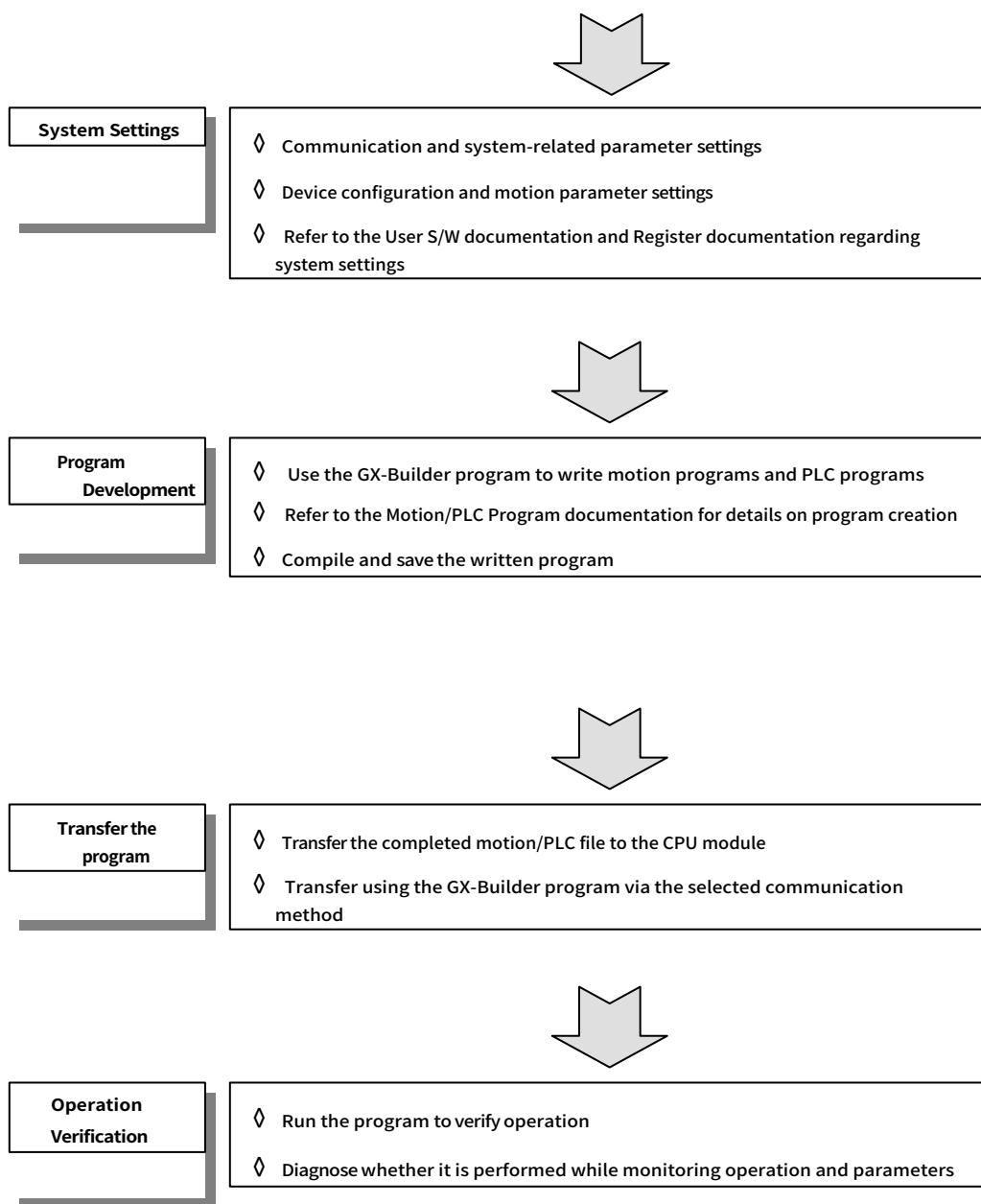
4.1 System Operation Sequence

Generally, it is used in the following sequence.

Refer to the respective installation and operation manuals for detailed procedures for each system operation step.

■ System Operation Sequence





4.2 System Installation

This section explains the operation of the GX-S200/S400 system step by step and provides a simple system configuration example to facilitate understanding of the system installation procedure.

After fully understanding the overall system operation steps, configure the system by referring to the detailed installation and operation methods for each process.

Please note that the test examples are provided to illustrate the system operation stages and may differ from the actual system.

4.2.1 Installation Preparation

The installation preparation phase involves gathering the necessary components for system configuration before proceeding to the installation stage.

In this phase, creating an overall system architecture diagram and a list of components for the system you intend to implement enables efficient progress through the installation preparation process.

4.2.1.1 Creating the Configuration Diagram

- Create a configuration diagram for the system to be implemented.
- The system configuration diagram schematically depicts the entire system centered around the backplane, showing the modules to be installed on the backplane, along with any expansion modules or other devices connected to each module.
- Creating the diagram facilitates easy understanding of the overall structure of the system to be implemented.

4.2.1.2 Creating a Component List

Based on the created schematic diagram, prepare the modules to be used, the various cables used with the modules, and other related equipment.

4.2.2 Module Installation

This step involves installing each module onto the backplane.

Install the modules to be used in their respective slots based on the created configuration diagram.

Refer to the installation method for the specific module and install it carefully.

■ Back-Plane Slot Locations

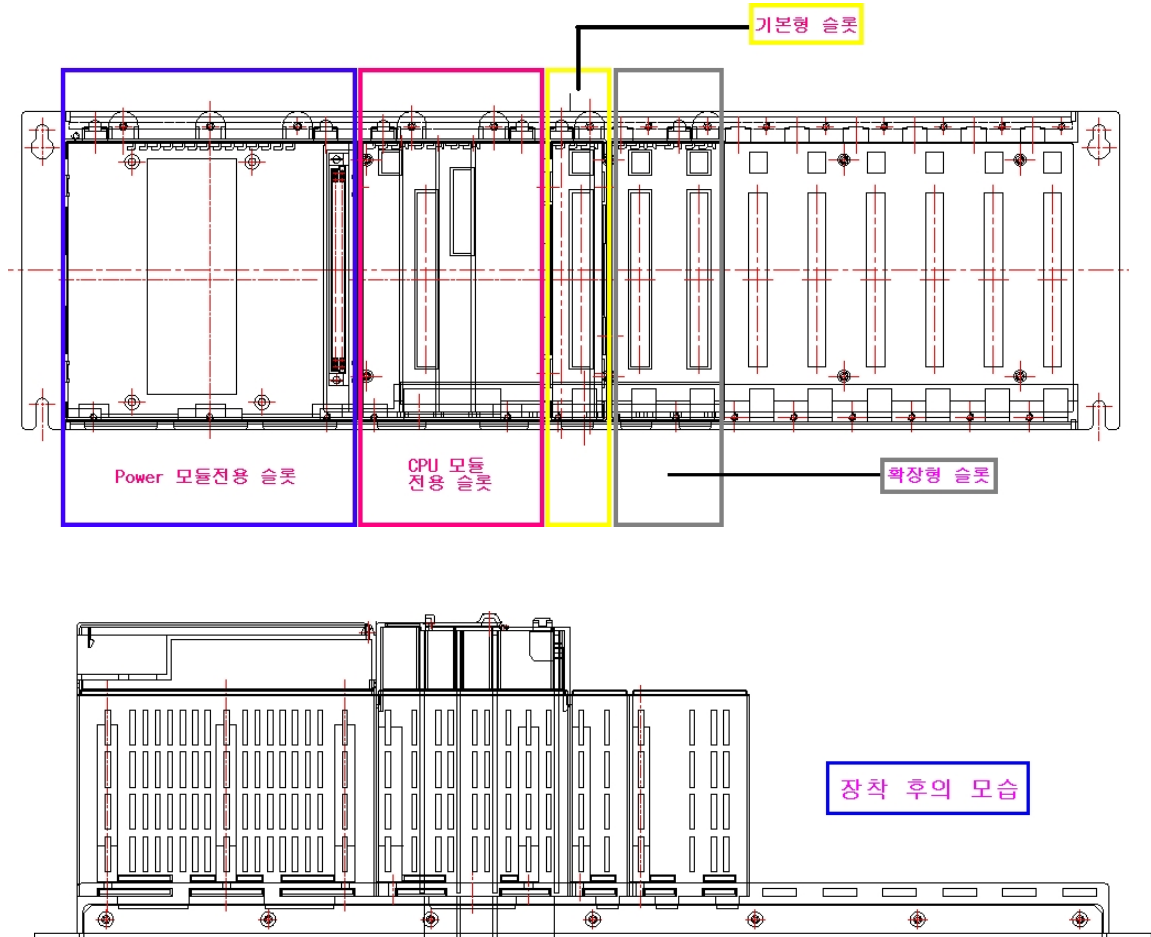
The names of each part indicated in the above diagram are as follows.

Number	Name	
①	Main Unit Installation	Secure S200, S400 to control panel or other PANEL
②	Power Module DIN Connector	S200, S400 Power Connector
②	EtherNET	EtherNET Connector
③	USB	USB Connector
④	MPG Connector	MPG Connector
⑤	RS232 Connector	RS232 Serial Connector
⑥	Axis Port	AXIS1 ~AXIS4 Axis Port
⑦	IN 16/ OUT 16	INPUT 16, OUTPUT 16 Connector
⑧	INPUT 32	INPUT 32 connector
⑨	OUTPUT 32	OUTPUT 32 connector

■ Module Installation

The location of the connection connectors installed on the front of the Back-Plane module is as follows. The Power module is always installed in the first slot, as shown in the following figure.

The CPU module should also be installed in the second slot. Other modules can be installed in either the first or second slot, depending on the slot size, as they share the same specifications.



4.2.3 Wiring

4.2.3.1 Module Wiring

This step involves wiring work for modules installed on the backplane.

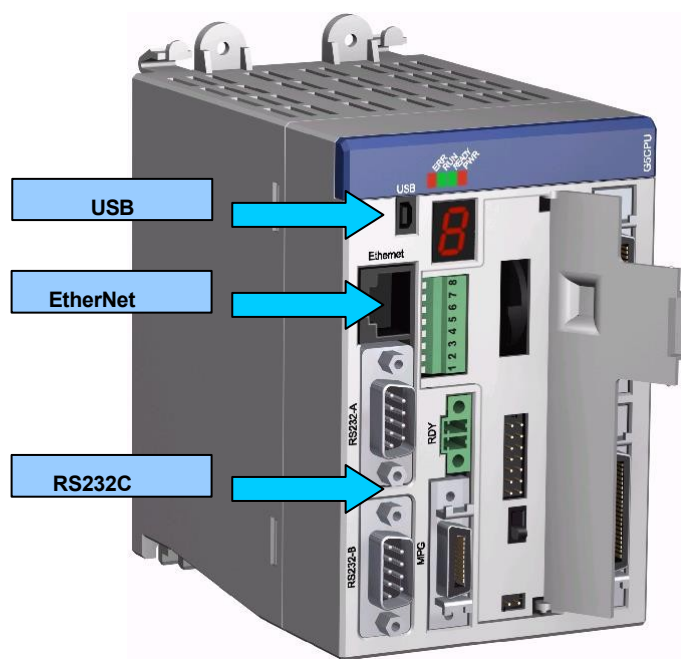
Perform wiring tasks for each module, including establishing communication connections with devices running the User S/W, connecting power to the power module, and performing I/O wiring.

For detailed information on wiring tasks, refer to the module wiring diagrams and module wiring methods in the [Maintenance Manual].

4.2.3.2 Communication Connection with User S/W

The User S/W includes a Setup program containing system settings, parameter settings, and motion/PLC program editors, as well as various types of HMI. Establish a communication connection with the equipment where this User S/W is installed.

Communication connections are established as follows, depending on the selected communication type (RS232C, USB, EtherNet).



4.2.4 Initial Setup

This step involves inspecting the modules installed/wired in the previous step and applying power to verify initial operation.

Before applying power, perform memory initialization using the DIP switches on the CPU module. Refer to [Chapter 3: Basic System Operation] for the memory initialization method.

Initial Power-On (Normal Operation)

	Operation Status	CPU LED Status	7 Segment
1	When initial power is applied, all CPU LEDs and the 7-segment display light up.		
2	CPU boot process begins		
3	CPU boot complete. CPU operating normally.		

DIP Switch Settings

Normal mode status	Full memory initialization
Program memory initialization	Battery Backup memory initialization

4.2.5 Project Creation

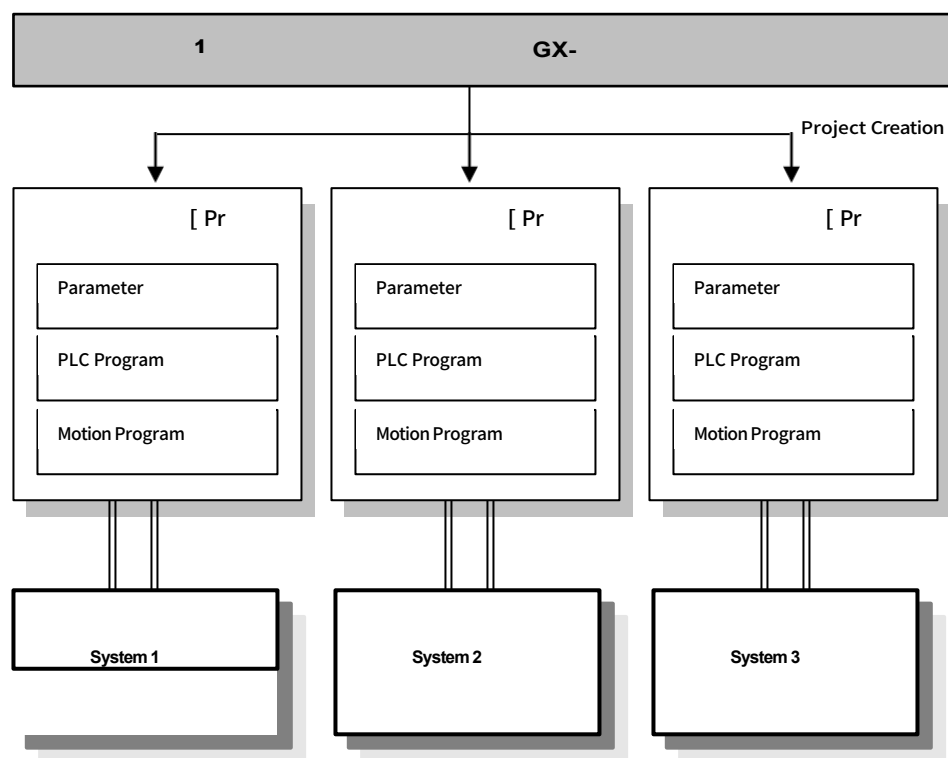
Once the basic installation and preparation of the GX Series are complete, run the GX-Builder program to create a project for managing the system you intend to implement. Through this project, you can perform tasks such as system configuration and program creation.

This section briefly explains how to create a project in the GX-Builder program.

For detailed information about the GX-Builder program, refer to the GX-Builder manual.

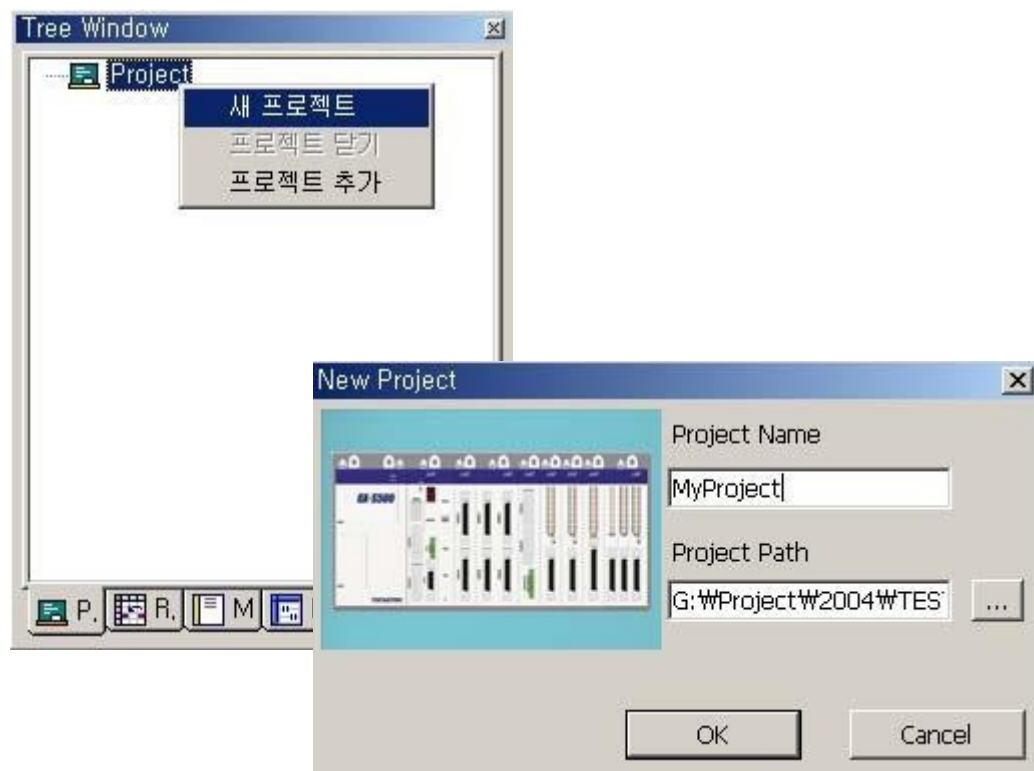
4.2.5.1 Project Overview

- A project is a system unit that integrates all tasks involved in controlling and monitoring systems operated using the GX-S200/S400.
- The GX-Builder program enables tasks related to project creation, management, and editing.
- Different systems with distinct characteristics can be configured for each project created.



4.2.5.2 Project Creation

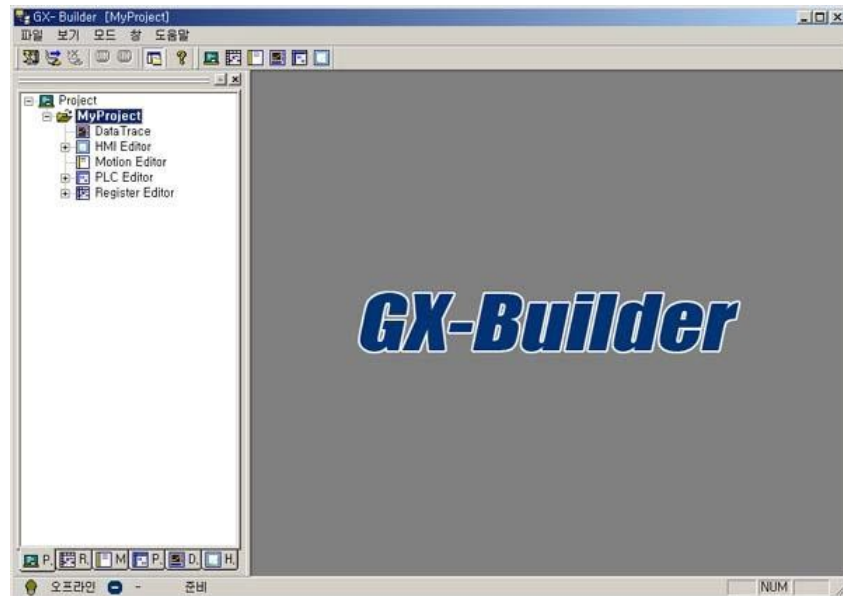
- The GX-Builder program is pre-installed by default.
- For details on installing GX-Builder, please refer to the GX-Builder manual.
- Selecting [File] - [New Project] from the menu on the GX-Builder Program Main screen opens the New Project Creation window. Alternatively, right-clicking the [Project] item in the tree view opens a pop-up menu. Selecting [New Project] from this menu opens the following New Project Creation window.



- Enter the information for the project you wish to create.
- Enter the project name, project folder path, user permission information, etc.
- After completing all inputs and selecting the [Confirm] button, the project will be automatically created.

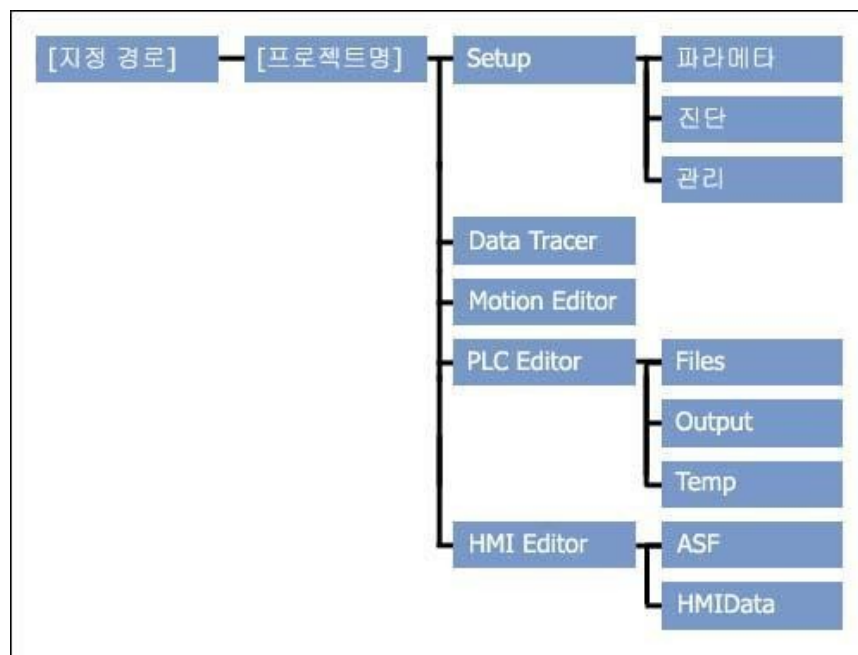
The next screen is the Main screen of the GX-Builder program, displaying the created project. The left Tree Window of GX-Builder shows the list of registered projects, and projects created within the program are automatically registered.

Selecting a project item opens only that specific project.

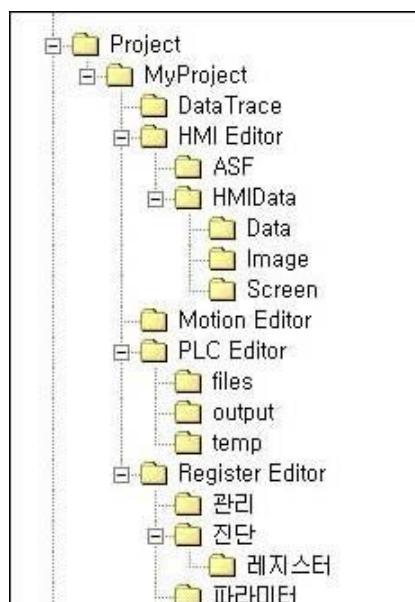


4.2.5.3 Project Folder Structure

- When you create a new project, folders with the following structure are created in the specified project path.
- Folders are created according to the types of Editors included in the GX-Builder program, and files edited in each Editor are created and saved in the corresponding folder.



- The following shows an example of a folder created when a new project is created.



4.2.6 System Settings

This step involves configuring system-related settings, device settings, and motion parameters using the Setup Editor integrated into the GX-Builder program.

4.2.6.1 Communication Settings

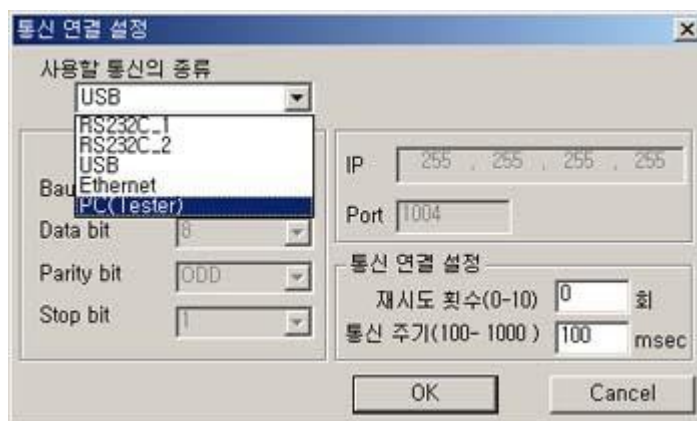
Communication settings involve configuring parameters related to communication between the PC where the GX-Builder program is installed and the GX system..

■ Communication Type

Communication Type	Configuration Parameters
RS232C / 422 / 485	Baud Rate
	Data Bit
	Parity Bit
	Stop Bit
USB	IP Address
EtherNet	IP Address
	Port No.

■ Communication Settings Window

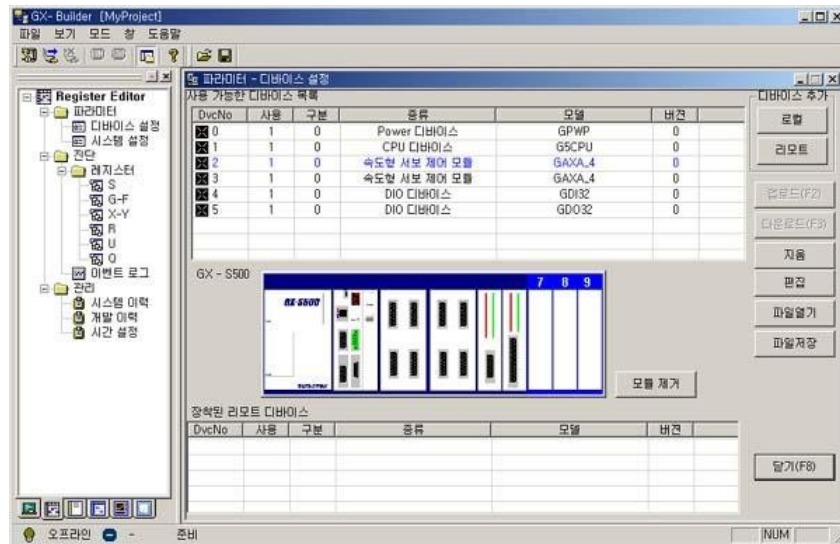
Select [Communication Settings] from the menu on the GX-Builder program Main screen to open the following Communication Settings window.



4.2.6.2 Device Settings

Open the Device Settings window in the Register Editor and configure the Local/Remote Device. Set the Slot number and Device type to match the modules installed on the GX system's backplane.

Open the Device Settings Editor and configure settings in Text/Graphic mode.

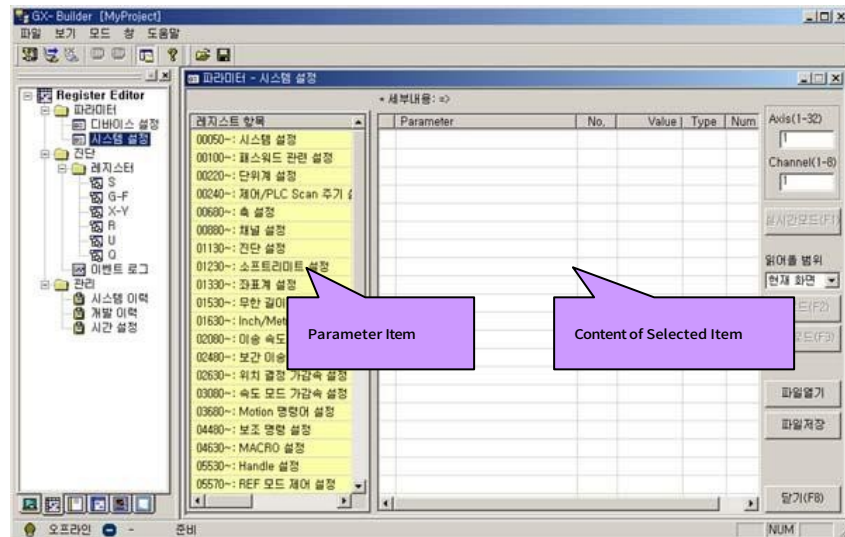


Device settings are divided into Local Device settings and Remote Device settings. The number of configurable devices is 16 Local Devices and 112 Remote Devices, allowing a total of 128 devices to be configured.

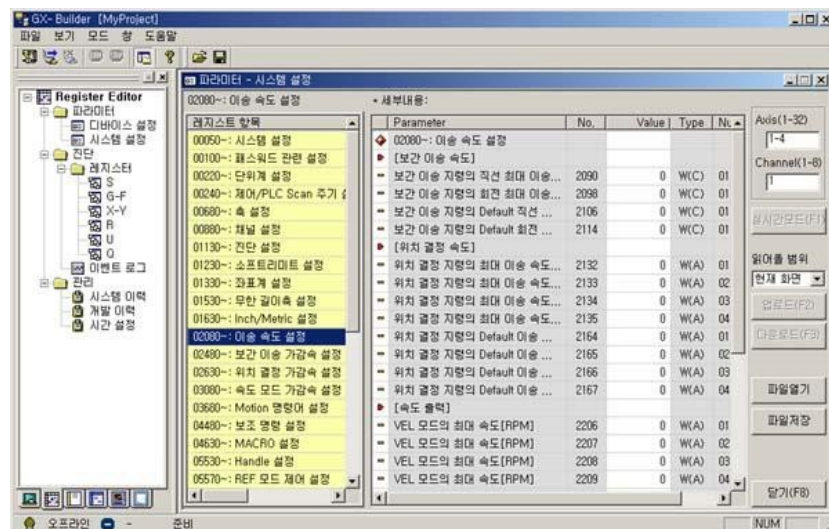


4.2.6.3 Parameter Settings

This step involves setting parameters related to system operation and motion control.

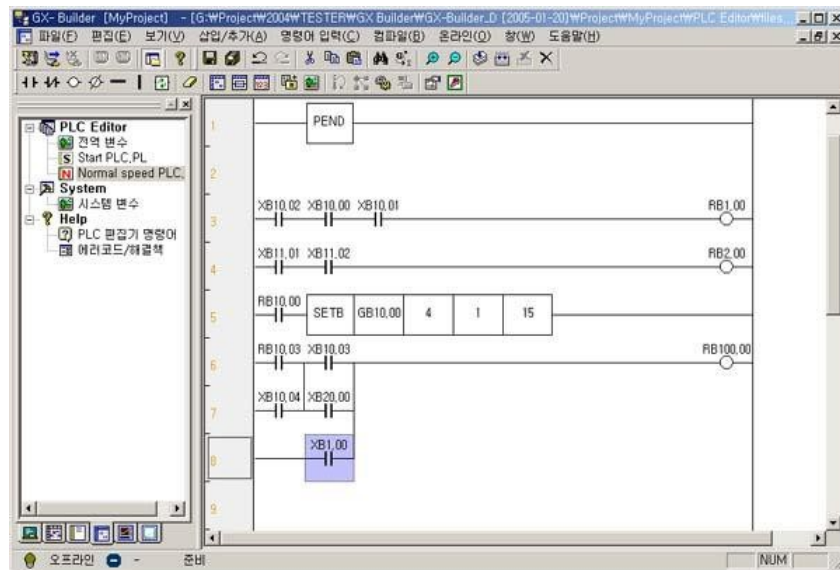
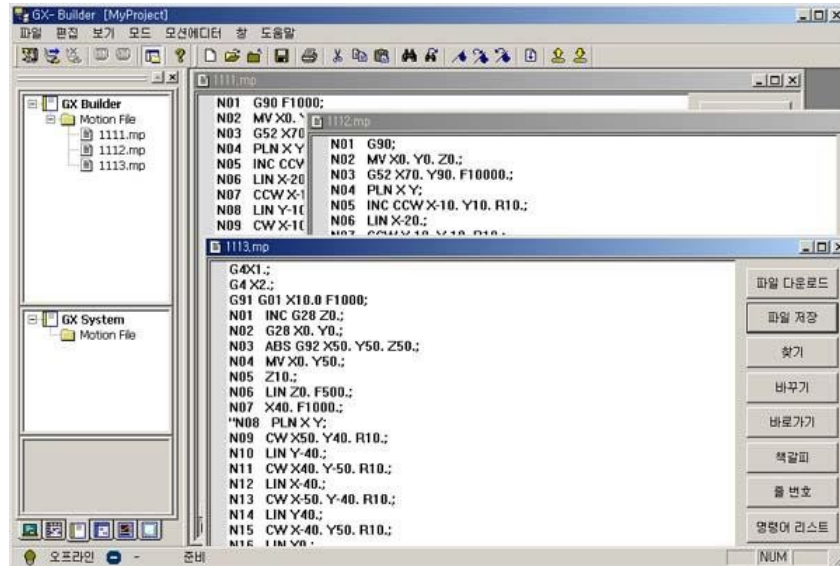


To view the detailed items of a parameter, click a specific item in the parameter item list to see its detailed items. If communication is not connected, no content will appear.



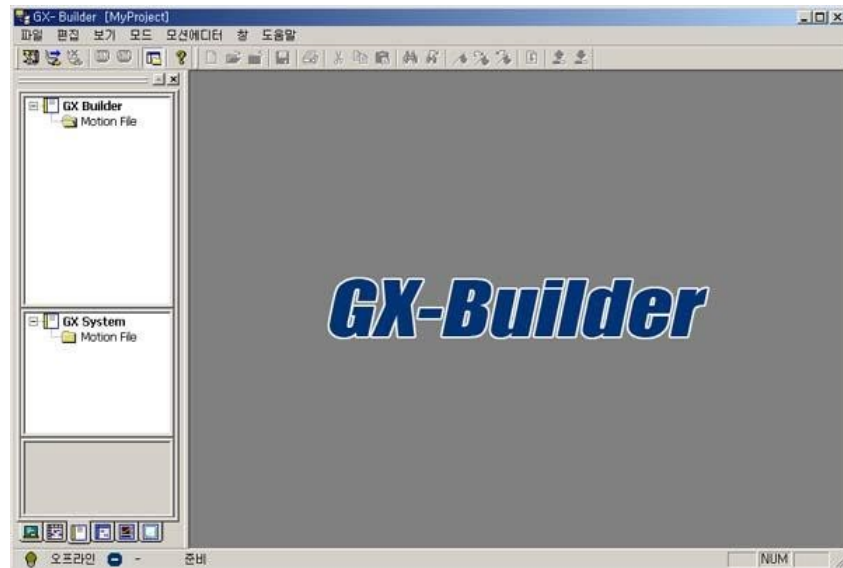
4.2.7 Programming

Use the Motion Program Editor and PLC Program Editor provided in the GX-Builder program to write motion/PLC programs that perform the desired tasks.

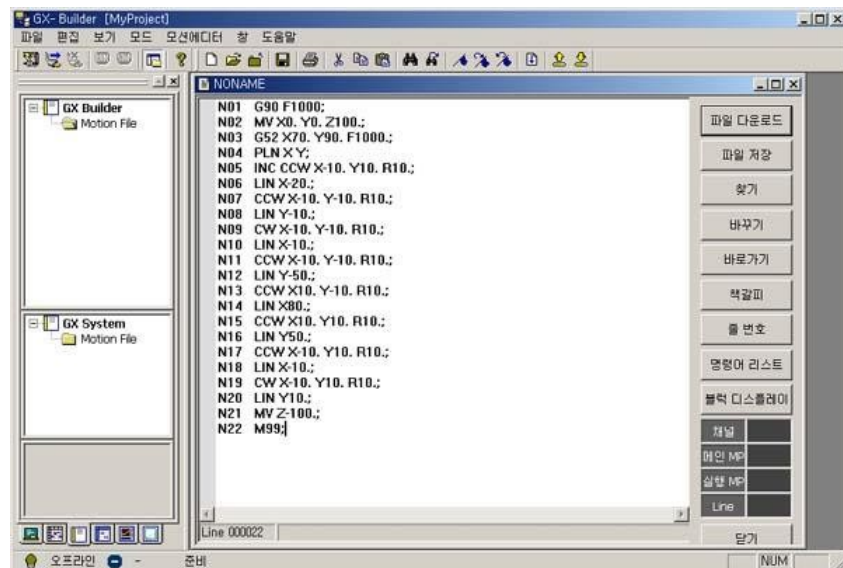


4.2.7.1 Writing a Motion Program

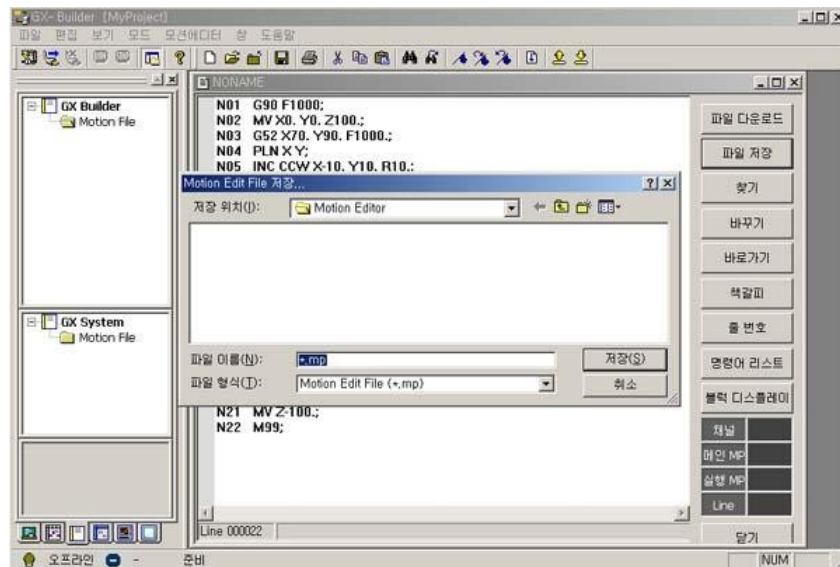
■ When you navigate to the Motion Editor from the Tree Window in the GX-Builder program, the following Editor screen appears.



To open the work window for writing motion programs, right-click the [Motion File] item in the Local area of the Tree Window and select [New File]. The work window provides basic editing functions for developing motion programs and also supports special functions using the command buttons on the right side of the work window.

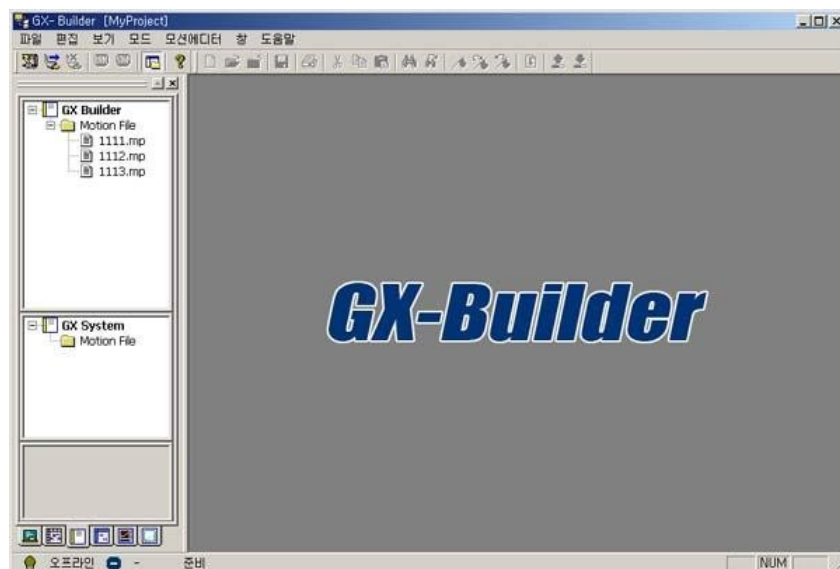


Once program creation is complete, use the file save button on the Toolbar or Menu bar to save the file. Selecting the save button opens the following file save window.



After entering the file name and clicking the [Save] button, you can see the motion file saved in the Local area list of the Tree Window.

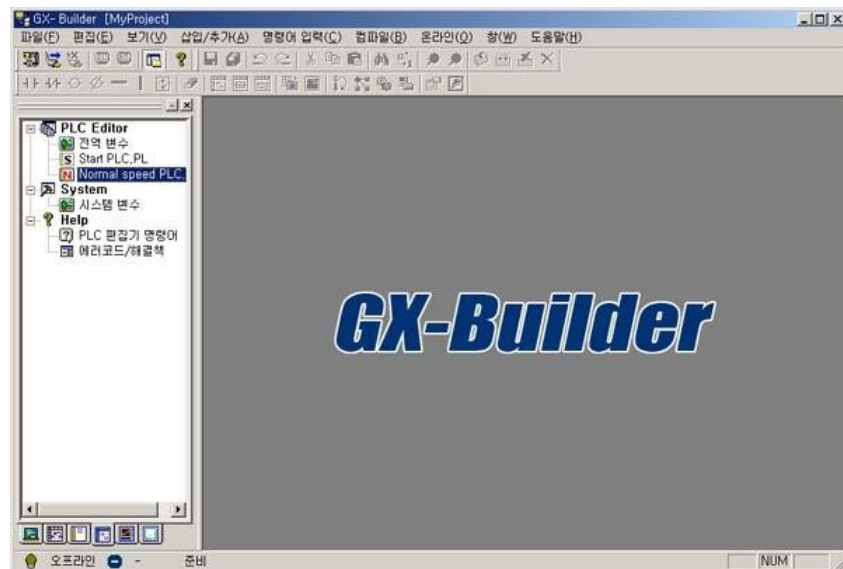
The following figure shows the list of motion files stored in the Local area list.



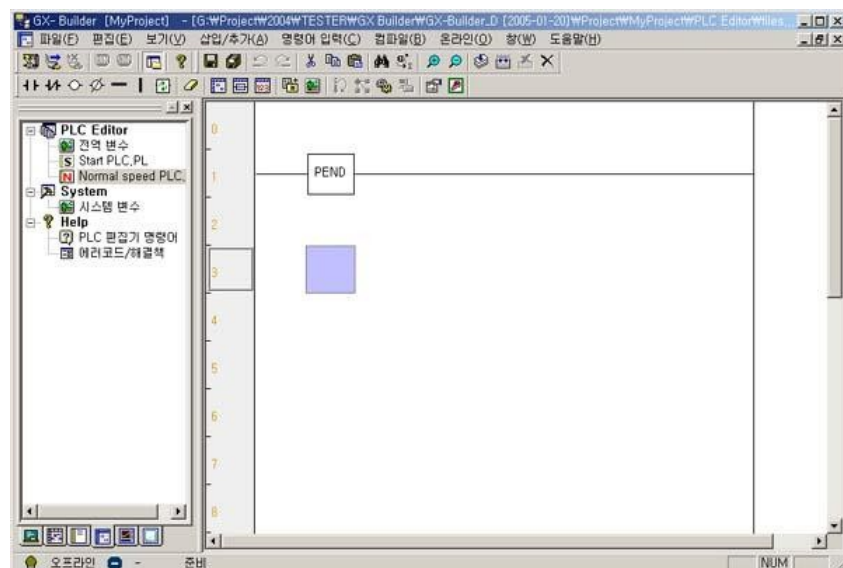
4.2.7.2 Writing the PLC Program

■ Opening a PLC File

Navigating to the PLC Editor in the Tree Window of the GX-Builder program displays the following Editor screen.



Examining the items in the Tree Window of the PLC Editor reveals that it consists of a total of 5 PLC channels. Selecting the channel you wish to edit opens the corresponding file.



■ PLC Command Input

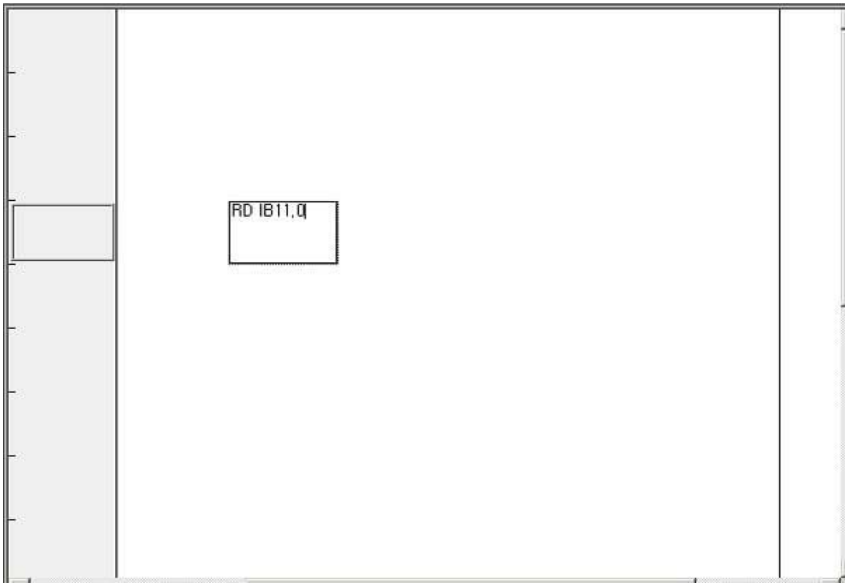
There are two methods for inputting PLC commands: directly entering the command or using the command input window. Choose the method that is easier to edit and write your desired program.

◇ Direct Command Input (In-Place Editing)

Select the cell where you want to enter the command, then use the keyboard to input the command directly. Both basic commands and function commands can be entered this way.

This method is used when you know the command name and format and want to input it quickly.

For basic commands, entering the basic command along with the register designation generates the corresponding Ladder for that command.

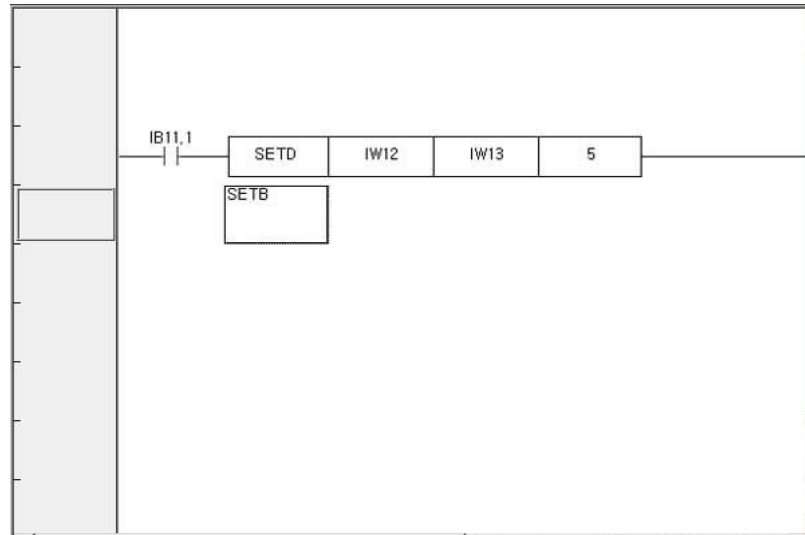


If the register designation for a command is incorrect or the data types of the operators differ, the input will not be accepted. Additionally, you can modify the entered command by clicking on the generated Ladder.

During in-place editing, the input format and delimiters are as follows:

Input Format	CommandArgument1Argument2...Argument n; Comment			
Delimiter	Space () arguments		: Separates commands and arguments Comma (,) : Separates	
	Semicolon (;)		: Separates arguments from comments	

When entering a function command, simply input the command itself to generate the corresponding Ladder. Then, input the required arguments into the generated Ladder.

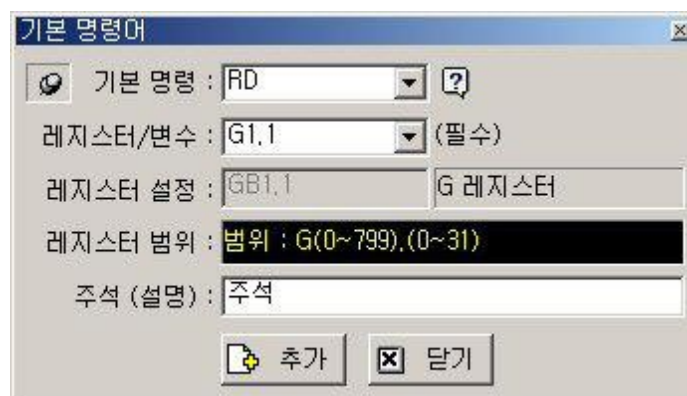


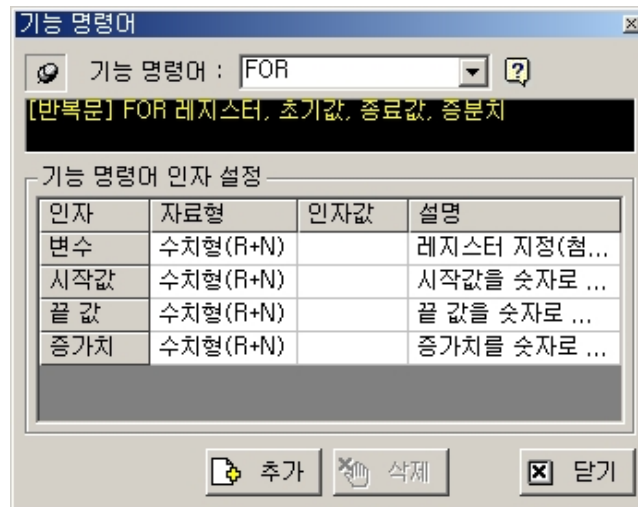
If the data type or specification of an operator is incorrect, it will not be entered. Also, as with basic commands, you can modify the entered command by clicking on the generated Ladder.

◇ Using the Command Input Window

Selecting [Basic Command Input Window] or [Function Command Input Window] under [Command Input] in the Menu Bar opens the following command input window.

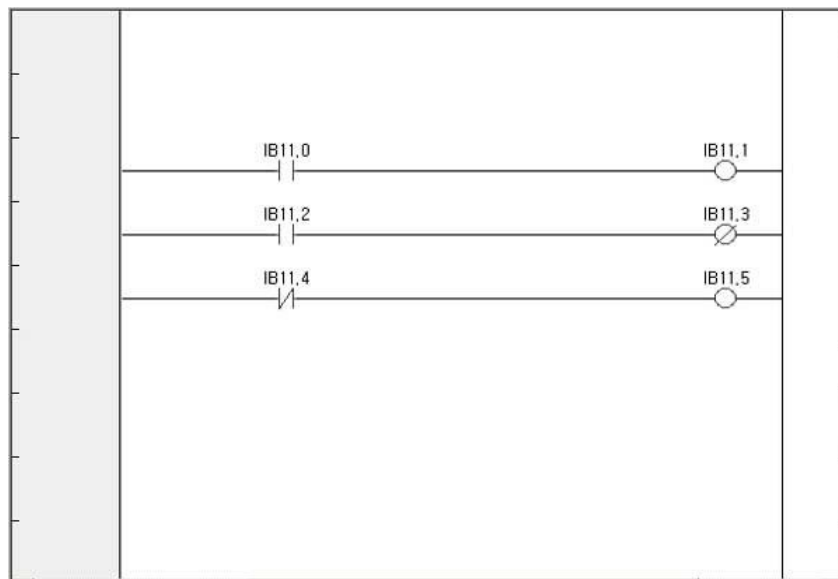
Clicking the [Ladder Command] icon on the Tool Bar also opens the same command input window.





Using the command input window allows you to easily enter commands because it displays the command list and information about each command. However, commands are only entered by specifying the register and corresponding operator for the command.

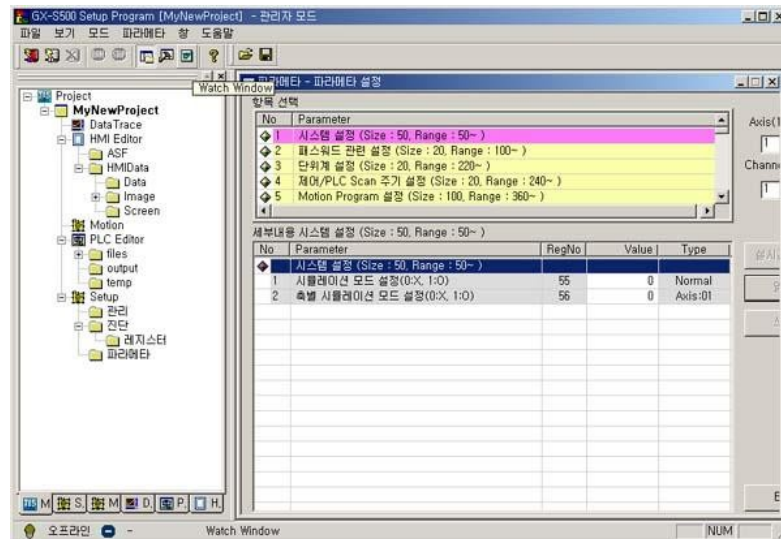
The following is a basic example of writing a program using both direct input and the command input window.



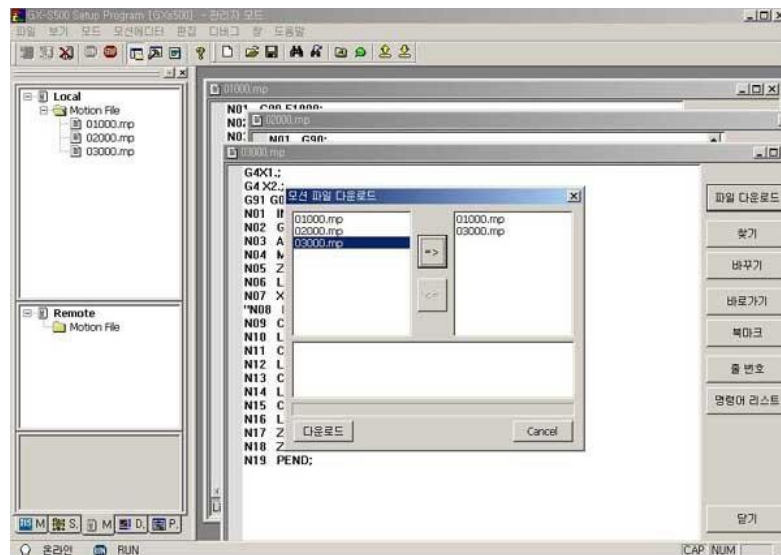
4.2.8 Program Transfer

To apply the parameters set in the GX-Builder program and the motion/PLC program written in the editor to the GX system, you must transfer them from each editor to the GX system. This is called file/register download. Detailed information about downloading is described in the GX-Builder manual.

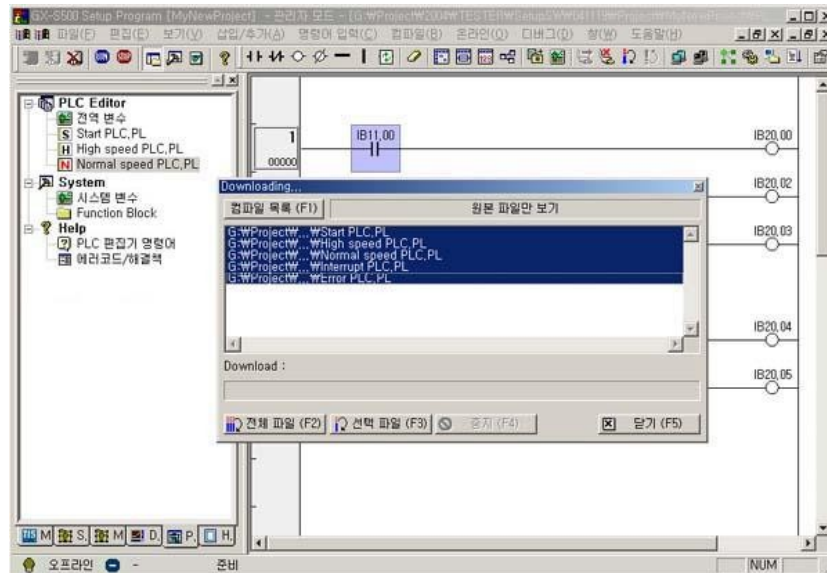
Downloading the configured parameters is performed in the parameter settings window, and independent downloads occur based on the selected items.



Downloading motion files is performed in the Motion Editor of the GX-Builder program, and independent downloads of all or individual files are performed via the Download Manager.



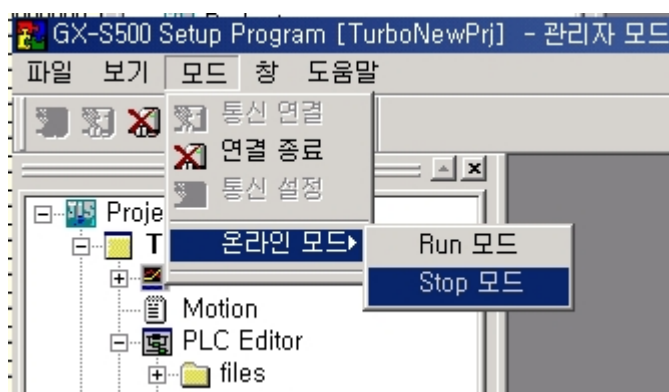
Downloading PLC files is performed in the PLC Editor of the GX-Builder program, and independent downloads for all files or individual files are performed via the Download Manager.



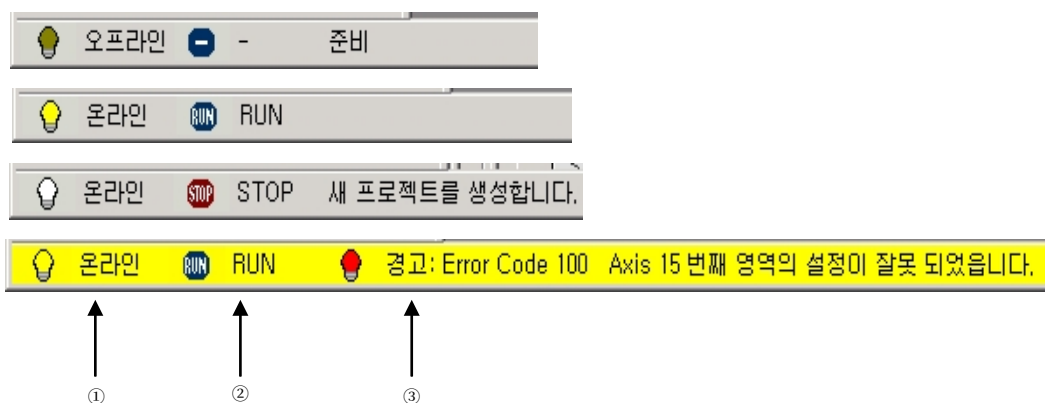
4.2.9 Operation Verification

Operate the system by switching the GX-S200/S400's operating mode.

The system's operating mode can be changed using the icons in the program's Menu Bar or ToolBar.



When a connection is active, you can view information about the current connection status, the system's current mode state, and other error messages via the Status Bar.



Module Number	Name
①	Indicates online/offline status
②	Indicates whether the machine is operating when online
③	Program displays help for various controls and event messages when errors occur.

Control Functions

This section describes the control functions implemented through the GX Series and explains the internal signals and parameter setting methods for each control.

5.1 System Settings

5.2 Preparing for Operation

5.3 Operating Mode

5.4 MP Driving Mode

5.5 MC Driving Mode

5.6 Status Output

5.7 Synchronous Control

5.8 External Positioning

5.9 Position Latch Functi
on

5.10 Position Compare Function

5.1 System Settings

5.1.1 Unit System Settings

■ Set Unit System

Select the s y s t e m 's setting units from metric or inch. Set according to the target equipment.

■ Position/Speed Default Unit

Set the default unit for configuring system registers or issuing position/speed commands from the PLC.

■ Parameter

Parameter	Formy
P220.00	Unit system setting (0:Metric, 1:Inch)
P227	Set Positioning Base Unit [PU]
P228	Set speed base unit [FU]

- Position Default Unit Setting Value

Setting Value	mm	deg	inch
0	0.001	0.001	0.0001
1	10	10	1
2	1	1	0.1
3	0.1	0.1	0.01
4	0.01	0.01	0.001
5	0.001	0.001	0.0001
6	0.0001	0.0001	0.00001

- Speed Default Unit Setting Value

Setting Value	mm/min	deg/min	inch/min
0	1	1	0.1
1	10	10	1
2	1	1	0.1

3	0.1	0.1	0.01
4	0.01	0.01	0.001
5	0.001	0.001	0.0001
6	0.0001	0.0001	0.00001

■ Example

- Entering the value 1000 into a register with the location unit (a register whose unit is [PU]) has the following meaning:

Setting Unit System	Position Default Unit	Meaning of Setting Value
P220.00 = 0	P227 = 0	$1000 * 0.001 = 1 \text{ mm}$
	P227 = 6	$1000 * 0.0001 = 0.1 \text{ mm}$
P220.00 = 1	P227 = 0	$1000 * 0.0001 = 0.1 \text{ Inch}$
	P227 = 6	$1000 * 0.00001 = 0.01 \text{ Inch}$

- Entering the value 1000 into the register for the speed unit (the register with the unit [FU]) has the following meaning:

Set unit system	Speed Base Unit	Meaning of the setting value
P220.00 = 0	P228 = 0	$1000 * 1 = 1000 \text{ mm/min}$
	P228 = 3	$1000 * 0.1 = 100 \text{ mm/min}$
P220.00 = 1	P228 = 0	$1000 * 0.1 = 100 \text{ Inch/min}$
	P228 = 3	$1000 * 0.01 = 10 \text{ Inch/min}$

■ Note

In the motion program, you can specify commands in actual units without adhering to the default position/speed units. If P220.00 = 0, the motion command and its meaning are as follows:

- "MV X100.0 F2000" → Move the X-axis to the 100.0mm position at a speed of 2000mm/min

5.1.2 Axis Setting

Set the axis to be used. Axes can be categorized as linear axes or rotary axes based on the type of movement.

The order for setting the axis is as follows.

- Select the axis to use.
- Set the movement type by distinguishing between linear axes and rotary axes.

■ Parameter

Parameter	Formy
P680	Axis Usage Setting
P681	Axis Type (0: Linear, 1: Rotational)

5.1.3 Channel setting

For complex equipment, it can be divided and categorized by function. For example, if the equipment is divided into three parts: loading, processing, and unloading, controlling these three parts as separate equipment can be convenient for commissioning or maintenance. The function applicable here is the [Channel Function]. Each part can be assigned to a separate channel.

The characteristics of the channel function are as follows:

- Channel-specific emergency stop and reset functions
- Channel-specific motion program execution function
- Assign axis names per channel
- Channel-specific diagnostics: If an alarm occurs, only the corresponding channel stops operation

The channel setup sequence is as follows.

- Select the channel to use.
- Select and set the axes to be used for each channel.
- Set the axis name for each axis

■ Parameter

Parameter	Formy
P885	Select whether to use the control channel
P886 ~ P893	Select axis per channel
P894 ~ P965	Set the axis number corresponding to the axis name per channel

■ Note

Axes not included in a channel operate independently of other channels.

5.1.4 Example of Axis and Channel Settings

The method for setting axes and channels for equipment with 2 channels and 8 axes is as follows.

■ Equipment Configuration

Channel	Axis Number	Axis Name	Axis Type
1	1	X	Straight
	2	Y	Line
	3	Z	Line
	4	A	Rotation
2	5	X	Straight
	6	Y	Line
	7	Z	Line
	8	U	Straight line

■ Example of Parameter Settings

Parameter	Setting Value	For My
P680	00000000-00000000-00000000-11111111	Axis Enable Setting
P681	00000000-00000000-00000000-00001000	Axis Type
P885	00000000-00000000-00000000-00000011	Control Channel Usage Selection
P886	00000000-00000000-00000000-00001111	Axis Selection per Channel (1 Channel)
P887	00000000-00000000-00000000-11110000	Channel-Specific Axis Selection (2 Channels)
P894	1 (X)	Set Axis Number Corresponding to Axis Name per Channel (1 Channel)
P895	2 (Y)	
P896	3 (X)	
P897	4 (A)	
P898	0 (B)	
P899	0 (C)	
P900	0 (U)	
P901	0 (V)	
P902	0 (W)	
P903	5 (X)	Set the axis number corresponding to the axis name per channel (2 channels)
P904	6 (Y)	
P905	7 (Z)	

P906	0 (A)	
P907	0 (B)	
P908	0 (C)	
P909	8 (U)	
P910	0 (V)	
P911	0 (W)	

5.1.5 Soft-limit Function

This function sets the axis travel limits in the parameters and monitors the axis to ensure it does not exceed the set travel limits from the control device. If a command exceeding the travel limit is input to the control device, it decelerates and stops the axis before it exceeds the travel limit, preventing machine damage caused by operator error.

The soft-limit setting sequence is as follows.

- Set the soft-limit range in the parameters.
- Enable the soft limit function in the parameters.
- Perform a return to origin.

■ Parameter

Parameter	For My
P1237	-Direction soft limit enable/disable (0: Disabled, 1: Enabled)
P1238	+Direction soft limit enable status (0: Disabled, 1: Enabled)
P1239 ~ P1270	-Direction soft limit position [PU]
P1271 ~ P1302	Direction Soft Limit Position [PU]

■ Warning

The soft limit function will not operate until the origin return is complete.

5.1.6 Coordinate System Settings

The coordinate systems used within the control device can be categorized as [Machine Coordinate System] and [Workpiece Coordinate System], both of which utilize a right-handed Cartesian coordinate system.

■ Machine Coordinate System

The machine coordinate system is defined around the machine's origin, and the origin position is fixed. Position commands in MC operation mode are based on the machine coordinate system.

■ Workpiece Coordinate System

The workpiece coordinate system is defined at an offset position relative to the machine coordinate system and can be set at any location. Position setting methods include [setting via motion commands] and [setting via parameters]. Generally, position commands in MP operation mode are based on the workpiece coordinate system.



■ Parameter

Parameter	Formy
P1330	Enable/Disable Automatic Work Coordinate System
P1333 ~ P1364	Automatic Work Coordinate System Position Setting

■ Note

For details, refer to Chapter 4. Coordinate Systems in the GX Series Motion Program Manual.

5.1.7 Infinite Length Axis

The infinite length axis function controls an axis that continuously moves in one direction. The

characteristics of the infinite length axis are as follows.

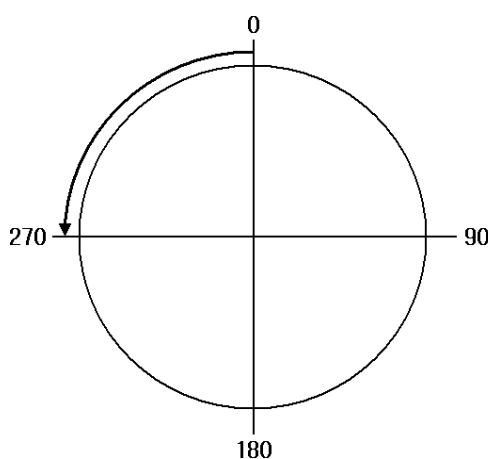
- If the axis position indication exceeds the value set in the parameter, it automatically resets to zero.
- Short-distance transfer is possible through parameter settings.
- The operating mode can be separately specified for incremental mode and absolute mode based on parameter settings.

Parameter

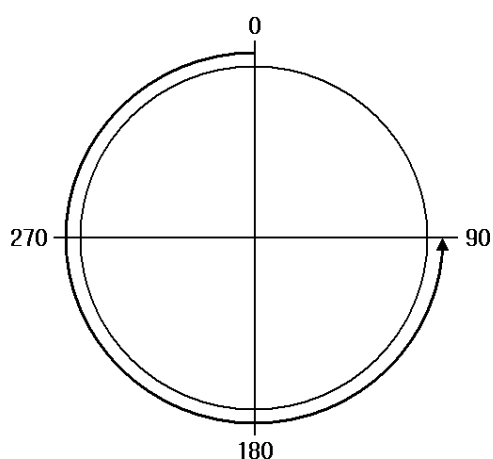
Parameter	Formy
P1530	Infinite length command u s a g e (0:X, 1:0)
P1533	Absolute command transfer method (0: Command code direction, 1: Near-distance)
P1534	Absolute command destination determination (0: Normal method, 1: Absolute value method)
P1535	Incremental Command Transfer Method (0: As Command, 1: Near Range)
P1538 ~ P1569	Reset Position for Coordinates of Infinite-Length Axis [PU]

Example

- When setting an absolute command of -90 from the 0 position, the axis movement is as follows:

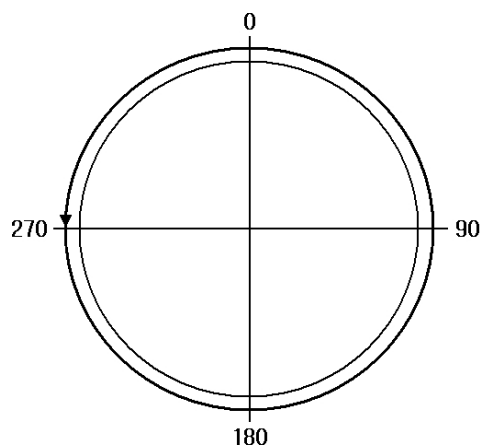


<When P1533 = 0 and P1534 = 0>

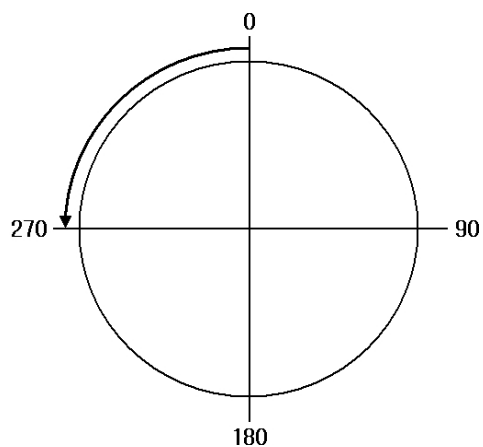


<When P1533 = 0 and P1534 = 1>

- When issuing an incremental command of -450 from the 0 position, the axis movement is as follows.



<When P1535 = 0>



<When P1535 = 1>

5.1.8 Inch/Metric Conversion Function

This function selects whether to use Metric or Inch as the command unit. The Inch/Metric conversion methods are [Conversion via Motion Commands] and [Conversion via Parameters].

The scope of application for the Inch/Metric conversion function is as follows.

- Position command and traverse speed command for MP drive mode (parameter selection)
- MC operating mode position command, traverse speed command (parameter selection)
- Automatic operation coordinate system position value (parameter selection)

■ Parameter

Parameter	Formy
P1634	Command Unit System (0: Metric, 1: Inch)
P1640	Unit System S w i t c h i n g i n M P M o d e (0:X, 1:O)
P1644	Automatic work coordinate system position value unit system conversion status (0:X, 1:O)
P1652	POS mode absolute command unit system conversion status (0:X, 1:O)
P1653	POS mode incremental command unit system conversion status (0:X, 1:O)
P1656	JOG mode unit conversion status (0:X, 1:O)
P1659	Unit conversion status in HAN mode (0:X, 1:O)
P1670	Location Display Unit System Switching Status (0:Off, 1:On)

■ Note

Switching between Inch/Metric remains effective even after rebooting the controller.

5.1.9 Transfer Speed Command

■ MP (Motion Program) Operation Mode

In MP operation mode, there are two methods for setting the feed rate: [How to issue an F command] in the motion program and [How to set the default feed rate] in the parameters. If both methods are used, the [F command] takes precedence and is applied.

■ MC (Motion Command) Operation Mode

In MC operation mode, the feed rate can only be set via the G register by default. As an exception, in REF mode, the feed rate for each step is set in the parameters.

■ Transfer Speed Limit

If the transfer speed exceeds the maximum transfer speed, it is automatically limited to the maximum transfer speed.

■ Parameter

Parameter	Formy
P2090 ~ P2097	Maximum linear feed rate for interpolation feed commands [FU]
P2098 ~ P2105	Maximum feed rate for rotary interpolation commands [FU]
P2106 ~ P2113	Default Linear Feed Rate for Interpolation Feed Command [FU]
P2114 ~ P2121	Default Rotational Feed Rate for Interpolation Feed Commands [FU]
P2132 ~ P2163	Maximum feed rate for positioning commands [FU]
P2164 ~ P2195	Default feed rate for positioning commands [FU]
P2206 ~ P2237	Maximum Speed in VEL Mode [RPM, 0.01%]
P2287	Feed Override Enabled in MP Mode (0:X, 1:0)
P2290	Feed Override applied in MC mode (0:X,;0)

5.1.10 Acceleration/Deceleration Function

The acceleration/deceleration function automatically generates a smooth speed profile within the control unit, enabling smooth operation without mechanical shock even in sections where the transfer speed fluctuates.

The acceleration/deceleration function is summarized below.

Command Classification	Acceleration/Deceleration Method	Acceleration/Deceleration Profile
Interpolation Transfer Command	Pre-Interpolation Acceleration/Deceleration Method	Asymmetric Linear
	Post-Interpolation Acceleration/Deceleration Method	Symmetrical Linear, S-curve, Exponential
Position Determination Command	Pre-interpolation acceleration/deceleration method	Asymmetric linear
	Post-interpolation acceleration/deceleration method	Symmetrical straight line, S-curve, exponential
Speed Output Command	–	Asymmetric linear

■ Pre-interpolation acceleration/deceleration

This function calculates acceleration/deceleration profiles and then performs interpolation. Even when acceleration/deceleration times increase, no shape distortion occurs. After interpolation, it is also possible to pass the profile through the acceleration/deceleration filter again to smooth the acceleration/deceleration profile.

■ Interpolation-Based Acceleration/Deceleration

This function generates acceleration/deceleration profiles after interpolation. While it enables smoother operation compared to pre-interpolation acceleration/deceleration, it has the drawback that shape distortion increases proportionally to the acceleration/deceleration time.

■ Parameter

Parameter	Formy
P2484	Interpolation Transfer Acceleration/Deceleration Selection (0: After Interpolation, 1: Before Interpolation)
P2492 ~ P2499	Interpolation Transfer Pre-Interpolation Acceleration Time [TU]
P2500 ~ P2507	Interpolation Transfer Deceleration Time Before Interpolation [TU]
P2526 ~ P2533	Interpolation Transfer Pre-Interpolation Acceleration/Deceleration Filter Correction Factor [TU]
P2534 ~ P2541	Interpolation Transfer Pre-Interpolation BIAS Speed [FU]
P2552 ~ P2559	Interpolation Transfer Post-Interpolation Acceleration/Deceleration Profile (0: Linear, 1: S-curve, 2: Exponential)
P2562 ~ P2569	Interpolation Transfer Post-Interpolation Acceleration/Deceleration Time [TU]
P2578 ~ P2585	Linear axis exponential acceleration/deceleration BIAS speed [FU] after interpolation transfer
P2586 ~ P2593	Interpolation Transfer Post-Interpolation Rotary Axis Exponential Acceleration/Deceleration BIAS Speed [FU]
P2634	Positioning Acceleration/Deceleration Selection (0: After Interpolation, 1: Before Interpolation)
P2642 ~ P2673	Positioning Acceleration Time Before Interpolation [TU]
P2674 ~ P2705	Positioning Deceleration Time Before Interpolation [TU]
P2870 ~ P2901	Acceleration/Deceleration Profile After Positioning Interpolation (0: Linear, 1: S-curve, 2: Exponential)
P2904 ~ P2935	Acceleration/Deceleration Time After Position Determination Interpolation [TU]
P2968 ~ P2999	Exponential acceleration/deceleration BIAS speed after position determination interpolation [FU]
P3089 ~ P3120	Acceleration time in speed mode [TU]
P3121 ~ P3152	Speed Mode Deceleration Time [TU]

■ Note

In HAN mode, acceleration/deceleration after interpolation is always applied.

5.1.11 Motion Command Functions

Motion commands include position determination, linear interpolation, circular interpolation (helical interpolation), SKIP interpolation, and dwell. By writing a motion program, complex motions can be implemented simply.

The types and characteristics of motion commands are as follows.

Command Classification	Internal Use
Positioning Command	Transfers to the commanded position in the shortest time.
Linear Interpolation Command	It moves along a straight trajectory to the designated position.
Circular Arc Interpolation Command	Moves along an arc or helical trajectory.
SKIP Command	Moves in the same manner as straight-line interpolation. When a SKIP signal is input, the remaining command amount is canceled.
External Position Determination	Moves in the same manner as position determination. When an external position determination signal is input, the remaining command is canceled and the unit moves by the incremental distance set in the parameters.
Position override	It moves in the same manner as position determination. The target position can be changed without stopping the movement.
Dwell Command	It pauses for the set duration.

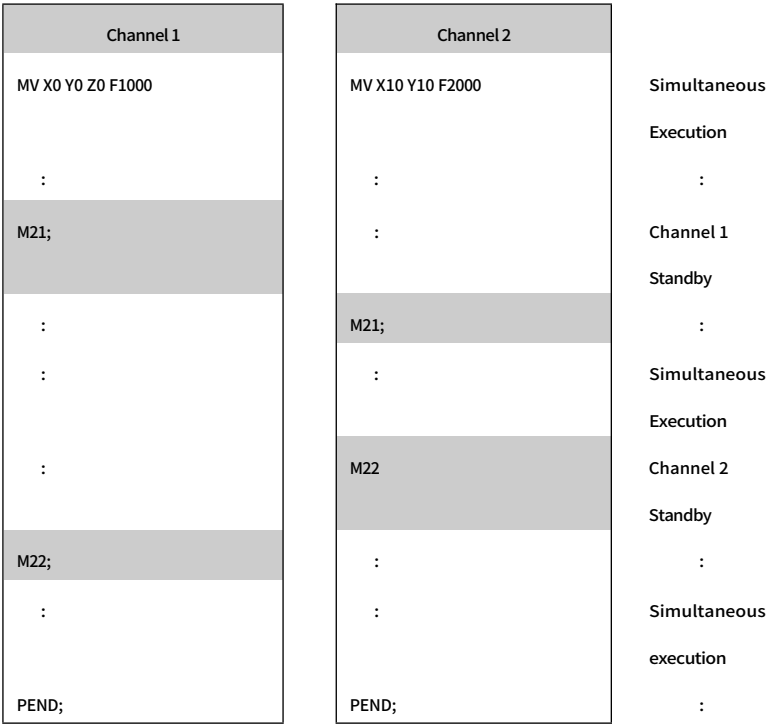
■ Note

For details, refer to the GX Series Motion Program Manual.

5.1.12 Synchronous M Code

When using two or more channels, you can control the execution sequence of each motion program using Synchronous M codes.

During MP operation, when a Synchronous M code is issued on one channel, execution of the next block begins only after the same M code is issued on the other channel.



Parameter

Parameter	Formy
P4545	Use of Synchronous M Code
P4548	Synchronous M code start number
P4549	Synchronous M code end number

Example

If you set 21 as the start number and 25 as the end number for Synchronous M Code, you can use 21, 22, 23, 24, and 25 as Synchronous M Codes.

5.1.13 Non-Buffered M Code

If registered as a non-buffered M code in the parameter, subsequent commands will not be buffered until the M code completes execution. Additionally, the control device's reference position will be updated to the actual position.

■ Parameter

Parameter	For My
P4510 ~ P4529	Buffering-Free M Code

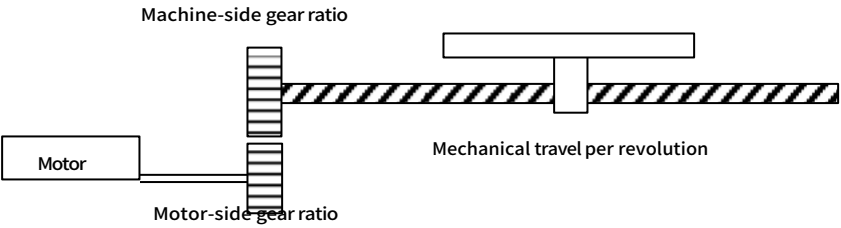
■ Note

If an offset occurs between the commanded position and the actual position of the motion program due to MC control intervention during MP control, using non-buffered M code can eliminate the offset.

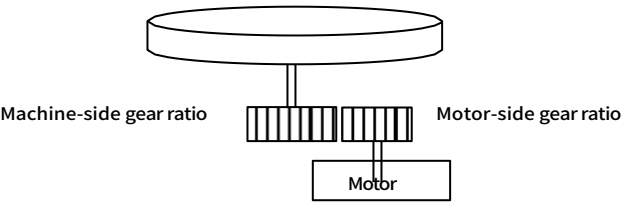
5.1.14 Electronic Gear

The electronic gear function is used in the control unit to convert between the [command unit] and the [control unit]. It assumes and configures complex machine mechanisms as simplified equivalent mechanical mechanisms.

■ Equivalent Mechanical Mechanism for Linear Axes



■ Equivalent mechanical mechanism of the rotating shaft



■ Parameter

Parameter	For internal use
P6424 ~ P6455	Motor-side Gear Ratio
P6456 ~ P6487	Machine-side gear ratio
P6488 ~ P6519	Machine Movement Per Motor Revolution [PU]

5.1.15 Control Mode Setting

Control modes include Position Control Mode and Speed Output Mode. You must set which mode to use for control on each axis. For axes performing position control, enable Position Control Mode. For axes performing speed output control (VEL mode), enable Speed Output Mode.

■ Parameter

Parameter	Formy
P6724	Position Control Mode Setting (0:X, 1:O)
P6729	Speed Output Mode Setting (0:X, 1:O)

■ Note

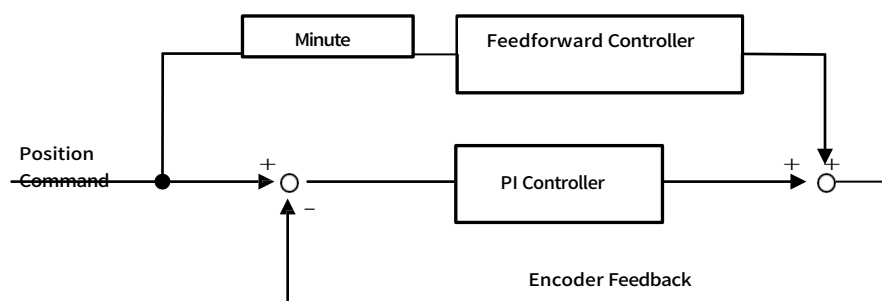
Axes with all control modes set to invalid will not be controlled.

■ Warning

For safety, set unused control modes to invalid for axes (e.g., main axis) that use only one mode from [Position Output Mode] or [Speed Output Mode].

5.1.16 Position Controller

5.1.15.1 Speed-type servo control



■ P gain (Proportional gain)

P gain determines the responsiveness of the position loop. Setting it higher increases the machine's responsiveness. However, be cautious as setting it too high may cause the machine to vibrate or run out of control.

The standard settings by equipment type are as follows.

Category	Target Equipment	Gain (0.1/S)
High Rigidity	Direct-coupled Structure Equipment	500~700
Medium Rigidity	Equipment using reducers or with long ball screw lengths	300~500
Low Rigidity	Equipment using belts	100~300

■ I gain (Integral gain)

I gain is the gain that shortens the position correction time in the position loop. Setting it smaller shortens the position correction time, while setting I gain larger reduces the effect of I control. Be careful not to set it too small, as this can cause the machine to vibrate.

■ Feedforward Gain

Using the feedforward control function can further enhance the machine's responsiveness. Be cautious, as setting it too high may cause mechanical noise or vibration. Generally, set it after the position loop gain (P, I gain) is determined, ensuring the gain value does not exceed 80%.

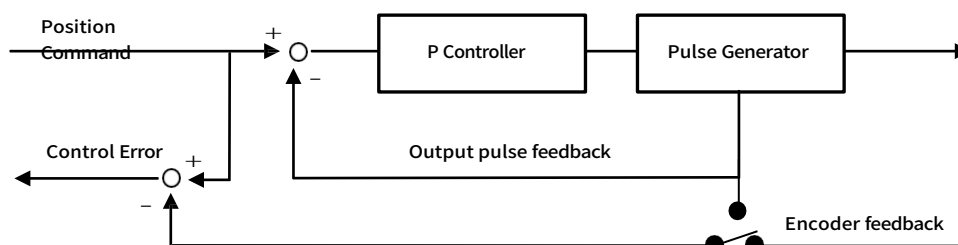
■ Tracking Error Detection Function

Tracking error refers to the [difference between the position command value and the encoder feedback]. Tracking error and control error are used interchangeably. The tracking error decreases as the responsiveness of the position control loop increases. If the tracking error exceeds the tolerance set in the parameters, a [Tracking Error Alarm] will occur.

■ Parameter

Parameter	Formy
P6889 ~ P6920	Position Control Speed Feed-forward Gain [0.1%]
P6988 ~ P7019	Position Control P Gain [0.1/S, 0.1%]
P7020 ~ P7051	Position Control I gain[msec]
P7052 ~ P7083	Position Control Integral Maximum Value
P7186 ~ P7217	Position Control Stopping Tolerance [PU]
P7218 ~ P7249	Position Control Error Tolerance During Movement [PU]

5.1.15.1 Position-type Servo Control (Stepper Motor Control)



■ P Gain (Proportional Gain)

This gain determines the responsiveness of the pulse output. Setting it higher increases responsiveness. Be sure to set it to 1000 (100%) or lower. Note that setting it too high may cause vibration during stopping.

The standard settings for each motor type are as follows.

Category	Gain (0.1%)
Stepper Motor	100
Microstep Motor	100~200
Servo Motor	200~300

■ Tracking error detection function

When no encoder is present, output pulse feedback is used instead of an encoder to calculate control error. All other operations are identical to those of speed-type servo control.

■ Parameter

Parameter	For internal use
P6988 ~ P7019	Position Control P gain [0.1/S, 0.1%]
P7186 ~ P7217	Position Control Stop Error Tolerance [PU]
P7218 ~ P7249	Position Control Error Tolerance During Movement [PU]

5.1.17 Absolute encoder

Using an absolute encoder allows detection of the machine position even when the control unit's power is OFF. Therefore, after completing a single origin return, subsequent operation is possible immediately without requiring another origin return when restarting the control unit.

■ Parameter

Parameter	Formy
P8828	Encoder Type (0: INC, 1: ABS)

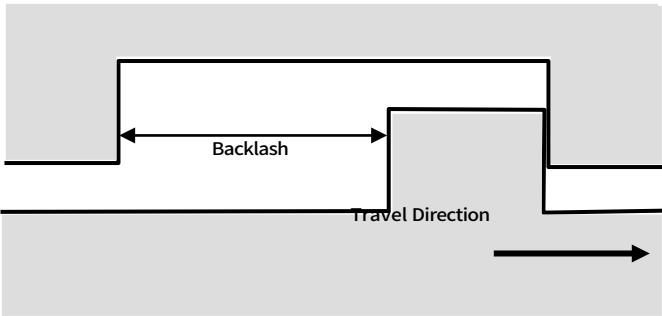
■ Note

For the origin return of axes using absolute encoders, refer to the description of REF mode.

5.1.18 Error Compensation

5.1.18.1 Backlash Compensation

The backlash compensation function compensates for backlash occurring in ball screws, gears, etc. The backlash compensation function operates only when the origin return is complete, and compensation is performed in the opposite direction of the origin return.



■ Parameter

Parameters	Formy
P9464	Backlash Compensation
P9465 ~ P9496	Backlash Compensation Value [PU]

5.2 Ready to Run

5.2.1 Emergency Stop

When an Emergency Stop signal is input, it rapidly stops the moving axis and resets the system.

■ Signal

Signal	Address	My Use
<u>GEMG</u>	GB5.00	All Emergency Stop Signal
<u>GCHEMG</u>	GB5.08 ~GB5.15	Channel-Specific Emergency Stop Signal
<u>GAXEMG</u>	GB6.00~GB6.31	Axis-specific emergency stop signal

5.2.2 Reset

When the Reset signal is input, the axis in motion decelerates to a stop and the system enters the Reset state. During Reset, the Reset status signal is output. For the GRSTS signal, it remains active for 200 milliseconds even after the GRST signal goes LOW.

■ Signal

Signal	Address	Internal Use
GRST	GB5.01	System Reset Signal
FRSTS	FB5.01	System Reset Confirmation Signal
GCHRST	GB5.16~GB5.23	Channel-Specific Reset Signal
FCHRSTS	FB5.16~FB5.23	Channel-specific Reset Acknowledgment Signal
GAXRST	GB7.0~GB7.31	Axis-specific Reset signal
FAXRSTS	FB7.0~FB7.31	Axis-specific Reset confirmation signal

■ Note

The controller can be reset using the mode switch on the controller CPU module. In this case, it operates identically to GRST.

5.2.3 Ready

Notifies the PLC that the control device is ready to operate. The conditions for

the Ready signal going Low are as follows.

- The control device has not completed booting.
- Emergency Stop state
- When the servo motor is in the Off state (due to factors such as an alarm)
- If the servo motor is not in the Ready state

■ Signal

Signal	Address	My Address
FRDY	FB8.00	All Ready Signal
FCHRDY	FB8.08~FB8.15	Channel-specific Ready Signal
FAXRDY	FB9.00~FB9.31	Axis-specific ready signal

■ Note

When the servo off signal goes high and the servo motor turns off, the [Overall Ready Signal] and [Channel Ready Signal] remain active, and only the [Axis Ready Signal] goes low. The ready signal can also be used for brake control on the vertical axis.

5.2.4 Servo Off

Generally, when a servo motor is forcibly held stationary by a brake or similar device, this is used to prevent the servo motor from overloading or to place the servo motor in a free-running state. Furthermore, since the control device continues to detect the encoder even while the servo is off, there is no need to perform a return-to-origin operation again.

■ Signal

Signal	Address	My Use
GSVOFF	GB12.00~GB12.31	Axis-specific servo OFF signal

5.2.5 Over-travel (OT)

Notifies the control unit that the control axis has reached the equipment's travel limit. When the OT signal is input, the corresponding axis will stop rapidly.

■ Signal

Signal	Address	Address Use
<u>GPOT</u>	GB20.00~GB20.31	+ Direction OT Signal
<u>GNOT</u>	GB21.00~GB21.31	- Direction OT Signal

5.2.6 Axis Interlock (IT)

This function prohibits the movement of the control axis. When the Axis Interlock signal is input, the axis currently moving will decelerate and stop. Furthermore, an axis stopped by this signal will resume movement according to the remaining commands once the signal returns to Low.

■ Signal

Signal	Address	My Use
<u>GPIT</u>	GB25.00~GB25.31	+ Direction Axis Interlock
<u>GNIT</u>	GB26.00~GB26.31	- Direction Axis Interlock

5.3 Operating Mode

Operation modes are broadly categorized into MP operation mode and MC operation mode. MP operation mode is set based on channels, while MC operation mode is set based on axes.

■ MP (Motion Program) Drive Mode

This is an automatic driving mode that executes motion programs created by the user. To use MP driving mode, at least one channel must exist.

■ MC (Motion Command) Drive Mode

This is a manual driving mode that uses G-register signals from the PLC to command axis movement. MC driving mode includes POS, JOG, HAN, REF, and VEL modes.

Drive Mode	Command Mode	Details
MP Drive Mode	MEM	Memory Drive Mode
MC Drive Mode	POS	Position Command Mode
	JOG	Jog (Constant Speed Command) Mode
	HAN	Handle (MPG) Mode
	REF	Return to Origin Mode
	VEL	Speed Output Mode

■ Drive Mode Overlay Function

This function allows simultaneous c o m m a n d i n g in both MP and MC driving modes for a single axis.

■ Example

- How to select Channel 1 and Channel 3 in MEM mode

G050	0	0	0	0	0	0	0	1
	0	0	0	0	0	0	0	1
	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0

- How to operate Axis 1 and Axis 3 in JOG mode, and Axis 2 and Axis 4 in HAN mode

G200	0	0	0	0	0	0	1	0
	0	0	0	0	0	1	0	0
	0	0	0	0	0	0	1	0
	0	0	0	0	0	1	0	0

5.4 MP Driving Mode

5.4.1 MEM Mode

MEM Drive Mode is a drive mode that downloads and executes a motion program created by the user using motion commands inside the controller. The motion program can be executed per channel, and the motion program can be selected from the PLC using the MCALL command.

■ Transfer Speed Override

Override can be applied to the commanded traverse speed. Traverse speed override is set separately for interpolation traverse commands and positioning commands.

■ Signal

- Mode Selection Signal

Signal	Address	My Use
GMEM	GB50.00~GB50.31	Mem Mode Selection Setting
FMEMS	FB50.00~FB50.31	Mem Mode Selection Confirmation Signal

- Control Signal

Signal	Address	My Use
GMPST	GB65.00~GB65.07	Start-up Settings
FMPSTS	FB65.00~FB65.07	Operation Start Confirmation Signal
<u>GMPFH</u>	GB65.16~GB65.23	Temporary Stop Setting
FMPFHS	FB65.16~FB65.23	Temporary Stop Confirmation Signal
GMPSB	GB65.24~GB65.31	Single Block Stop Setting
FMP SBS	FB65.24~FB65.31	Single Block Stop Confirmation Signal
GMPCFOR	GW80 ~ GW87	Interpolation Transfer Command Transfer Speed Override (0.01%)
GMPPFOR	GW88 ~ GW95	Positioning command transfer speed override (0.01%)

The status signal output according to the MP operating status is as follows.

Signal	During Operation	Operation Ended	Temporary Stop	Single Block
FMPSTS	HIGH	LOW	HIGH	HIGH
FMPFHS	LOW	HIGH	HIGH	LOW
FMP SBS	LOW	HIGH	LOW	HIGH

■ Note

- MEM mode is maintained during MEM operation.
- If the mode turns off during MEM operation, the vehicle decelerates and stops. Operation resumes when GMPST input is received again.

5.4.2 Skip Function

The Skip function interrupts processing of the currently executing block and proceeds to process the next block. The motion commands to which the Skip function applies are Skip and Dwell.

The Skip operation is as follows.

- Perform axis movement using the SKIP command.
- While the axis is moving, change the SKIP signal from Low to High.
- The system terminates execution of the current block and executes the next block.

■ Signal

Signal	Address	My Use
GSKIP1	GB70.00~GB70.07	SKIP1 and Dwell Command Skip Setting
GSKIP2	GB70.08~GB70.15	SKIP2 and Dwell Command Skip Settings

■ Note

The skip causes an offset between the motion program command position and the actual position, and the tool returns to the motion program command position from the block commanded to the absolute position (excluding arc blocks).

5.4.3 Auxiliary Code (Aux Code)

Auxiliary codes are functions used to control pneumatic/hydraulic equipment, spindle control, clamps, and auxiliary machine operations. When an M/S Code is commanded in the motion program, the corresponding code is sent to the PLC. Once the PLC completes processing, the control device proceeds to process the next block.

M/S Code Signals

(FMNO, FSNO)

Start Signal

(FMST, FSST)

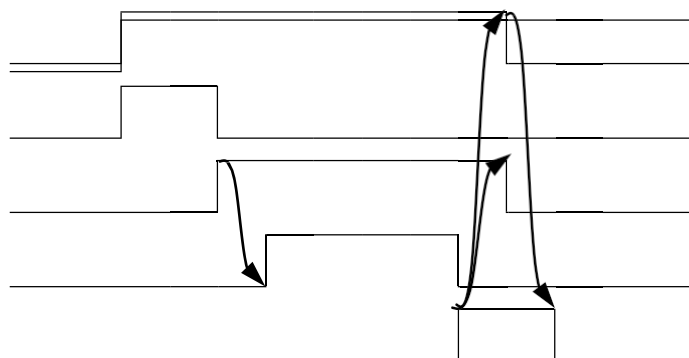
Axis Transfer Operation

Axis Transfer Complete
Signal (FMEOP, FSEOP)

PLC Processing

Completion Signal

(GMF, GSF)



■ Signal

Signal	Address	My Use
GMF	GB100.00~GB100.07	M Code Complete
FMST	FB100.00~FB100.07	M Code Start Signal
FMEOP	FB101.00~FB101.07	Axis Travel Complete Signal
FMNO	FW102 ~ FW109	M Code No. Signal
FMD	FW110 ~ FW117	M Code Data Signal
GSF	GB100.08~GB100.15	S Code Complete
FSST	FB100.08~FB100.15	S Code Start Signal
FSEOP	FB101.08~FB101.15	S Code End Signal
FSNO	FW134 ~ FW141	S Code No. Signal
FSD	FW142 to FW149	S Code Data Signal

■ Example

- If you write M100 in the motion program, FMNO outputs 0 and FMD outputs 100. (S Code is the same)
- If you write M2=100 in the motion program, FMNO outputs 2 and FMD outputs 100. (S Code is the same)
- If you write POS X100 F1000 S5000 in the motion program (), it operates with the following timing.

S Code signal (FSD=100)

(FSNO)

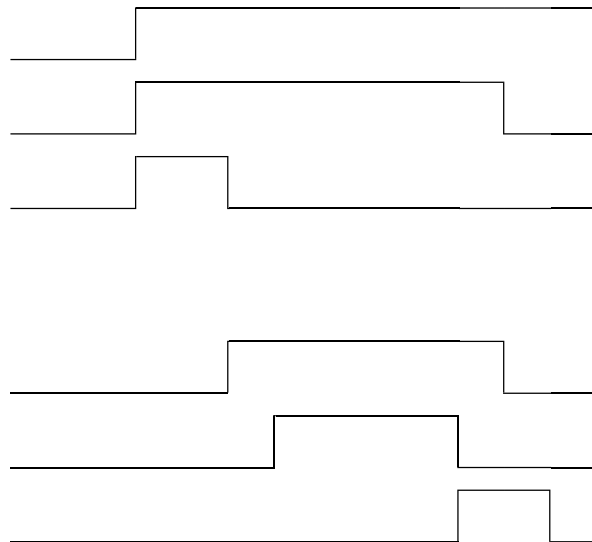
Start signal
(FSST)

Axis Transfer (Transfer to X100)

Axis Transfer Complete
Signal (FSEOP)

PLC Processing (Main Spindle 5000
RPM)

Completion Signal
(GSF)



5.5 MC Operation Mode

5.5.1 POS Mode

5.5.1.1 Basic Operation

POS mode is a mode that moves the axis to the commanded position and speed. Command methods include absolute command and incremental command.

The operation sequence for POS mode is as follows.

- Select POS mode.
- Set the position command value, speed command value, and position command method.
- Input the start signal (GMCST). The controller starts axis movement when this signal transitions from Low to High.
- The controller transmits the operation complete signal (FMCF) after completing the axis movement. The operation complete signal remains High until the operation start signal goes Low.

To cancel a command during axis movement, set the operation stop signal (GMCAB) to High.

POS Mode Selection

(GPOS)

POS Mode Status

(FPOSS)

Position, Speed, Command Mode

(GMCP, GMCF, GMCABS)

Operation Start

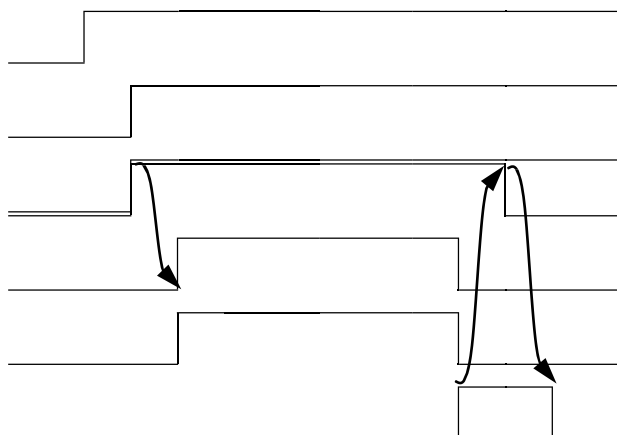
Signal (GMCST)

Axis Transfer Operation

Operation in
Progress Signal
(FMCBUSY)

Operation Complete Signal

(FMCF)



(Caution) Be sure to set GMCP and GMCABS in the PLC before setting GMCS.

■ In-Position Check Skip Function

The In-Position Check Skip function enables the control device to transmit the motion complete signal (FMCF) without performing an In-Position check. This can be applied to prevent speed changes due to acceleration/deceleration when issuing consecutive position commands. Ensure acceleration/deceleration after interpolation is always enabled.

■ User Interpolation Mode

This mode allows the user to perform path interpolation directly, while the controller only performs position control. The controller's acceleration/deceleration function can still be used in this case. Ensure that acceleration/deceleration after interpolation is enabled.

■ Transfer Speed Override Function

Override can be applied to the commanded travel speed. Travel speed override can be set differently for each axis.

■ Signal

— Mode Selection Signal

Signal	Address	My Use
GPOS	GB200.00	MC Mode Selection Signal
FPOSS	FB200.00	MC Mode Status Signal

— Control Signal

Signal	Address	Internal Use
GMCST	GB350.00~GB350.31	Start Operation
FMCF	FB350.00~FB350.31	Operation Complete Signal
GMCAB	GB351.00~GB351.31	Transfer Halt Signal (Command Cancel Signal)
FMCBUSY	FB355.00~FB355.31	Operation in Progress Signal
GMCABS	GB345.00~GB345.31	Position Command Mode (0: Incremental Command, 1: Absolute Command)
GMCP	GW215 ~ GW246	Position Command [PU]
GMCF	GW247 ~ GW278	Movement Speed [FU]
GMCFOR	GW360 ~ GW391	Transfer Speed Override [0.01%]

GPIS	GB396.00~GB396.31	In-position Skip Signal
GPUI	GB397.00~GB397.31	User Interpolation Mode Signal

■ Parameter

Parameter	For My
P2287	Feed Override applied in MP mode (0:X, 1:0)
P2290	Apply Feed Override in MC Mode (0:X, 1:0)
P5290	User Interpolation Mode Enable (0:X, 1:0)

■ Note

- POS mode is maintained during POS operation.
- If the mode is canceled during POS operation, deceleration and stop occur, and the command is also canceled.

5.5.2 JOG Mode

5.5.2.1 Basic Operation

JOG mode is a mode that continuously moves the axis at the commanded speed. The travel speed can be changed even during travel.

The operation sequence for JOG mode is as follows.

- Select JOG mode.
- Set the speed command value and command direction.
- Input the start signal (GMCST) for the operation (). The control device starts axis movement when this signal is High.

It is possible to change the axis travel direction during operation using the command direction signal. To stop axis travel, set the start operation signal to Low.

JOG Mode Selection

(GJOG)

JOG Mode Status

(FJOGS)

Travel Speed

(GPCF)

Travel Direction

(GPCDIR)

Motion Start Signal

(GMCST)

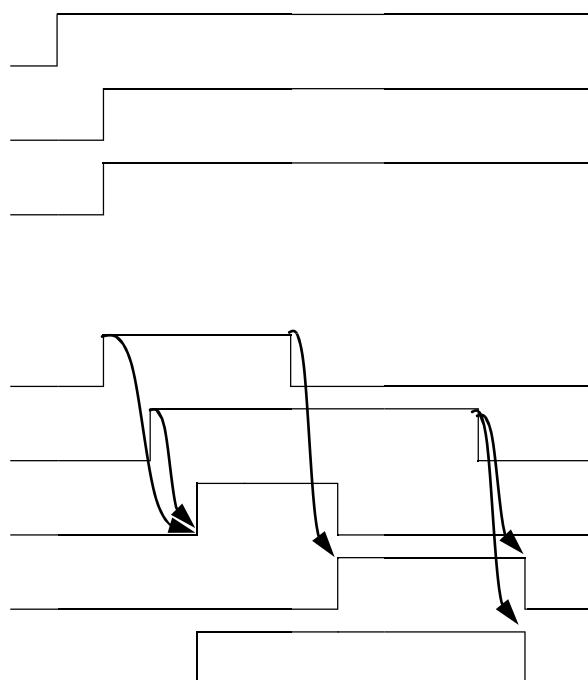
+ Direction Axis Feed Motion

– Direction Axis Feed Motion

Motion

Operation in Progress Signal

(FMCBUSY)



■ Signal

- Mode selection signal

Signal	Address	My Use
GJOG1	GB200.01	MC Mode Selection Signal
FJOGS1	FB200.01	MC Mode Status Signal

- Mode control signal

Signal	Address	My Use
GMCST	GB350.00~GB350.31	Operation Start
FMCBUSY	FB355.00~FB355.31	Operation in Progress Signal
GMCDIR	GB346.00~GB346.31	Operation direction selection signal (0: +direction, 1: -direction)
GMCF	GW247 ~ GW278	Travel Speed [FU]

■ Note

During JOG mode, the feed rate override is invalid.

5.5.3 HAN Mode

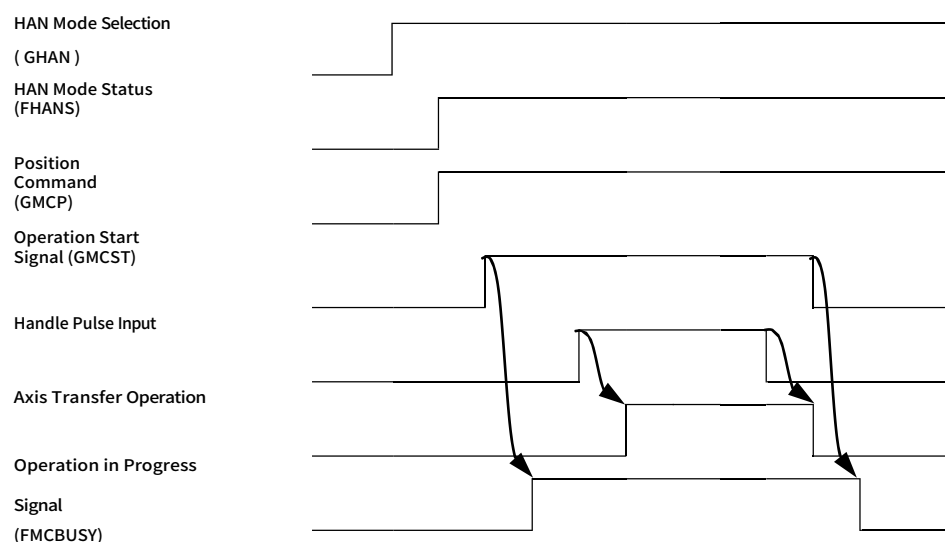
5.5.3.1 Basic Operation

HAN mode moves the axis a distance corresponding to the number of pulses input from the Handle (MPG) attached to the controller. The default command unit corresponding to the input pulse is based on the value set in the position command value (GMCP). (Command distance = Input pulse × GMCP).

The operation sequence for HAN mode is as follows.

- Select HAN mode.
- Set the position command value (default command unit).
- Input the start signal (GMCST). The controller waits for a handle input when this signal is High.
- When a pulse is input from the handle, the axis moves.

The basic command unit can be changed even during movement.



■ Command Basic Unit Determination

Command units can be specified using either integer or fractional formats. While the integer format is the standard method, the fractional format should be used for more precise commands. In fractional mode, the lower 16 bits of the GMCP serve as the denominator, and the upper 16 bits serve as the numerator. The maximum setting values for both the denominator and numerator are 32767. For example, to set 1/100, configure the GMCP to 0x00010064.

■ Signal

- Mode Selection Signal

Signal	Address	My Use
GHAN1	GB200.02	MC Mode Selection Signal
FHANS1	FB200.02	MC Mode Status Signal

- Mode Control Signal

Input Signal	Address	Internal Use
GMCST	GB350.00~GB350.31	Start Operation Signal
FMCBUSY	FB355.00~FB355.31	Operation in Progress Signal
GMCP	GW215 ~ GW246	Position Command (Basic Command Unit) [PU]

■ Parameter

Parameter	Formy
P5475	Handle #1 for the corresponding axis
P5476	Corresponding axis for Handle #2
P5477	Handle #3 corresponds to the axis
P5482	Command unit setting method (0: integer, 1: fraction)
P5508	Handle Pulse multiplication rate (0: x1, 1: x4)
P5511	Handle A/B Phase Signal Polarity (0: Positive Logic, 1: Negative Logic)

■ Warning

Setting the position command value too high causes the axis to move at extremely high speeds, posing significant danger. Always set it to 0.1mm or 0.01inch or less.

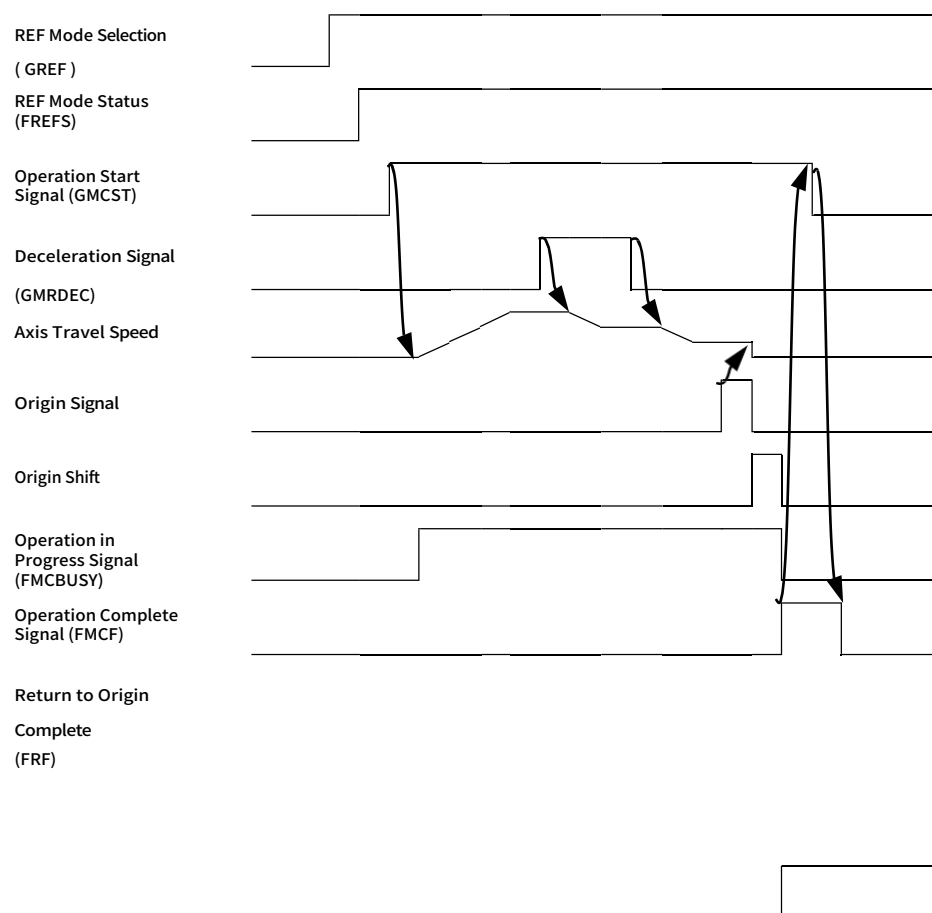
5.5.4 REF Mode

5.5.4.1 Default Operation

REF mode is the mode for setting the machine's origin. Origin return consists of three stages: rapid traverse, first deceleration, and second deceleration. Notably, the traverse direction can be set for each stage, enabling the implementation of various origin return methods. It also supports various additional functions.

The operation sequence for REF mode is as follows:

- Select REF mode.
- Input the start signal (GMCST). When this signal transitions from Low to High, the control device initiates axis movement at rapid speed.
- When the deceleration signal (GRDEC) goes High, it moves at the first deceleration speed.
- When the deceleration signal (GRDEC) goes low, the axis moves at the secondary deceleration speed and stops when it encounters the origin signal.
- The axis moves by the origin return shift distance.
- The control device sets the current position as the origin and sends a completion signal (FMCF).
- The completion signal remains high until the operation start signal goes low.
- The origin return completion status signal (FRF) is sent as High.



■ Transfer Speed Override Function

Transfer speed override can be applied during Stage 1 (Rapid Transfer). Transfer speed override can be set differently for each axis.

■ Signal

- Mode Selection Signal

Signal	Address	My Use
GRES	GB200.03	MC Mode Selection Signal
FREFS	FB200.03	MC Mode Status Signal

- Mode Control Signal

Signal	Address	Internal Use
GM CST	GB350.00~GB350.31	Start Operation
FM CF	FB350.00~FB350.31	Operation Complete Signal
FMC BUSY	FB355.00~FB355.31	Operation in Progress Signal
GMR DEC	GW215 ~ GW246	Deceleration signal (Deceleration sensor signal)
GM CFOR	GW390 ~ GW391	Transfer Speed Override [0.01%]

■ Parameter

Parameter	Formy
P5609 ~ P5640	Rapid Return to Origin Speed [FU]
P5641 ~ P5672	Home Return Primary Deceleration Speed [FU]
P5673 ~ P5706	Home Return Secondary Deceleration Speed [FU]
P5707 ~ P5708	Shift Transfer Amount After Home Return Signal (Phase C) Detection [PU]
P5739 ~ P5770	Shift Transfer Speed After Detecting Home Return Signal (Phase C) [FU]

■ Note

If a return-to-origin operation is performed again while the previous one is complete, canceling during the return-to-origin process will leave the operation incomplete.

5.5.4.2 Current Position Home Setting Function

This function allows you to set the current position as the origin without performing a return-to-origin operation.

The operation sequence is as follows.

- Set the [Current Position Origin Designation Function] to Enabled in the parameters.
- Move the axis to the position where you want to set the origin.
- Select REF mode.
- Input the start operation signal (GMCST).
- The controller sends the completion signal (FMCf) after setting the current position as the origin. The completion signal remains high until the start signal goes low.
- The origin return complete status signal (FRF) is sent as High.

■ Parameter

Parameter	Formy
P5589	Current Position Home Setting Function

■ Note

The origin return shift amount is also valid in the [Current Position Origin Setting Function].

5.5.4.3 Retry Function After OT Signal Detection

This function moves the axis back to the position before the deceleration sensor when an OT signal is encountered during origin return operation, then retries the origin return.

The operation sequence is as follows.

- Set [Retry Function After OT Signal Detection] to Enabled in the parameters.
- Select REF mode.
- Input the operation start signal (GMCST). The controller starts rapid axis movement when this signal changes from Low to High.
- If an OT signal for that direction is input during rapid movement, the axis moves to a position before the deceleration sensor.
- It automatically retries the origin return.

■ Parameter

Parameter	Formy
P5603	After OT detection, select retry function (0:X, 1:O)

5.5.4.3 Absolute encoder origin return

When using an absolute encoder, operation is possible without performing a home return after powering on the control device. However, a home return must be performed after initial operation or maintenance such as disconnecting the motor.

The origin return operation when using an absolute encoder is as follows.

- Set the encoder type to absolute encoder in the parameters.
- Enable the current position origin designation function (P5589) in the parameters.
- Turn OFF the controller power.
- Initialize the encoder data stored internally in the servo drive.
- Turn on the controller power.
- Move the axis to the origin position.
- Switch to REF mode and perform origin return.

■ Parameter

Parameter	Formy
P8825	Encoder Usage
P8828	Encoder Type (0: INC, 1: ABS)
P5589	Current Position Origin Designation Function

■ Note

If memory is initialized, perform origin return again.

5.5.4.4 Additional Function

■ Home Return Function Without Deceleration Signal (Deceleration Sensor)

This function is used to return axes without deceleration sensors to the origin. When this function is enabled, origin signal detection begins immediately when the start signal (GMCST) changes from Low to High.

■ Home Return Function Without Using Home Signal

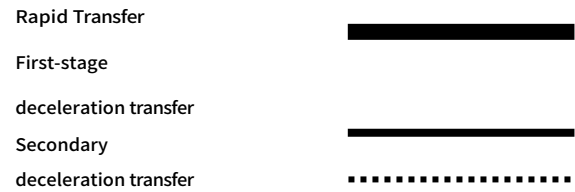
This function is used to return an axis without an origin signal to its origin. When this function is used, origin return is performed using a deceleration signal. This function is applied to stepper motors that do not use encoders.

■ Parameter

Parameter	Formy
P5592	Deceleration sensor usage (0: On, 1: Off)
P5597	Use of origin signal (0:0, 1:X)

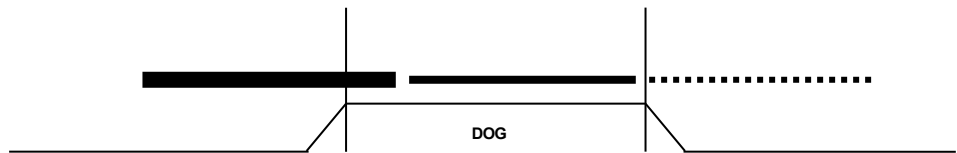
5.5.4.5 Examples of various origin return applications

The origin return method based on the origin return direction setting in the parameters is as follows.



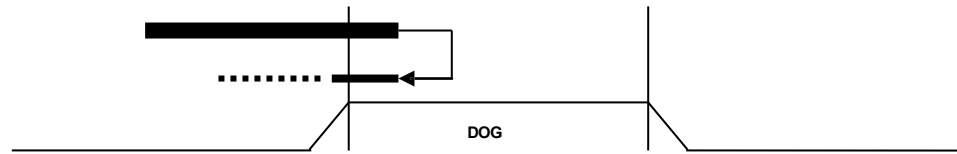
■ + Direction Home Return Method

Rapid Travel Speed	First Deceleration Speed	Secondary deceleration speed
5000 mm/min	200 mm/min	50 mm/min



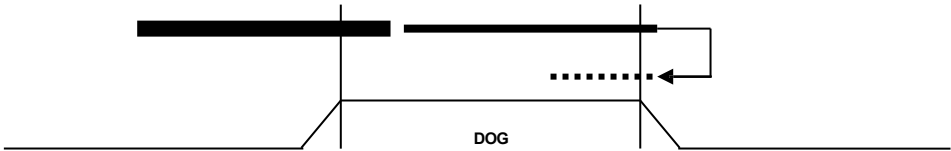
■ Direction Change Origin Return Method 1

Rapid Transfer Speed	Primary Deceleration Speed	Secondary Deceleration Speed
5000 mm/min	-200 mm/min	-50 mm/min



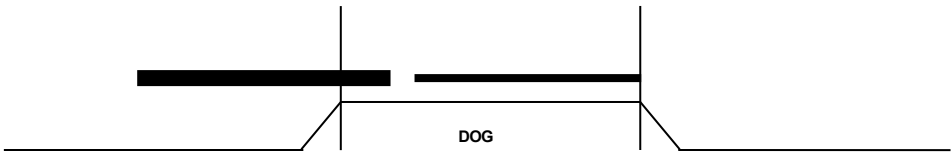
■ Direction Change Home Return Method 2

Rapid traverse speed	Primary deceleration speed	Secondary Deceleration Speed
5000 mm/min	200 mm/min	-50 mm/min



■ When not using the origin sensor

Rapid traverse speed	Primary deceleration speed	Secondary deceleration speed
5000 mm/min	200 mm/min	-



5.5.5 VEL Mode

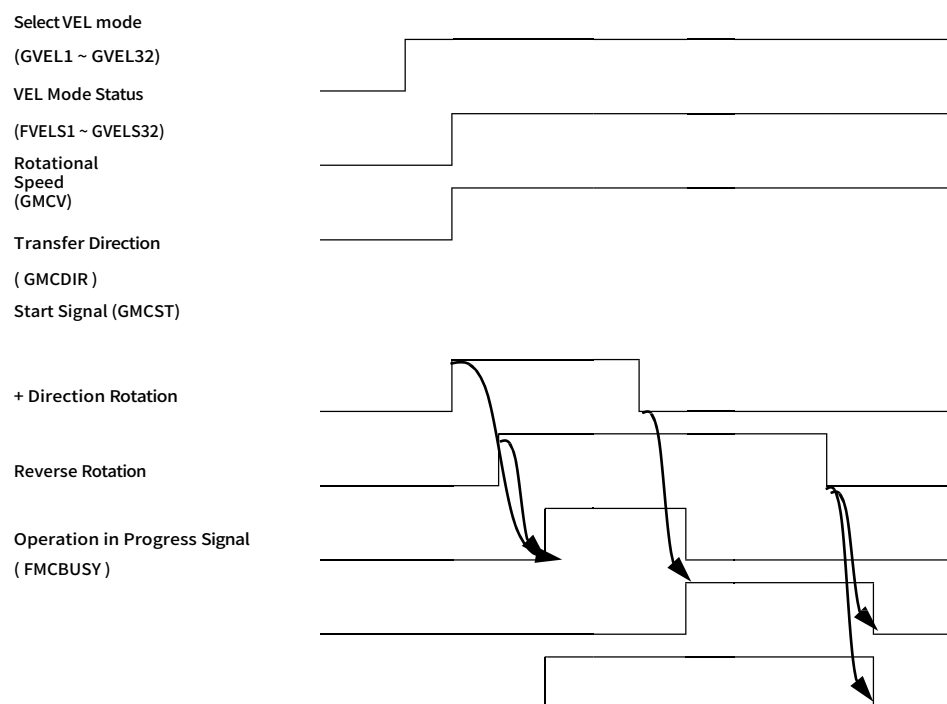
5.5.5.1 Basic Operation

VEL mode is a mode that rotates the axis at the commanded rotational speed. It is typically used for spindle control. The rotational speed can be changed even during rotation.

The operation sequence for VEL mode is as follows:

- Select VEL mode.
- Set the speed command value (RPM) and command direction.
- Input the start signal (GMCST) for the operation . The control device starts shaft rotation when this signal is High.

During operation, the axis rotation direction can be changed using the rotation direction signal. To stop axis movement, set the start signal to Low.



■ Rotation Speed Override Function

You can apply an override to the commanded rotation speed. The rotation speed override can be set differently for each axis.

■ Signal

- Mode Selection Signal

Signal	Address	My Use
GVEL	GB200.04	MC Mode Select Signal
FVELS	FB200.04	MC Mode Status Signal

- Mode Control Signal

Signal	Address	Internal Use
GMCST	GB350.00~GB350.31	Operation Start
FMCF	FB350.00~FB350.31	Operation Complete Signal
FMCBUSY	FB355.00~FB355.31	Operation in Progress Signal
GMCDIR	GB346.00~GB346.31	Operation direction selection signal (0: +direction, 1: -direction)
GMCV	GW279 ~ GW310	Rotational Speed [RPM]

5.6 Status Output

5.6.1 Transfer Status Signal

This function outputs information about the axis's transfer direction. When the axis is stopped, both the + direction and - direction transfer status signals are Low.

■ Signal

Signal	Address	Address Address
FPMV	FB500.00~FB500.31	+ Direction Transfer Status Signal
FNMV	FB501.00~FB501.31	- Directional Transfer Status Signal

5.6.2 In-position signal

This function outputs whether the position is within the range set in the parameters after the axis has completed its movement.

■ Signal

Signal	Address	My Use
FINP	FB510.00~FB510.31	In-Position Status Signal

■ Parameter

Parameter	Formy
P9120	In-Position Check Mode (0: Encoder Signal, 1: INP Signal): Position-Type Servo
P9123 ~ P9154	In-Position Range
P9155	In-Position Allowable Delay Time [msec]

5.6.3 Speed Reach Signal

The speed reach output is sent when the commanded rotational speed is reached after the axis begins rotation. Determination of whether the speed has been reached is based on the value set in the parameter.

■ Signal

Signal	Address	Address Use
FCSAGR	FB513.00~FB513.31	Command Speed Reach Signal

■ Parameter

Parameter	Formy
P9225 ~ P9256	Command Speed Reach Status Judgment Range [0.01%]
P9289	Speed Reach Diagnosis Function Enabled/Disabled
P9291	Speed Reach Allowable Delay Time [msec]

■ Note

This function is only valid in VEL mode.

5.6.4 0 Speed Reach Signal

When the axis reaches zero speed (0 RPM) after stopping rotation, a zero speed reach signal is transmitted. Determination of whether zero speed has been reached is based on the value set in the parameter.

■ Signal

Signal	Address	Address Use
FZSAGR	FB512.00~FB512.31	0 Speed Reached Signal

■ Parameter

Parameter	Formy
P9257 ~ P9288	0 Speed Reach Status Determination Range [RPM]

■ Note

Function is only valid in VEL mode.

5.7 Synchronous Control

5.7.1 Basic Operation

This function controls an axis driven by two or more servo motors.

It consists of one master axis and multiple slave axes. It is typically used in precision stages or gantry-type equipment where one axis is driven by two linear motors.

The characteristics of the synchronous control function are as follows.

- One axis can be driven by two or more servos.
- Synchronization errors between the master and slave axes are compensated.
- Synchronous control can be enabled or disabled via the PLC.
- It diagnoses whether the synchronization error exceeds the allowable value set by parameters, preventing machine damage.

■ Synchronization Control Factor

Set this when the movement distances of the master and slave axes are not identical but maintain a fixed ratio. The movement distance of the slave axis is determined as follows:

Slave axis travel distance = Synchronous Control Factor × Master axis travel distance

■ Master Axis Actual Position Synchronization Function

Synchronization control is typically performed with the [Master Axis Command Position], but synchronization control with the [Master Axis Actual Position] is also possible. In this case, the synchronization error compensation control loop does not operate, but the synchronization control error detection function is supported identically. Note that the synchronization control error is identical to the tracking error; therefore, set the synchronization control tolerance value considering this point.

■ Synchronization Control Error Detection Function

Synchronization control error refers to the [control deviation between the master axis and the slave axis]. If this exceeds the synchronization error tolerance set in the parameters, a [synchronization error alarm] will occur.

■ Signal

Signal	Address	My Use
GSYN	GB550.00~GB550.31	Synchronous Control Select Signal
FSYNS	FB550.00~FB550.31	Synchronous Control Status Signal

■ Parameter

Parameter	Formy
P9822	Synchronous Control Enable (0:X, 1:O)
P9823	Synchronous Control Function Selection (0: Boot, 1: G Register Signal)
P9825	Synchronous control target selection (0: Command position, 1: Actual position)
P9828 ~ P9859	Synchronous Control Master Axis Setting
P9862 ~ P9893	Synchronous Control Factor Denominator
P9894 ~ P9925	Synchronous Control Factor Numerator
P9962 ~ P9993	Synchronous Control Error Tolerance [PU]

■ Note

The master axis must be set to a faster axis number than the slave axis.

5.7.2 Synchronous Control Gain Setting

The synchronous control gain is the P gain used to minimize the synchronous control error between the master axis and the slave axis. Setting it higher improves tracking performance, but be cautious as excessive gain can cause the machine to vibrate.

■ Parameter

Parameter	Formy
P9928 ~ P9959	Synchronous Control P Gain [0.1%]

■ Note

When synchronizing with the [actual position of the master axis], the synchronous error compensation loop does not operate, so the synchronous control P gain is not applied.

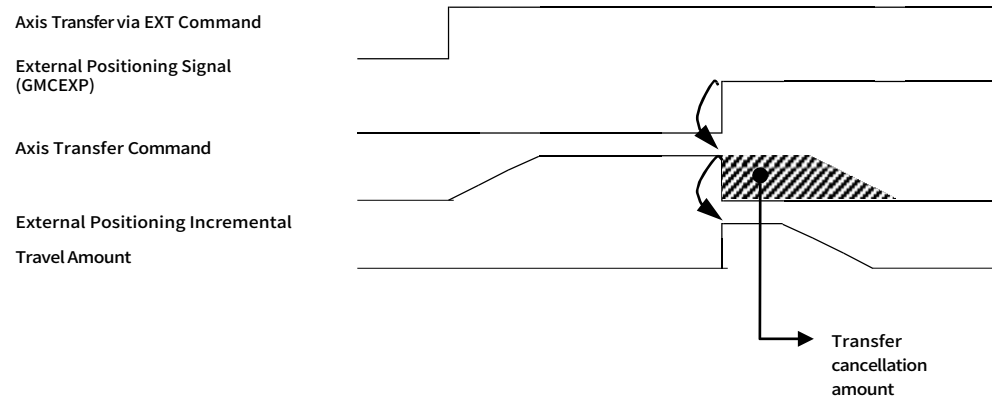
5.8 External Position Determination

5.8.1 Default Operation

[External Positioning Function] cancels remaining commands and moves by the [External Positioning Incremental Travel Amount] set in the parameters when the [External Positioning Signal] goes High during travel. External positioning is supported only in MEM mode and POS mode.

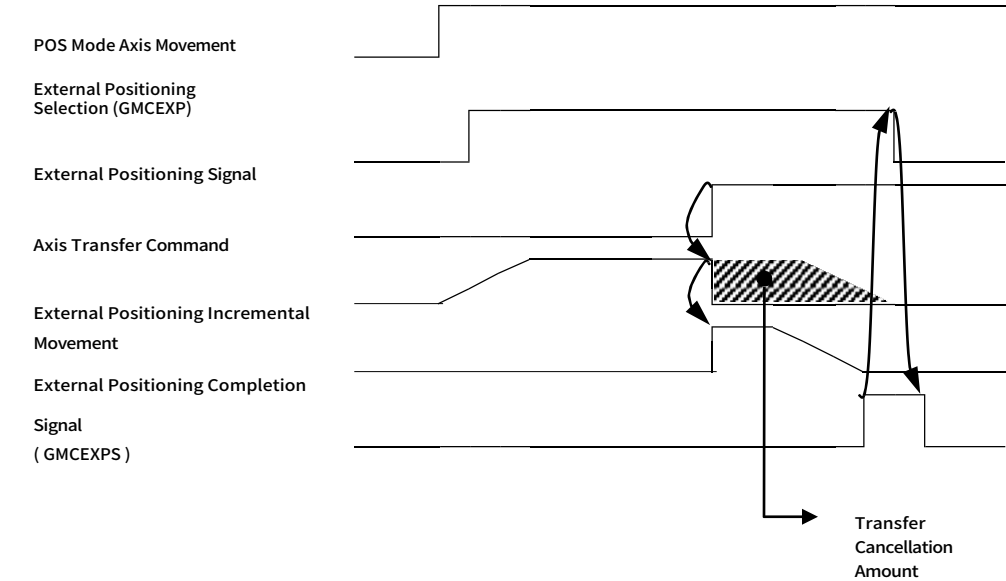
The operation sequence in MEM mode is as follows:

- Issue axis commands using the EXT command.
- When the [External Positioning Signal] goes High, cancel the remaining axis travel amount and change the target position to the "Current Position + External Positioning Incremental Transfer Amount".
- Move the axis to the target position. (If the remaining distance is less than the distance required to stop or if the direction is opposite, decelerate and stop immediately, then move to the target position.)



The operation sequence in POS mode is as follows.

- Set the external position determination selection signal to High.
- Select POS mode and perform axis movement.
- Subsequent operations are identical to MEM mode.



■ Signal

Signal	Address	My Use
GMCEXP	GB558.00~GB558.31	External Positioning Selection
FMCEXPS	FB558.00~FB558.31	External Positioning Complete Signal (MC Drive Mode)

■ Parameter

Parameter	ForInternal
P10520	External Positioning Enable (0:X, 1:O)
P10527	External Positioning Signal Register Selection (0:R, 1:X)
P10528 ~ P10559	External Position Determination Signal Register Address
P10560	External Positioning Signal Polarity
P10568 ~ P10599	External Positioning Incremental Transfer Amount [PU]

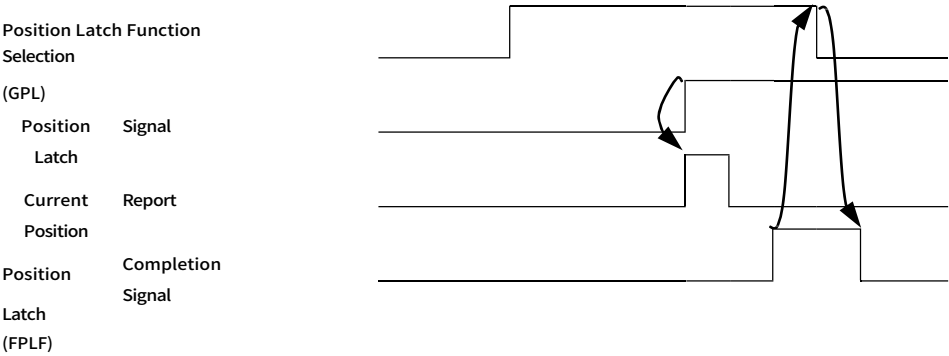
5.9 Position Latch Function

5.9.1 Basic Operation

The [Position Latch Function] reports the mechanical position of the corresponding axis to the [Latch Position] item in the S-register when a latch signal is input.

The operation sequence is as follows.

- Set the [Position Latch Select Signal] to High.
- When the [Position Latch Signal] goes High, the current position of the corresponding axis is reported to the S register, and a position latch complete signal is output.



■ Signal

Signal	Address	My Use
GPL	GB561.00~GB561.31	Position Latch Selection
FPLF	FB561.00~FB561.31	External Latch Complete Signal (MC Operation Mode)

■ Parameter

Parameter	For internal
P10720	PL Function Enable Status (0:X, 1:O)
P10727	PL Signal Register Selection (0:R, 1:I)
P10728 ~ P10759	PL Signal Register Address
P10760	PL Signal Polarity

■ Register

Parameter	Formy
S1482 ~ S1513	Latch Position [PU]

■ Note

To re-execute the position latch operation after the position latch is complete, the position latch function select signal (GPL) must be changed from LOW to HIGH again.

5.10 Position Compare Function

5.10.1 Basic Operation

The [Position Compare function] outputs a signal externally when the axis reaches the set position.

The output and setting method are as follows.

Output Format	Position Setting Method
Positioning Method	Incremental Position Setting
	Absolute Position Setting
Zone Method	Absolute Position Setting

■ Positioning Mode

The position mode is a function that sets multiple positions as parameters and outputs the corresponding signal externally when the axis passes through the set position. The output signal remains High within a certain range. Position setting modes include [Incremental Position Setting Mode] and [Absolute Position Setting Mode]. [Incremental Position Setting Mode] should be used when the intervals between set positions are constant.

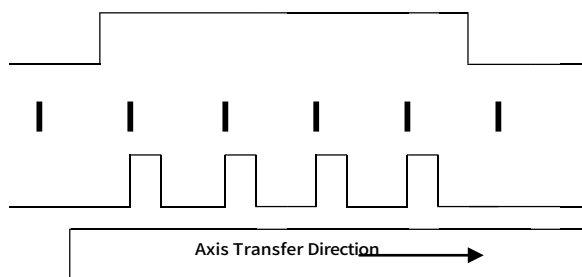
The operation sequence is as follows:

- Set the reference position in the parameter.
- Input the [Position Comparison Selection Signal] as High and move the axis.
- When the axis reaches the set position, the [Position Comparison Signal] outputs High. If the axis moves beyond the position set in the parameter, the signal outputs Low.

Position Comparison
Function Selection
(GPC)
Setting Location

Position Comparison
Output

Axis Transfer Operation



■ Area Mode

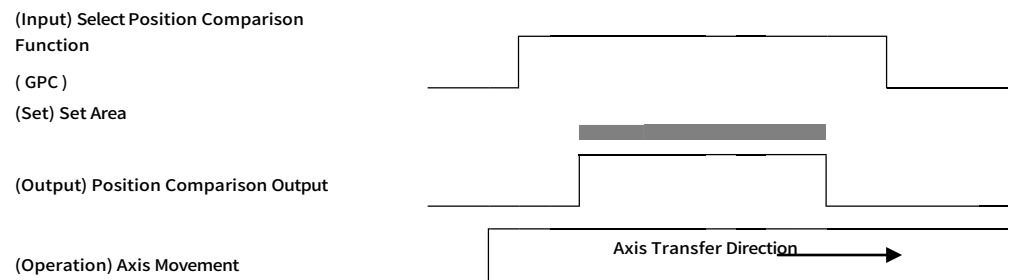
The area mode is a function that sets an area in the parameters and outputs a signal externally when the axis is positioned within the set area.

The operation sequence is as follows.

Set the area to be compared in the parameters.

Input the [Position Comparison Select Signal] as High and move the axis.

When the axis reaches the set zone, the [Position Comparison Signal] outputs High. When the axis moves outside the zone set in the parameter, the signal outputs Low.



■ Signal

Signal	Address	In Use
GPC	GB565.00~GB565.31	Location Comparison Function Selection

■ Parameter

Parameter	For My
P10820	PC Function Enabled
P10827	PC Signal Register Selection (0:R, 1:I)
P10828 ~ P10859	PC Signal Register Address
P10860	PC Signal Polarity
P10868	PC Function Mode (0: Position Mode. 1: Range Mode)
P10869	Position Setting Mode (0: Incremental. 1: Absolute)
P10872 ~ P10903	Output Signal Hold Distance [PU]
P10906 ~ P10937	Number of Incremental Mode Sections
P10938 ~ P10969	Incremental Mode Start Position [PU]
P10970 ~ P11001	For incremental mode, the incremental distance [PU]
P11004 ~ P11035	For the absolute method, the start index of the position data table
P11036 ~ P11069	For the absolute method, the end index of the position data table
P11070 ~ P11101	– Direction position for the area method
P11102 ~ P11133	Area method – direction position

6

Communication Functions

This section describes the communication features of the GX-S200/S400. The GX-S200/S400 communication enables the GX Series to communicate with PCs, other GX Series units, or third-party PLCs using communication methods such as RS232, Ethernet, USB, and RS422/485.

6.1 Specifications

6.1.1 Functional Specifications

6.1.2 H/W Specifications

6.2 Communication Configuration

6.2.1 PC : GX

6.2.2 GX: GX

6.2.3 Third-party PLC: GX

6.3 Operation Settings

6.3.1 Parameter Settings

6.3.2 Wiring Diagram

6.4 Protocol

6.4.1 T-BUS

6.4.2 Third-Party Protocol

6.5 Third-Party HMI

6.5.1 Graphic Panel

6.5.2 GX Settings

6.5.3 Wiring Diagram

6.5.4 Register Mapping

6.5.5 PLC Processing

6.1 Specifications

6.1.1 Functional Specifications

Item	Item	Internal Communication
Communication Configuration	PC: GX,	GX: GX, Third-party PLC: GX communication is possible.
		Supports USB, RS232, RS422/485, and Ethernet communication, enabling 1:1, 1:N, and N:M communication.
Communication Expandability		Modular design allows adding communication modules as needed to connect with multiple devices.
Communication Settings		The communication ports on the CPU module have default parameters applied. To configure parameters, use the Setup program for software-based settings.
Communication Method		Operates as a Slave by default, but can function as a Master using PLC commands as needed.
Communication Protocol		Communicates via a proprietary protocol (T-BUS) by default. Additionally, it incorporates protocols for Mitsubishi's MELSEC and LG's Master-K, enabling communication with these devices.

6.1.2 H/W Specifications

The CPU module includes one USB port, two RS232 ports, and one Ethernet port as standard. Users can expand communication ports by adding serial communication boards or Ethernet communication boards.

■ CPU Module

Item	Item	Internal Purpose	
USB	Features	USB 1.1, CLIENT TYPE.	
	Speed	Up to 10Mbps	
	Quantity	1	
Ethernet	Features	Supports 4 TCP/UDP ports each	
	Speed	Up to 10 Mbps	
	Number	1	
RS232	Features	Default 38400 bps, 8 Data bits, No Parity, 1 Stop bit	
		Baud Rate	2400, 4800, 9600, 19200, 38400 bps
		Parity	None, Even, Odd
	Speed	Up to 38400 bps	
	Number	2	

Serial Module

Port	Port	Number	Purpose
RS232	Features	Default 38400 bps, 8 Data bits, No Parity, 1 Stop bit	
		Baud Rate	2400, 4800, 9600, 19200, 38400 bps
		Parity	None, Even, Odd
	Speed	Up to 38400 bps	
	Number	2	
RS422/485	Features	Default 38400 bps, 8 Data bits, No Parity, 1 Stop bit	
	Speed	Up to 38,400 bps	
	Number	1	

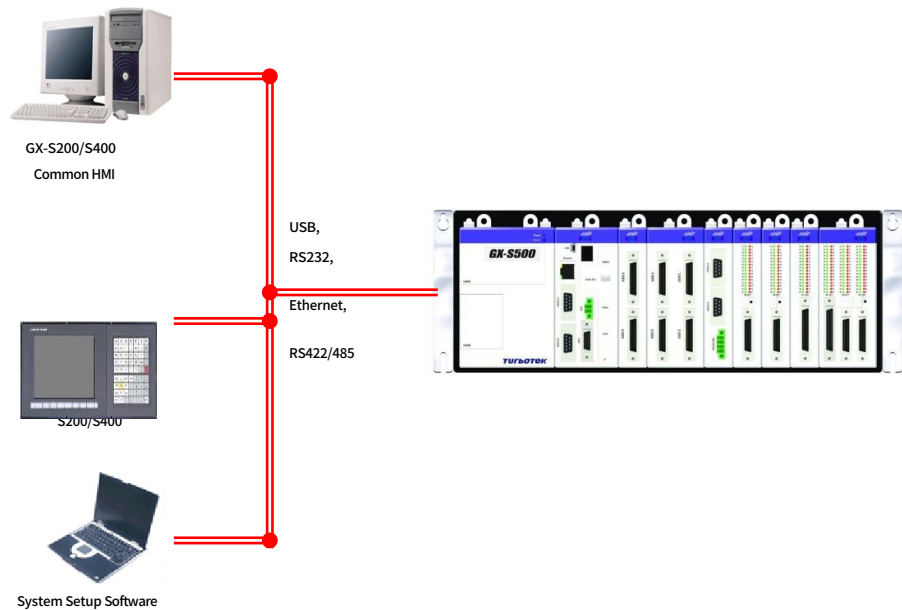
Ethernet module

Port	Port	Number	Use
Ethernet	Features	Supports 4 TCP/UDP ports each	
	Speed	Up to 10Mbps	
	Number	1	

6.2 Communication Configuration

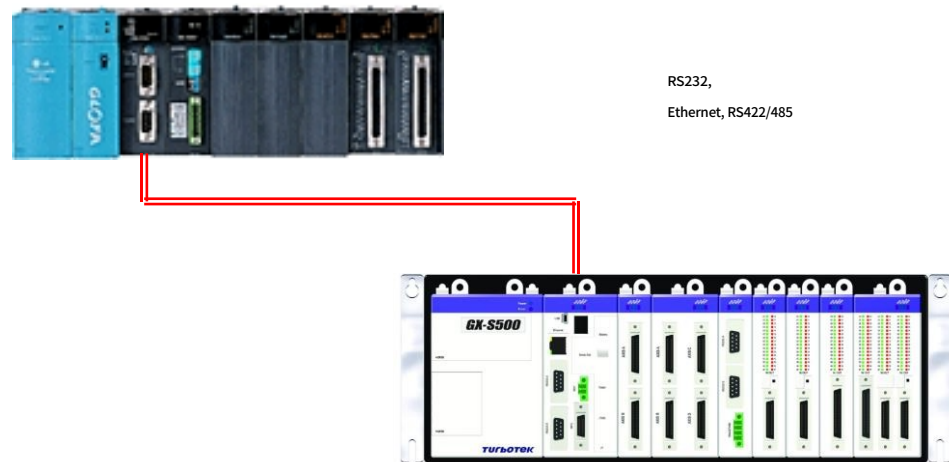
Communication is broadly categorized into 3 types PC : GX, GX: GX, T h i r d -party PLC: GX.

6.2.1 PC: GX Series



The user software and GX Series communicate via dedicated T-BUS protocol over communication ports such as USB, RS232, and Ethernet. User software includes GX-Builder and HMI. These programs configure GX Series parameters, modify or monitor register values, and upload/download motion or PLC programs. Additionally, users can create custom communication message frames using T-BUS to communicate directly with the GX Series.

6.2.2 Third-party PLCs: GX-S200/S400



The GX-S200/S400 communicates with third-party PLCs using their communication protocols via ports such as RS232, Ethernet, and RS422/485. Protocols compatible with third-party PLCs include the Melsec PLC Protocol and the Master-K PLC Protocol, both of which are implemented within the GX.

Before communication, configure parameters such as the communication port and protocol for the GX-S200/S400 using the GX-Builder program. When the third-party PLC requests to read/write GX registers, the GX-S200/S400 responds accordingly.

Refer to Sections 6.3.1 and 6.4.2 for information on parameters and register mapping.

6.3 Operation Settings

The GX Series communication system operates as a Slave by default. That is, once the user sets the communication parameters and connects the cables, the GX communicates by automatically responding to requests from the Host.

For T-BUS, if the GX Series attempts to communicate as the Host, it can use PLC commands to issue a communication request.

6.3.1 Parameters

Communication-related parameters are set in the D Register. This register can be divided into a Local area for setting the GX Series device and a Remote area for setting the communication target device connected to the GX Series device via communication.

The D Register allows Local to use only numbers 0 through 19, while Remote has no restrictions. Up to 254 devices can be registered. Each device has an offset in units of 100.

Device Number	Assigned Register Number
0	0-99
1	100-149
2	200-249
:	:
:	:
253	25300-25399
254	25400-25499

Parameters can be set using the GX-Builder program.

■ Local Area

◇ CPU Module

Parameter	No	Bit	Size	Description
[USB Port]				
Timeout	32	–	1W	Input in 1-millisecond increments
[Ethernet Port]				
Communication ID	39	–	1W	Country Code
IP Address	40	–	1W	GX's IP Address
Port number	41	–	1W	GX's Port Number
Subnet Mask	42	–	1W	Subnet Mask
Gateway	43	–	1W	Gateway Address
[RS232_1 Port]				
Master/Slave	51	0	1B	1:Master, 0:Slave
Communication Protocol	52	0	8B	※1.
Communication Protocol Number	52	8	24B	※2.
Baud Rate	54	0	8B	※3.
Data Bit	54	8	8B	Fixed at 8 bits, no configuration required
Parity Bit	54	16	8B	※4.
Stop Bit	54	24	8B	Fixed at 1 bit, no configuration required
Timeout	57	–	1W	Input in 1-millisecond increments
[RS232_2 Port]				
Master/Slave	62	0	1B	1:Master, 0:Slave
Communication Protocol	63	0	8B	※1.
Communication Protocol Number	63	8	24B	※2.
Baud rate	65	0	8B	※3.
Data Bit	65	8	8B	Fixed at 8 bits, no configuration required
Parity Bit	65	16	8B	※4.
Stop Bit	65	24	8B	Fixed at 1 bit, no configuration required
Timeout	68	–	1W	Input in 1-millisecond increments

◇ Serial Module

Parameter	No	Bit	Size	Description
Device Usage	2	0	1B	1: Enabled, 0: Disabled
Device Classification	3	0	1B	1:Local, 0:Remote
Device Type	7	–	1W	15 : Serial Communication
Slot No.	13	–	1W	Enter the slot number where the local device is plugged in
[RS232_1 Port]				
Master/Slave	27	0	1B	1:Master, 0:Slave
Communication Protocol	28	0	8B	※1.
Communication Protocol Number	28	8	24B	※2.
Baud Rate	30	0	8B	※3.
Data Bit	30	8	8B	Fixed at 8 bits, no configuration required
Parity Bit	30	16	8B	※4.
Stop Bit	30	24	8B	Fixed at 1 bit, no configuration required
Timeout	33	–	1W	Input in 1-millisecond increments
[RS232_2 Port]				
Master/Slave	38	0	1B	1:Master, 0:Slave
Communication Protocol	39	0	8B	※1.
Communication Protocol Number	39	8	24B	※2.
Baud Rate	41	0	8B	※3.
Data Bit	41	8	8B	Fixed at 8 bits, no configuration required
Parity Bit	41	16	8B	※4.
Stop Bit	41	24	8B	Fixed at 1 bit, no configuration required
Timeout	44	–	1W	Input in 1-millisecond increments
[RS422/485]				
422/485	49	0	1B	1:422, 0:485
Master/Slave	50	0	1B	1:Master, 0:Slave
Communication Protocol	51	0	8B	※1.
Communication Protocol Number	51	8	24B	※2.
Communication ID	52	–	1W	
Baud Rate	53	0	8B	※3.
Data Bit	53	8	8B	Fixed at 8 bits, no configuration required
Parity Bit	53	16	8B	※4.

Stop Bit	53	24	8B	Fixed at 1 bit, no configuration required
Timeout	56	-	1W	Input in 1-millisecond increments

◇ Ethernet Module

Parameter	No	Bit	Size	Description
Device Usage	2	0	1B	1: Enabled, 0: Disabled
Device Classification	3	0	1B	1:Local, 0:Remote
Device Type	7	–	1W	16: Ethernet Communication
Slot No.	13	–	1W	Enter the slot number where the local device is inserted
[Ethernet Port]				
Communication ID	29	–	1W	Country Code
IP Address	30	–	1W	GX's IP Address
Port Number	17	–	1W	GX's Port Number
Subnet Mask	18	–	1W	Subnet Mask
Gateway	19	–	1W	Gateway Address
[Communication Diagnosis]				
Timeout	23	–	1W	Input in 1-millisecond increments

◇ Remote area

Parameter	No	Bit	Size	Description
Device Usage	1	0	1B	1: Enabled, 0: Disabled
Device Classification	1	1	1B	1:Local, 0:Remote
Device Type	2	–	1W	100 : Set as Unknown Device
Communication Device No.	7	–	1W	Number of the connected local device
Communication Port No.	8	–	1W	Port number of the connected local device
Protocol Type	16	–	1W	0: T-Bus, 1: Third-party
Protocol No	17	–	1W	※3.
Node ID (Area Code)	20	–	1W	Node ID of Remote Device
IP Address	27	–	1W	Remote Device's Ethernet IP Address
Port Number	28	–	1W	Remote Device's Ethernet Port Address Server
Server/Client	29	0	1B	1:Server, 0:Client
TCP/UDP	29	1	1B	1: TCP, 0: UDP

※1.

Value	Description
1	T-BUS
2	Modicon's Modbus
3	Mitsubishi's Melsec
4	LG Industrial Systems' Master-K
100	User

※2.

Value	Content	Value	Content
0x01	3C Frame	0x____01	Type 1
0x02	4C Frame	0x____02	Type 2
0x03	3E Frame	0x____03	Type 3
-	-	0x____04	Type 4

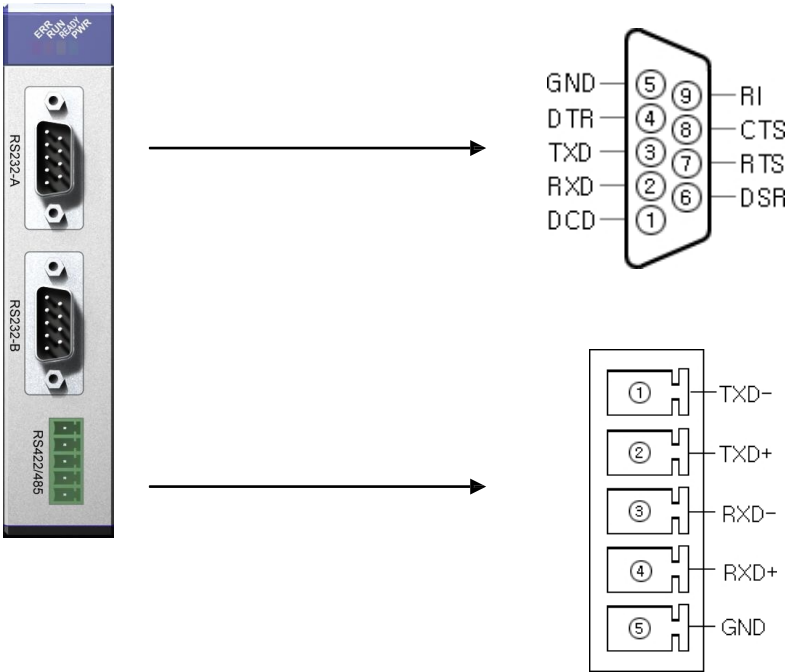
※3.

Value	Content
0	38400 bps
1	2400 bps
2	4800 bps
3	9600 bps
4	19200 bps

※4.

Value	Description
0	None Parity
1	Odd Parity
2	Even Parity

6.3.2 Wiring Diagram



◇ RS232

GX			Communication Device	
Pin	Signal		Signal	Pin
1	CD		CD	1
2	RXD		RXD	2
3	TXD		TXD	3
4	DTR		DTR	4
5	SG		SG	5
6	DSR		DSR	6
7	RTS		RTS	7
8	CTS		CTS	8
9	RI		RI	9

◇ RS422

GX			Communication Device
Pin	Signal		Signal
1	TXD-		TXD-
2	TXD+		TXD+
3	RXD-		RXD-
4	RXD+		RXD+
5	GND		GND

◇ RS485

GX			Communication Device
Pin	Signal		Signal
1	TXD-		TRXD-
2	TXD+		TRXD+
3	RXD-		GND
4	RXD+		
5	GND		

6.4 Protocol

The GX Series supports the proprietary T-BUS protocol and third-party protocols.

6.4.1 T-BUS

T-BUS is the dedicated protocol for the GX Series, defining the messages exchanged for register read/write and program upload/download operations. Communication is fundamentally structured such that the Master issues requests and the Slave responds. The message frame structure is as follows. (Numbers in parentheses indicate data length in bytes. Data is represented as hexadecimal values).

Basic Frame Structure

◇ Request Frame

ENQ (1)	Node ID (1)	Command Code (1)	Data Length (2)	Data (Maximum 1024)	EOT (1)	Check Sum (1)
------------	----------------	---------------------	--------------------	---------------------	------------	------------------

◇ Normal Processing Response Frame

ACK (1)	Node ID (1)	Command Code (1)	Data Length (2)	Data (Maximum 1024)	ETX (1)	Check Sum (1)
------------	----------------	---------------------	--------------------	---------------------	------------	------------------

◇ Response Frame for Abnormal Processing

NAK (1)	Node ID (1)	Command Code (1)	Data Length (2)	Error Code (2)	ETX (1)	Check Sum (1)
------------	----------------	---------------------	--------------------	----------------	------------	------------------

Node ID

In GX Series communication, when using RS422/485 as the Physical Layer, this corresponds to the ID applied to each node.

■ Control Character Table

Code	Code	Hex	Description	Name
	ENQ	0x05	A code included in the header, standing for Enquire, placed at the start of a request frame.	
	ACK	0x06	A code included in the header, standing for Acknowledge, placed at the start of a normal response frame.	
	NAK	0x15	A code included in the header, standing for Not Acknowledge, and appears at the start of an abnormal response frame.	
	EOT	0x04	The code inserted into the tail, standing for End of Transmission, is placed at the end of the request frame.	
	ETX	0x03	A code inserted into the tail, standing for End of Text, placed at the end of the response frame.	

■ Command Code Table

Code Code	Hex	Description Command
Register Write	0x51	A command to write to one or more TMC registers. Requests are made using the register name, number, and value.
Register Read	0x52	A command to read one or more TMC registers. Requested by specifying the register name and number.
Register Sequential Write	0x53	A command that writes register values with consecutive numbers within a register of the same name. Requested by specifying the register name, number, length, and value.
Register Sequential Read	0x54	A command to read the values of registers with consecutive numbers within a register of the same name. Requested using the register name, number, and length.
File Download Start	0x61	A command to notify the start of sending a program to the opposing TMC. It informs the file name and file size.
File Download	0x62	Sends the program contents.
File download complete	0x63	Indicates the end of program download.
File Upload start	0x64	When receiving a program from the opposing TMC, this command signals the start. It provides the file name.
File Upload	0x65	Receiving program content.
File Upload complete	0x66	Notifies that program upload is complete.
File deletion	0x67	Command to delete a program from the other party's TMC. Provides the file name.
File Stop upload/download	0x68	Stops communication during file upload/download.

Request frame data area structure

Register	Set	Na																	
Register Write	<table><tr><th>Reg. Count</th><th>Reg. Name</th><th>Registration Number</th><th>Reg. Value</th><th>Reg. Name</th><th>Reg. Number</th><th>Reg. Value</th><th>...</th><th>Reg. Value</th></tr><tr><td>(1)</td><td>(1)</td><td>(2)</td><td>(4)</td><td>(1)</td><td>(2)</td><td>(4)</td><td>(4)</td></tr></table>	Reg. Count	Reg. Name	Registration Number	Reg. Value	Reg. Name	Reg. Number	Reg. Value	...	Reg. Value	(1)	(1)	(2)	(4)	(1)	(2)	(4)	(4)	
	Reg. Count	Reg. Name	Registration Number	Reg. Value	Reg. Name	Reg. Number	Reg. Value	...	Reg. Value										
	(1)	(1)	(2)	(4)	(1)	(2)	(4)	(4)											
Maximum number of registers: 146																			
Register read	<table><tr><th>Reg. Count</th><th>Reg. Name</th><th>Registration Number</th><th>Reg. Name</th><th>Registration Number</th><th>...</th><th>Registration Number</th></tr><tr><td>(1)</td><td>(1)</td><td>(2)</td><td>(1)</td><td>(2)</td><td>(2)</td></tr></table>	Reg. Count	Reg. Name	Registration Number	Reg. Name	Registration Number	...	Registration Number	(1)	(1)	(2)	(1)	(2)	(2)					
	Reg. Count	Reg. Name	Registration Number	Reg. Name	Registration Number	...	Registration Number												
	(1)	(1)	(2)	(1)	(2)	(2)													
Maximum number of registers: 146																			
Register continuous write	<table><tr><th>Reg. Count</th><th>Reg. Name</th><th>Reg. Start Number</th><th>Reg. Value</th><th>Reg. Value</th><th>Reg. Value</th><th>Reg. Value</th><th>...</th><th>Reg. Value</th></tr><tr><td>(1)</td><td>(1)</td><td>(2)</td><td>(4)</td><td>(4)</td><td>(4)</td><td>(4)</td><td>(4)</td></tr></table>	Reg. Count	Reg. Name	Reg. Start Number	Reg. Value	Reg. Value	Reg. Value	Reg. Value	...	Reg. Value	(1)	(1)	(2)	(4)	(4)	(4)	(4)	(4)	
	Reg. Count	Reg. Name	Reg. Start Number	Reg. Value	Reg. Value	Reg. Value	Reg. Value	...	Reg. Value										
	(1)	(1)	(2)	(4)	(4)	(4)	(4)	(4)											
Maximum number of registers: 255																			
Register continuous read	<table><tr><th>Reg. Count</th><th>Reg. Name</th><th>Reg. Start Number</th></tr><tr><td>(1)</td><td>(1)</td><td>(2)</td></tr></table>	Reg. Count	Reg. Name	Reg. Start Number	(1)	(1)	(2)												
	Reg. Count	Reg. Name	Reg. Start Number																
	(1)	(1)	(2)																
Maximum number of registers: 255																			
File download start	<table><tr><th>Program Type</th><th>Prog. Name</th><th>Prog. Size</th></tr><tr><td>(1)</td><td>(4)</td><td>(4)</td></tr></table>	Program Type	Prog. Name	Prog. Size	(1)	(4)	(4)												
	Program Type	Prog. Name	Prog. Size																
	(1)	(4)	(4)																
Types: Motion and PLC Program (2 types)																			
File Download	<table><tr><th>Frame Number</th><th>Prog. Content</th></tr><tr><td>(2)</td><td>(Maximum 1022)</td></tr></table>	Frame Number	Prog. Content	(2)	(Maximum 1022)														
	Frame Number	Prog. Content																	
	(2)	(Maximum 1022)																	
Programs are sent in segments of up to 1022 bytes each																			

File Download Complete	None		
File upload start	<table><tr><td>Prog. Type (1)</td><td>Prog. Name (32)</td></tr></table>	Prog. Type (1)	Prog. Name (32)
Prog. Type (1)	Prog. Name (32)		
File Upload	None		
File Deletion	<table><tr><td>Prog. Type (1)</td><td>Prog. Name (32)</td></tr></table>	Prog. Type (1)	Prog. Name (32)
Prog. Type (1)	Prog. Name (32)		
File upload/download stop	None		

■ Response Frame Data Area Structure

Code	Dr	Code	Name
Register Write	None		Maximum number of registers: 146
Register Read	Reg Reg Reg Reg Reg Count Value		Maximum number of registers: 255
No consecutive register writes	None		Maximum number of registers: 255
Register sequential read	Reg Reg Reg Reg Reg Count Value		Maximum number of registers: 255
File download start	Frame number (2)		
File download	None		

File download complete	None		
File upload start	Prog. Size (4)		
File Upload	<table><tr><td>Frame No. (2)</td><td>Prog. Content (Maximum 1022)</td></tr></table>	Frame No. (2)	Prog. Content (Maximum 1022)
Frame No. (2)	Prog. Content (Maximum 1022)		
File upload complete	None		
File Deletion	None		
Communication Cancellation	None		

6.4.2 Third-party Protocol

■ LG Industrial Systems' Master-K

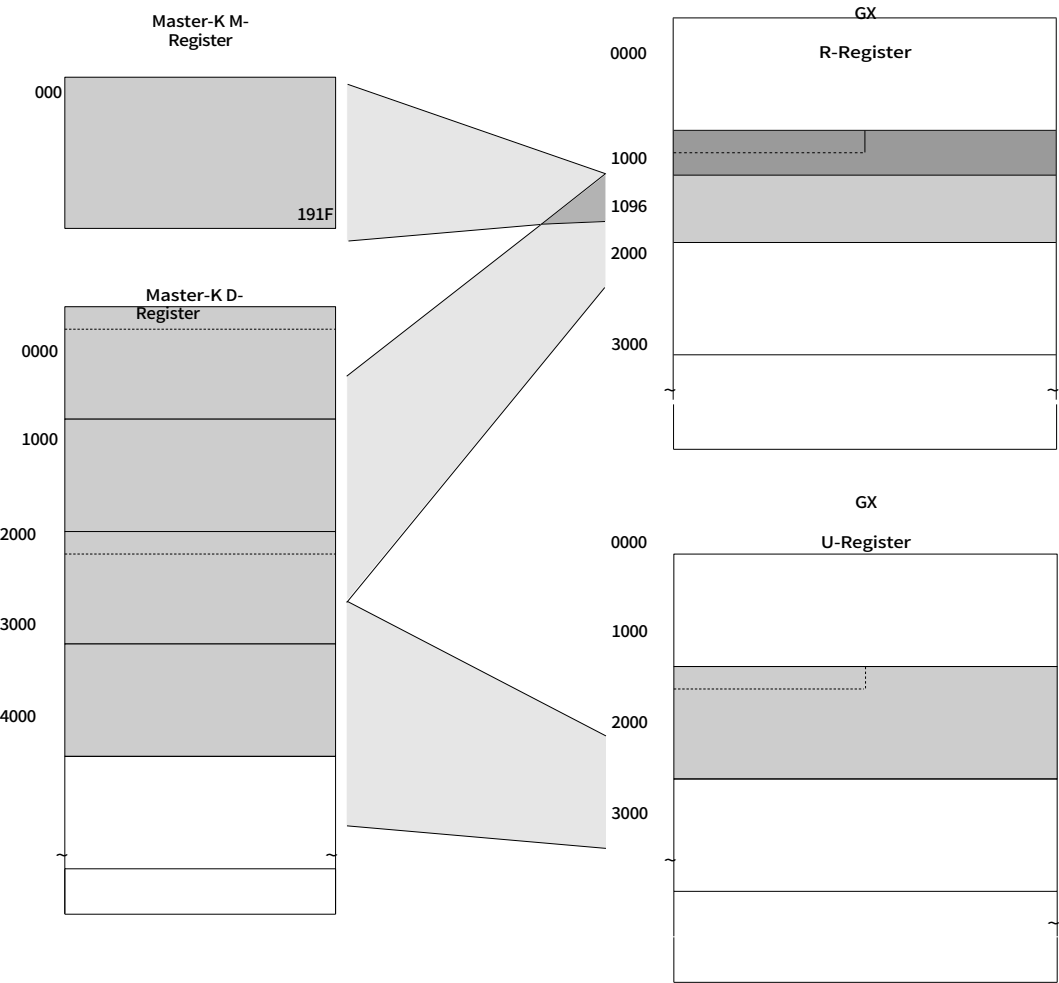
◇ Protocol

Target Module	Available Communication Frames	Communication Data Code
RS232 Module (G6L-CUEB)	CNET Dedicated Protocol	ASCII Code
Ethernet Module (G6L-EUTB)	ENET Dedicated Protocol	ASCII Code

◇ Supported Commands

Item Item		My Use
Individual Read	Bit	Rea d s a bit-type device
	Word	Read a word-type device
Individual Write	Bit	Writing data to a bit-type device
	Word	Writing data to a word-type device
Sequential Read	Word	Reading consecutive devices in Word format
Sequential write	Word	Writing data to a contiguous device of type Word

◇ Register Mapping



GX (1 word = 32 bits)	LG Industrial Systems' Master-K (1 word = 16 bits)
R1000 ~ R1999 (1000 words)	M0000 ~ M0191F (192 words) D0000 ~ D1999 (2000 words)
U1000 ~ U1999 (1000 words)	D2000 ~ D3999 (2000 words)

■ Mitsubishi's Melsec

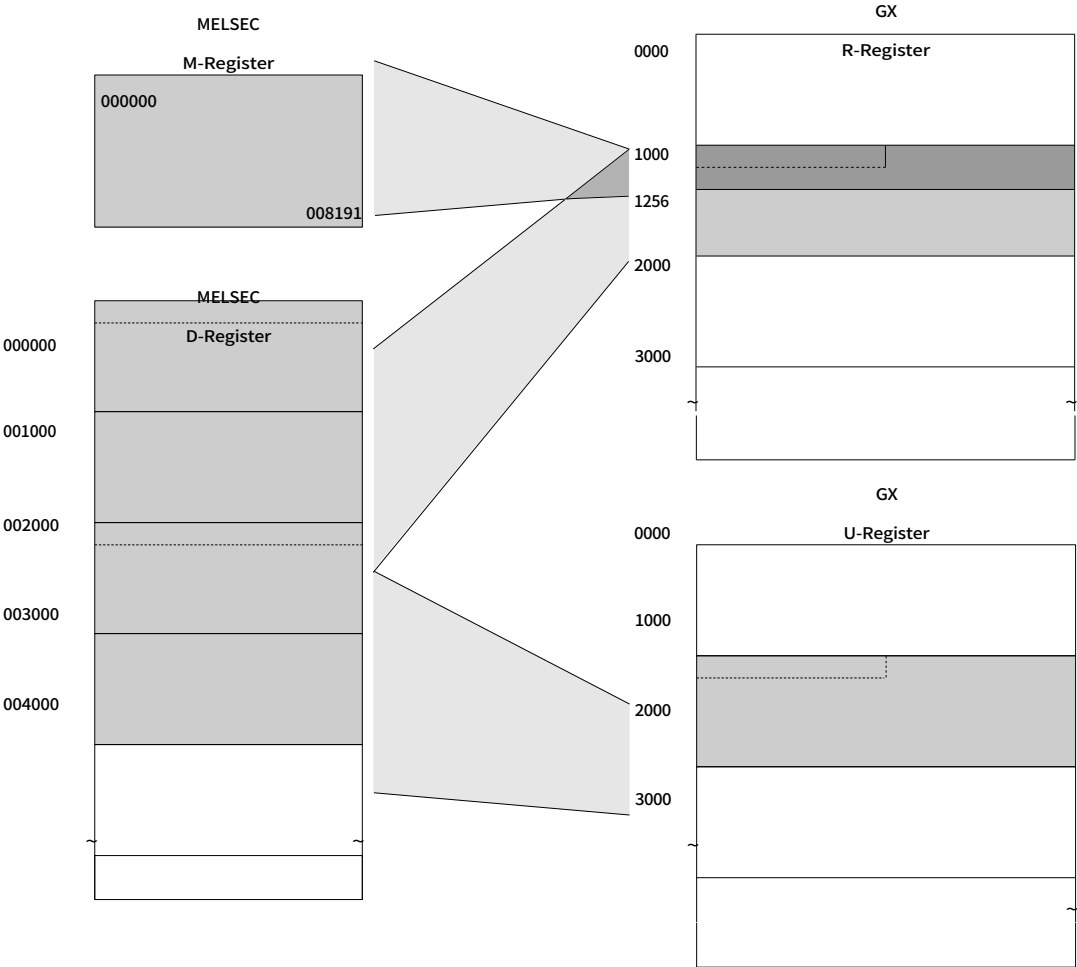
◇ Protocol

Target Module	Available Communication Frames		Communication Data Code
Q Series C24	QnA-Compatible 3C Frame	Formats 1–4	ASCII Code
	QnA-Compatible 4C Frame		
Q Series E71	QnA-Compatible 3E Frame		ASCII Code

◇ Supported Commands

Item			My Use
De vic e Me m ory	Batch Read	Bit	Reads bit devices (X, Y, M, etc.) one point at a time.
		Word	Reads bit devices (X, Y, M, etc.) in 16-bit units.
			Read word devices (D, R, T, C, etc.) one point at a time.
	Batch Write	Bit	Writes bit devices (X, Y, M, etc.) in 1-point increments.
		Word	Write bit devices (X, Y, M, etc.) in 16-bit units.
			Word devices (D, R, T, C, etc.) are used in 1-point increments.
	Random Read	Word	Read bit devices (X, Y, M, etc.) in 16-point or 32-point increments by randomly assigning devices and device numbers.
			Read word devices (D, R, T, C, etc.) in 1-point or 2-point increments by randomly specifying the device and device number.
	Random Write	Bit	Writes to bit devices (X, Y, M, etc.) in 1-point increments, randomly assigning the device and device number.
		Word	Write to bit devices (X, Y, M, etc.) in 16-point increments, randomly assigning the device and device number.
			Writes to word devices (D, R, T, C, etc.) in 1-point or 2-point increments, randomly assigning the device and device number.

◇ Register Mapping



GX (1 word = 32 bits)	Mitsubishi's Melsec (1 word = 16 bits)
R1000 ~ R1999 (1000 words)	M000000 ~ M008191 (256 words) D000000 ~ D001999 (2000 words)
U1000 ~ U1999 (1000 words)	D002000 ~ D003999 (2000 words)

6.5 Third-Party HMI

By connecting third-party HMI modules to the GX system for communication, you can exchange and diagnose desired information via third-party protocols.

This manual describes the interface between the GX system and third-party HMI modules, specifically the Proface/M2I products' GP (Graphic Panel).

6.5.1 Graphic Panel

Proface's GP can connect to the GX system using the MELSEC protocol and the Master-K protocol, while M2I's GP connects using the Master-K protocol.

When creating a Graphic Panel project, configure the target device as follows.

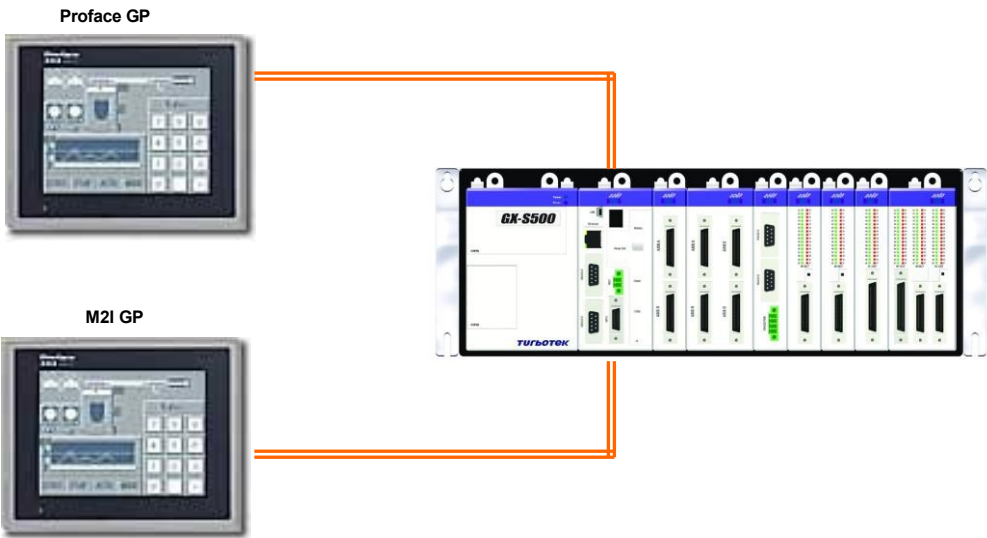
■ Proface GP Settings

	Target Device	Protocol Type
Mitsubishi	MELSEC-QnA (LINK)	3C Frame Type 4
LG	MASTER-K **S SERIES CNET	

■ M2I GP Settings

	Target Device	Protocol Type
LG	LG: Master-K (LINK)	

※ The protocol type is fixed within the GP, so no separate configuration is required.



6.5.2 GX System Settings

Configure the GX system's communication port to match the protocol setting status of the GP module you intend to use.

Open the device parameter settings window in the GX Builder register editor and configure the following settings for the two RS232C ports of the GX CPU module.

■ GX RS232C Port Settings

Item	Description
Master/Slave	Always set the GX system as Slave
Communication Protocol	Set to the same protocol as GP (Melsec / MK)
Communication Protocol Number	When the protocol is set to Melsec (3C Frame Type4)
Baud rate	Maximum 38400 (2400/4800/9600/19200/38400)
Data Bit	8 bits
Parity Bit	None
Stop Bit	1 bit

Device NO.1 세부 설정

디바이스 종류 선택

항목	선택
디바이스 종류	CPU 디바이스
제품 모델 번호	G5CPU
제품 버전	0

내용 갱신

디바이스 상세 내용

No	항목	값	Bit	크기
	[RS232_1 포트]			
151	Master/Slave	Slave	0	1B
152	통신 프로토콜	Melsec	0	8B
152	통신 프로토콜 번호	3CFrame Type4	8	24B
154	Baudrate	38400	0	8B
154	Data Bit	8	8	8B
154	Parity Bit	NONE	16	8B
154	Stop Bit	1	24	8B
156	TimeOut	0	0	1W
	[RS232_2 포트]			

Hex

적용

확인

닫기

6.5.3 Wiring Diagram

The following is the wiring diagram for the RS232C connection cable.

■ Proface

Proface (25-pin)			GX (9-pin)	
Pin	Signal		Signal	Pin
1	FG		CD	1
2	TXD	→	RXD	2
3	RXD	→	TXD	3
4	RS	→	DTR	4
5	CS	→	SG	5
6	NC		DSR	6
7	SG	→	RTS	7
8	CD		CTS	8
20	ER		RI	9

■ M2I

Proface (9-pin)			GX (9-pin)	
Pin	Signal		Signal	Pin
1	CD		CD	1
2	RXD	→	RXD	2
3	TXD	→	TXD	3
4	DTR		DTR	4
5	SG	→	SG	5
6	DSR		DSR	6
7	RTS		RTS	7
8	CTS		CTS	8
9	RI		RI	9

6.5.4 Register Mapping

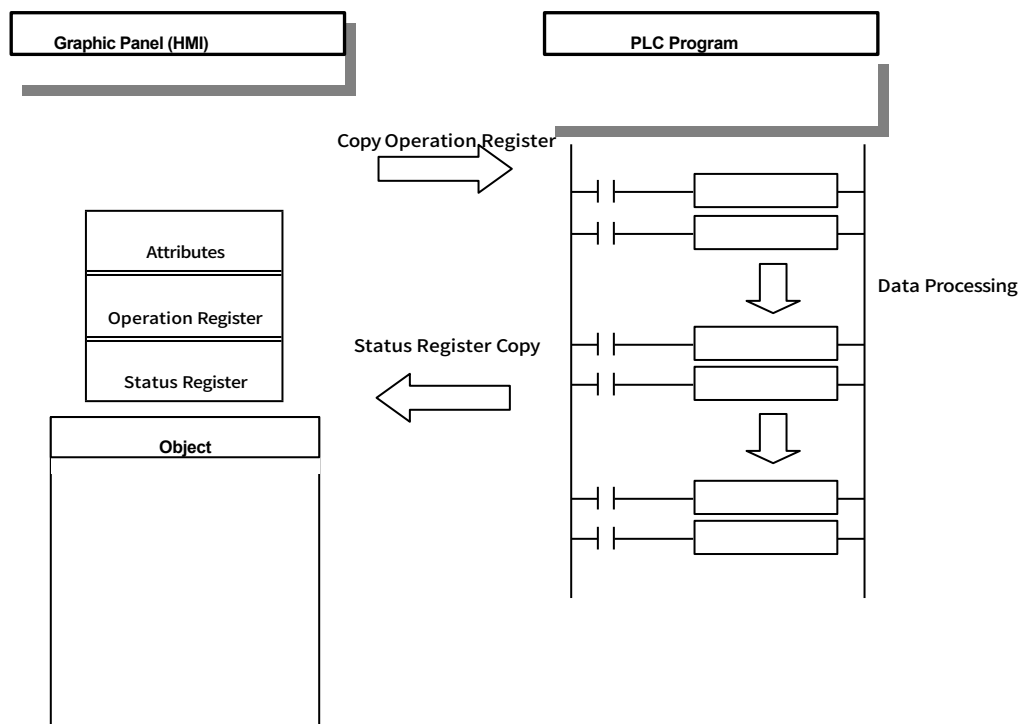
The register mapping is identical to that of third-party protocols. Entering data into the corresponding R register will display the value in the mapped register.

6.5.5 PLC Processing

To display values on Objects configured in third-party HMI screens connected to the GX system, the PLC must perform the following data processing.

Assign an action register and a status register to each object. Use the action register when an object such as a button is pressed, and use the status register to indicate the result or data value.

Inserting a program within the PLC that reads the data value from the action register, processes the data, and then stores the result value in the status register will ensure the GP operates normally.



Remote Device Settings

This document describes various remote devices used in the GX system and explains how to configure each device.

7.1 DeviceNet Device Configuration

7.1.1 s Configuration in GX-Builder

7.1.2 SyCon Program Overview

7.1.3 DeviceNet System Configuration Using the SyCon Program

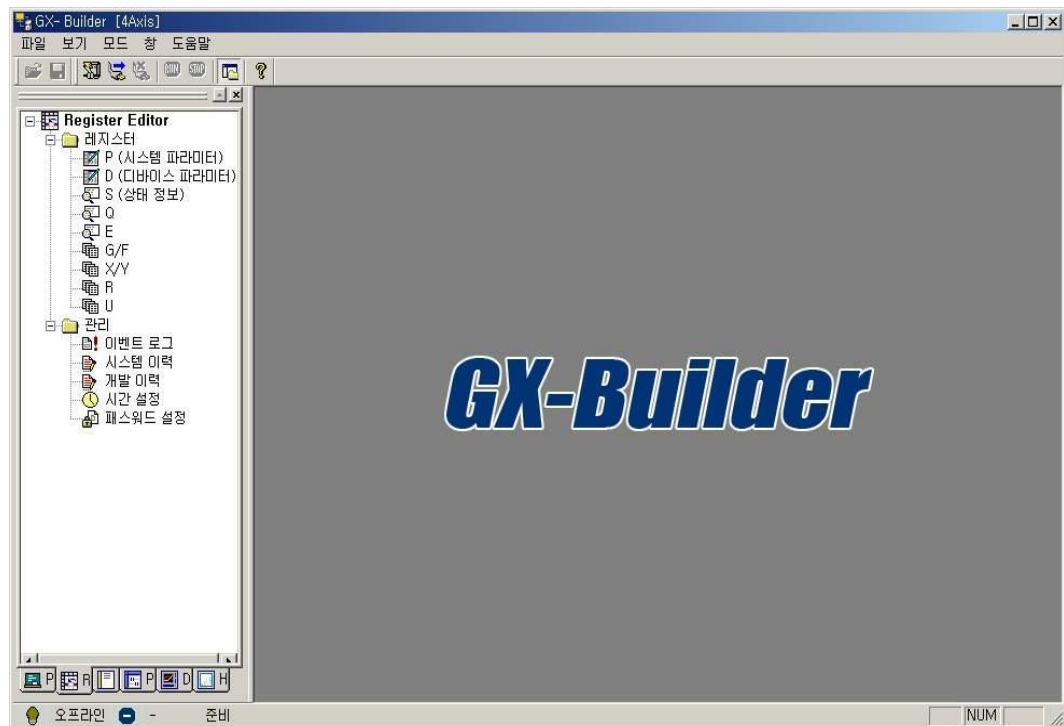
7.1 DeviceNet Device Configuration

7.1.1 Configuration in GX-Builder

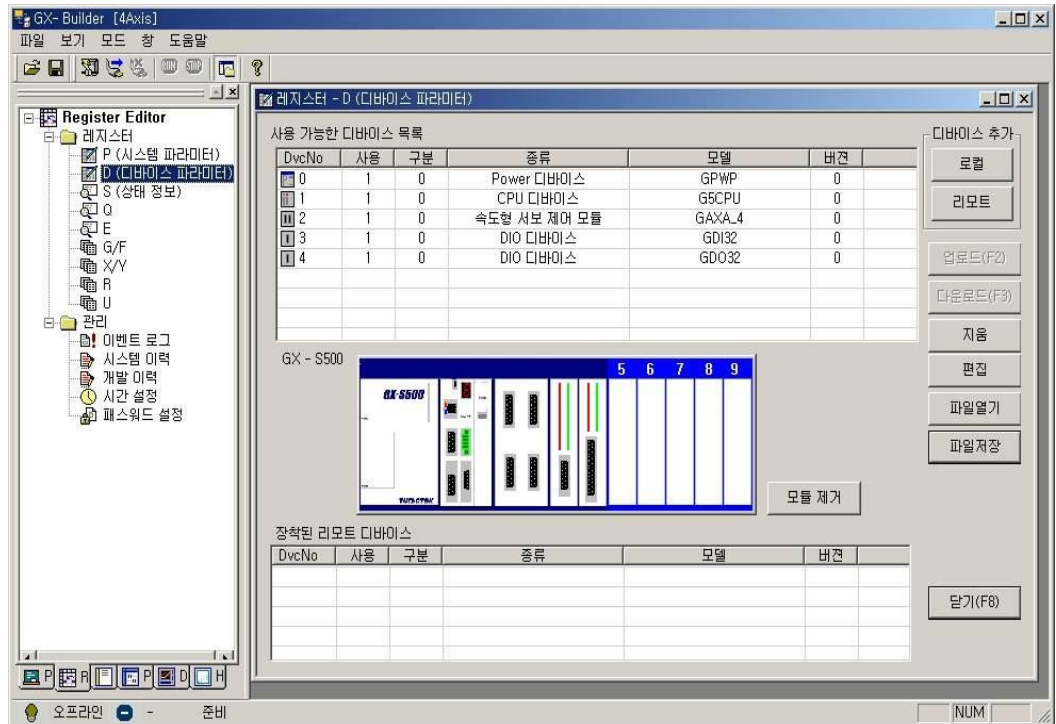
Insert the DeviceNet Master module into GX as a single module. To utilize it, the following two settings must be configured via the GX-Builder program:

- (1) DeviceNet Master Module Settings – Local Device Configuration
- (2) DeviceNet Slave Device Configuration – Remote Device Configuration

To configure the DeviceNet Master module, create a project and display the Register Editor as follows: either select it from the Project Manager appearing in the workspace, or select the Register tab from the Module Selection tab.



Within the register menu displayed in the workspace, you can configure the DeviceNet Master module using the Local Device Settings under Device Configuration via D (Device Parameters), and configure the DeviceNet Slave device using the Remote Device Settings. To do this, double-click D (Device Parameters) in the Register Editor of the workspace to open the corresponding window.



7.1.1.1 Master Configuration

You can add a DeviceNet Master module using the Local Device Settings. To do this, Click the button to add a new device to the list of available devices.

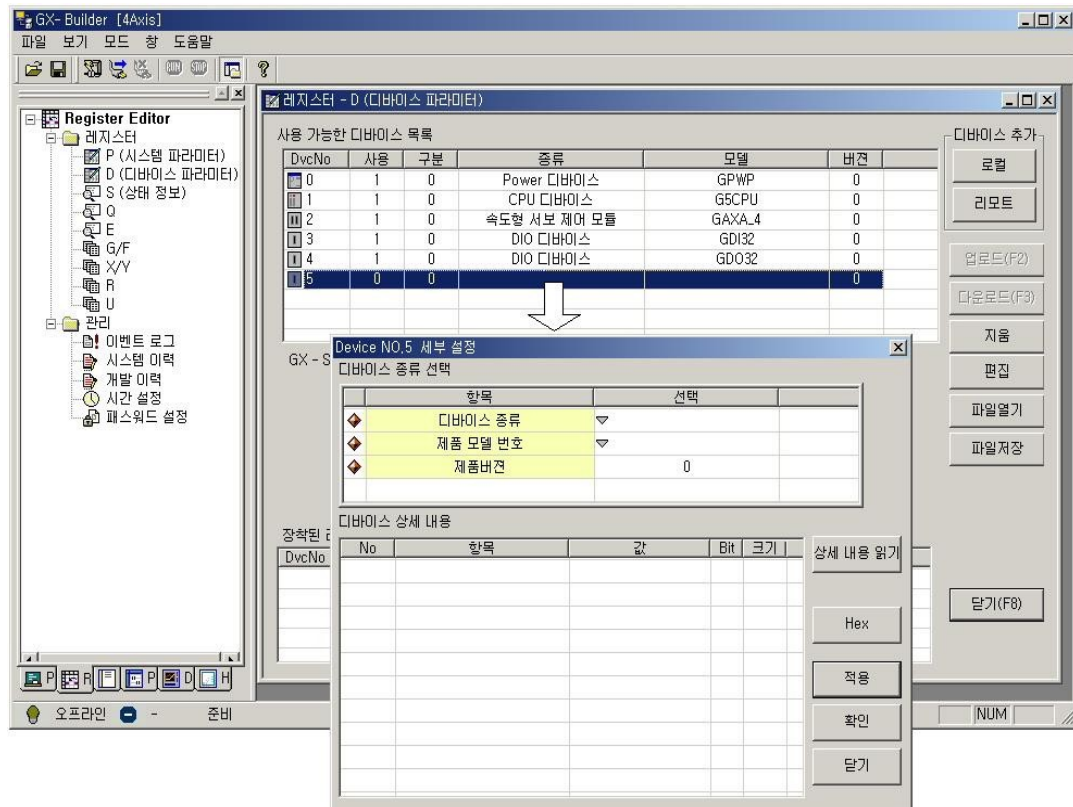
Local

The module appears. To apply the module added through detailed settings as a DeviceNet Master module,

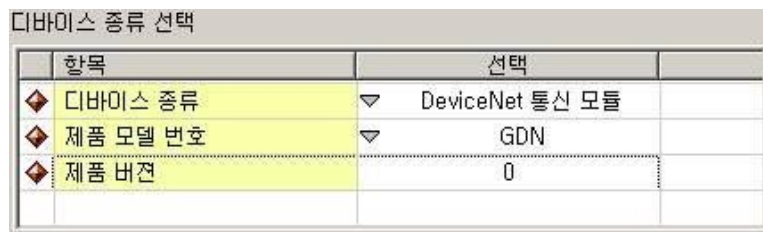
Double-click the relevant content or select the

Edit

select the button.



The detailed settings window that appears is divided into the "Select Device Type" and "Device Details" sections. First, use the dropdown button in the "Select Device Type" section to choose the device type and product model number as follows.



Now, click the

Read Details

Click the button.

The detailed information items displayed for configuring the DeviceNet Master module are as shown in the following table.

Category	Item	Meaning	
Communication Settings	Module Search Time [msec]	The time required for the DeviceNet Master module to search for the connected to the Slave device during initial boot or after a reset Time required	
	Comm ID (Country Code) unicati on	The country code of the Master module, identical to the ID assigned via the SyCon program, the fieldbus configurator and must be the same value	
Communication Information	Conne Slave Information (Subordinate) ction	DeviceNet Master Slave device	Stored in the module's internal Flash 32 lower ID information
	Conne Slave information (upper) ction	DeviceNet Master Slave device	ID information stored in the module's internal Flash Top 32 ID information
	Conne Slave operation status ction (subordinate)	DeviceNet Master Slave devices	Connected to the module normally ID information for the bottom 32 in Korea
	Conne Slave Operation Status (Upper) ction	DeviceNet Master Slave devices	Connected to the module normally The top 32 ID information

Among the above contents, the user must configure the communication settings items: "Module Search Time [msec]" and "Communication ID (Country Code)". Here, "Module Search Time [msec]" should be set to a sufficient duration (typically 6000~8000) because the longer the number of Slave devices connected to the DeviceNet Master module, the longer the Master module may take to search for Slave devices.

Additionally, the "Communication ID (Local Number)" must be set to match the MAC ID of the Master module designated via the SyCon program.

The information regarding the IDs of Slave devices discovered by the DeviceNet Master module via the SyCon program is stored in the DeviceNet Master module's Flash memory. Among the communication information items, the "Connected Slave Information" item reads this Flash content and informs the user of the IDs of the Slave devices currently configured on the DeviceNet Master module, using the bit positions corresponding to those IDs.

That is, if three Slave devices are connected with assigned IDs of 2, 16, and 32, the "Connected Slave Information (Lower)" field represents information for the lower 32 IDs and thus holds the value 0x0104.

while the "Connected Slave Information (Upper)" field indicates information for the upper 32 IDs, thus holding the value 0x0001. Additionally, among the communication information items, the "Connected Slave Operation Status" field indicates the IDs of the Slave devices currently communicating with the DeviceNet Master module during operation.

디바이스 상세 내용

No	항목	값	Bit	크기
	[통신 설정]			
529	모듈 검색 소요시간[msec]	6000	0	1W
538	통신 ID(자국 번호)	0	0	1W
	[통신 정보]			
572	연결 Slave 정보(하위)	0	0	1W
573	연결 Slave 정보(상위)	0	0	1W
574	연결 Slave 동작 상태(하위)	0	0	1W
575	연결 Slave 동작 상태(상위)	0	0	1W

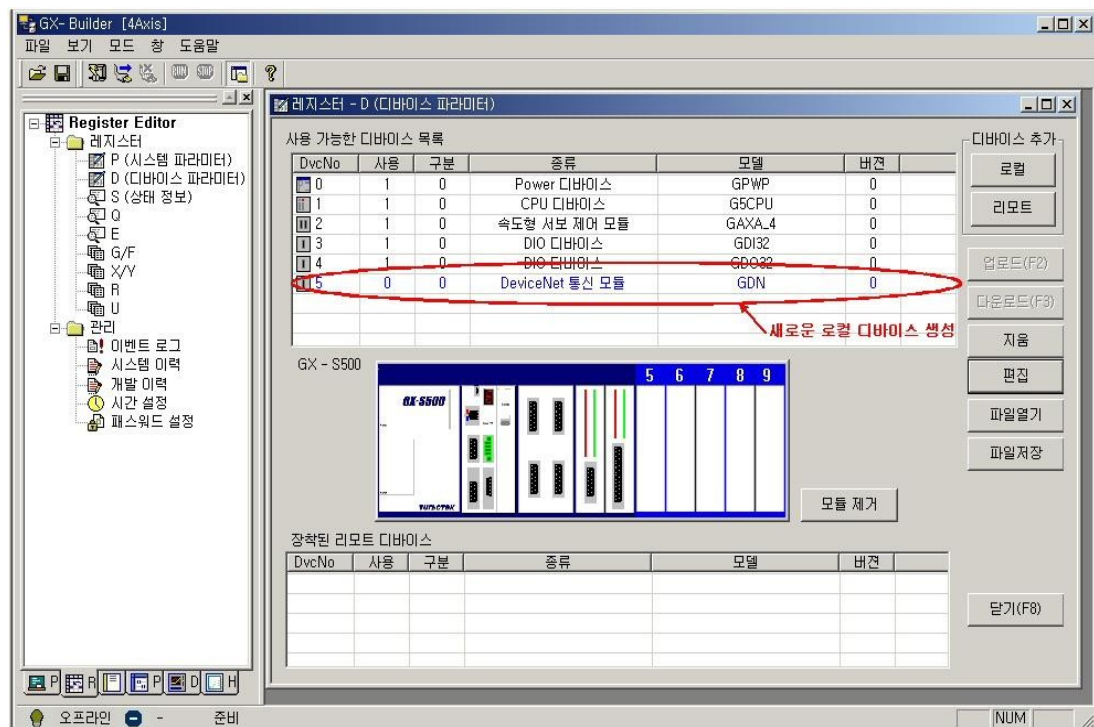
After setting the "Module Search Time [msec]" and "Communication ID (Country Code)"

After setting these items, clicking the button applies the changes.

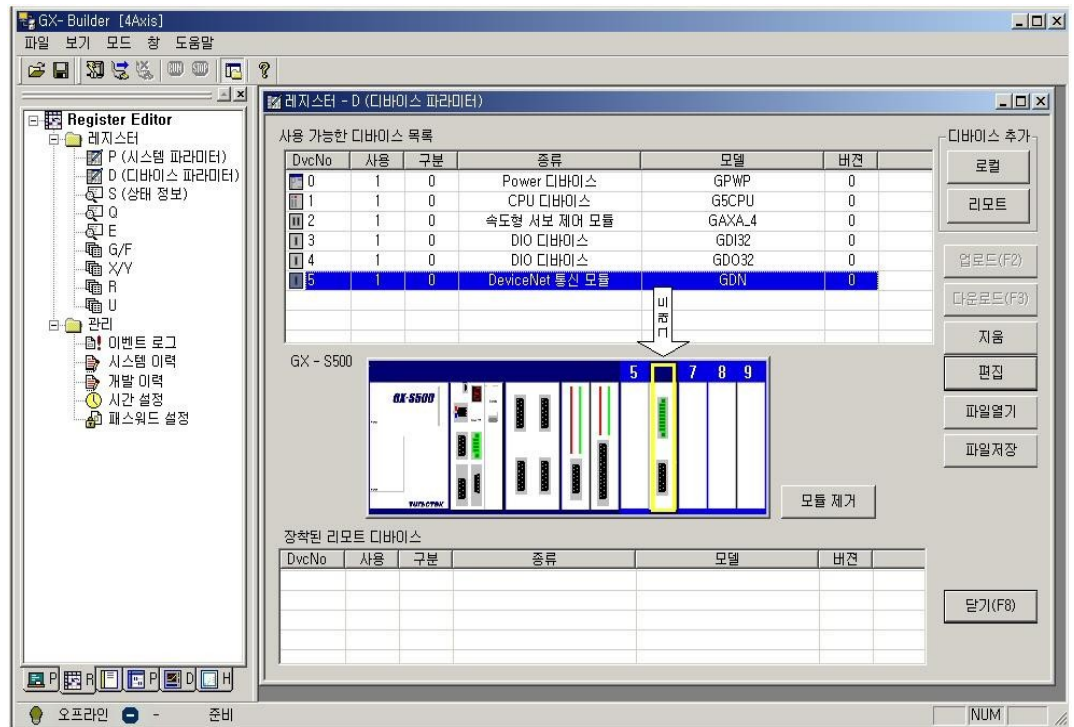
Apply

Confirm

Clicking this button applies the configured settings and closes the dialog box.



The newly applied local device appears in the list of available devices as shown above. Users can configure the DeviceNet Master module by dragging and dropping it to the slot location where the actual DeviceNet Master module will be installed.



7.1.1.2 Slave Settings

You can add DeviceNet Slave devices through Remote Device Settings. This

Remote

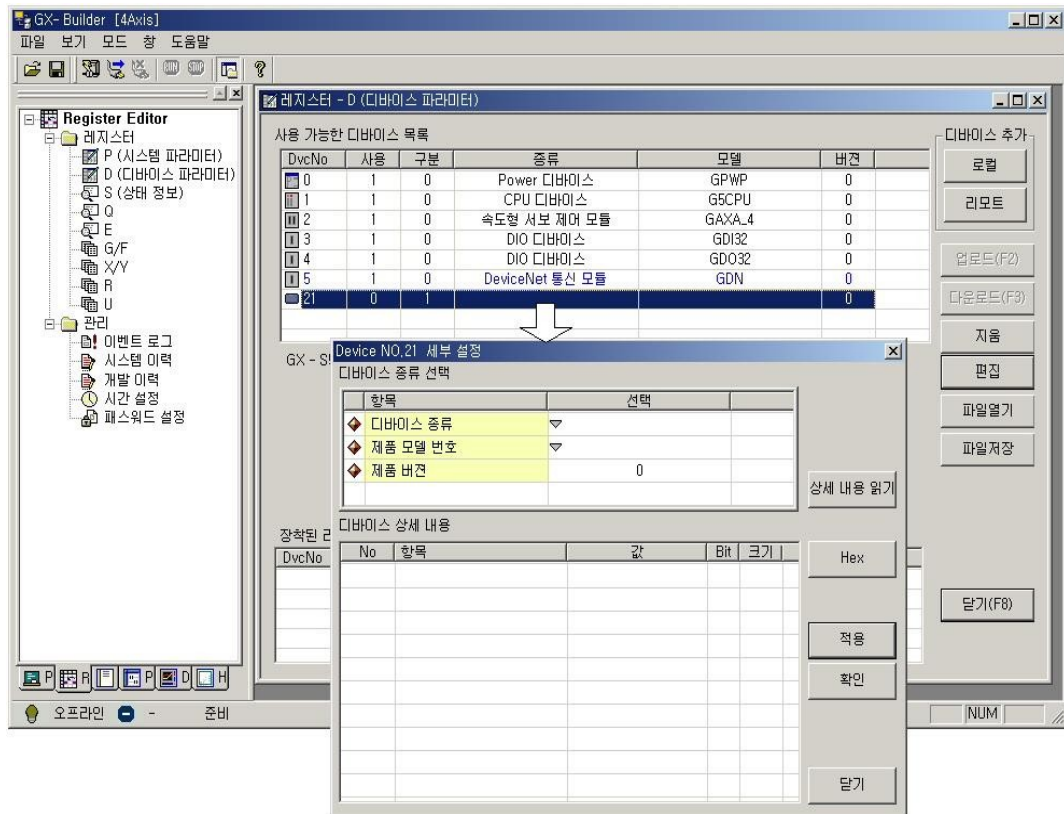
Click the button to add the new device to the list of available devices.

. To use the module as a DeviceNet Slave module, double-click the entry to access the detailed settings.

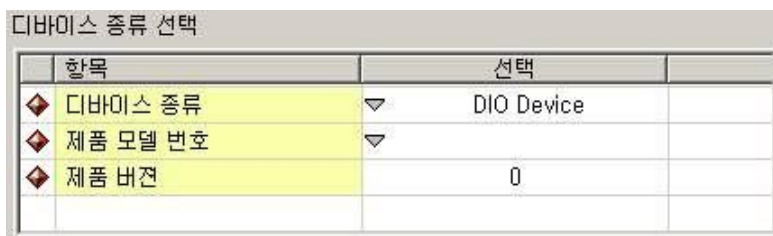
, double-click the entry or select the

Edit

button.



The detailed settings window that appears is divided into the "Select Device Type" and "Device Details" sections. First, use the dropdown button in the "Select Device Type" section to choose the device type and product model number.



Among the above, if the DeviceNet Slave device is a digital input/output device, select "DIO Device"; if it is an analog input/output device, select "AIO Device".

Now, click the button for detailed device settings.

[Read more details](#)

Click the button.

The detailed items displayed for configuring the DeviceNet Slave device are as shown in the following table.

Item	Meaning
Communication (I/F) Method	Select the communication I/F method connecting to the Master module
Master Communication Device No.	Communication device number for the Master device to which the Slave device will be connected Device Number
Master Communication Port No.	When the Master module has multiple communication ports The port number through which communication is performed. Always 1
Device ID (Area Code)	The area code of the slave device, designated by the switch number designated by the mode switch
Number of DI contacts	If the selected device includes digital input contacts Number of input contacts
X Register number corresponding to DI	Number assigned to the X Register for the input contact
Number of DO contacts	If the selected device includes digital output contacts Number of output contacts
Y Register number corresponding to DO	Number assigned to the Y Register of the output contact
Number of AI channels	Number of AI channels Number of input channels
X Register number corresponding to AI	Number assigned to the X Register for analog inputs
Number of channels for AO	If the selected device includes analog output channels Number of output contacts
Y Register number corresponding to AO	Number assigned to the Y Register for analog outputs

For the "Communication (I/F) Method" item in the above content, since the current setting is for a DeviceNet Slave device, select "DeviceNet". The "Master Communication Device No" item is an internally assigned number within the GX. It indicates the number of the Master device with which the Slave device being edited operates, referring to any communication Master module configured in the GX. This number is automatically assigned to the Master device with which the edited Slave device will communicate via drag-and-drop. The value for the "Device ID (Area Code)" field must be set to match the value configured via the rotary switch for the area code setting on the Slave devices.

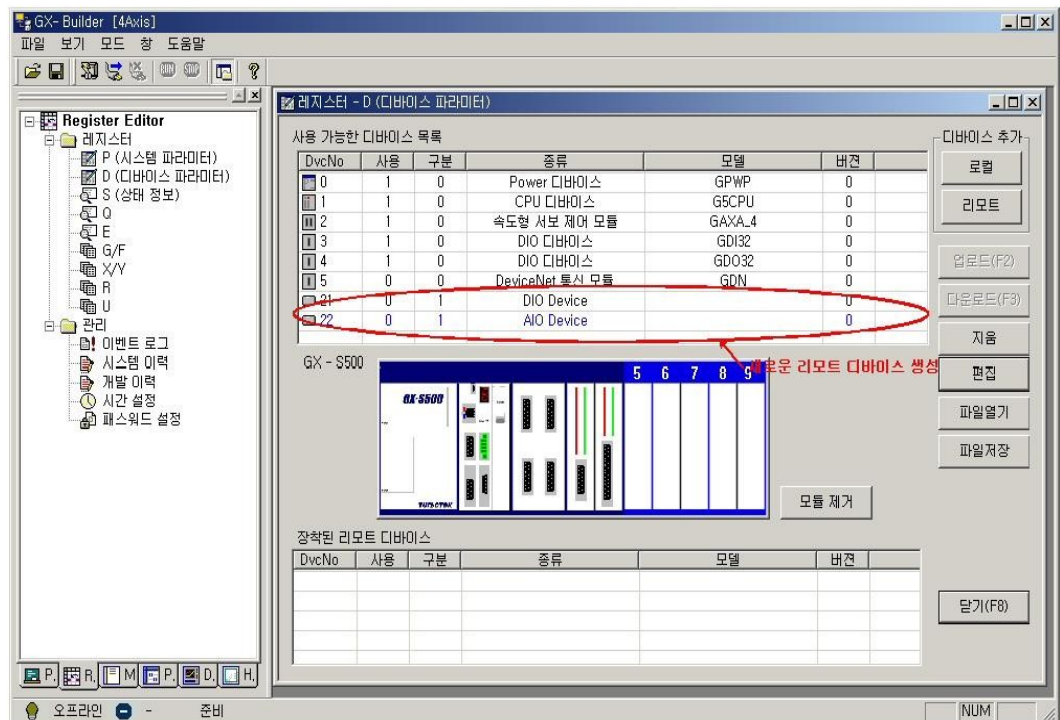
If the module being edited is a device related to digital input/output contacts, content related to DI/DO appears. If it is a device related to analog input/output contacts, content related to AI/AO appears. Complete the settings for the relevant content. When sequentially assigning corresponding X/Y register numbers for DI/DO and AI/AO, each GX register is 32 bits, and X/Y register numbers are assigned as new numbers for each Slave device. For example, if one Slave device has an ID number of '1' and consists of 24 input contacts and 40 output contacts, and another Slave device has an ID number of '2' and consists of 5 analog outputs (16-bit), and X/Y register numbers are set consecutively, it would be as follows.

Settings for the Slave device with ID number '1'	
Device ID (Area Code)	1
Number of DI Contacts	24
X Register Number Corresponding to DI	M (any I Register number)
Number of DO contacts	40
DO corresponding Y Register number	N (arbitrary O Register number)
Settings for Slave device with ID number '2'	
Number of AI channels	0
X Register number corresponding to AI	- (No corresponding value)
Number of Channels on AO	5
Y Register number corresponding to AO	N+2

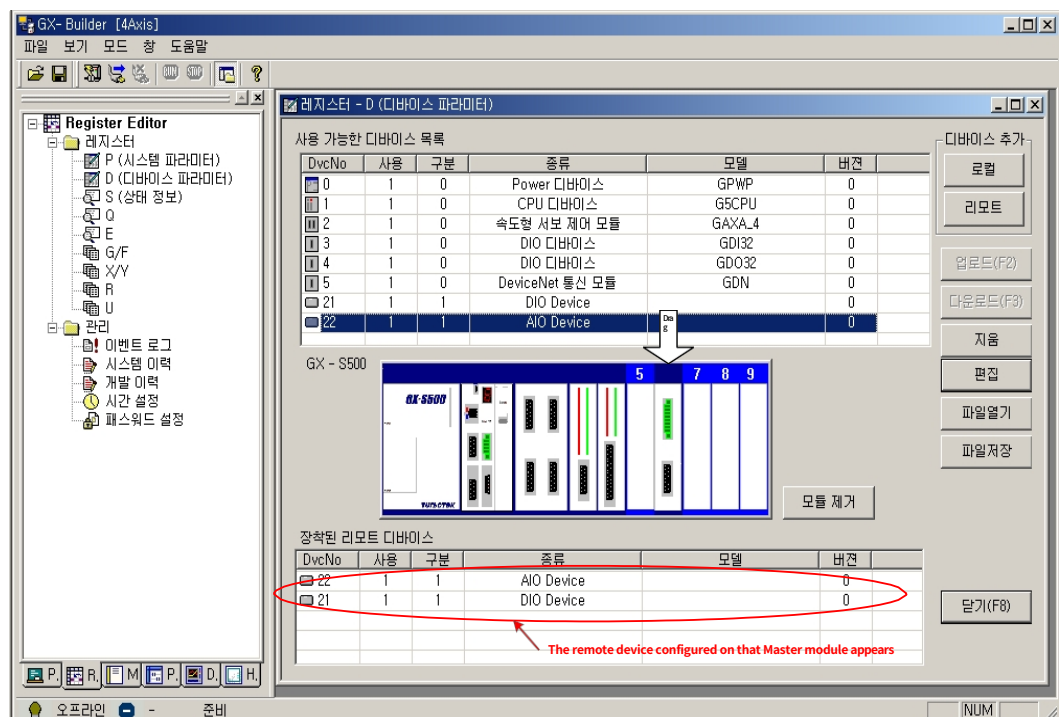
디바이스 상세 내용					디바이스 상세 내용				
No	항목	값	Bit	크기	No	항목	값	Bit	크기
2108	통신(I/F) 방식	DeviceNet	0	1W	2208	통신(I/F) 방식	DeviceNet	0	1W
2113	Master 통신 디바이스 No.	5	0	1W	2213	Master 통신 디바이스 No.	5	0	1W
2114	Master 통신 Port No.	1	0	1W	2214	Master 통신 Port No.	1	0	1W
2125	Device ID(국번)	1	0	1W	2225	Device ID(국번)	2	0	1W
2130	DI 접점 수	24	0	1W	2230	AI 채널 수	0	0	1W
2131	DI에 해당하는 X Register 번호	10	0	1W	2231	AI에 해당하는 X Register...	0	0	1W
2134	DO 접점 수	40	0	1W	2234	AO 접점 수	5	0	1W
2135	DO에 해당하는 Y Register 번호	20	0	1W	2235	AO에 해당하는 Y Register...	22	0	1W

After configuring the details for the two slave devices as above, Apply click the button to apply the changes and close the dialog box. Confirm When you click the button, the configured settings are applied immediately.

The newly applied remote device appears in the list of available devices as follows:



Users can configure the DeviceNet Slave module by dragging and dropping it to the slot location where the Master module is installed for the actual DeviceNet Slave device to communicate with.



7.1.2 SyCon Program Overview

7.1.2.1 SyCon Program

SyCon is a general fieldbus configurator that allows users to configure fieldbus systems such as PROFIBUS, InterBus, CANOpen, SDA, DeviceNet, ControlNet, AS-Interface, and others.

7.1.2.2 System Specifications

- 486/Pentium-class or higher processor
- Windows 95 (with SP1), Windows 98, Windows NT 4.0 (with SP3) or later
- Disk space: 30 to 80 MB
- RAM: Minimum 16MB

7.1.2.3 Installation

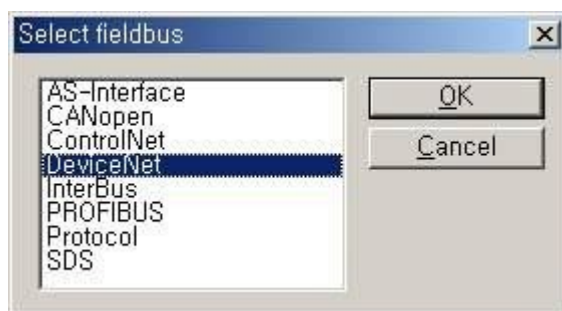
Insert the installation CD to automatically launch the installer. If the installer does not start, run the Autorun.exe file from the CD's root directory. Select YES or NO for the components you wish to install (SyCon System Configurator, OPC Server, license code), then select Next>.

During the installation process, you must enter your user name, company name, and license code. If the license code is not entered, the installed SyCon program will operate as a demo version, recognizing up to two slave devices. All other functions provided by SyCon will be unavailable. Select the fieldbus you wish to install. The required GSD or EDS files will be copied to your computer, and the SyCon program will be installed.

7.1.3 Configuring a DeviceNet System Using the SyCon Program

7.1.3.1 Project Creation: Starting DeviceNet Configuration

To create a new configuration, select the File>New menu. A list of applicable fieldbus systems will appear. Select DeviceNet.



7.1.3.2 EDS (Electronic Data Sheet) File

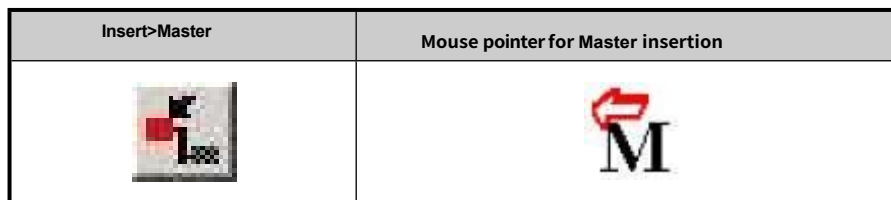
An EDS file is an external file containing the configurable parameters for a specific device. When the SyCon program starts, all EDS files stored in the EDS directory are automatically scanned, and the device names are saved in the internal list. During system configuration, data for a specific device is retrieved directly from the EDS file.

If the internal list does not contain information for a specific DeviceNet device, the corresponding EDS file can be copied to the EDS directory using the File>Copy EDS menu.

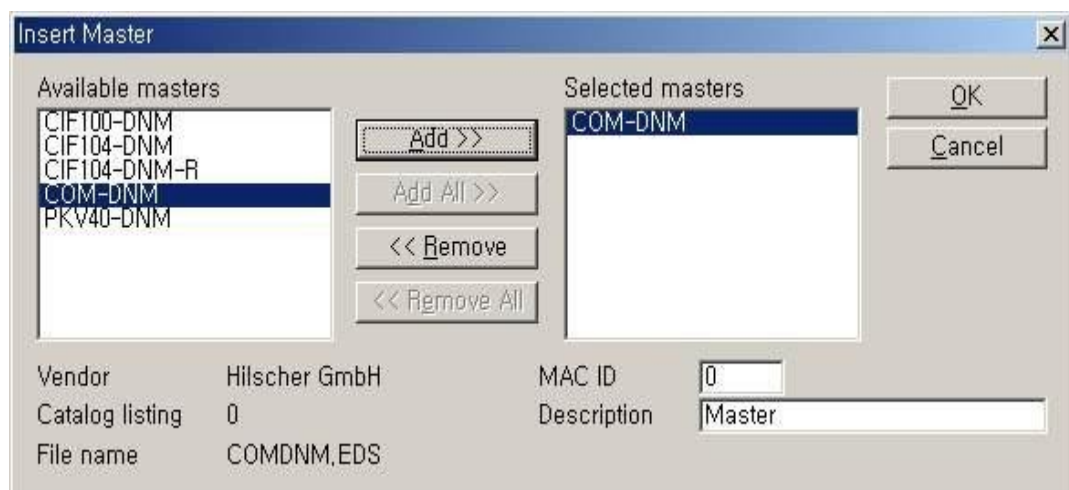
7.1.3.3 Master

■ Inserting a Master

To insert a Master module into the system being configured, select the Insert>Master menu or select the following symbol from the menu bar.



The mouse pointer automatically changes to the shape of the mouse pointer for inserting a Master. When you specify the position to insert the Master, the following dialog box opens to select the Master.



Select "COM-DNM" from the "Available masters" box on the left and click the Add>> button. The selected Master will appear in the "Selected masters" section on the right. Here, you can change the MAC ID for the Master and describe the Master's characteristics in the Description field, limited to 32 characters, which will appear in the main window. Click the OK>> button to apply the settings.



If a user attempts to add another Master while one is already applied, the SyCon program displays a message window asking whether to replace the existing Master.



■ Master Device Assignment

In a typical DeviceNet Master module, the system configurator selects the Settings>Device Assignment menu to configure Device Assignment settings, thereby determining which of the following three methods will be used to communicate with the Slave device:

◇ CIF Device Driver→

CIF Serial Driver xml-ph-0000@deepl.internal

◇ CIF Serial Driver→ Communicates with the Slave device via a serial connection.

◇ CIF TCP/IP Driver→ Communicates with the Slave device via a TCP/IP connection.

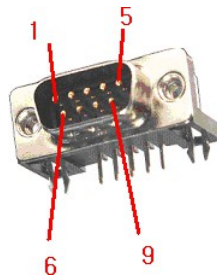
To communicate with the DeviceNet Master module, a GX module, the user must select the CIF Serial Driver.



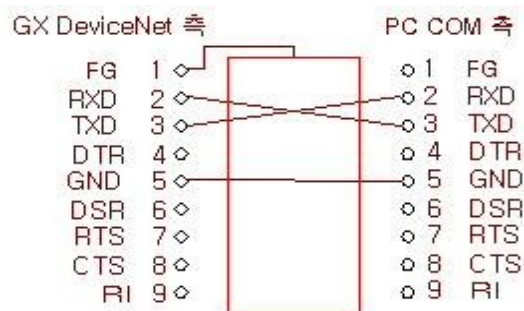
For this purpose, the PC and COM interface must be connected to the GX's DeviceNet module, and the connector specifications are as follows.

Configuration Name	PIN Number	Connector Type			Cable Type
		Module Side	Cable Side	Manufacturer	
PC COM	9	DSUB 9 P MALE	DSUB 9 P FEMALE		
Module COM	9	DSUB 9 P MALE	DSUB 9 P FEMALE		

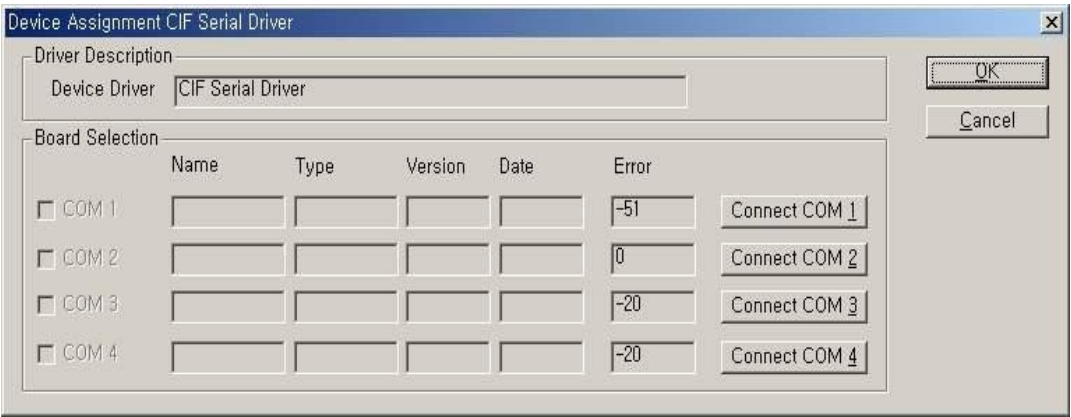
The pin assignments and cable connections for the communication ports on the PC side (COM1, COM2) and the GX DeviceNet module side are as follows.



Number	Signal Name	Direction	Description
1	/CD	Input	Received line signal detector
2	RxD	Input	Received data
3	TxD	Output	Transmitted data
4	/DTR	Output	Data terminal ready
5	GND		Signal ground
6	/DSR	Input	Data set ready
7	/RTS	Output	Request to send
8	/CTS	Input	Clear to send
9	RI	Input	Ring indicator

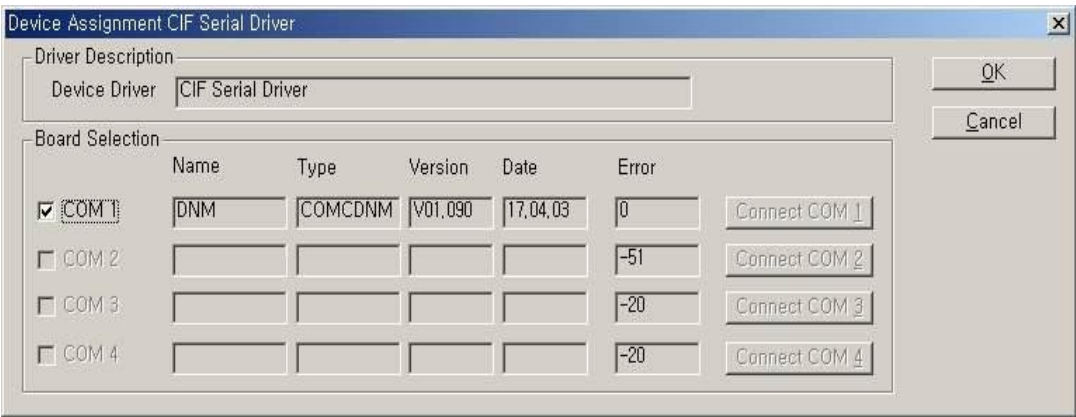


GX DeviceNet module and interface to perform Serial Driver supports COM1 through COM4.



For connection to the COM port of the PC connected to the GX DeviceNet module, configure the corresponding COM port.

Selecting **Connect COM x>>** , the system configurator will connect to the specified COM port via and waits for a response. Once the connection to the device is established, the system configurator The device's firmware information is displayed. If there is no response, a timeout error (-51) occurs.

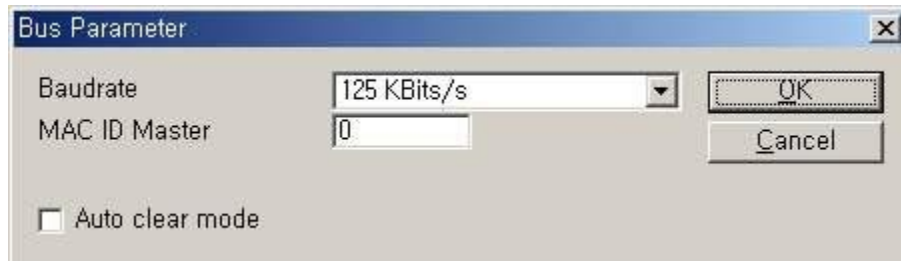


Select the checkbox for the connected device and click the OK>> button.



■ Bus Parameters

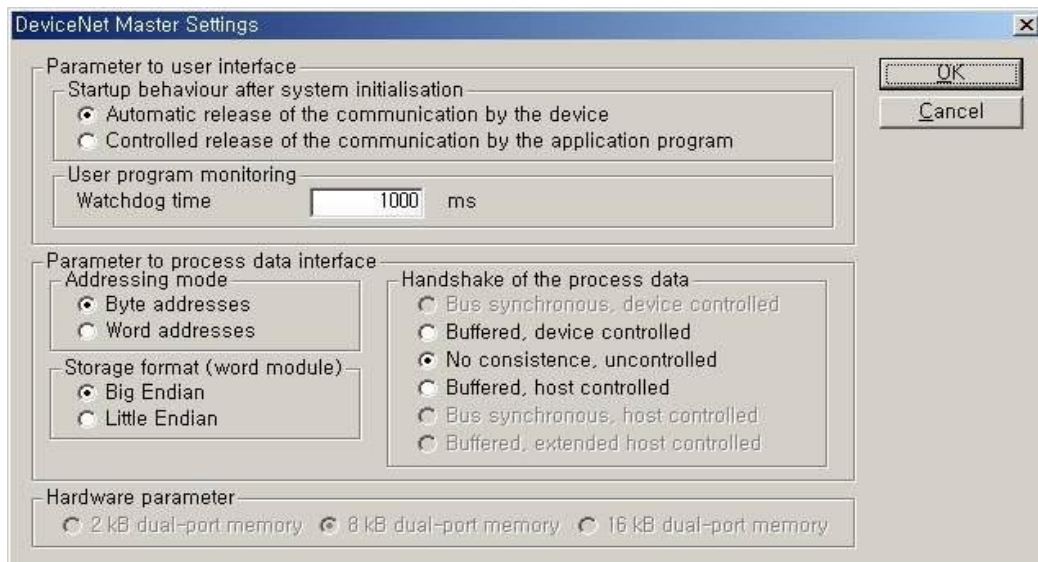
To determine the communication speed (Baudrate) for communication between the DeviceNet Master module and Slave module, select the Settings>Bus Parameters menu with the Master module selected. The communication speeds supported by the GX DeviceNet module through this Bus Parameters window are 125kbit/s, 250kbit/s, and 500kbit/s. Select the desired communication speed and click the OK>> button.



The communication speed of the Slave is automatically set by the detecting Master to the communication speed configured here.

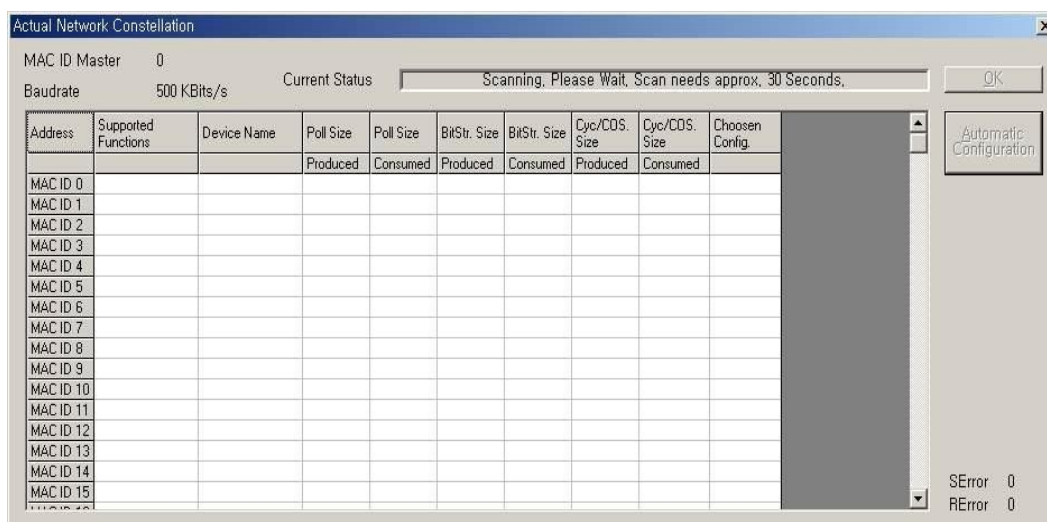
■ Master Default Parameter Settings

Double-click the Master module displayed in the main window, or select the Master in the main window with the mouse and access the Settings>Master Settings menu to configure parameters related to the Master's operation as set by the user. These parameters do not directly relate to the DeviceNet configuration. For using the GX DeviceNet module, the default values can be used as is.

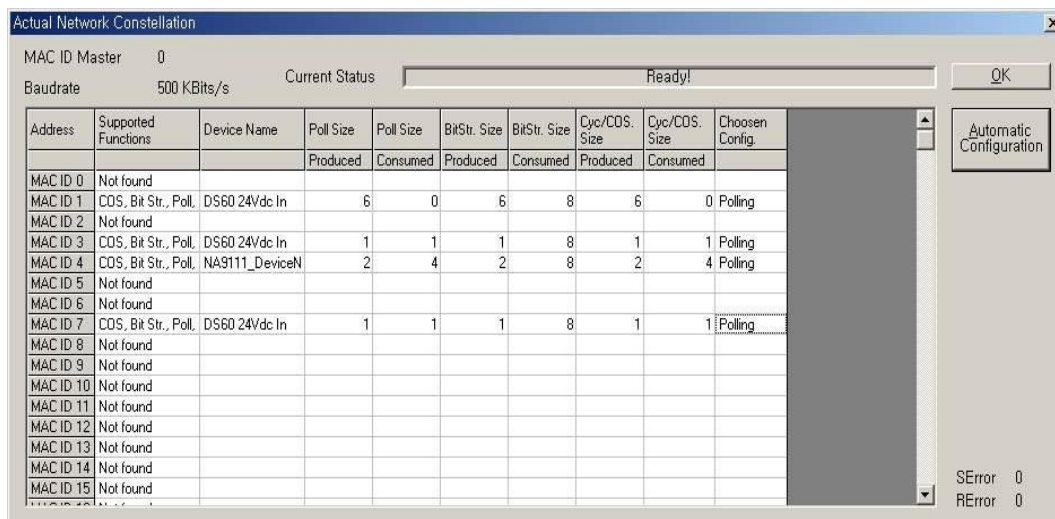


Automatic Network Scan

If you have configured the DeviceNet Master as described above, use the Automatic Network Scan to search the DeviceNet network, i.e., to configure Slave devices. This allows users to quickly configure the network, with detailed parameter settings for Slave devices possible later. To do this, select the Online>Automatic Network Scan menu, which displays the following window.



In the "Current Status" section of this window, the user can observe that the network is initially scanned for approximately 30 seconds. Once the network scan completes, the "Current Status" displays the "Ready!" message, and the discovered Slave devices appear in the table under their respective MAC IDs.



The example shown in the figure above displays slave devices found at MAC IDs 1, 3, 4, and 7. Each column represents the following information:

Column	Variable	of	Supported
Supported functions		Functions supported by the device	(Poll, Bit strobe, Cyclic/change of state)
Device Name		Name of the detected device	
Poll Size Produced		Number of data bytes for the poll connection (input)	
Poll Size Consumed		Number of data bytes for the poll connection (output)	
BitStr. Size Produced		Number of data bytes for the bit strobe connection (input)	
BitStr. Size Consumed		Number of data bytes for the bit strobe connection (output)	
Cyc/COS. Size Produced		Number of Data Bytes for Cyclic/COS Connection (Input)	
Cyc/COS. Size Consumed		Number of data bytes for Cyclic/COS connection (output)	
Chosen Config		Supported features of the device selected by the user, Select the appropriate function desired by the user	

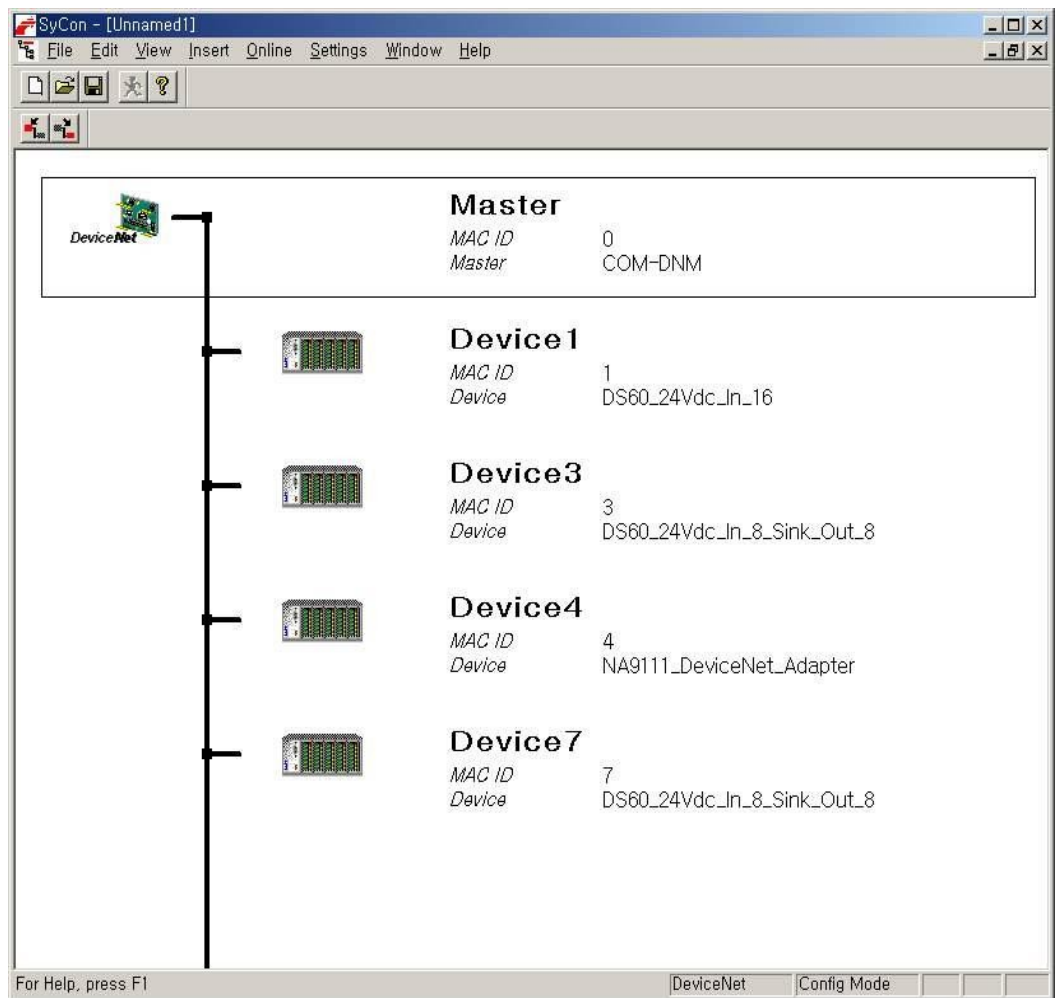
The user selects the appropriate function supported by the Slave device from the "Chosen Config" item in each column. To apply the selected content to the user's system configuration, select the Automatic Configuration button. The following message box appears.



Yes (Y)

Select the button and click the OK>> button in the Actual Network Constellation window to apply the settings.

The information applied here must be referenced when configuring the GX DeviceNet Slave (Remote IO).



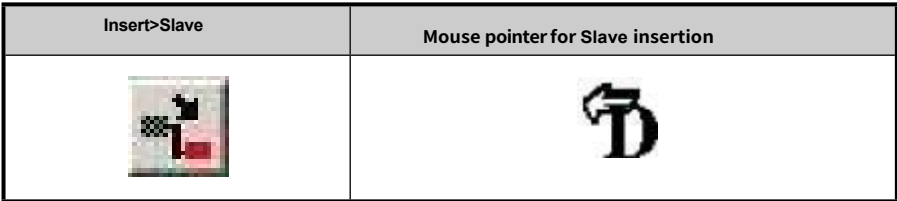
If the user does not wish to use the Automatic Network Scan menu and instead wants to manually configure Slave devices connected to the network, refer to the following section (Slave).

7.1.3.4 Slave

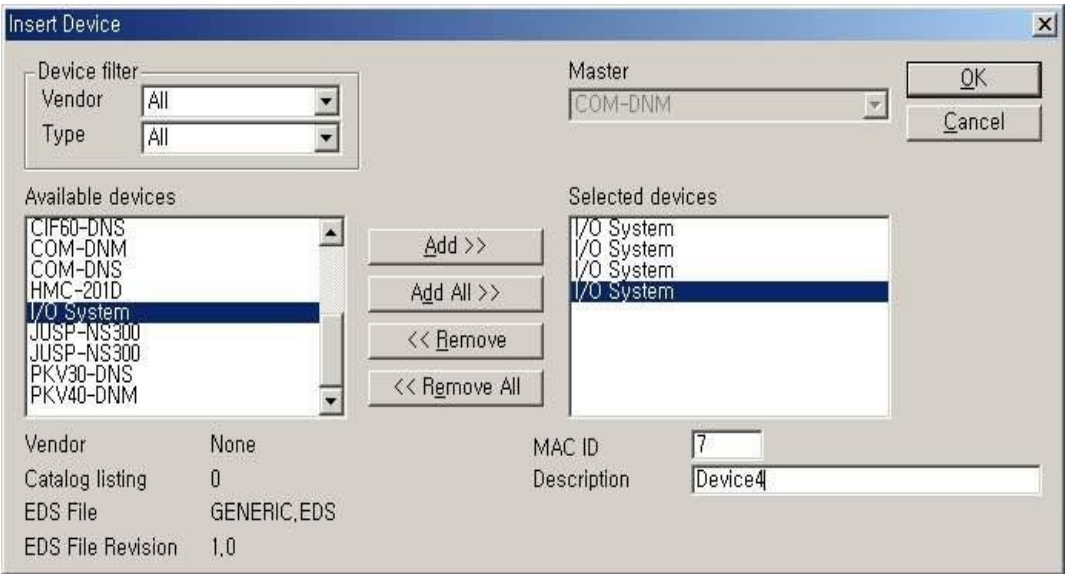
This section is referenced when manually configuring the system for Slaves after configuring the Master, without using the automatic network scan, or when changing the detailed parameters of the Slave based on the scanned information.

■ Inserting a Slave

To insert a Slave device into the system being configured, select the Insert>Slave menu or choose the following symbol from the menu bar.



The mouse pointer automatically changes to the mouse pointer shape for Slave insertion. Once you specify the position to insert the Slave, the following dialog box opens to select the Slave.



The "Available devices" box on the left displays all Slave devices contained in the EDS directory. Selecting any device allows you to view additional information about the selected device at the bottom of the box. Double-click the device you wish to configure or use the Add>> button to add it to the "Selected devices" box on the right.



When adding each Slave module, ensure it has a unique MAC ID that does not overlap with others. This MAC ID must match the MAC ID of the actual Slave device in the system configuration. The content displayed in the Description field is the description for each Slave device and will appear in the main window. Select the OK>> button to apply the settings.

■ Slave Configuration

To configure any device, select the device with the mouse, then choose Settings>Device Configuration. Alternatively, double-click the device with the mouse to display the following device configuration dialog box.

The Device Configuration dialog box contains the following sections and controls:

- MAC ID:** A text field containing the value '1'.
- File name:** A text field containing the value 'GENERIC.EDS'.
- Description:** A text field containing the value 'Device1'.
- Activate device in actual configuration:** A checked checkbox.
- Actual device:** A dropdown menu showing '1 / I/O System'.
- Actual chosen I/O connection:** Radio buttons for Poll (selected), Bit strobe, Change of state, and Cyclic.
- UCMM check:** An unchecked checkbox.
- Group:** A dropdown menu showing 'Group 3'.
- Connection Object Instance Attributes:**
 - Expected packet rate: 200
 - Production inhibit time: 10
 - Watchdog timeout action: Timeout (dropdown)
 - Fragmented Timeout: 1600 ms
 - Produced connection size: 0
 - Consumed connection size: 0
- Parameter Data...** button.
- Available predefined connection data types:**

Data type	Description	Data length
BYTE ARRAY	Input	1
BYTE ARRAY	Output	1
- Add to configured I/O data** button.
- Configured I/O connection data and its offset address:**

Data type	Description	I Type	I Len	I Addr	O Type	O Len	O Addr
- Delete configured I/O data** button.
- Symbolic Names** button.

Through this Device Configuration dialog box, users can assign logical addresses from the Process Data Image on the Master to the device's I/O. The settings configured here are essential reference information when configuring the GX DeviceNet Slave (Remote IO).

The Description and MAC ID fields contain information set when inserting the Slave into the system and can be modified here.

The checkbox for "Activate device in actual configuration" determines whether the device of the node currently being configured will be connected when communicating with the Master. That is, if the device of the node being created is not currently physically configured but will be configured for use in the future, you can uncheck this box and proceed with the configuration.

◇ **MAC ID:** The device's network address, used to identify multiple devices connected to the current DeviceNet network. It must be a value within the range 0 to 63.

◇ **Actual chosen IO Connection:** Select the appropriate type of I/O connection between the DeviceNet Master module and Slave devices (Poll, Bit strobe, COS (Change of Status), Cyclic). Since each Slave device supports different types of I/O connections, when configuring the DeviceNet network system yourself, you must verify which type of I/O connection the Slave device supports.

◇ **Connection Object Instance Attributes:** Configure the connection attributes with the device.

◇ **UCMM Check:** Configure only on modules intending to use the UCMM message format.

◇ **Parameter Data:** Click the " " button to display the parameter data for the " " device's " " parameter.

A dialog box appears to edit the parameter data. The basic parameters for each device are configured in the EDS file. If the appropriate EDS is loaded, you can view the parameter data corresponding to the device.

◇ **Available predefined connection data types:** Apply the I/O configuration of the corresponding Slave device. All data transmitted over DeviceNet is represented in byte form, regardless of its actual format. Therefore, determine the number of bytes required for input data and the number of bytes required for output data from the respective Slave manual, and add the corresponding number of bytes to the configured I/O data.

button.

If the device contains 3 bytes of input (output) data, you can configure a total of 3 bytes of input (output) data by clicking the button 3 times. However, you can simply add a single input (output) data point or directly enter the byte count by clicking the corresponding position in I Len (O Len).

The location addresses within the Process Image Data in the Master corresponding to each byte are applied sequentially starting from the device with the lowest MAC ID. If the user wishes to determine this address directly, disable Settings>Auto Addressing.

The following is a general configuration example for the above content.

Device Configuration

MAC ID: 7 File name: GENERIC.EDS
 Description: ExampleDevice7
☒ Activate device in actual configuration

Actual chosen IO connection:
☒ Poll ☐ Bit strobe ☐ Change of state ☐ Cyclic ☐ UCMM check Group 3

Connection Object Instance Attributes
 Expected packet rate: 200 Production inhibit time: 10
 Watchdog timeout action: Timeout Fragmented Timeout: 1600 ms
 Produced connection size: 1 Consumed connection size: 1

Available predefined connection data types

Data type	Description	Data length
BYTE ARRAY	Input	1
BYTE ARRAY	Output	1

Add to configured I/O data

Configured I/O connection data and its offset address

Data type	Description	I Type	I Len.	I Addr.	Q Type	Q Len.	Q Addr.
BYTE ARRAY	Input	IB	1	9			
BYTE ARRAY	Output				QB	1	5

Delete configured I/O data
 Symbolic Names

OK Cancel
 Actual device: 7 / I/O System
 Parameter Data...

Slave Default Parameter Settings

DeviceNet Slave Settings

Parameter to user interface

Startup behaviour after system initialisation

☒ Automatic release of the communication by the device
☐ Controlled release of the communication by the application program

User program monitoring

Watchdog time: 1000 ms

Parameter to process data interface

Handshake of the process data

☐ Bus synchronous, device controlled
☐ Buffered, device controlled
☒ No consistence, uncontrolled
☐ Buffered, host controlled
☐ Bus synchronous, host controlled
☐ Buffered, extended host controlled

☐ AutoBaud

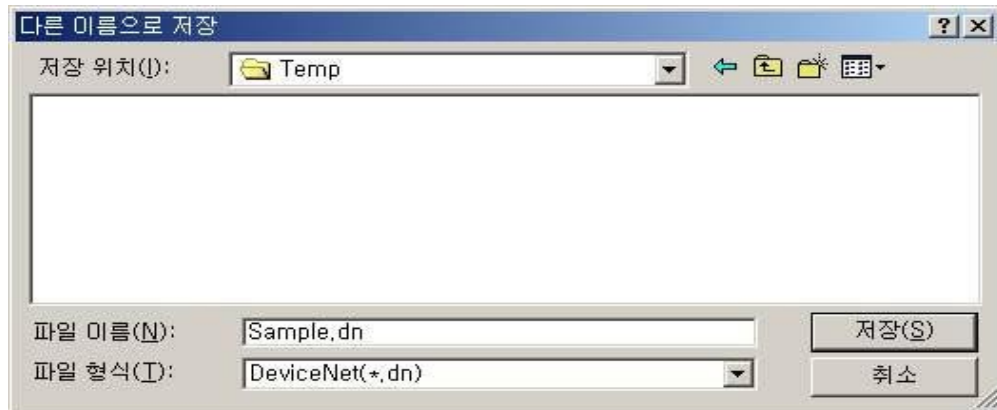
OK Cancel

Through the "Device Settings" item for the DeviceNet Slave, users can configure parameters determining how the Slave device will operate on the user interface. This menu is only available for Hilscher devices.

To configure the DeviceNet Slave settings, select the desired Slave with the mouse, then choose the Settings>Device Settings menu. This will display the dialog box shown above. Select the appropriate options to match the desired Slave behavior.

7.1.3.5 Saving the Project

To save the project configured above, select the File>Save menu or the File>Save As menu. Specify the appropriate directory and project name where you wish to save the project, then save it.



7.1.3.6 Downloading the Configured System Content

To download the configured content to the GX DeviceNet Master module via the above process, select the Online>Download menu. A warning message will appear stating that ongoing communication on the DeviceNet will stop. The user must confirm that stopping the currently communicating content will not affect the system.

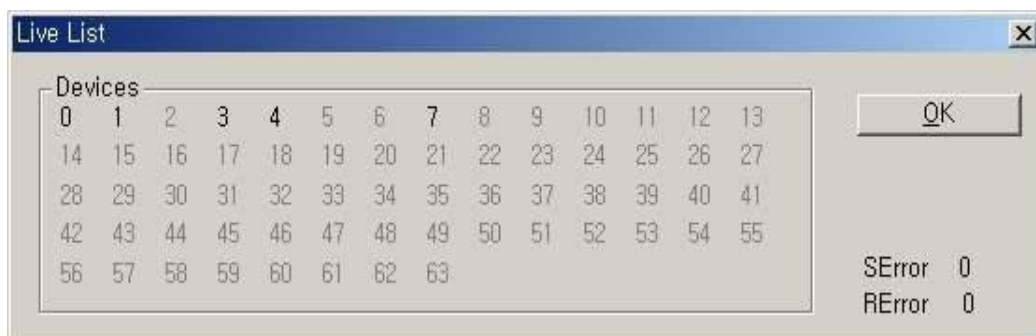


Downloading the configured system will now begin.



7.1.3.7 Simple Diagnostic Test

To perform a quick check of the downloaded system configuration, select the Online>Live List menu. This allows users to detect the MAC IDs of all Slave devices currently deployed and operating on the network. Verify this information against the downloaded content.



As shown in the example above, the MAC IDs of devices currently connected to the network are displayed in black, while those not connected appear in gray.

Register

8.1 Register Overview

8.2 P Register

8.3 D Register

8.4 S Register

8.5 Q Register

8.6 E Register

8.7 G/F Register

8.8 X/Y Register

8.9 R Register

8.10 U Register

8.1 Register Overview

8.1.1 Configuration

8.1.1.1 Register Configuration and Description

■ Register Configuration and Description

Classification	Name	Description
System	P	System Parameters
	D	Device Parameters
	S	Status Information Parameters
	E	System Event Information (Alarm/Warning)
	U	Motion Program Global Variables
	#	Motion Program Local Variables
	Q	System Control Data
PLC	G/F	PLC & System Interface Data
	X/Y	Input/Output Data
	R	PLC Program Global Variables
	#	PLC Program Local Variables

8.1.2 Data Type

8.1.2.1 Data Type Description

■ Data Type and Scope

Data Type	Data Range
Bit (32 Bit)	32 Bit
Word (32 Bit)	-2,147,483,648 ~ 2,147,483,647
Float (32 Bit)	3.4E ± 38 (7 digits)

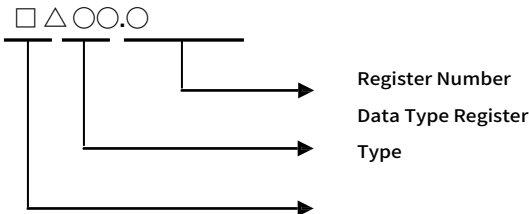
8.1.2.2 Description of Data Types Supported per Register

■ Data types supported per register

Classification	Name	Supported Data Format
System	P	Bit, Word, Float
	D	Bit, Word
	S	Bit, Word
	E	Bit, Word
	U	Bit, Word, Float
	#	Bit, Word, Float
	Q	Bit, Word
PLC	G/F	Bit, Word, Float
	X/Y	Bit, Word
	R	Bit, Word, Float
	#	Bit, Word, Float

8.1.3 Display Method

■ Command Data Structure



■ Data Type

	Data Type	Display Symbol
Register Type	BIT	B
	WORD (32-bit)	W
	FLOAT (32-bit)	F
Constant Type	WORD constant	CW
	FLOAT Type Constant	CF

■ Usage Example

Usage Example	Description
GB 100.10	G Register 100, 10-bit
IW 50	Word-type data in I register 50
GF 200	Float data at register 200 in G register

8.1.4 Parameter Display Format

1) P, D, S, Q, E registers

①	②		
		③	
[Format]	④		
mat			
[Short]	⑤		
t			
[Scope]	⑥		
pe			
[Usage]	⑦		
ge]			

Register Notation Explanation

- ① Indicates the starting number of the parameter.
 - * - After modifying parameters, changes take effect only after selecting Stop->Run.
 - ** - After changing parameters, they will be applied after turning the system power Off->On.
- ② Indicates the parameter's termination number.
- ③ Displays the parameter title.
- ④ Indicates the parameter format (data type (Bit, Word), application type (General, CH, Axis)).
- ⑤ Specifies the parameter unit (Time, Length, Speed).
- ⑥ Specifies the parameter range.
- ⑦ Specifies parameter characteristics and descriptions.

[Notation Example]

**P55			
Full Simulation Mode			
[Type]	Bit type, General		
type [Unit]			
[Range]	0 or 1		
[Usage]			
- 0: Not set.			
- 1: Set.			

2) G/F Register

①	Bit 07	Bit 06	Bit 05	Bit 04	Bit 03	Bit 02	Bit 01	Bit 00
	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 09	Bit 08
	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16
	Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24

Register Notation Explanation

- ① Indicates the register number.
- ② Bits 00 through 31 denote each bit position.
- ___ If marked as shown, it indicates a B contact.

[Marking Example]

G005							GRST	GEMG
	<u>GCEMG8</u>	<u>GCEMG7</u>	<u>GCEMG6</u>	<u>GCEMG5</u>	<u>GCEMG4</u>	<u>GCEMG3</u>	<u>GCEMG2</u>	<u>GCEMG1</u>
	GCHRST8	GCHRST7	GCHRST6	GCHRST5	GCHRST4	GCHRST3	GCHRST2	GCHRST1
F005							FRSTS	
	FCHRSTS8	FCHRSTS7	FCHRSTS6	FCHRSTS5	FCHRSTS4	FCHRSTS3	FCHRSTS2	FCHRSTS1

8.2 P Register (System Parameters)

8.2.1 Configuration

■ Configuration

Description	System Parameter
Data Type	Bit, Word, Float
Structure	
1. Default Settings	① Operation Mode: Simulation Mode
	② Unit System Settings
	③ Task Cycle Settings
2. Control Channel and Axis Settings	① Number of Channels Setting
	② Channel Axis Settings
	③ Axis Selection and Form Setting
	④ Soft Limit Settings
	⑤ Coordinate System Settings
	⑥ Infinite Length Axis Setting
	⑦ Inch/Metric Setting
3. Setting the Transfer Speed	① Settings for Interpolation Feed and Positioning
4. Acceleration/Deceleration Settings	① Setting for Interpolation Feed and Positioning
5. MP Settings	① Positioning, Linear Interpolation, Circular Interpolation Settings
	② Auxiliary code settings
	③ MACRO, Sub-Program Settings
	④ Tool compensation settings
6. MC Settings	① Handle Settings
	② Origin Setting
7. Electronic Gear Ratio Settings	① Gear Ratio Setting
8. Controller Settings	① Position, Speed, Torque Controller Settings
9. Servo Settings	① Motor Settings
	② Encoder Settings
10. In-Position Setting	① In-position Setting
	② Speed Setting
11. Calibration Settings	① Backlash Setting
	① Synchronous Control Settings

12. Extended Function Settings	② External Position Determination Settings
	③ Position Latch Settings
	④ Position Compare Settings
13. Data Table Settings	① Data Table Settings

8.2.2 Detailed Description

 System Settings

**P55	
Full Simulation Mode	

- [Type] Bit Type, General Type
- [Range] 0 or 1 (Default: 0)
- [Usage]
- 0: Not set.
 - 1: Set.
 - This means applying the entire GX system in simulation mode.

*P56	
Axis-by-Axis Simulation Mode	

- [Formula Type] Bit type, General type
- [Range] 0 or 1 (Default: 0)
- [Usage] Purpose]
- 0: Not set.
 - 1: Set.
 - This means applying only the axis in simulation mode in the GX system.

 Unit System Settings

*P220	
Set Unit System	

[Type] Bit Type, General Type
[Range] 0 or 1 (Default: 0)
[Usage]]

- 0 : Metric
- 1: Inch
- Affects the display of PLC MC commands, S registers, P registers, and D registers.

※ Refer to Section 5.1.1 on setting the control function unit system.

*P227	
Minimum Position Unit	

[Type Type] Word Type, General Type
[Scope] 0 to 6 (Default: 0) [Usage] Purpose]

Setting Value	mm	deg	inch
0	0.001	0.001	0.0001
1	10	10	1
2	1	1	0.1
3	0.1	0.1	0.01
4	0.01	0.01	0.001
5	0.001	0.001	0.0001
6	0.0001	0.0001	0.00001

[Example]

- When the value 1000 is entered into P1239 and S400 respectively, it is applied as follows.

	Parameter Setting Value		
Unit	0	2	4
mm	$1000 * 0.001 = 1$	$1000 * 1 = 1000$	$1000 * 0.01 = 10$
inch	$1000 * 0.0001 = 0.1$	$1000 * 0.1 = 100$	$1000 * 0.001 = 1$

When entering 1000 with a setting value of 0 and the unit as mm, the minimum position unit becomes 1mm.

*P228	
Minimum speed unit	

[Type] Word type, General type

[Scope] 0 to 6 (Default: 0) [Usage] Purpose]

Setting Value	mm/min	deg/min	inch/min
0	1	1	0.1
1	10	10	1
2	1	1	0.1
3	0.1	0.1	0.01
4	0.01	0.01	0.001
5	0.001	0.001	0.0001
6	0.0001	0.0001	0.00001

[Example Example]

- When the value 1000 is entered into P2090 and S536 respectively, it is applied as follows.

	Parameter Setting Value	
Unit	0	5
mm/min	$1000 * 1 = 1000$	$1000 * 0.001 = 1$
inch/min	$1000 * 0.1 = 100$	$1000 * 0.0001 = 0.1$

If you enter 1000 and the setting value is 0 with the unit being mm/min, the minimum speed unit becomes 1000 mm/min.

Control / PLC Scan Cycle Setting

*P245	
Interpolation Cycle	
[Type]	Word Type, General Type
[Scope]	msec, from 4 msec to 32 msec (Default: 8) [
	]

*P247	
Position Control-H Cycle	
[Bro]	Word type, General type
[범]	msec, from 1 msec to 16 msec (Default: 4) [
	]

*P248	
Position Control-N Cycle	
[Type Type]	Word Type, General Type
[Range age]	msec, from 1 msec to 16 msec (Default: 4) [
	]

*P251	
PLC-N Cycle	
[Type]	Word Type, General Type
[Scope]	msec, from 4 msec to 32 msec (Default: 16) [
	]

*P253	
Fieldbus Communication Cycle	
[Format]	Word Type, General Type
[Type]	msec, from 4 msec to 32 msec (Default: 16) [
	Purpose]

 PLC Program Settings

*P495	
PLC-H Maximum Execution Steps	
[Bro Type]	Word type, General type [범 Type]
[Usage]	Use]

*P496	
PLC-N L1 Maximum Execution Steps	
[Formula]	Type] Word Type, General Type [Range]
[	Use]

*P497	
PLC-N L2 Maximum Number of Execution Steps	
[Formula]	Type] Word Type, General Type [Range]
[	Use]

 Axis Setting

*P680	
Axis Usage Setting	

[Type Type] Bit type, Axis type
[Range] 0 or 1 (Default: 0) [
]
- 0: Do not use the axis.
- 1: Use axis.
※ Refer to Chapter 5 Control Function Axis Settings (5.1.2)

*P681	
Axis Type	

[Type] Bit Type, Axis Type
[범 We] 0 or 1 (Default: 0)
[Usage]
- 0: Linear axis
- 1: Rotation axis

 Channel Setting

*P885	
Select whether to use control channel	

[Type Type] Bit type, CH type
[Range] 0 or 1 (Default: 0)
[Usage]]
- 0: Disabled.
- 1: Enabled.
※ Refer to Chapter 5 Control Function Channel Settings (5.1.3)

*P886	P893
Axis Usage Status per Channel	

[Type] Word type, CH type
[Type] 0 or 1 (Default: 0) []
- 0: Disabled.
- 1: Enabled.

P894	P965
Setting the axis number corresponding to the axis name per channel	

[Type] Word type, CH type
e

[Type] 1 to 31 (Default: 0)
e

[Usage]

Channel	Register Number	Axis Name	Channel	Register Number	Axis Name
1	P894	X	2	P903	X
	P895	Y		P904	Y
	P896	Z		P905	Z
	P897	A		P906	A
	P898	B		P907	B
	P899	C		P908	C
	P900	U		P909	U
	P901	V		P910	V
	P902	W		P911	W
...	...	...	...	...	...

[Yes Example]

- Channel 1 is set to 1, 2, 3, 4, and Channel 2 is set to 5, 6, 7, 8 as follows.

Channel	Register Number	Axis Number Setting
1	P894	1
	P895	2
	P896	3
	P897	4
2	P907	5
	P908	6
	P909	7
	P910	8

 Soft Limit Setting

P1237	
- Direction Soft Limit Enabled	

[Type] Bit type, Axis type
[Range] 0 or 1 (Default: 0)
[Description]]
- 0: Disabled.
- 1: Enabled.
※ Refer to Chapter 5 Control Function Soft Limit Function section (5.1.4)

P1238	
+ Use of directional soft limit	

[Type] Bit type, Axis type
e
[Range] 0 or 1 (Default: 0)
ge
[Usage] Use]
ge
- 0: Disabled.
- 1: Enabled.

P1239	P1270
- Direction Soft Limit Position	

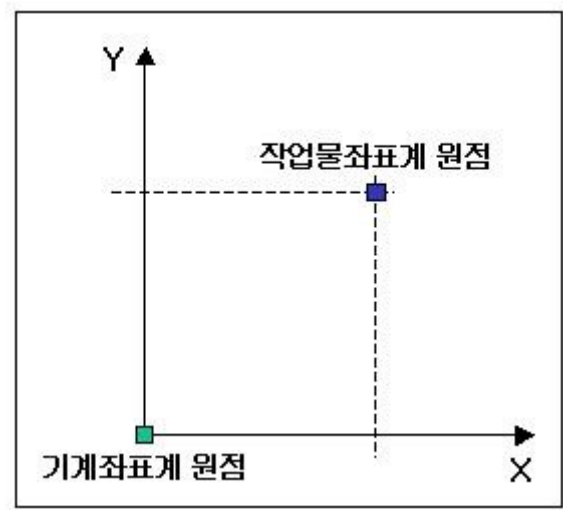
[Type] Word type, Axis type [
] PU
[Scope] -2³¹ to 2³¹⁻¹ (Default: -1000000)
[Usage] Use]

P1271	P1302
+ Direction Soft Limit Position	

[Type Type] Word type, Axis type
[Single] PU
[Scope] -2³¹ to 2³¹⁻¹ (Default: +1000000)
[Use]

● Coordinate System Settings

P1330	
Use of Automatic Work Coordinate System	
[Type] Bit type, Axis type	
[Range] 0 or 1 (Default: 0)	
[Description]]	
- 0: Disabled.	
- 1: Enabled.	
※ Refer to Chapter 5 Control Function Coordinate System Settings (5.1.5)	



P1333	P1364
Automatic Work Coordinate System Position Setting	
[Formula Type] Word type, Axis	
type [Single] PU	
[Scope] -2 ³¹ to 2 ³¹⁻¹ (Default: 0) [In	
Purpose]	
- [Parameter] This parameter defines the offset position relative to the machine coordinate system for each axis when using the [Method].	



Infinite Length Axis Setting

*P1530	
Use of infinite length command	

[Type] Bit type, Axis type

[Range] 0 or 1 (Default: 0) []

— 0: Disabled.

— 1: Enabled.

※ Refer to Chapter 5 Control Function Soft Infinite Transfer Axis Setting Section (5.1.6)

P1533	
Absolute Command Transfer Method	

[Type] Bit Type, Axis Type

[Type] 0 or 1 (Default: 0) []

Purpose]

— 0: Command code direction

— 1: Near Range

P1534	
Absolute Command Endpoint Determination Method	

[Formula] Bit type, Axis type

[범 We] 0 or 1 (Default: 0)

[Usage] Purpose]

— 0: Normal

— 1: Absolute value of command value

P1535	
Incremental command transfer method	

[Type] Bit type, Axis type

[Type] 0 or 1 (Default: 0) []

— 0: As instructed

— 1: Near

*P1538	P1569
Infinite-length axis coordinate Reset position	

[Type] Word type, Axis type [
] PU
[Scope] -2³¹ to 2³¹⁻¹ (Default: 0) [In
Use]

[Example]
- To set 360° on the rotation axis as the Reset position for an infinite-length axis, configure as follows:
Enter 359999 for the corresponding axis in P1538 through P1569. It will increase
from 0° to 359.999° before resetting.

Inch/Metric Setting

P1634	
Command Unit System	

[Type] Bit Type, General Type
e

[Range] 0 or 1 (Default: 0)

ge

[Usage]

ge

— 0: Metric

— 1: Inch

※ Refer to Chapter 5 Control Functions, Inch/Metric Conversion Section (5.1.9)

P1640	
Unit System Switching in MP Mode	

[Type] Bit type, CH type

[범 We] 0 or 1 (Default: 0)

[Usage] Use]

— 0: Do not switch.

— 1: Switch.

— If unit system switching is set to 0 in MP mode, it will be affected by the set unit system and not by the command unit system.

P1644	
Automatic Work coordinate system position value unit system conversion status	

[Type] Bit type, Axis type

[Range] 0 or 1 (Default: 0) []

— 0: Do not switch.

— 1: Switch.

P1652	
Whether to switch units in POS mode absolute commands	

[Type] Bit type, Axis type

[범 We] 0 or 1 (Default: 0)

[Usage] Use]

— 0: Do not switch.

– 1: Switch.

P1653	
Whether to switch units in POS mode incremental commands	

[Type] Bit type, Axis type
e
[Range] 0 or 1 (Default: 0)
ge
[Usage]

- 0: Do not switch.
– 1: Switch.

P1656	
Whether to switch units in JOG mode	

[Type] Bit type, Axis type
e
[Range] 0 or 1 (Default: 0)
ge
[Usage]

- 0: Do not switch.
– 1: Switch.

P1659	
Whether to switch units in HAN mode	

[Type] Bit type, Axis type
e
[범 We] 0 or 1 (Default: 0)
[Usage]

- 0: Do not switch.
– 1: Switch.

P1670	
Position Display Unit Conversion Status	

[Type] Bit type, General type
e
[Range] 0 or 1 (Default: 0)
ge

[Usage]

- 0: Do not switch.
- 1: Switch.

Transfer Speed Setting

Interpolation Transfer Speed

P2090	P2097
Maximum linear feed rate for interpolation feed commands	

[Type] Word type, CH type

[Type] FU

[Scope] From 0 to 2^{31-1} (Default: 10000) []

– If commanded above the set speed, it is limited to the maximum linear travel speed.

※ Refer to Chapter 5 Control Functions, Travel Speed Command section (5.1.8)

P2098	P2105
Maximum rotation feed rate for interpolation feed commands	

[Type] Word type, CH type

[Single] FU

[Scope] From 0 to 2^{31-1} (Default: 360000) []

Purpose]

– If commanded above the set speed, rotation is limited to the maximum feed rate.

P2106	P2113
Default linear feed rate for interpolated feed commands	

[Type] Word type, CH type

[Type] FU

[범 We] From 0 to 2^{31-1} (Default: 1000) [Usage]

Purpose]

– Used when F command is not possible during interpolation transfer.

P2114	P2121
Default rotation feed rate for interpolation feed commands	

[Type] Word type, CH type

[Type] FU

[Scope] From 0 to 2^{31-1} (Default: 36000) []

Purpose]

- Used when F commands are not possible during interpolation transfer.



Positioning Speed

P2132	P2163
Maximum feed rate for position determination commands	

[Type] Word type, CH type

[Single] FU

[범 We] From 0 to 2^{31-1} (Default: 10000) [Purpose]

- If commanded above the set speed, it is limited to the maximum transfer speed.

P2164	P2195
Default feed rate for position determination commands	

[Type Type] Word type, CH type

[Type] FU

[Scope] From 0 to 2^{31-1} (Default: 1000) [Purpose]

- Used when the F command is not possible in the position determination command.



Speed Output

P2206	P2237
Maximum speed in VEL mode	

[Type] Word type, Axis type

[Type] RPM

[범 We] From 0 to 2^{31-1} (Default: 1000) [Purpose]

- If commanded above the set speed, it is limited to the maximum speed.

 Transfer Speed Override

P2287	
Feed Override applied in MP mode	

[Type Type] Word Type, CH Type
[Type] 0 or 1 (Default: 0) []
— 0: Not applied.
— 1: Apply.

P2290	
Apply Feed Override in MC Mode	

[Type] Bit type, Axis type
[Range] 0 or 1 (Default: 0)
[Description]]
— 0: Not applied.
— 1: Apply.

Interpolation Transfer Acceleration/Deceleration Settings

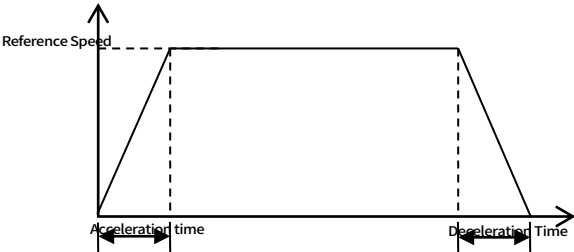
P2484	
Acceleration/Deceleration Selection	

[Type Type] Bit Type, CH Type
[Range] 0 or 1 (Default: 0)
[Usage]

- 0: After interpolation.
- 1: Before interpolation.

※ Refer to Chapter 5 Control Function Acceleration/Deceleration Section (5.1.9)

Pre-interpolation acceleration/deceleration



- The reference speed is the maximum linear feed rate of the interpolation feed command.
- P2090 ~ P2097 (8 CH)

P2492	P2499
Acceleration time	

[Type] Word type, CH type
[Single] msec
[범 Level] 0 to 2³¹⁻¹ (Default: 500) [Usage]
Purpose]

P2500	P2507
Deceleration Time	

[Type Type] Word type, CH type
[Single] msec
[Range] 0 to 2³¹⁻¹ (Default: 500) []

P2526	P2533
Acceleration/Deceleration Filter Constant	
[Type] e	Word type, CH type
[Type] e	msec
[Range] ge	(Default: 0)
[]	
- A filter that smooths acceleration/deceleration profiles.	

 After interpolation, acceleration/deceleration

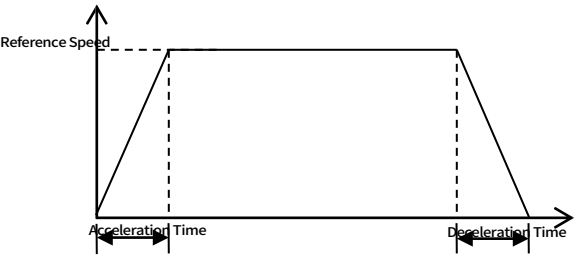
P2552	P2559
Acceleration/Deceleration Profile	
[Type]	Word type, CH type
[Type]	0 to 2 (Default: 0)
[Usage]	
– This feature generates acceleration/deceleration profiles after interpolation. While it enables smoother driving compared to pre-interpolation acceleration/deceleration, it has the disadvantage of causing shape distortion when acceleration/deceleration time increases. – In HAN mode, acceleration/deceleration after interpolation is always applied. – 0 : Straight line shape – 1: S-curve shape – 2: Exponential shape	

P2562	P2569
Acceleration/Deceleration Time	
[Type] e	Word type, CH type
[Type] e	msec
[Range] ge	(Default: 500)
[]	

P2578	P2585
Linear Axis Exponential Acceleration/Deceleration BIAS Speed	
[Formula] Type] Word type, CH type [Type] FU [범 Level] 0 to 2 ³¹⁻¹ (Default: 0) [Usage] Purpose]	

P2586	P2593
Rotary Axis Indexed Acceleration/Deceleration BIAS Speed	
[Type Type] Word type, CH type [Type] FU [Scope] 0 to 2 ³¹⁻¹ (Default: 0) [Usage Purpose]	

 Positioning Acceleration/Deceleration Settings



- The reference speed is the maximum feed rate for the position determination command.
- P2132 ~ P2163 (32 Axis)

P2634	
Acceleration/Deceleration Selection	

- [Type] Bit Type, Axis Type
- [범 We] 0 or 1 (Default: 0)
- [Usage]
- 0: After interpolation
 - 1: Before interpolation

 Pre-interpolation acceleration/deceleration

P2642	P2673
Acceleration time	

- [Type] Word type, Axis type
- [Single] msec
- [Range] 0 to 2³¹⁻¹ (Default: 500) [Purpose]

P2674	P2705
Deceleration Time	
[Type Type]	Word type, Axis type
[Single]	msec
[Range]	0 to 2 ³¹⁻¹ (Default: 500) []

P2772	P2803
Acceleration/Deceleration Filter Constant	
[Type Type]	Word type, Axis type
[Type]	msec
[범 Level]	0 to 2 ³¹⁻¹ (Default: 0) [Usage]
Purpose]	

 Acceleration/Deceleration After Interpolation

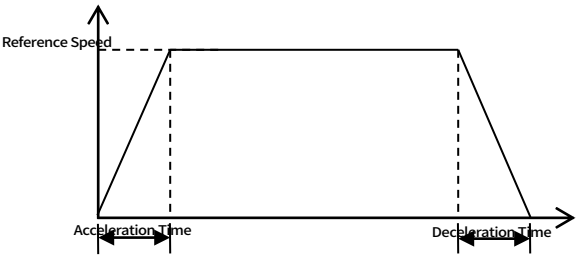
P2870	P2901
Acceleration/Deceleration Pattern	
[Type]	Word type, Axis type
[Type]	0 to 2 (Default: 0)
[Usage]]	
- 0: Straight shape	
- 1: S-curve shape	
- 2: Exponential shape	

P2904	P2935
Acceleration/Deceleration Time	
[Typ]	Word type, CH type
e	
[Typ]	msec
e	
[Ran]	(Default: 500)
ge	
[]	

P2968	P2999
Exponential acceleration/deceleration BIAS speed	

[Type Form Word-type, CH-type
e ula] FU
[Unit Top]
[범 Top] (Default: 0)
[Usage]
ge]

 Speed Mode Acceleration/Deceleration Settings



- The reference speed is the maximum speed in Vel mode.
- P2206 ~ P2237 (32 Axis)

P3089	P3120	
Acceleration time		

[Type] Word type, Axis type
e
[Unit] msec
[Range] (Default: 2000)
ge
[]

P3121	P3152	
Deceleration Time		

[Type Type] Word type, Axis type
e
[Type] msec
e
[Range] (Default: 2000)
ge
[]

 Motion Command Settings

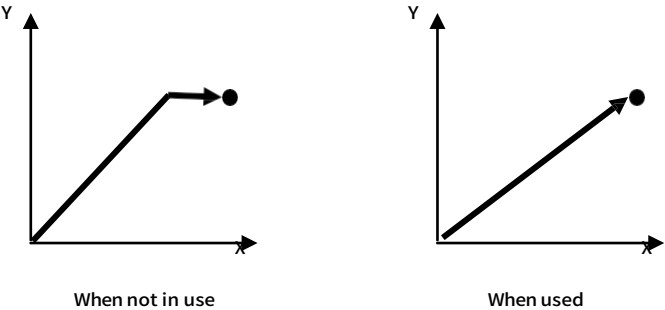
 Position Determination Command

P3702	
Interpolation Positioning Enable	

[Type] Bit type, CH type
[범 We] 0 or 1 (Default: 0)
[Usage] Use]

- 0: Disabled.
- 1: Enabled.

※ Refer to Chapter 5 Control Function Motion Command Function Section (5.1.10)



 Arc interpolation command

P3743	
Whether to regenerate arc center point when issuing I, J, K commands	

[Type] Bit type, CH type
[Range] 0 or 1 (Default: 0)
[Usage]]

- This parameter determines whether to generate an arc alarm when the difference between the two radii at the start and end points exceeds the tolerance during I, J, K arc commands, or to regenerate the arc center point instead.
- 0: Alarm triggered.
- 1: Regenerate arc center point

P3744	P3751
Arc Tolerance Setting	

[Type Form Word Type, CH Type
e ula] pU

[Unit Top]

[범 Top] (Default: 1)

[Usa Purpo
ge se]

- This parameter sets the tolerance for the difference between the two radii at the start and end points when issuing an arc command. A larger value may result in reduced precision at the end of the arc.

P3752	
Use of Orbital Transfer Speed Limitation Function	

[Type] Bit type, CH type

[Range] 0 or 1 (Default: 0)

[Usage] Purpose]

- This function limits the commanded speed by calculating the allowable acceleration and allowable speed based on the commanded speed for circular motion.
- 0: Disabled.
- 1: Enabled.

P3753	P3760
Arc radius to which the arc transfer speed limit function is applied	

[Type Form Word type, CH type
e ula] pU

[Unit Top]

[범 Top] (Default: 0)

[Usa Purpo
ge se]

- Enter the radius of the arc to which the arc machining speed limit function will be applied.

P3761	P3768
Maximum speed for arc feed rate limit function	

[Type] Form Word Type, CH Type
e ula] FU
[Unit] [Top]
[범] (Default: 0)
[Usage]
ge]
- For arcs with radii less than or equal to the radius entered from P3753 to P3760, the maximum cutting speed is limited.
The cutting speed for the arc radius is calculated as follows:
a = Maximum Speed²/ Command Arc Radius F
= (a * Command Arc Radius)^{1/2}

P3769	P3776
Minimum speed for arc feed rate limit function	

[Type] Form Word Type, CH Type
e ula] FU
[For] [Top]
mula
[범 Top] (Default: 0)
[Usage]
ge]
- When using the circular machining speed limit function, the speed is restricted to at least the minimum cutting speed entered from P3769 to P3776.

 Teaching Command

P4027	P4034
Teaching Axis Selection	

[Type] Bit type, CH type
[Type] 0 or 1 (Default: 0)
[Usage]
- 0: Not selected.
- 1: Selected.

P4035	P4042
Number of Teaching Data	

[Type] Word type, CH type
[Type] 1 to 5 (Default: 0)
[Internal Purpose]

- This parameter sets the number of controllable data points.

P4043	
Teaching Register Address	

[Format Type] Bit type, CH type
[범 We] 0 or 1 (Default: 0)
[Usage] Use]

- 0 : U register
- 1 : R register

P4044	P4051
Teaching Register Number	

[Type] Word Type, CH Type
[Type] Refer to the Register range selected in P4043 (Default: 0) [
Purpose]

- This parameter sets the number of the Teaching Register address selected in P4043.

 Auxiliary Command Setting

 M Code

P4510	P4529
Non-Buffered M Code	

[Type] Word type, General
type [Range] (Default: 0)
[Description]

- This parameter inputs the M code number that performs the axis traverse completion check.
When the M code number entered here is commanded, the axis travel completion check is performed after the M code execution. The default value is 0, and in this case, it is ignored.

※ Refer to Section 5.1.12 on non-buffered M codes in Chapter 5 Control Functions.

P4545	
Synchronous M Code Usage	

[Type] Bit type, CH type
[Range] 0 or 1 (Default: 0)
[Usage]]

- To wait for Channel 1 and Channel 2 during machining, control them using M codes. By issuing a synchronous M code in each channel's machining program, you can make both channels wait at the commanded block. During automatic operation, if a synchronous M code is issued on one channel, it waits for the same M code to be issued on the other channel before starting execution of the next block.

- 0: Not used.
- 1: Enabled.

※ Refer to Chapter 5 Control Functions Synchronous M Code Function section (5.1.11)

Ex] Synchronous M Code Usage Example

Channel 1		Channel 2	
MV X0 Y0 Z0 F1000;		MV X0 Y0 Z0 F1000;	
:		:	
M21;		:	
:		M21;	
:		:	
:		M22;	
M22;		:	
:		:	
PEND;		PEND;	
		Program termination	

P4548	
Synchronous M Code Start Number	

[Type] Word Type, General
Type [Range] (Default: 0)
[Usage] Use

— Set to not buffer, differing from M Code.

P4549	
Synchronous M Code Termination Number	

[Type] Word Type, General
Type [Scope] (Default: 0)
[Usage] Use

— Set differently from M Code that does not buffer.

 MACRO Settings

P4650	4689
-------	------

Register ML Code Command

[Type] Word Type, General
Type [Scope] (Default: 0)
[Purpose]

- ML = Motion Language
- This parameter inputs ML code to enable calling motion programs (from 9010.MP to 9029.MP). The input ML code sequentially calls programs from 9010.MP to 9029.MP. ML is registered using two registers.

Example] To register ML = "ABCDEFGH", register as follows.

- P4650 = "ABCD"
- P4651 = "EFGH"

P4690	4709
-------	------

G Code Command Registration

[Type] Word Type, General
Type [Scope] (Default: 0)
[Usage] Purpose]

- This parameter inputs the G code number to enable calling the motion program (from 9010.MP to 9029.MP) via G code. The input G code sequentially calls programs from 9010 to 9029.
- Input G codes 0, 1, 2, and 3 are ignored.

P4710	P4729
M Code Command Registration	

[Bro Type] Word type, General
 type [범Type] (Default: 0) [Usage]
 Purpose]

- This parameter inputs the M code number to enable calling motion programs (from 9010.MP to 9029.MP) via M code. The entered M codes sequentially call programs from 9010 to 9029.
- Input M codes 0 and 30 are ignored.

[Example]

- Comparing methods to call 9029.MP from 9010.MP using ML, G Code, and M Code

MP Number	ML (Motion Language)	G Code	M Code
9010	ABCDEFGH	100	100
9011	IJKLMNOP	101	101

Using ML ABCDEFGH or entering G Code 100 or M Code 100 will call 9010.MP.

Handle Settings

P5475	P5477
Handle #1 to #3 corresponds to the axis	

[Type Type] Bit type, Axis type

[Type] 0 or 1 (Default: 1) [Purpose]

- Handle mode is a mode that moves the axis by a distance corresponding to the number of pulses input from the Handle (MPG) attached to the control device. The default command unit corresponding to the input pulse is based on the value set in the position command value (GMCP).

(Command Increment Distance = Input Pulse × GMCP)

※ Refer to Chapter 5 Control Functions, HAN Mode section (5.5.3)

Caution	Increasing the position command value causes the axis to move at extremely high speeds, posing significant danger. Be sure to set it to 0.1 mm or 0.01 inch or less.
---------	---

P5482	
Command Default Unit Setting Method	

[Type Type] Bit Type, Axis Type

[Range] 0 or 1 (Default: 0)

[Description] Purpose]

- 0: Integer
- 1: Fraction

P5487	
Maximum Speed Exceeded Command Pulse Processing Method	

[Type] Bit Type, Axis Type

[Range] 0 or 1 (Default: 0)

[Description]

- 0: Ignore
- 1: Process

P5508	
Handle Pulse Multiplication Rate	

[Formula] Bit type, Axis type

[Range] 0 or 1 (Default: 0)
[Description]]

- 0 : x1
- 1: x4

P5511	
Handle Pulse Phase A/B Signal Polarity	

[Type] Bit type, Axis type
[Range] 0 or 1 (Default: 0) [
]
- 0: Positive logic
- 1: Negated Logic

REF Mode Control Setting

P5589	
Current Position Origin Designation Function	

[Type] Bit Type, Axis Type

[Range] 0 or 1 (Default: 0)

[Description]]

- This feature allows you to set the current position as the origin without performing a return-to-origin operation.
- 0: Disabled
- 1: Enabled.

※ Refer to Chapter 5 Control Functions REF Mode section (5.5.4)

Caution	When using an ABS encoder, setting this to 0 will cause the system to perform a return to origin operation.
---------	---

P5592	
Use of deceleration sensor (origin dog)	

[Type Type] Bit type, Axis type

[Range] 0 or 1 (Default: 0) []

- 0: Enabled.
- 1: Disabled.

P5594	
Deceleration Sensor (Home Position Dog) Signal Polarity	

[Bro Type] Bit type, Axis type

[Scope] 0 or 1 (Default: 0) []

- 0 : Positive polarity
- 1: **Negative polarity**

P5597	
Use of origin signal (C(Z) phase, ORG)	

[Type] Bit type, Axis type

[Range] 0 or 1 (Default: 0)

[Description]]

- 0: Enabled.
- 1: Disabled.

P5603	
OT detection, retry function selection	

[Type] Bit type, Axis type
[Range] 0 or 1 (Default: 0) [
]

- This function moves the axis back to the position before the deceleration sensor when an OT signal is encountered during origin return operation, and then retries the origin return.

- 0: Not selected.
- 1: Selected.

P5609	P5640
Rapid Return to Origin Speed	

[Type Form Word Type, Axis Type
e ula] FU
[Sing Top]
le
[범 Top] (Default: 1000)
[Usa]
ge]

- - Entering a speed sets the origin return direction to the - direction.

P5641	P5672
Home Return Primary Deceleration Speed	

[Type Form Word Type, Axis Type
e ula] FU
[Type Top]
e
[범 Top] (Default: 200)
[Usa]
ge]

- - When you input the speed, the direction for the first deceleration during origin return is set to the negative direction.

P5673	P5704
Secondary deceleration speed for origin return	

[Type Form Word Type, Axis Type
e ula] FU

[Unit Top]

[범 Top] (Default: 50)

[Usage] Purpose]

- - Entering a speed sets the direction for the second deceleration during origin return to the - direction.

P5707	P5738
After detecting the origin return signal (Phase C), the shift transfer amount	

[Typ Form Word type, Axis type
e ula] pU
[Typ Top]
e
[범 Top] (Default: 0)
[Usa Use]
ge

P5739	P5770
After detecting the origin return signal (Phase C), the shift transfer speed	

[Typ Form Word type, Axis type
e ula] FU
[Typ Top]
e
[범 Top] (Default: 100)
[My Use]

 Set origin position

P5872	P5967
2nd / 3rd / 4th Origin Position	

[Typ Form Word Type, Axis Type
e ula] pU

[Sing Top]

le

[범 Top] (Default: 0)

[Usa Use]
ge

- P5872 ~ P5903: Second origin position
- P5904 ~ P5935: Third origin position
- P5936 ~ P5967: Fourth origin position

P5968	P5999
Origin Position Recognition Range	

[Typ Type Word Type, Axis Type
e] PU

[Typ Top]

e

[범 Top] (Default: 1)

[Usa]
ge

- If set to 0, it is set to the minimum unit of the ± position.
- If set to 1, it is set to 1 * 0.001 = 0.001 mm.

 Electronic Gear Ratio Setting

*P6424	P6455
Motor-side Gear Ratio	

[Type Type] Word type, Axis type
[Range] 1 to 2 ³¹⁻¹ (Default: 1) [Usage]
Purpose]
※ Refer to Chapter 5 Control Functions Electronic Gear Section (5.1.13)

*P6456	P6487
Machine-side Gear Ratio	

[Type Type] Word type, Axis type
[Type] 1 to 2 ³¹⁻¹ (Default: 1) [Usage]
Use]

*P6488	P6519
Mechanical Movement Per Motor Revolution	

[Formula] Type] Word type,
Axis type [Single]
PU
[범 Level] 1 to 2 ³¹⁻¹ (Default: 10) [Usage]Purpose]
- This parameter is only set for linear axes.

Control Mode Setting

*P6724	
Position Control Mode Support	
[Type]	Bit Type, Axis Type
[Range]	0 or 1 (Default: 0) []
– 0: Not supported. – 1: Supported.	
Note	Axes using MEM, POS, JOG, HAN, or REF modes must be set to Enabled.

*P6729	
Speed Output Mode Support	
[Type Type]	Bit type, Axis type
[Range]	0 or 1 (Default: 0)
[Description]]	
– 0: Not supported. – 1: Supported.	
Note	Axes using VEL mode must be set to Enabled.

 Controller Settings

P6889	P6920
Position Control Speed Feed-forward Gain	

[Type] Word type, Axis type
e

[Type] 0.1%
e

[Range] 0 to 2000 (Default: 0)
ge

[Usage Purpose]

- Using feed-forward gain can improve responsiveness. Set it after configuring the position loop gain (P, I).

※ Refer to Chapter 5, Position Controller section (5.1.14)

Caution	Be cautious: Setting the feed-forward gain value too high may cause mechanical noise or vibration. Generally, do not set it above 80%.
---------	---

P6988	P7019
Position Control P Gain	

[Formula] Word type, Axis type

[Type] 0.1 / S
e

[Range] 0 to 32767 (Default: 200)
ge

[Purpose]

Caution	Recommended setting range Refer to the User Manual for insertion
---------	--

P7020	P7051
Position Control I gain	

[Type Type] Word type, Axis type
e

[Type] msec

e
[Ran] 0 to 32767 (Default: 0)
ge
[Usa Purpo
ge] se]

P7052	P7083
Position Control Integral Maximum Value	

[For Type] Word type, Axis type
mula
[Typ]
e
[Sco] 0 to 32767 (Default: 32767)
pe
[Usa Purpo
ge] se]

P7186	P7217
Position Control Stop Tolerance	

[Formula Type] Word type, Axis
type [Single] PU
[범 Level] 0 to 2³¹⁻¹ (Default: 1000) [Usage]
Purpose]

Caution	If the control error exceeds the limit, a stop tracking error alarm will occur.
---------	---

P7218	P7249
Position Control Movement Tolerance	

[Type Type] Word type, Axis type
[Single] PU
[Scope] 0 to 2³¹⁻¹ (Default: 10000) [Usage]
Purpose]

Warning	If the control error exceeds the limit, a tracking error alarm will occur.
---------	--

 Servo Motor/Drive Settings

*P8327	
Servo Motor Control Mode	
[Type]	Bit Type, Axis Type
[Range]	0 or 1 (Default: 0) []
— 0: Speed-type	
— 1: Position-type	

*P8330	
Motor rotation direction	
[Bro Type]	Bit type, Axis type
[Scope]	0 or 1 (Default: 0) []
Purpose]	
— Used to change the axis traverse direction.	
— 0: Forward direction	
— 1: Reverse direction	

*P8335	P8366
Motor Maximum Speed Setting	
[Type Type]	Word type, Axis type
[Single]	RPM, FU
[Range]	1 to 2 ³¹⁻¹ (Default: 5000) [Purpose]

*P8367	P8398
Motor Control Resolution Setting (Position Type)	

[Type] Word Type, Axis Type
[Range] 1 to 2³¹⁻¹ (Default: 2048) [Usage]

- This parameter sets the number of servo motor input pulses corresponding to one motor rotation command.

*P8507	P8538
Servo motor input for maximum speed	

[Type] Word type, Axis type
[Type] mV
[Range] 1 to 2³¹⁻¹ (Default: 10000) [

*P8605	
Servo Motor Pulse Input Method (Position Type)	

[Type] Bit Type, Axis Type
e
[Range] 0 or 1 (Default: 0)
ge
[Usage]
ge

- 0: CW / CCW
- 1: P + Dir

*P8747	
Allowable delay time for servo ready signal	

[Bro Type] Word type, General
type [Single] msec
[Range] 0 to 2³¹⁻¹ (Default: 2000) [Purpose]

Caution	If the Servo ready signal is not input for the set time, a Servo ready alarm will occur.
---------	--

*P8748	
Servo Ready Signal Polarity	

[Type] Bit type, Axis type
[Range] 0 or 1 (Default: 0) []
- 0 : Positive polarity
- 1: Negative polarity

*P8751	
Servo alarm signal polarity	

[Type] Bit type, Axis type
[Range] 0 or 1 (Default: 0) []
- 0 : + Polarity
- 1: Negative polarity

 Encoder Settings

*P8825	
Encoder Enable	

[Type] Bit Type, Axis Type
[Range] 0 or 1 (Default: 0)
[Description]
- 0: Enabled.
- 1: Disabled.

*P8828	
Encoder Type	

[Type] Bit type, Axis type
[Range] 0 or 1 (Default: 0) [
]
- 0 : INC Type
- 1 : ABS Type

*P8833	
Encoder A/B Phase Output Direction	

[Type] Bit Type, Axis Type
[Range] 0 or 1 (Default: 0)
[Description]
- 0: Forward
- 1: Reverse

Caution	If the settings are incorrect, the servo motor may run out of control, and a stop tracking error or tracking error alarm may occur.
---------	---

*P8874	P8905	
Encoder Resolution (After Multiplication, Resolution)		
[Type]	Word type, Axis type	
[Type]	4 to 2 ³¹⁻¹ (Default: 2048) [Purpose]	

*P8910	P8941	
ABS Encoder Manufacturer		
[Mod el]	Word Type, Axis Type	
[Range]	0 to 2 (Default: 0)	
[Usage]	Purpose]	
	— 0: Not specified.	
	— 1: Yaskawa	
	— 2: Mitsubishi	

*P9022		
Encoder Disconnection Diagnosis Function Enable/Disable		
[Type]	[Form ula]	Word Type, Axis Type
[범 Level]	0 or 1 (Default: 0)	
[Usage]	Use]	

*P9025		
Encoder Phase Alarm Diagnostic Function Enabled		
[Type]	[Type]	Word type, Axis type
		e
[Type]	0 or 1 (Default: 0)	
	e	
[Usage]	Purpose]	
	ge se]	

 Position/Speed/Torque Reach Setting

*P9120	
In-position Check Mode (Position Type)	
[Type Type]	Word Type, Axis Type
[Type Type]	0 or 1 (Default: 0) [Usage]]
— 0: Encoder feedback	
— 1: Servo In-position signal	
※ Refer to Chapter 5 Control Function In-Position Signal section (5.6.2)	

*P9123	P9154
In-position range	
[Type]	Word type, Axis type
[Single]	PU
[Scope]	-2^{31} to 2^{31-1} (Default: 0) [Usage
Use]	

*P9155	
In-position Allowable Delay Time	
[Formula]	Type] Word type,
General type [Single]	
msec	
[Range]	-2^{31} to 2^{31-1} (Default: 0) [Usage]
Use]	

Caution	An alarm will sound if the set time is exceeded.
---------	--

*P9225	P9256
0 Speed Agree Range	
[Type Type]	Word type, Axis type
[Single]	0.01%
[Range]	-2^{31} to 2^{31-1} (Default: 10) [Use]
※ See Chapter 5 Control Functions 0 Speed Reach Signal section (5.6.4)	

Caution	Applies only in Vel mode.
---------	---------------------------

P9257	P9288
-------	-------

Command Speed Reach Status Determination Range

[Type Type] Word type, Axis type
[Single] PU
[Scope] -2³¹ to 2³¹⁻¹ (Default: 1) [Usage Use]

※ Refer to Chapter 5 Control Function Speed Reach Signal section (5.6.3)

Caution	Applies only in Vel mode.
---------	---------------------------

P9289	
-------	--

Enable/Disable Speed Reach Diagnostic Function

[Type Type] Bit type, Axis type
[Single]
[Range] 0 or 1 (Default: 0) [Usage]
 — 0: Disabled.
 — 1: Enabled.

P9291	
-------	--

Speed Attainment Allowable Delay Time

[Type] Word type, General type
e
[] msec
[Range] (Default: 0)
ge
[]

 Calibration Settings

*P9464	
--------	--

Backlash Compensation

[Type] Bit type, Axis type
[Type] 0 or 1 (Default: 0)
[Description Purpose]

- 0: No correction.
- 1: Calibrate.

*P9465	P9496
--------	-------

Backlash Compensation Value

[Type] Word type, Axis type
[Unit] PU
[Scope] 0 to 2³¹⁻¹ (Default: 0) [Usage Purpose]



Synchronous Control Setting

*P9822	
Synchronous Control Enable	

[Type] Bit type, Axis type

[범 We] 0 or 1 (Default: 0)

[Usage] Use]

— 0: Disabled.

— 1: Enabled.

※ Refer to Chapter 5 Control Functions Synchronous Control section (5.7)

*P9823	
Synchronous Control Function Selection	

[Type] Bit Type, Axis Type

[Range] 0 or 1 (Default: 0)

[Description]]

— 0: Always synchronous control

— 1: Synchronous control On/Off via G register signal

P9825	
Synchronous Control Target Selection	

[Type] Bit type, Axis type

[Range] 0 or 1 (Default: 0) []

— 0: Command Location

— 1: Actual Position

*P9828	P9859
Synchronous Control Master Axis Setting	

[Type] Word Type, Axis Type

[Range] 0 to 32 (Default: 0)

[Usage] Purpose]

Note	Ensure the Master axis number is set in sequence before the Slave axis number
------	---

P9862	P9893
Synchronous Control Sync. Factor Denominator	

[Formula type] Word type, Axis type

[Type] e

[Scope] (Default: 0)

[Integral Use]

P9894	P9925
Synchronization Control Sync. Factor Numerator	

[Formula type] Word type, Axis type

[Type]

[Scope] (Default: 0)

[Integral Use]

P9928	P9959
P gain in dynamic control	

[Formula type] Word type, Axis type

[Type] 0.1%

[Range] From 0 to 1000 (Default: 200)

[Usage]

P9962	P9993
Synchronous Control Error Tolerance	

[Formula type] Word type, Axis

[Type] PU

[Scope]	From 0 to 2^{31-1}	(Default: 100)
[Usage]	Purpose]	

- If the set value is exceeded, a synchronous tracking error alarm will occur.

 External Positioning Setting

*P10520	
External Positioning Enable	

- [Type] Bit type, Axis type
[Range] 0 or 1 (Default: 0)
[Description]]
- 0: Disabled.
 - 1: Enabled.
- ※ See Section 5.8 for details on determining the external location of control functions.

P10527	
External Positioning Signal Register Selection	

- [Type] Bit Type, Axis Type
[Range] 0 or 1 (Default: 0)
[Description]]
- 0 : R register
 - 1 : X register

P10528	10559
External Position Determination Signal Register Address	

- [Type] Word type, Axis type
[Range] From 0 to 2³¹⁻¹ (Default: 0)
[Usage] Purpose]

P10560	
--------	--

External Position Determination Signal Polarity

[Type] Bit Type, Axis Type
[범 We] 0 or 1 (Default: 0)
[Usage]

- 0: Positive logic
- 1 : Negation Logic

P10568	10599
--------	-------

External Position Determination Incremental Transfer Amount

[Formula] Word type, Axis
type [Single] PU
[Scope] -2³¹ to 2³¹⁻¹ (Default: 0) [In
Use]

 Position Latch Setting

*P10720	
PL Function Enable	

- [Type] Bit Type, Axis Type
- [Range] 0 or 1 (Default: 0)
- [Description]
- The PL function sets the current position of the corresponding axis in the machine coordinate system to the [Latch Position] entry in the S register when the Latch signal is input.
 - 0: Disabled.
 - 1: Enabled.
- ※ Refer to Chapter 5 Control Functions External Latch Function (5.9)

P10727	
PL Signal Register Selection	

- [Type] Bit Type, Axis Type
- [Range] 0 or 1 (Default: 0)
- [Description]
- 0 : R register
 - 1 : X register

P10728	P10759
PL Signal Register Address	

- [Type] Word type, Axis type
- [범 We] From 0 to 2³¹⁻¹ (Default: 0)
- [Usage]
- Example] When RB100.03 is entered, P10727 is set to 0
- P10728: 10003 is set.

P10760	
PL Signal Polarity	

[Type Type] Bit type, Axis type
[Range] 0 or 1 (Default: 0) [
]
 — 0: Positive logic
 — 1 : Negation Logic

 Position comparison function

*P10820	
PC Function Usage Status	

[Type] Bit Type, Axis Type
[Type] 0 or 1 (Default: 0) []
— 0: Disabled.
— 1: Enabled.
※ See Chapter 5, External Comparison Function section (5.10)

P10827	
PC Signal Register Selection	

[Type] Bit Type, Axis Type
[Range] 0 or 1 (Default: 0) []
— 0 : R register
— 1 : X register

P10828	P10859
PC Signal Register Address	

[Type] Word type, Axis type
[Single] PU
[Scope] From 0 to 2³¹⁻¹ (Default: 0)
[Usage] Purpose]
Example] When RB100.03 is entered, P10727 is
set to 0
P10728: 10003

P10860	
PC Signal Polarity	
[Type] Bit type, Axis type	
e	
[Range]	0 or 1 (Default: 0)
ge	
[My Use]	
– 0: Positive logic	
– 1: Negation Logic	

P10868	
PC Function Mode	
[Form] Bit Type, Axis Type	
m	
[Range]	0 or 1 (Default: 0)
ge	
[Usage]	
ge	
– 0 : Position mode	
– 1 : Area mode	

P10869	
Position Setting Method	
[Type] Bit Type, Axis Type	
e	
[Range]	0 or 1 (Default: 0)
ge	
[Usage]	
ge	
– 0: Incremental mode	
– 1: Absolute mode	

P10872	P10903
Output Signal Hold Distance	
[Type Form Word Type, Axis Type	
e ula] pU	
[Type Top]	
e	

[범 Top] (Default: 0)

[My PC]

- The PC signal is maintained at High0 until the shaft moves beyond the set distance after the signal is output externally.

P10906	P10937
Number of Incremental Modes Sections	

[Type] Word type, Axis type

[Type] -2^{31} to 2^{31-1} (Default: 0) [

Use]

P10938	P10969
Start position of incremental method	

[Type Type] Word type, Axis type

[Unit] PU

[Scope] -2^{31} to 2^{31-1} (Default: 0) [In

Use]

P10970	P11001
For incremental mode, incremental distance	

[Type Type] Word type, Axis type

[Unit] PU

[범 We] From 0 to 2^{31-1} (Default: 0)

[Usage] Purpose]

P11004	P11035
For absolute mode, the start index in the data table	

[Typ Type] Word type, Axis type
e

[Typ Range From 0 to 1999 (Default: 0)

e]

[Usa Use]

ge

P11036	P11067
For absolute mode, the end index in the data table	

[For Type] Word type, Axis type
mat]

[Typ Range From 0 to 1999 (Default: 0)

e]

[Usa Use]

ge

P11070	P11101
Area Mode - Direction Position	
[Formula type]	[Type] Word type, Axis type [Single] PU
[범 We]	-2 ³¹ to 2 ³¹⁻¹ (Default: 0) [Use]

P11102	P11133
Area Mode + Direction Position	
[Type Type]	Word type, Axis type
[Single]	PU
[Scope]	-2 ³¹ to 2 ³¹⁻¹ (Default: 0)
[Use] Use]	

 Communication Data Table

P10820	P11319
Communication Data Table	
[Type]	Word type, Axis type
[Unit]	PU
[Scope]	-2 ³¹ to 2 ³¹⁻¹ (Default: 0)
[Use] Use]	

 Motion Data Table

P11320	P13319
Data Table	
[Type]	Word type, Axis type
[Single]	PU
[범 We]	-2 ³¹ to 2 ³¹⁻¹ (Default: 0) [
Use]	
- Pitch Error Correction [PU]	
- Electronic Cam[PU]	

8.3 D Register (Device Parameter)

8.3.1 Configuration

■ Configuration

Description	Device Parameter
Data Type	Bit, Word

■ Device Category

No	InternalUse
0	System Settings
1	CPU Settings
2 ~ 20	Local Device
21 ~ 255	Remote Device

■ Zone Classification

Device No	Register No	Device Settings
0	0 ~ 99	System Settings
1	100 ~ 199	CPU Settings
2	200 ~ 299	Local Device 1 Settings
3	300 to 399	Local Device 2 Settings
...	...	...
20	2000 ~ 2099	Local Device 18 Settings
21	2100 ~ 2199	Remote Device 1 Settings
22	2200 ~ 2299	Remote Device 2 Settings
...	...	...
255	25500 ~ 25599	Remote Device 234 Settings

8.3.2 Detailed Description

(1) System Settings

D27.0	
Power Monitoring Status	
[Type]	Bit Type
e	
[Range]	0 or 1
ge	
[Usage Purpose]	
se]	

**D28	
Power Outage Tolerance Time	
[Type]	Word Type
e	
[Short]	msec
t	
[Range]	
ge	
[]	

**D29	
Power Outage (Momentary Power Outage) Detection Time	
[Format]	Word Type
mat]	
[]	msec
[범 [범]	
[Purpose Purpose]	
pose se]	
]	

(2) CPU Settings

 Model Settings

**D6	
Product Model Number	
[Type]	Word Type
[Single Type]	

[Range]

[Usage] Use]

**D7	
Product Version	
[Model]	Word Type
[Single]	
[Scope]	
[Usage] Use]	

 Handle Port

**D13	
Handle Count	
[Type]	Word Type
e	
[Unit]	
[Sco]	
pe	
[Usa Use]	
ge	

**D14	
Handle #1 Pulse	
[For]	Word Type
m	
[Sing Scope	
le]	
[Sco]	
pe	
[Usa Use]	
ge	

**D15	
Handle #2 Pulse	
[For]	Word Type
m	
[Sing Scope	
le]	
[Sco]	
pe	
[Usa Use]	
ge	

**D16	
Handle #3 Pulse	
[Form]	Word Type
[Single]	

[Range]

[Usage] Use]

 USB Port

**D32	
Timeout	

[Bro] Word type

[Single Scope]

[Scope]

[Usage]]

 Ethernet Port

**D39	
Communication ID (Country Code)	

[Typ] Word Type

e

[Shor Scope

t]

[Sco]

pe

[Usa]

ge

**D40	
IP Address	

[For Forma Word Type

mat t]

[Shor Scope

t]

[Sco]

pe

[Usa Use]

ge

**D41	
-------	--

Port Number

[Type] Word Type
e
[Sing Scope
le]
[Sco]
pe
[Usa Use]
ge

**D42	
Subnet Mask	

[For] Word Type
mat
[Unit Type]
[Sco]
pe
[Usa Use]
ge

**D43	
Gateway	

[Typ Type] Word Type
e
[Ter Scope
m]
[Sco]
pe
[Usa]
ge

 RS232_1 Port

**D51.0	
Master / Slave	

[Typ] Bit Type
e
[Sing]
le

[Range]
ge
[Usage]
ge

**D52.0	D52.7
Communication Protocol	

[Format] Bit type
mat

[Short Scope]
t

[Range] 1, 2, 3, 4, 100 (Default:)
ge

[My Use]

- 1: T-Bus
- 2: Modbus
- 3: Melsec
- 4: MK
- 100: User

**D52.8	D52.31
Communication Protocol Number	

[Type] Bit type
e
[Short]
t
[Range]
ge
[Usage]

**D54.0	D54.7
Baud Rate	

[Format] Bit
[Unit]
[Range]
ge
[Usage]

**D54.8	D54.15
Data Bit	

[Format] Bit Type
mat
[Unit]
[Range]
ge
[Usage]

**D54.16	D54.23
Parity Bit	

[Format] Bit Type
m
[
[Range]

ge
[Usa Use]
ge

**D54.24	D54.31
Stop Bit	

[For] Bit Type
mat
[Unit]
[Ran]
ge
[Usa Use]
ge

**D57	
Timeout	

[For] Word
mat
[Typ Scope
e]
[Sco]
pe
[In Drago
n]

 RS232_2 Port

**D62.0	
Master / Slave	

[Typ] Bit Type
e
[Typ]
e
[Ran]
ge
[Usa Use]
ge

**D63.0	D63.7
---------	-------

Communication Protocol

[For] Bit
mat

[Shor Scope

t]

[Ran] 1, 2, 3, 4, 100 (Default:)

ge

[Usa Use]

ge]

- 1: T-Bus
- 2: Modbus
- 3: Melsec
- 4: MK
- 100: User

**D63.8	D63.31
---------	--------

Communication Protocol Number

[Type] Bit type

[Unit]

[Range]

[Usage Use]

**D65.0	D65.7
Baud Rate	

[For] Bit Type
mat]

[Unit]

[Ran]

ge
[Usa Use]
ge

**D65.8	D65.15
Data Bit	

[For] Bit Type
mat

[Unit]

[Ran]

ge
[Usa Use]
ge

**D65.16	D65.23
Parity Bit	

[Typ [Form Bit Type
e ula]

[Shor]
t

[Ran]

ge
[Usa Use]
ge

**D65.24	D65.31
Stop Bit	

[For] Bit Type
mat

[Unit]

[Ran]

ge
[Usa Use]
ge

**D68	
Timeout	

[For] Word
mat
[Typ Scope
e]
[Sco]
pe
[Usa]
ge

(3) Local / Remote Device Common Settings

**D2.0	
Device Usage	
[Type] Bit	
Type [Single]	
[Range]	
[Usage]	

**D3.0	
Device Classification	
[Type] Bit Type	
[Unit]	
[Range] 0 or 1 (Default:)	
[Usage]	
- 0 : Local Device	
- 1 : Remote Device	

(4) Local Device

■ Common Area

D7	
Device Type	
[Bro] Word type	
[Single Scope]	
[Scope] 3, 4, 5, 6, 7, 15, 16, 20, 21, 25, 26, 28 (Default:)	
[My Use]	
- 3: Speed servo	
- 4: Position-type servo	
- 5: DIO	
- 6: AIO	
- 7: Counter	
- 15: Serial Communication	

- 16: Ethernet Communication
- 20: T-Link
- 21: Machatrolink
- 25: T-Net
- 26: DeviceNet
- 27: Profibus
- 28: CC-Link

**D10	
Product Model Number	

[Type] Word Type
[Single]
[범 We] 1 to 5 (Default:)
[My Use]

- 1: GSC, GEN, GDN, GPB, GCC, GML, GAXA2, GAXP2, GDI32, GAI08, GCNT4
- 2: GAXA4, GAXP4, GDO32, GCNT8

- 3 : GDI64
- 4: GDO64

- 5: GDIO32

**D11	
Product Version	

[Type] Word Type
[Single]

[Scope]

[Usage] Use]

**D13	
Slot No.	

[Type] Word Type
[Single Type]

[Scope]

[Usage] Use]

■ Individual Area

● Velocity Type Axis Device

**D25	
Number of Ports	
[Type Type]	Word Type
[Single]	
[Range]	
[Usage]	Use]

**D26	D33
The axis number corresponding to port n	
[Type]	Word type
[Single]	
[Range]	
[Usage]	]

● Position Type Axis Device

**D25	
Number of Ports	
[Type Type]	Word Type
[Single]	
[Range]	
[Usage]	Use]

**D26	D33
Axis number corresponding to port n	
[Formula type]	Word
[Single]	
[Range]	
[Usage]	]

 DIO Device

**D25	
DI Contact Count	

[Type] Word Type
e
[Sing]
le
[Sco]
pe
[Usa Use]
ge

**D26	
X Register Number Corresponding to DI	

[For] Word Type
mat
[Typ]
e
[Sco]
pe
[Usa Use]
ge

**D29	
DO Contact Count	

[Type] Word Type
e
[Sing]
le
[Sco]
pe
[Usa Use]
ge

**Number of Y Registers	
Number of Y Registers corresponding to DO	

[Type] Word type
e

[Sing Scope
le]
[범 Unit]
[Pur Purpo
pose se]

 AIO Device

**D25	
AI Contact Count	
[Type]	Word Type

[Single Scope]
[Scope]
[Usage Use]

**D26	
X Register Number Corresponding to AI	
[Format]	Word
type [Single]	
[Range]	
[Usage]	Use

**D29	
Number of AO Contacts	
[Type]	Word Type
e	
[Single]	
le	
[Scope]	
pe	
[Usage Use]	
ge	

**Number of Y Registers	
Number of Y Registers corresponding to AO	
[Type]	Word Type
e	
[Unit]	
[Scope]	
pe	
[Usage]	
ge	

 Serial COM Device

 RS232_1 Port

**D27.0	
Master / Slave	
[Type]	Bit
Type [Single]	

[Range] 0 or 1 (Default:)
[In Use]

- 0 : Slave
- 1 : Master

D28.0	D28.7
Communication Protocol	
[Type]	Bit type
[Unit]	
[Range]	1, 2, 3, 4, 100 (Default:)
[Usage]	Use]
- 1: T-Bus - 2: Modbus - 3: Melsec - 4: MK - 100: User	

**D28.8	D28.31
Communication Protocol Number	
[Type]	Bit type
[Unit]	
[Range]	
[Usage]	

**D30.0	D30.7
Baud Rate	
[For]	Bit Type
mat]	
[Unit]	
[Ran]	1 to 11 (Default:)
ge	
[My Use]	
- 1: 110 - 2: 300 - 3: 600 - 4: 1200 - 5: 2400 - 6: 4800 - 7: 9600	

- 8: 19200
- 9: 38,400
- 10: 57600
- 11: 115200

**D30.8	D30.15
Data Bit	

[Type] Bit
Type [Unit]
[Range] 1 to 4 (Default:)
[My Use]
- 1 : 5
- 2 : 6
- 3 : 7
- 4 : 8

**D30.16	D30.23
Parity Bit	

[For] Bit Type
m
[]
[Ran] 1 to 5 (Default:)
ge
[My Use]
- 1 : None
- 2: Even
- 3: Odd
- 4: Mark
- 5 : Space

**D30.24	D30.31
Stop Bit	

[Typ] Bit Type
e
[Unit]
[Ran] 1 to 3 (Default:)
ge
[My Use]
- 1 : 1

- 2 : 1.5
- 3 : 2

**D33	
Timeout	
[Type]	Word type
[Single]	
[Scope]	
[Usage]	

 RS232_2 Port

**D38.0	
Master / Slave	
[Type]	Bit
Type [Single]	
[Range]	0 or 1 (Default:)
[In Use]	
- 0 : Slave	
- 1 : Master	

**D39.0	D39.7
Communication Protocol	
[Type]	Bit type
[Unit]	
[범 We]	1, 2, 3, 4, 100
[Usage]	Use]
- 1: T-Bus	
- 2: Modbus	
- 3 : Melsec	
- 4: MK	
- 100: User	

**D39.8	D39.31
Communication Protocol Number	

[Type] Bit type
[Unit]
[Range]
[Usage]

**D41.0	D41.7
Baud Rate	

[For] Bit
mat
[Unit]
[Ran] 1 to 11 (Default:)
ge
[My Use]

- 1: 110
- 2: 300
- 3: 600
- 4: 1200
- 5: 2400
- 6: 4800
- 7: 9600
- 8: 19200
- 9: 38400
- 10: 57600
- 11: 115200

**D41.8	D41.15
Data Bit	

[Type] Bit
Type [Unit]
[Range] 1 to 4 (Default:)
[My Use]

- 1 : 5
- 2: 6
- 3 : 7
- 4: 8

**D41.16	D41.23
Parity Bit	
[Form]	Bit
Type [Unit]	
[Range]	1 to 5 (Default:)
[My Use]	
- 1 : None	
- 2 : Even	
- 3 : Odd	
- 4 : Mark	
- 5 : Space	

D41.24	D41.31
Stop Bit	
[Type]	Bit Type
e	
[Unit]	
[Range]	1 to 3 (Default:)
ge	
[My Use]	
- 1 : 1	
- 2 : 1.5	
- 3 : 2	

**D44	
Timeout	
[Type]	Word type
[Single]	
[Scope]	
[Usage]	

 RS422/485 Port

**D49.0	
422 / 485	
[Type]	Bit Type

[Unit]

[Range]

[Usage Use]

**D50.0	
Master / Slave	

[Type] Bit

Type [Single]

[Range] 0 or 1 (Default:)

[In Use]

- 0 : Slave
- 1 : Master

**D51.0	D51.7
Communication Protocol	

[Type] Bit type

[Unit]

[범 We] 1, 2, 3, 4, 100

[Usage Use]

- 1: T-Bus
- 2: Modbus
- 3: Melsec
- 4: MK
- 100: User

D51.8	D51.31
Communication Protocol Number	

[Type] Bit type

[Unit]

[Range]

[Usage]

D52	
Communication ID (Country Code)	

[Format Format] Word Type

[Short Scope]

[Scope]

[Usage Use]

**D53.0	D53.7
Baud Rate	

[Format]] Bit
type[Unit]
[Range] 1 to 11 (Default:)
[Usage Use]

- 1: 110
- 2: 300
- 3: 600
- 4: 1200
- 5: 2400
- 6: 4800
- 7: 9600
- 8: 19200
- 9: 38400
- 10: 57600
- 11: 115200

**D53.8	D53.15
Data Bit	

[Type] Bit
Type[Unit]
[Range] 1 to 4 (Default:)
[My Use]

- 1 : 5
- 2: 6
- 3 : 7
- 4: 8

**D53.16	D53.23
Parity Bit	

[Form] Bit
Type [Unit]
[범 We] 1 to 5 (Default:)
[My Use]

- 1 : None
- 2 : Even
- 3 : Odd
- 4 : Mark
- 5 : Space

**D53.24	D53.31
Stop Bit	

[Typ] Bit Type
e
[Unit]
[Ran] 1 to 3 (Default:)
ge
[My Use]
- 1 : 1
- 2 : 1.5
- 3 : 2

**D56	
Timeout	

[Type] Word type
[Single]
[Scope]
[Usage]]

D80	D99
Communication Buffer	

[Format]]
Word type [Single
]
[Range]
[Usage]]

 Ethernet COM Device

 Communication Settings

**D29	
Telecom ID (Country Code)	

[Type] Word
e Format
[Short Scope
t]
[Scope]
pe
[Usage Use]
ge

**D30	
IP Address	

[For Format Word Type
mat t]
[Short Scope
t]
[Scope]
pe
[Usage Use]
ge

**D31	
Port Address	

[For Format Word Type
mat t]
[Single Scope
le]
[Scope]
pe
[Usage Use]
ge

**D32	
Subnet Mask	

[For] Word Type
mat
[Unit Type]
[Sco]
pe
[Usa Use]
ge

**D33	
Gateway	

[Type Type] Word Type
[Single Scope]
[Scope]
[Usage] Use]

 System-exclusive

D80	D99
Communication Buffer	

[Type] Word type
[Single]
[Range]
[Usage]]

 DeviceNet Device

 Basic Settings

**D29	
Module Search Time	

[Type] Word Type
e
[Single] msec
le
[Range]
ge
[]

**D31	
Scan Time	

[Bro] Word
[Single] msec
e
[Range]
e
 []

Comm Setting

unicat s

ion

**D38	
Communication ID (Country Code)	

[Type Forma Word Type
e t]
[Short Scope
t]
[Scope]
pe
[Usage Use]
ge

 System Exclusive

**D72	
Connection Slave Information	
[Type]	Word Type
[Single]	
[Scope]	

[Usage] Use]

**D73	
Connection Slave Operation Status	

[Type Format]Word Type
[Single]

[Range]

[Usage]]

(5) Remote Device

 Common Area

**D7	
Device Type	
[Type]	Word Type
[Single]	
[Scope]	3, 4, 5, 6, 7, 100
[Usage]	Use]
- 3: Speed Servo - 4: Position-type servo - 5: DIO - 6: AIO - 7: Counter - 100: Common	

**D8	
Communication (I/F) Method	
[Type]	Word Type
[Single]	
[Range]	1, 2, 3, 4, 5, 10, 11, 15, 16, 17
[Usage]	Use]
- 1: RS232C Communication - 2: RS422 Communication - 3: RS485 Communication - 4: Ethernet Communication - 5: USB Communication - 10: T-Link - 11: Machatrolink - 15: DeviceNet - 16: Profibus - 17: CC-Link	

**D10	
Product Model Number	

[Type] [Form] Word
[e] [ula] format
[Single] Scope
[Scope]]
[Scope]
[Usage] Use]
[ge]

**D11	
Product Version	

[Type]] Word Type
[e]
[Single] Scope
[Scope]]
[Scope]
[Usage] Use]
[ge]

**D13	
Master Communication Device No.	

[Type]] Word Type
[e]
[Single] Scope
[Scope]]
[Scope]
[Usage] Use]
[ge]

**D14	
Master Communication Port No.	

[Type] [Form] Word Type
[e] [t]
[Single]]
[le]

[Scope]

pe

[Usage Purpose]

ge se]

■ Individual Area

 DIO Device

**D25	
Device ID (Area Code)	

[Type] Word Type
e
[Short Type]
t
[Scope]
pe
[Usage Use]
ge

**D30	
Number of DI contacts	

[Type] Word Type
e
[Single]
le
[Scope]
pe
[Usage Use]
ge

**D31	
X Register number corresponding to DI	

[Format] Word Type
mat
[Type]
e
[Scope]
pe
[Usage Use]
ge

**D34	
DO Contact Count	

[Type] Word Type
e
[Sing [Top]
le
[범 Wei]
[Usa Use]
ge]

**D35	
DO corresponding Y Register number	

[Format] Word
type [Single]

[Range]

[Usage]]

 AIO Device

**D25	
Device ID (Area Code)	

[Mod] Word Type
el

[Shor Type]
t

[Sco]
pe
[Usa Use]
ge

**D30	
AI Channel Count	

[Typ Forma Word Type
e t]

[Sing Scope
le]

[Sco]
pe
[Usa Use]
ge

**D31	
X Register number corresponding to AI	

[For] Word Type
mat]

[Typ Scope
e]

[Sco]
pe
[Usa Use]
ge

**D34	
Number of AO Contacts	

[Type] Word Type
e
[Sing]
le
[Sco]
pe
[Usa Use]
ge

**D35	
Y Register Number Corresponding to AO	
[Format]	Word
Type [Single]	
[Range]	
[Usage]]	

 General Communication Device

 Ethernet Method

D29	
Connection Waiting Time	
[Typ]	Word
e	
[Sing]	msec
le	
[Ran]	
ge	
[]	

**D36	
Server / Client Selection	
[For]	Word Type
mat	
[Sing Scope	
le]	
[Sco]	
pe	
[Usa Use]	
ge	

**D37	
Master / Slave	
[Typ]	Word Type
e	
[Sing Scope	
le]	
[Sco]	0 or 1 (Default:)
pe	
[Usa Use]	

ge]

- 0 : Slave
- 1 : Master

**D38.00	D38.07
----------	--------

Communication Protocol	
------------------------	--

[Form]

Bit

Type [Unit]

[Range]

1, 2, 3, 4, 100

[Usage

Use]

– 1: T-Bus

– 2: Modbus

– 3: Melsec

– 4: MK

– 100: User

**D38.08	D38.23
----------	--------

Communication Protocol Number	
-------------------------------	--

[Bro]

Word type

[Single Scope]

[Scope]

[Usage]

Use]

**D39	
-------	--

Communication ID	
------------------	--

[For

Forma

Word Type

mat

t]

[Sing

Scope

le

]

[Sco

]

pe

[Usa

]

ge

**D40	
-------	--

IP Address	
------------	--

[For

Forma

Word Type

mat

t]

[Shor

Scope

t

]

[Sco

]

pe

[Usa

Use]

ge

**D41	
Port Number	
[Type] Word Type	
[Scope]	
[Usage]	

**D42.0	
TCP / UDP	
[Form]	
[Unit]	
[Range] 0 or 1 (Default:)	
[Usage]	
- 0 : TCP	
- 1 : UDP	

**D47	
Communication Timeout	
[Type] Word Type	
[Unit]	
[Scope]	
[Usage] Use	

**D48	
Retransmission Count	
[Format] Word Type [Single]	
[Range]	
[Usage] Use	

8.4 S Register

8.4.1 Configuration

■ Configuration

Description		Status Data
Data Type		Bit, Word
Structure		
Non-Storage Area	1. System Information	① Product ID
		② Product Version (System Version)
		③ Date/Time
		④ Permissions
	2. MP Operation Information	① Program Execution Time
		② MP Operation Execution Information
		③ Interpolation Command Speed
		④ Position Determination Command Speed
		⑤ Dwell
	3. MC Drive Information	① Handle Pulse
		② Home Return Information
	4. Drive Information	① Current Coordinates
		② Status Coordinates
		③ Remaining distance
		④ Coordinate System Offset
		⑤ Actual Speed
	5. Control Information	① Control deviation quantity (position, torque, speed)
		② Synchronization control deviation
		③ Servo information (encoder, servo axis I/O, control output 1/2)
Storage Area	1. System Information	① System Operation Information
	2. MP Operation Information	① MP Operation Execution Information (Executed Program, Executed Blocks)
	3. Operation Information	① Machine Coordinates
	4. Control Information	① Encoder

8.4.2 Detailed Description

8.4.2.1 Non-Storage Area

 System Information

S15		System H/W Version
[Model] Word Type, Standard Type [Scope] [Purpose Use]		
S16		System S/W Version (BCD)
[Bro Type] Word type, General type [범 Type] [Usage] Use]		
S18		System Mode
[Format Type] Word Type, General Type [Type] 0 to 2 (Default:) [Usage] - 0 : Boot - 1 : Stop - 2 : Run		
S21		System Permission Levels
[Type] Word Type, General Type [Scope] [Purpose Use]		

S24	
System Date (BCD)	

[Format Format]
Word format, General format
[Description]

-

Example] When it is Thursday, November 25, 2004, it is written as follows.

	November 25, 2004	November	25	Thursday
0x	04	11	25	04

- Numeric value for date input

- 2. Monday: 01
- 3. Tuesday: 02
- 4. Wednesday: 03
- 5. Thursday: 04
- 6. Friday: 05
- 7. Saturday: 06
- 8. Sunday: 07

S25	
System Time (BCD)	

[Type] Word Type, General
Type [Usage]

-

Example] When it is 10:06:10, it is displayed as follows.

	10:06 min10 seconds			
0x	00	10	06	10

S27	
System Timer (32-bit)	
[Type]	Word type, General
type [Purpose]	

S32	
System Mode Switch Status	
[Type]	Word Type, General
Type [Description]	

 MP Operation Information

S170	S201
Skip Position	

[Mod] Word Type, Axis Type
el
[Sing] PU
le
[Sco]
pe
[My Use]

S212	S219
Dwell Count	

[Typ Type] Word type, CH type
e
[Typ] msec
e
[Ran]
ge
[Use]

 Driving Information

S400	S431	Current Position (Workpiece Coordinate System)	
[Type]		Word type, Axis type	
e			
[Sing]		PU	
le			
[Sco]			
pe			
[Inte Use]			
rnal			
S432	S463	Coordinate System Offset (IPR)	
[For]		Word type, Axis type	
mula			
[Unit]		PU	
[Sco]			
pe			
[Inte Use]			
rnal			
S528	S535	Current Vector Transfer Speed	
[For Type]		Word type, CH type	
mula			
[Type]		FU	
e			
[Sco]			
pe			
[Inte Use]			
rnal			
S536	S567	Current Transfer Speed	
[Type Type]		Word type, Axis type	
e			
[Type]		FU, RPM	
e			

[Range]
ge
[]

S568	S599
Current Speed	

[Type Type] Word type, Axis type
e
[Single] RPM
le
[Range]
ge

[My Use]

S600	S631
Current Torque	

[Typ Type] Word type, Axis type
e

[Sing] 0.01%

le

[Ran]

ge

[Usa]

ge]

S632	S663
Remaining Distance	

[Typ Type] Word type, Axis type
e

[Unit] PU

[Sco]

pe

[Inte Use]

rnal

 Control Information

S900	S931
Feedback Position	

[Type Type] Word Type, Axis Type
e
[Sing] PU
le
[Sco]
pe
[Inte Use]
rnal

S932	S963
Feedback Transfer Speed	

[Type Type] Word type, Axis type
e
[Typ] FU
e
[Sco]
pe
[Inte Use]
rnal

S964	S995
Feedback Speed	

[Type Type] Word type, Axis type
e
[Sing] RPM
le
[Ran]
ge
[]
- RPM is an average value.

S996	S1027
Feedback Torque	

[Type Type] Word Type, Axis Type
e

[Sing] 0.01%
le
[Ran]
ge
[Usa Use]
ge]

S1028	S1059
Control Deviation Vehicle	

[Bro Type] Word type, Axis type

[Single] PU

[Scope]

[Internal Purpose]

 I/O Information

 Handle

S1220	S1222
Handle Input Pulse	
[Type Type] Word Type, General Type [Single]	
[Range]	
[Usage]	

 Axis Control Input/Output

S1223	S1264
Control Output 1	
[Type] Word type, Axis type [Single]	
[Range]	
[Usage]	

S1265	S1296
Control Output 2	
[Type] Word type, Axis type e	
[Pin]	
[Scope]	
pe	
[Usage Use]	
ge	

S1297	S1328
Control Input	
[Type Type] Word type, Axis type e	
[Pin Scope]	
[Scope]	
pe	
[Usage]	
ge	

 Axis DIO-I (Servo)

S1331	
Servo Ready Signal	

[Type Type] Bit Type, Axis
Type [Pin]
[범 We] 0 or 1
[Usage] Use]

- 0 : Positive polarity
- 1: **Negative** polarity

S1332	
Servo Alarm Signal	

[Type] Bit type, Axis type
[Single]
[Range] 0 or 1
[Usage]

- 0 : Positive polarity
- 1: **Negative** polarity

S1333	
Servo In-position Signal	

[Type] Bit type, Axis type
[Single Level]
[Range] 0 or 1
[Usage]

- 0 : Positive polarity
- 1: **Negative** polarity

S1334	
Servo Phase C Signal	

[Type] Bit Type, Axis
Type [Single]
[범 Wei] 0 or 1
[Usage] Use]

- 0 : Positive polarity
- 1: **Negative** polarity

S1339	
Servo On Signal	
[Type]	Bit type, Axis type
[Single]	
[Range]	0 or 1
[Usage]	
- 0 : Positive polarity	
- 1: Negative polarity	

S1340	
Servo Reset Signal	
[Type]	Bit type, Axis type
[Pin Level]	
[Range]	0 or 1
[Usage]	
- 0 : Positive polarity	
- 1: Negative polarity	

S1341	
Servo Pulse Clear Signal	
[Type]	Bit type, Axis type
[Single]	
[범 We]	0 or 1
[Usage]	Use]
- 0 : Positive polarity	
- 1: Negative polarity	

 Axis DIO-II (General)

S1351	
ORG	
[Type]	Bit Type, Axis Type

[Type Scope]

[Range] 0 or 1

[Usage Use]

- 0 : Positive polarity
- 1: **Negative** polarity

S1352	
Latch Signal	

[Type] Bit type, Axis type

[Single]

[Range] 0 or 1

[Usage]

- 0 : Positive polarity
- 1: **Negative** polarity

S1357	
Position Comparison Signal	

[Type] Bit type, Axis type

[Single]

[범 We] 0 or 1

[Usage] Use]

- 0 : Positive polarity
- 1: **Negative** polarity

 Extension Function Information

 Synchronous Control

S1410	S1441
Synchronous Control Derailment Vehicle	

[Type] Word type, Axis type
pe
[Type] PU
pe
[Scope]
pe
[Internal Use]
ern
al Latch



S1482	S1513
Latch Position	

[Type] Word type, Axis type
e
[Pin] PU
[Scope]
pe
[Internal Purpose]
mal se]

8.4.2.2 Storage Area

 System Information

S2010	
System Uptime Date (BCD)	
[Type Format] Word format, General format [Short]	
[Range]	
[Usage]	

S2011	
System Start Time (BCD)	
[Format] Format]	
Word Type, General Type	
[Type]	
[범 We]	
[Purpose] Use]	

S2012	
System Total Operating Days (BCD)	
[Format Format]	
Word Type, General Type	
[Single]	
[Range]	
[Usage]	

S2013	
System Total Uptime (BCD)	
[Format] Format]	
Word Type, Standard Type	
[Single]	
[Range]	
[Usage]	

MP Driving Information

S2066	S2073
Main MP Number	

[Type] Word Type, CH Type
e
[Sing Scope
le]
[Sco]
pe
[Usa Use]
ge

S2074	S2081
Execution MP Number	

[Type Type] Word Type, CH Type
e
[Sing Scope
le]
[Sco]
pe
[Usa Use]
ge

S2082	S2089
Execution MP Call Phase	

[For] Word type, CH type
m
[Scope
]
[Sco]
pe
[Usa Use]
ge

S2090	S2097
Execution MP Block Number	

[Type [Form Word Type, CH Type
e ula]

[Sing Scope

le]

[Sco]

pe

[Usa Use]

ge

S2098	S2105
Execution MP Block Search Information	
[Format] Word Type, CH Type [Single Scope]	
[Scope]	
[Usage]	

S2122	S2377
Execution MP Modal Information	
[Format] Word Type, CH Type [Single Scope]	
[Scope]	
[Usage]	

 Driving Information

S2550	S2581
Current Position (Machine Coordinate System)	
[Type] Word type, Axis type [Single] PU	
[Scope]	
[Usage Use]	

 Control Information

S2750	S2781
Encoder Pulse	
[Type] Word Type, Axis Type [Single] PU	
[Scope]	
[Usage Use]	

 ABS Encoder Information

S2955	
Home Return Completion Status	

[For] Bit type, Axis type
mat]

[Typ]
e
[범 We]

S2956	
Return to Origin Direction	

[Typ Type] Bit type, Axis type
e
[Typ Level]
e
[Ran] 0 or 1 (Default: 0)
ge
[]

- 0: Positive direction
- 1: - direction

S2961	S2992
Encoder Offset Pulse	

[Type] Word type, Axis type
[Single] PU
[Scope]

8.5 Q Register

8.5.1 Configuration

■ Configuration

Description	System Control Data
Data Type	Bit, Word
Structure	
1. System Mode Control	① System Mode
	② System Reset
2. System Data Control	① System Data Control Command Processing Start Signal
	② System Data Control Command Code
	③ System Data Control Data 0 ~ 3
3. MP Program Selection	① MP Program Number

8.5.2 Detailed Description

 System Mode Control

Q5	
System Mode	
[Type]	Word Type, General
Type [Single	Scope]
[Scope]	0, 1, 2
[Usage	Purpose]
- 0 : Run, 1 : Stop, 2 : Shutdown	

Q7	
System Reset	
[Format]	Word type,
General type [Single	Scope]
[Scope]	
[Usage]	

 System Data Control

Q20	
System Data Control Command Processing Start Signal	
[Format	Type] Word type,
General type [Single	]
[Range]	
[My Use]	

Q21	
System Data Control Command Code	
[Format	Format]
	Word type, General type [
	Scope]
[Sc	] 1 or 2 [
	Use]
— 1. System Privilege Level Approval Request	
	:: Data 0 : Password
— 2. System Date/Time Change	
	:: Data 0 : Data, Ex] 0x20040611 (2004/6/11, BCD format)
	:: Data 1: Time, Ex] 0x12251234 (12:25:12:34, BCD format)

Q22	Q25	
System Data Control Data 0 ~ 3		
[Format	Format]	
	Word Type, General Type	
[Single	]	
[Range	]	
[Usage	]	
— Q22: System Data Control Data 0		
— Q23: System Data Control Data 1		
— Q24: System Data Control Data 2		
— Q25: System Data Control Data 3		

Q45	Q52	
MP Program Number		
[Type	]	Word Type, Channel Type
[Single	Scope]	
[Scope	]	
[Usage	]	

8.6 E Register

8.6.1 Configuration

■ Configuration

Description	System Event (Alarm/Warning)
Data Type	Bit, Word
Structure	
1. Common Area	
2. System Area	
3. Communication Area	
4. Motion Program Area	① Common Area
	② CH1 Area
	③ CH2 to CH8 Area
5. Motion Control Area	① Common Area
	② CH1 Area
	③ CH2 to CH8 Area
	④ Axis1 Area
	⑤ Axis2 to Axis32 Area
6. PLC Program Area	① Common Area
	② S-PLC Area
	③ H-PLC Area
	④ N-PLC Area
	⑤ I-PLC Area
	⑥ E-PLC Area
7. External I/O Area	
8. User Area	

8.6.2 Detailed Description



Information Area

Block	Min	Size	No	My Use
Information Area		1	5	Event Status
		1	8	Event Code
		1	9	Event Data 1
		1	10	Event Data 2
		1	11	Event Data 3
		1	12	Event Data 4

※ Event Status

[Event Status] data consists of 1 word, representing 32 bits of status data. The event status data in the common area allows you to check the event occurrence status by event type. For example, if bit 0 is set to 1, a SYS event has occurred; if bit 8 is set to 1, a USER event has occurred.

		EIO	PLC	MC	MP	COM	SYS
							USER

Events are categorized into the following 7 items.

※ Event Classification

Abbreviation	Description	Scope	Code
① SYS	System	25 ~	100 ~
② COM	Communication	40 ~	200 ~
③ MP	Motion Program	65 ~	300 ~
④ MC	Motion Control	110 ~	500 ~
⑤ PLC	PLC Program	325 ~	700 ~
⑥ EIO	External Input/Output, Device	360 ~	900 ~
⑦ USER	User Event	420 ~	2000 ~

System Area

Unit	Min	Size	No	My	Use
Information Area		1	25		Event Status
System Area		2	26		Event Table
		1	28		Event Code
		1	29		Event Data1
		1	30		Event Data 2

※ Event Table

[Event Table] Data consists of 2 words, representing 32-bit status data. This data indicates the occurrence status for each code within System area events.

Events in the System area are assigned starting from code 100, and the event occurrence status can be determined based on the value of the corresponding bit.

			...	103	102	101	100
					...	109	108
					...	116	115
					...	:	:



Communication Area

District	Minute	Size	No	Internal Use
Information Area		1	40	Event Status (General Area)
		8	42	Event Status (Communication Area)
General Area		1	50	Event Code
		2		Reservation Area
Communication Area		1	53	Event Code
		1	54	Event Data1
		1	55	Event Data 2
		1	56	Event Data 3

※ Event Status (General Area)

[Event Status] data consists of 1 word, representing 32 bits of status data. It allows you to check the occurrence status of common communication events.

							COMM

※ Event Status (Communication Area)

[Event Status] data consists of 8 words, representing 32-bit status data.

It indicates the status of events occurring on up to 4 ports per device, allowing you to check the status of up to 32 devices.

Device 1 ->					Port4	Port3	Port2	Port1
Device 2 ->					Port4	Port3	Port2	Port1
Device 3 ->					Port4	Port3	Port2	Port1
:					:	:	:	:



Motion Program Area

Circle	Min	Size	Type	Internal	Use
Information Area		1	65	Event Status (General Area)	
		1	66	Event Status (Channel Area)	
General Area		1	67	Event Code	
CH1 Area		1	70	Event Code	
		1	71	Event Data 1 (CH Number)	
		1	72	Event Data 2 (File Name)	
		1	73	Event Data 3 (Line Number)	
CH2 ~ CH8 Range		:	:	:	

※ Event Status (General Area)

[Event Status] Data consists of 1 word, representing 32 bits of status data. It allows you to check the occurrence status of common events in the motion program.

							COMM

※ Event Status (Channel Area)

[Event Status] data is composed of 1 word, representing 32 bits of status data. It allows you to check the event occurrence status for each channel in the motion program.

For example, if the value of bit 3 is 1, it indicates an event has occurred on channel 4.

CH8	CH7	CH6	CH5	CH4	CH3	CH2	CH1

**Motion Control Area**

Circle	Min	Size	Type	Internal Use
Information Area		1	110	Event Status (General Area)
		1	111	Event Status (Channel Area)
		1	112	Event Information (Axis Area)
General Area		2	113	Event Table
		1	115	Event Code
CH1 Area		2	118	Event Table
		1	120	Event Code
		1	121	Event Data (CH Number)
CH2 ~ CH8 Range		:	:	:
Axis 1 Area		2	158	Event Table
		1	160	Event Code
		1	161	Event Data 1 (axis number)
		1	162	Event Data 2 (Alarm Number, etc.)
Axis2 ~ Axis32 Range		:	:	:

※ Event Status (General Area)

[Event Status] data consists of 1 word, representing 32 bits of status data. It allows you to check the occurrence status of common motion control events.

							COMM

※ Event Status (Channel Area)

[Event Status] data is composed of 1 word, representing 32 bits of status data. It allows you to check the event occurrence status for each channel in motion control.

For example, if bits 2 and 7 have a value of 1, it indicates that an event has occurred on channels 3 and 8.

CH8	CH7	CH6	CH5	CH4	CH3	CH2	CH1

※ Event Status (Axis Area)

[Event Status] Data consists of 1 word, representing 32 bits of status data. It allows you to check the event occurrence status per axis in motion control.

Axis8	Axis7	Axis6	Axis5	Axis4	Axis3	Axis2	Axis1
Axis16	Axis15	Axis14	Axis13	Axis12	Axis11	Axis10	Axis9
Axis24	Axis23	Axis22	Axis21	Axis20	Axis19	Axis18	Axis17
Axis32	Axis31	Axis30	Axis29	Axis28	Axis27	Axis26	Axis25

※ Event Table (General Area) – Code: 500 ~ 563

[Event Table] Data consists of 2Words, representing 32-bit status data. This data indicates the occurrence status for each code within the motion control area events.

General area events in the motion control area are assigned 64 codes starting from code 500, and the event occurrence status can be determined based on the value of the corresponding bit.

			...	503	502	501	500
					...	509	508
					...	516	515
					...	:	:

※ Event Table (Channel Area) – Code: 564 ~ 627

[Event Table] data consists of 2Words, representing 32-bit status data.

Events in the channel area of the motion control region are assigned codes starting from code 564 through 64 codes. The event occurrence status can be determined based on the value of the corresponding bit.

			...	567	566	565	564
					...	573	572
					...	581	580
					...	:	:

※ Event Table (Axis Area) – Code: 628 ~ 691

[Event Table] data consists of 2Words, representing 32-bit status data. This data indicates the occurrence status for each code within the motion control area events.

Axis area events in the motion control area are assigned codes starting from code 628, totaling 64 codes. The event occurrence status can be determined based on the value of the corresponding bit.

			...	631	630	629	628
					...	637	636
					...	645	644
					...	:	:



PLC Program Area

Block	Min	Size	Type	Internal	Use
Information Area		1	325		Event Status (General Area)
		1	326		Event Status (Channel Area)
General Area		1	327		Event Code
S-PLC CH Area		1	330		Event Code
		1	331		Event Data 1 (CH Number)
		1	332		Event Data 2 (File Name)
		1	333		Event Data 3 (Step Number)
N-PLC CH Area		1			Event Code
		1			Event Data 1 (CH Number)
		1			Event Data 2 (File Name)
		1			Event Data 3 (Step Number)

※ Event Status (General Area)

[Event Status] Data consists of 1 word, representing 32 bits of status data. It allows you to check the occurrence status of common events in the PLC program.

							COMM

※ Event Status (Channel Area)

[Event Status] data is composed of 1 word, representing 32 bits of status data. It allows you to check the event occurrence status for each channel in the PLC program.

For example, if the value of bit 1 is 1, it indicates that an event has occurred in the Normal PLC.

						PLC-N	PLC-S

※ PLC Channel

- ① PLC-S - Start PLC
- ② PLC-N - Normal Speed PLC

**External I/O Area**

Block	Min	Size	Type	My Use
Information Area		1	360	Event Status (General Area)
		1	361	Event Status (Device Area)
Device Area		1	374	Event Code
		1	375	Event Data1
		1	376	Event Data 2
		1	377	Event Data 3
		1	378	Event Data 4

※ Event Status (Device Area)

[Event Status] data consists of 1 word, representing 32 bits of status data.

It allows checking the event occurrence status per device for external I/O and displays information for up to 20 devices.

DEV8	DEV7	DEV6	DEV5	DEV4	DEV3	DEV2	DEV1
DEV16	DEV15	DEV14	DEV13	DEV12	DEV11	DEV10	DEV9
				DEV20	DEV19	DEV18	DEV17

User Area

Group	Min	Size	Type	My	Use
Information Area		1	420		Event Status
		4	421		Event Table
		4	425		Event Table (Event Log Area)
		1	429		Event Code
		1	430		Event Data1
		1	431		Event Data 2

※ Event Information (General Area)

[Event Information] Data consists of 1 word, representing 32-bit status data. It allows you to check the occurrence status of common events in the user area.

							COMM

※ Event Table – Code: 2000 ~ 2127

[Event Table] Data consists of 4Words, representing 32-bit status data.

General events in the user area are assigned 128 codes starting from code 2000, and the event occurrence status can be determined based on the value of the corresponding bit.

			...	567	566	565	2000
					...	573	572
					...	581	580
					...	:	:

※ Event Table (Event Log Area) – Code: 2128 ~ 2255

This data represents the occurrence status of events by code within the Event Log area of the user domain. A total of 128 codes are assigned, starting from code 2128.

			...	631	630	629	628
					...	637	636
					...	645	644
					...	:	:

8.7 G/F Register

8.7.1 Configuration

■ Configuration

Description	PLC & System Interface Data
Data Type	Bit, Word, Float
Structure	
1. Operation (Preparation, System Operation Control)	① Emergency, Reset (System, CH, Axis)
	② Ready (System, Channel, Axis)
	③ Servo Motor Control (Servo Motor Off, Servo Motor Reset)
	④ OT, IT
2. MP Operation (MP Basic Operation Functions)	① Mode (CH) Setting
	② Start, Feed Hold, Single Block, Skip (CH)
	③ Interpolation Transfer Override, Positioning Override
	④ Auxiliary Codes (M Code, S Code, T Code)
3. MC Operation (Basic MC Operation Functions)	① Mode (Axis) Setting
	② Mode Change Signal During Feed
	③ Position/Speed/Torque Command Values
	④ Start, Abort, Busy, Error
	⑤ MC Override
	⑥ Setting 1st to 4th Reference Points
	⑦ Home Return Complete Signal
	⑧ Origin Transfer Signal
	⑨ Origin Deceleration Signal
4. Operation Status	① Signal During Transfer in +, - Direction
	② Constant Speed Section
	③ In-position
5. Key (Extended) Functions (MP/MC Operation Expansion Features)	① Synchronous Control
	② External Position Determination Function
	③ Position Latch Function Selection
	④ Position Compare Function Selection

8.7.2 Detailed Description

Area	PLC → GX		PLC ← GX		Content
	Symbol	G Map	Symbol	F Map	
O p e r a t i o n	<u>GEMG</u>	GB5.00			System Emergency Settings
	GRST	GB5.01	FRSTS	FB5.01	System Reset Setting/Confirmation Signal
	<u>GCHEMG1</u> <u>GCHEMG8</u>	GB5.08 ~ GB5.15			Emergency Settings per CH
	GCHRST1 GCHRST8	GB5.16 ~ GB5.24	FCHRSTS1 FCHRSTS8	FB5.16 ~ FB5.24	CH Reset Setting/Confirmation Signal
	<u>GAXEMG1</u> <u>GAXEMG32</u>	GB6.00 ~ GB6.31			Axis-specific Emergency Setting
	GAXRST1 GAXRST32	GB7.00 ~ GB7.31	FAXRSTS1 FAXRSTS32	FB7.00 ~ FB7.31	Axis-specific Reset Setting/Confirmation Signal
			FRDY	FB8.00	System Ready Confirmation Signal
			FCHRDY1 FCHRDY8	FB8.08 ~ FB8.15	Channel-Specific Ready Confirmation Signal
			FAXRDY1 FAXRDY32	FB9.00 ~ FB9.31	Axis-specific Ready Confirmation Signal
	GSVOFF1 GSVOFF32	GB12.00 ~ GB12.31			Servo Motor Off Setting
	GSVRST1 GSVRST32	GB13.00 ~ GB13.31			Servo Motor Reset Setting
	<u>GPOT1</u> <u>GPOT32</u>	GB20.00 ~ GB20.31			Axis-specific + OT Settings
	<u>GNOT1</u> <u>GNOT32</u>	GB21.00 ~ GB21.31			Axis-specific - OT Settings
	<u>GPIT1</u> <u>GPIT32</u>	GB25.00 ~ GB25.31			Axis-specific + Interlock Settings
	<u>GNIT1</u> <u>GNIT32</u>	GB26.00 ~ GB26.31			Axis-specific - Interlock Settings
M P O p e r a t i o n	GMEM1	GB50.00	FMEM1	FB50.00	CH1 MP Mode Set/Check Signal
	GMEM2	GB50.04	FMEM2	FB50.04	CH2 MP Mode Set/Confirm Signal
	GMEM3	GB50.08	FMEM3	FB50.08	CH3 MP Mode Set/Confirm Signal
	GMEM4	GB50.12	FMEM4	FB50.12	CH4 MP Mode Set/Confirm Signal
	GMEM5	GB50.16	FMEM5	FB50.16	CH5 MP Mode Set/Confirm Signal
	GMEM6	GB50.20	FMEM6	FB50.20	CH6 MP Mode Set/Confirm Signal
	GMEM7	GB50.24	FMEM7	FB50.24	CH7 MP Mode Set/Confirm Signal
	GMEM8	GB50.28	FMEM8	FB50.28	CH8 MP Mode Set/Confirm Signal
	GMPST1 GMPST8	GB65.00 ~ GB65.07	FMPSTS1 FMPSTS8	FB65.00 ~ FB65.07	CH-specific MP Start Setup/Confirmation Signal
	<u>GMPFH1</u> <u>GMPFH8</u>	GB65.16 ~ GB65.23	FMPFHS1 FMPFHS8	FB65.16 ~ FB65.23	Channel-specific Feed Hold Set/Check Signal
	GMPSB1 GMPSB8	GB65.24 ~ GB65.31	FMPSBS1 FMPSBS8	FB65.24 ~ FB65.31	Single Block per Channel Set/Check Signal

n	GOBSKP1 GOBSKP8	GB69.00 ~ GB69.07			Operation Block Skip Setting per CH
	GSKP11 GSKP18	GB70.00 ~ GB70.07			CH-specific Skip 1 setting
	GSKP21 GSKP28	GB70.08 ~ GB70.15			CH-specific Skip 2 setting
	GMPCFOR	GW80 ~ GW87			CP Feed Override Setting per Channel
	GMPPFOR	GW88 ~ GW95			PP Feed Override Settings per Channel
	GMF1 GMF8	GB100.00 ~ GB100.07	FMST1 FMST8	FB100.00 ~ FB100.07	CH-specific M Code Setting/Confirmation Signal
	GSF1 GSF8	GB100.08 ~ GB100.15	FSST1 FSST8	FB100.08 ~ FB100.15	Channel-specific S Code setting/confirmation signal
	GTF1 GTF8	GB100.16 ~ GB100.23	FTST1 FTST8	FB100.16 ~ FB100.23	Channel-specific T Code setting/confirmation signal
			FMEOP1 FMEOP8	FB101.00 ~ FB101.07	CH-specific M Code EOP confirmation signal
			FSEOP1 FSEOP8	FB101.08 ~ FB101.15	CH-specific S Code EOP confirmation signal
			FTEOP1 FTEOP8	FB101.16 ~ FB101.23	CH-specific T Code EOP confirmation signal
			FMNO	FW102 ~ FW109	M Code No. per CH
			FMD	FW110 ~ FW117	CH-specific M Code Data
			FSNO	FW134 ~ FW141	S Code No. by CH
			FSD	FW142 ~ FW149	CH-specific S Code Data
MC O p e r a t i o n	GPOS1	GB200.00	FPOS1	FB200.00	Axis 1 MC Mode Set/Confirm Signal
	GJOG1	GB200.01	FJOG1	FB200.01	
	GHAN1	GB200.02	FHAN1	FB200.02	
	GREF1	GB200.03	FREF1	FB200.03	
	GVEL1	GB200.04	FVEL1	FB200.04	
	GTRQ1	GB200.05	FTRQ1	FB200.05	
	GPOS2	GB200.08	FPOS2	FB200.08	Axis 2 MC Mode Set/Confirm Signal
	GJOG2	GB200.09	FJOG2	FB200.09	
	GHAN2	GB200.10	FHAN2	FB200.10	
	GREF2	GB200.11	FREF2	FB200.11	
	GVEL2	GB200.12	FVEL2	FB200.12	
	GTRQ2	GB200.13	FTRQ2	FB200.13	
	GPOS3	GB200.16	FPOS3	FB200.16	Axis 3 MC Mode Set/Confirm Signal
	GJOG3	GB200.17	FJOG3	FB200.17	
	GHAN3	GB200.18	FHAN3	FB200.18	
	GREF3	GB200.19	FREF3	FB200.19	

	GVEL3	GB200.20	FVEL3	FB200.20	Axis 4 MC Mode Set/Confirm Signal
	GTRQ3	GB200.21	FTRQ3	FB200.21	
	GPOS4	GB200.24	FPOS4	FB200.24	
	GJOG4	GB200.25	FJOG4	FB200.25	
	GHAN4	GB200.26	FHAN4	FB200.26	
	GREF4	GB200.27	FREF4	FB200.27	
	GVEL4	GB200.28	FVEL4	FB200.28	
	GTRQ4	GB200.29	FTRQ4	FB200.29	
	GJOG5 GTRQ32	GW201 ~ GW207	FJOG5 FTRQ32	FW201 ~ FW207	Axis 5 ~ Axis 32 MC Mode Set/Check Signal
	GMCP	GW215 ~ GW246			MC Pos Command Settings per Axis
	GMCF	GW247 ~ GW278			MC Feed Command Settings per Axis
	GMCV	GW279 ~ GW310			MC Velocity Command per Axis Setting
	GMCT	GW311 ~ GW342			MC Torque Command per Axis Setting
	GMCABS1 GMCABS32	GB345.00 ~ GB345.31			Axis-Specific MC Position Command Type Setting
	GMCDIR1 GMCDIR32	GB346.00 ~ GB346.31			MC Command Direction per Axis Setting
	GMCST1 GMCST32	GB350.00 ~ GB350.31	FMCST1 FMCST32	FB350.00 ~ FB350.31	Axis-specific MC Start Setup/Confirmation Signal
	GM CAB1 GM CAB32	GB351.00 ~ GB351.31			MC Abort Setting per Axis
			FMCBUSY1 FMCBUSY32	FB355.00 ~ FB355.31	Axis-Specific MC Busy Confirmation Signal
			FM CER1 FM CER32	FB356.00 ~ FB356.31	MC Error Check Signal per Axis
	GMCFOR	GW360 ~ GW391			MC Feed Override Setting per Axis
	GPIS1 GPIS32	GB396.00 ~ GB396.31			Axis-specific In-position Check Skip
	GPUI1 GPUI32	GB397.00 ~ GB397.31			User interpolation mode signal per axis
			FRF1 FRF32	FB405.00 ~ FB405.31	Axis-specific MC origin return completion signal
	GRDEC1 GRDEC32	GB408.00 ~ GB408.31			Axis-specific MC origin return deceleration signal
			FRP1S01 FRP1S32	FB411.00 ~ FB411.31	Axis-specific MC First Origin Confirmation Signal
			FRP2S01 FRP2S32	FB412.00 ~ FB412.31	Axis-specific MC Second Origin Confirmation Signal
			FRP3S01 FRP3S32	FB413.00 ~ FB413.31	Axis-specific MC Third Origin Confirmation Signal
			FRP4S01 FRP4S32	FB414.00 ~ FB414.31	Axis-specific MC 4th origin confirmation signal

Op era tin g Sta tus			FPMV1 FPMV32	FB500.00 ~ FB500.31	Axis-specific + direction movement confirmation signal
			FNMV1 FNMV32	FB501.00 ~ FB501.31	Axis-specific - Direction Movement Confirmation Signal
			FINP1 FINP32	FB510.00 ~ FB510.31	Axis-Specific In-Position Confirmation Signal
			FCSAGR1 FCSAGR32	FB512.00 ~ FB512.31	Axis-Specific Command Speed Attained Confirmation Signal
			FZSAGR1 FZSAGR32	FB513.00 ~ FB513.31	Axis-specific 0 speed reach confirmation signal
Ex pa nsi on Fu nct ion	GSYN1 GSYN32	GB550.00 ~ GB550.31	FSYNS1 FSYNS32	FB550.00 ~ FB550.31	Axis-specific synchronous control setting/confirmation signal
	GMCEXP1 GMCEXP32	GB558.00 ~ GB558.31	FMCEXPS1 FMCEXPS32	FB558.00 ~ FB558.31	MC External Positioning per Axis Set/Check Signal
	GPL1 GPL32	GB561.00 ~ GB561.31	FPLS1 FPLS32	FB561.00 ~ FB561.31	Position Latch per Axis Set/Check Signal
	GPC1 GPC32	GB565.00 ~ GB565.31			Position Compare Setting per Axis

※ For details on operation, refer to Chapter 5, Control Function Operation Preparation section (P.5-22 ~ 5-26)

※ For details on MP Operation, refer to Chapter 5, Control Functions, MP Operation Mode section (P.5-29 ~ 5-33)

※ For details on MC Operation, refer to Chapter 5, Control Function MC Operation Mode section (P.5-34 ~ 5-50)

※ For details on the operating status, refer to Chapter 5, Control Function Status Output section. (P.5-51 ~ 5-54)

※ For details on extension functions, refer to Chapter 5: Control Functions - Synchronous Control, (P.5-55 ~ 5-62)

External Position Determination,

Position Latch Function, Position Comparison Function.



Run, Reset Command / Ready, Reset Status

G005							GRST	GEMG
	<u>GCHEMG8</u>	<u>GCHEMG7</u>	<u>GCHEMG6</u>	<u>GCHEMG5</u>	<u>GCHEMG4</u>	<u>GCHEMG3</u>	<u>GCHEMG2</u>	<u>GCHEMG1</u>
	GCHRST8	GCHRST7	GCHRST6	GCHRST5	GCHRST4	GCHRST3	GCHRST2	GCHRST1
F005							FRSTS	
	FCHRST8	FCHRST7	FCHRST6	FCHRST5	FCHRST4	FCHRST3	FCHRST2	FCHRST1

G006	<u>GAXEMG8</u>	<u>GAXEMG7</u>	<u>GAXEMG6</u>	<u>GAXEMG5</u>	<u>GAXEMG4</u>	<u>GAXEMG3</u>	<u>GAXEMG2</u>	<u>GAXEMG1</u>
	<u>GAXEMG16</u>	<u>GAXEMG15</u>	<u>GAXEMG14</u>	<u>GAXEMG13</u>	<u>GAXEMG12</u>	<u>GAXEMG11</u>	<u>GAXEMG10</u>	<u>GAXEMG9</u>
	<u>GAXEMG24</u>	<u>GAXEMG23</u>	<u>GAXEMG22</u>	<u>GAXEMG21</u>	<u>GAXEMG20</u>	<u>GAXEMG19</u>	<u>GAXEMG18</u>	<u>GAXEMG17</u>
	<u>GAXEMG32</u>	<u>GAXEMG31</u>	<u>GAXEMG30</u>	<u>GAXEMG29</u>	<u>GAXEMG28</u>	<u>GAXEMG27</u>	<u>GAXEMG26</u>	<u>GAXEMG25</u>
F006								

G007	GAXRST8	GAXRST7	GAXRST6	GAXRST5	GAXRST4	GAXRST3	GAXRST2	GAXRST1
	GAXRST16	GAXRST15	GAXRST14	GAXRST13	GAXRST12	GAXRST11	GAXRST10	GAXRST9
	GAXRST24	GAXRST23	GAXRST22	GAXRST21	GAXRST20	GAXRST19	GAXRST18	GAXRST17
	GAXRST32	GAXRST31	GAXRST30	GAXRST29	GAXRST28	GAXRST27	GAXRST26	GAXRST25
F007	FAXRSTS8	FAXRSTS7	FAXRSTS6	FAXRSTS5	FAXRSTS4	FAXRSTS3	FAXRSTS2	FAXRSTS1
	FAXRSTS 16	FAXRSTS 15	FAXRSTS 14	FAXRSTS 13	FAXRSTS 12	FAXRSTS 11	FAXRSTS 10	FAXRSTS9
	FAXRSTS 24	FAXRSTS 23	FAXRSTS 22	FAXRSTS 21	FAXRSTS 20	FAXRSTS 19	FAXRSTS 18	FAXRSTS 17
	FAXRSTS 32	FAXRSTS 31	FAXRSTS 30	FAXRSTS 29	FAXRSTS 28	FAXRSTS 27	FAXRSTS 26	FAXRSTS 25

G008								
F008								FRDY
	FCHRDY8	FCHRDY7	FCHRDY6	FCHRDY5	FCHRDY4	FCHRDY3	FCHRDY2	FCHRDY1

G009								
F009	FAXRDY8	FAXRDY7	FAXRDY6	FAXRDY5	FAXRDY4	FAXRDY3	FAXRDY2	FAXRDY1
	FAXRDY16	FAXRDY15	FAXRDY14	FAXRDY13	FAXRDY12	FAXRDY11	FAXRDY10	FAXRDY9
	FAXRDY24	FAXRDY23	FAXRDY22	FAXRDY21	FAXRDY20	FAXRDY19	FAXRDY18	FAXRDY17
	FAXRDY32	FAXRDY31	FAXRDY30	FAXRDY29	FAXRDY28	FAXRDY27	FAXRDY26	FAXRDY25

**Servo Motor Off**

G012	GSVOFF8	GSVOFF7	GSVOFF6	GSVOFF5	GSVOFF4	GSVOFF3	GSVOFF2	GSVOFF1
	GSVOFF16	GSVOFF15	GSVOFF14	GSVOFF13	GSVOFF12	GSVOFF11	GSVOFF10	GSVOFF9
	GSVOFF24	GSVOFF23	GSVOFF22	GSVOFF21	GSVOFF20	GSVOFF19	GSVOFF18	GSVOFF17
	GSVOFF32	GSVOFF31	GSVOFF30	GSVOFF29	GSVRST28	GSVOFF27	GSVOFF26	GSVOFF25
F012								

**Servo Motor Reset**

G013	GSVRST8	GSVRST7	GSVRST6	GSVRST5	GSVRST4	GSVRST3	GSVRST2	GSVRST1
	GSVRST16	GSVRST15	GSVRST14	GSVRST13	GSVRST12	GSVRST11	GSVRST10	GSVRST9
	GSVRST24	GSVRST23	GSVRST22	GSVRST21	GSVRST20	GSVRST19	GSVRST18	GSVRST17
	GSVRST32	GSVRST31	GSVRST30	GSVRST29	GSVRST28	GSVRST27	GSVRST26	GSVRST25
F013								

**P-Overtravel**

G020	<u>GPOT8</u>	<u>GPOT7</u>	<u>GPOT6</u>	<u>GPOT5</u>	<u>GPOT4</u>	<u>GPOT3</u>	<u>GPOT2</u>	<u>GPOT1</u>
	<u>GPOT16</u>	<u>GPOT15</u>	<u>GPOT14</u>	<u>GPOT13</u>	<u>GPOT12</u>	<u>GPOT11</u>	<u>GPOT10</u>	<u>GPOT9</u>
	<u>GPOT24</u>	<u>GPOT23</u>	<u>GPOT22</u>	<u>GPOT21</u>	<u>GPOT20</u>	<u>GPOT19</u>	<u>GPOT18</u>	<u>GPOT17</u>
	<u>GPOT32</u>	<u>GPOT31</u>	<u>GPOT30</u>	<u>GPOT29</u>	<u>GPOT28</u>	<u>GPOT27</u>	<u>GPOT26</u>	<u>GPOT25</u>
F020								

**N-Overtravel**

G021	<u>GNOT8</u>	<u>GNOT7</u>	<u>GNOT6</u>	<u>GNOT5</u>	<u>GNOT4</u>	<u>GNOT3</u>	<u>GNOT2</u>	<u>GNOT1</u>
	<u>GNOT16</u>	<u>GNOT15</u>	<u>GNOT14</u>	<u>GNOT13</u>	<u>GNOT12</u>	<u>GNOT11</u>	<u>GNOT10</u>	<u>GNOT9</u>
	<u>GNOT24</u>	<u>GNOT23</u>	<u>GNOT22</u>	<u>GNOT21</u>	<u>GNOT20</u>	<u>GNOT19</u>	<u>GNOT18</u>	<u>GNOT17</u>
	<u>GNOT32</u>	<u>GNOT31</u>	<u>GNOT30</u>	<u>GNOT29</u>	<u>GNOT28</u>	<u>GNOT27</u>	<u>GNOT26</u>	<u>GNOT25</u>
F021								

**P-Axis Interlock**

G025	<u>GPIT8</u>	<u>GPIT7</u>	<u>GPIT6</u>	<u>GPIT5</u>	<u>GPIT4</u>	<u>GPIT3</u>	<u>GPIT2</u>	<u>GPIT1</u>
	<u>GPIT16</u>	<u>GPIT15</u>	<u>GPIT14</u>	<u>GPIT13</u>	<u>GPIT12</u>	<u>GPIT11</u>	<u>GPIT10</u>	<u>GPIT9</u>
	<u>GPIT24</u>	<u>GPIT23</u>	<u>GPIT22</u>	<u>GPIT21</u>	<u>GPIT20</u>	<u>GPIT19</u>	<u>GPIT18</u>	<u>GPIT17</u>
	<u>GPIT32</u>	<u>GPIT31</u>	<u>GPIT30</u>	<u>GPIT29</u>	<u>GPIT28</u>	<u>GPIT27</u>	<u>GPIT26</u>	<u>GPIT25</u>
F025								

**N-Axis Interlock**

G026	GNIT8	GNIT7	GNIT6	GNIT5	GNIT4	GNIT3	GNIT2	GNIT1
	GNIT16	GNIT15	GNIT14	GNIT13	GNIT12	GNIT11	GNIT10	GNIT9
	GNIT24	GNIT23	GNIT22	GNIT21	GNIT20	GNIT19	GNIT18	GNIT17
	GNIT32	GNIT31	GNIT30	GNIT29	GNIT28	GNIT27	GNIT26	GNIT25
F026								

**MP Mode**

G050				GMEM2				GMEM1
				GMEM4				GMEM3
				GMEM6				GMEM5
				GMEM8				GMEM7
F050				FMEM2				FMEM1
				FMEM4				FMEM3
				FMEM6				FMEM5
				FMEM8				FMEM7

**MP Start, Hold, Single Block**

G065	GMPST8	GMPST7	GMPST6	GMPST5	GMPST4	GMPST3	GMPST2	GMPST1
	<u>GMPFH8</u>	<u>GMPFH7</u>	<u>GMPFH6</u>	<u>GMPFH5</u>	<u>GMPFH4</u>	<u>GMPFH3</u>	<u>GMPFH2</u>	<u>GMPFH1</u>
	GMPSB8	GMPSB7	GMPSB6	GMPSB5	GMPSB4	GMPSB3	GMPSB2	GMPSB1
F065	FMPST8	FMPST7	FMPST6	FMPST5	FMPST4	FMPST3	FMPST2	FMPST1
	FMPFHS8	FMPFHS7	FMPFHS6	FMPFHS5	FMPFHS4	FMPFHS3	FMPFHS2	FMPFHS1
	FMPSBS8	FMPSBS7	FMPSBS6	FMPSBS5	FMPSBS4	FMPSBS3	FMPSBS2	FMPSBS1

 MP Optional Block Skip

	GOBSKP8	GOBSKP7	GOBSKP6	GOBSKP5	GOBSKP4	GOBSKP3	GOBSKP2	GOBSKP1
G069								
F069								

 MP Skip

	GSKP18	GSKP17	GSKP16	GSKP15	GSKP14	GSKP13	GSKP12	GSKP11
G070	GSKP28	GSKP27	GSKP26	GSKP25	GSKP24	GSKP23	GSKP22	GSKP21
F070								

 MP CP Feed Override

G080 ~087	GMPCFOR(8Word)
F080 ~087	

 MP PP Feed Override

G088 ~095	GMPPFOR(8-word)
F088 ~095	



Auxiliary Code

G100	GMF8	GMF7	GMF6	GMF5	GMF4	GMF3	GMF2	GMF1
	GSF8	GSF7	GSF6	GSF5	GSF4	GSF3	GSF2	GSF1
	GTF8	GTF7	GTF6	GTF5	GTF4	GTF3	GTF2	GTF1
F100	FMEOP8	FMEOP7	FMEOP6	FMEOP5	FMEOP4	FMEOP3	FMEOP2	FMEOP1
	FSEOP8	FSEOP7	FSEOP6	FSEOP5	FSEOP4	FSEOP3	FSEOP2	FSEOP1
	FTEOP8	FTEOP7	FTEOP6	FTEOP5	FTEOP4	FTEOP3	FTEOP2	FTEOP1

G102 ~109	
F102 ~109	FMNO((8Word))
G110 ~117	
F110 ~117	FMD((8Word))

G134 ~141	
F134 ~141	FSNO((8Word))
G142 ~149	
F142 ~149	FSD((8Word))

G158 ~165	
F158 ~165	FTD((8Word))

 MC Mode

G200			GTRQ1	GVEL1	GREF1	GHAN1	GJOG1	GPOS1
			GTRQ2	GVEL2	GREF2	GHAN2	GJOG2	GPOS2
			GTRQ3	GVEL3	GREF3	GHAN3	GJOG3	GPOS3
			GTRQ4	GVEL4	GREF4	GHAN4	GJOG4	GPOS4
F200			FTRQ1	FVEL1	FREF1	FHAN1	FJOG1	FPOS1
			FTRQ2	FVEL2	FREF2	FHAN2	FJOG2	FPOS2
			FTRQ3	FVEL3	FREF3	FHAN3	FJOG3	FPOS3
			FTRQ4	FVEL4	FREF4	FHAN4	FJOG4	FPOS4

 MC POS Command / Status

G215 ~246	GMCP((32-word))
F215 ~246	

 MC Feed Command / Status

G247 ~278	GMCF((32Words))
F247 ~278	

**MC Velocity Command / Status**

G279 ~310	GMCV((32Word))
F279 ~310	

**MC Torque Command / Status**

G311 ~342	GMCT((32Words))
F311 ~342	

**MC Position Command Type**

G345	GMCABS8	GMCABS7	GMCABS6	GMCABS5	GMCABS4	GMCABS3	GMCABS2	GMCABS1
	GMCABS16	GMCABS15	GMCABS14	GMCABS13	GMCABS12	GMCABS11	GMCABS10	GMCABS9
	GMCABS24	GMCABS23	GMCABS22	GMCABS21	GMCABS20	GMCABS19	GMCABS18	GMCABS17
	GMCABS32	GMCABS31	GMCABS30	GMCABS29	GMCABS28	GMCABS27	GMCABS26	GMCABS25
F345								

**MC Command Direction**

G346	GMCDIR8	GMCDIR7	GMCDIR6	GMCDIR5	GMCDIR4	GMCDIR3	GMCDIR2	GMCDIR1
	GMCDIR16	GMCDIR15	GMCDIR14	GMCDIR13	GMCDIR12	GMCDIR11	GMCDIR10	GMCDIR9
	GMCDIR24	GMCDIR23	GMCDIR22	GMCDIR21	GMCDIR20	GMCDIR19	GMCDIR18	GMCDIR17
	GMCDIR32	GMCDIR31	GMCDIR30	GMCDIR29	GMCDIR28	GMCDIR27	GMCDIR26	GMCDIR25
F346								

**MC Start / State**

G350	GMCST8	GMCST7	GMCST6	GMCST5	GMCST4	GMCST3	GMCST2	GMCST1
	GMCST16	GMCST15	GMCST14	GMCST13	GMCST12	GMCST11	GMCST10	GMCST9
	GMCST24	GMCST23	GMCST22	GMCST21	GMCST20	GMCST19	GMCST18	GMCST17
	GMCST32	GMCST31	GMCST30	GMCST29	GMCST28	GMCST27	GMCST26	GMCST25
F350	FMCST8	FMCST7	FMCST6	FMCST5	FMCST4	FMCST3	FMCST2	FMCST1
	FMCST16	FMCST15	FMCST14	FMCST13	FMCST12	FMCST11	FMCST10	FMCST9
	FMCST24	FMCST23	FMCST22	FMCST21	FMCST20	FMCST19	FMCST18	FMCST17
	FMCST32	FMCST31	FMCST30	FMCST29	FMCST28	FMCST27	FMCST26	FMCST25

**MC Abort**

G351	GMCAB8	GMCAB7	GMCAB6	GMCAB5	GMCAB4	GMCAB3	GMCAB2	GMCAB1
	GMCAB16	GMCAB15	GMCAB14	GMCAB13	GMCAB12	GMCAB11	GMCAB10	GMCAB9
	GMCAB24	GMCAB23	GMCAB22	GMCAB21	GMCAB20	GMCAB19	GMCAB18	GMCAB17
	GMCAB32	GMCAB31	GMCAB30	GMCAB29	GMCAB28	GMCAB27	GMCAB26	GMCAB25
F351								

**MC Busy**

G355								
F355	FMCBUSY8	FMCBUSY7	FMCBUSY6	FMCBUSY5	FMCBUSY4	FMCBUSY3	FMCBUSY2	FMCBUSY1
	16	15	14	13	12	11	10	9
	24	23	22	21	20	19	18	17
	32	31	30	29	28	27	26	25

**MC Error**

G356								
F356	FM CER8	FM CER7	FM CER6	FM CER5	FM CER4	FM CER3	FM CER2	FM CER1
	FM CER16	FM CER15	FM CER14	FM CER13	FM CER12	FM CER11	FM CER10	FM CER9
	FM CER24	FM CER23	FM CER22	FM CER21	FM CER20	FM CER19	FM CER18	FM CER17
	FM CER32	FM CER31	FM CER30	FM CER29	FM CER28	FM CER27	FM CER26	FM CER25

**MC Feed Override**

G360 ~391	GMCFOR(32Word)
F360 ~391	

**MC POS**

G396	GPIS8	GPIS7	GPIS6	GPIS5	GPIS4	GPIS3	GPIS2	GPIS1
	GPIS16	GPIS15	GPIS14	GPIS13	GPIS12	GPIS11	GPIS10	GPIS9
	GPIS24	GPIS23	GPIS22	GPIS21	GPIS20	GPIS19	GPIS18	GPIS17
	GPIS32	GPIS31	GPIS30	GPIS29	GPIS28	GPIS27	GPIS26	GPIS25
F396								

G397	GPUI8	GPUI7	GPUI6	GPUI5	GPUI4	GPUI3	GPUI2	GPUI1
	GPUI16	GPUI15	GPUI14	GPUI13	GPUI12	GPUI11	GPUI10	GPUI9
	GPUI24	GPUI23	GPUI22	GPUI21	GPUI20	GPUI19	GPUI18	GPUI17
	GPUI32	GPUI31	GPUI30	GPUI29	GPUI28	GPUI27	GPUI26	GPUI25
F397								



MC Reference

G405								
F405	FRF8	FRF7	FRF6	FRF5	FRF4	FRF3	FRF2	FRF1
	FRF16	FRF15	FRF14	FRF13	FRF12	FRF11	FRF10	FRF9
	FRF24	FRF23	FRF22	FRF21	FRF20	FRF19	FRF18	FRF17
	FRF32	FRF31	FRF30	FRF29	FRF28	FRF27	FRF26	FRF25

G408	GRDEC8	GRDEC7	GRDEC6	GRDEC5	GRDEC4	GRDEC3	GRDEC2	GRDEC1
	GRDEC16	GRDEC15	GRDEC14	GRDEC13	GRDEC12	GRDEC11	GRDEC10	GRDEC9
	GRDEC24	GRDEC23	GRDEC22	GRDEC21	GRDEC20	GRDEC19	GRDEC18	GRDEC17
	GRDEC32	GRDEC31	GRDEC30	GRDEC29	GRDEC28	GRDEC27	GRDEC26	GRDEC25
F408								

G411								
F411	FRP1S8	FRP1S7	FRP1S6	FRP1S5	FRP1S4	FRP1S3	FRP1S2	FRP1S1
	FRP1S16	FRP1S15	FRP1S14	FRP1S13	FRP1S12	FRP1S11	FRP1S10	FRP1S9
	FRP1S24	FRP1S23	FRP1S22	FRP1S21	FRP1S20	FRP1S19	FRP1S18	FRP1S17
	FRP1S32	FRP1S31	FRP1S30	FRP1S29	FRP1S28	FRP1S27	FRP1S26	FRP1S25

G412								

F412	FRP2S8	FRP2S7	FRP2S6	FRP2S5	FRP2S4	FRP2S3	FRP2S2	FRP2S1
	FRP2S16	FRP2S15	FRP2S14	FRP2S13	FRP2S12	FRP2S11	FRP2S10	FRP2S9
	FRP2S24	FRP2S23	FRP2S22	FRP2S21	FRP2S20	FRP2S19	FRP2S18	FRP2S17
	FRP2S32	FRP2S31	FRP2S30	FRP2S29	FRP2S28	FRP2S27	FRP2S26	FRP2S25

G413								
F413	FRP3S8	FRP3S7	FRP3S6	FRP3S5	FRP3S4	FRP3S3	FRP3S2	FRP3S1
	FRP3S16	FRP3S15	FRP3S14	FRP3S13	FRP3S12	FRP3S11	FRP3S10	FRP3S9
	FRP3S24	FRP3S23	FRP3S22	FRP3S21	FRP3S20	FRP3S19	FRP3S18	FRP3S17
	FRP3S32	FRP3S31	FRP3S30	FRP3S29	FRP3S28	FRP3S27	FRP3S26	FRP3S25

G414								
F414	FRP4S8	FRP4S7	FRP4S6	FRP4S5	FRP4S4	FRP4S3	FRP4S2	FRP4S1
	FRP4S16	FRP4S15	FRP4S14	FRP4S13	FRP4S12	FRP4S11	FRP4S10	FRP4S9
	FRP4S24	FRP4S23	FRP4S22	FRP4S21	FRP4S20	FRP4S19	FRP4S18	FRP4S17
	FRP4S32	FRP4S31	FRP4S30	FRP4S29	FRP4S28	FRP4S27	FRP4S26	FRP4S25



+ Direction Move

G500								
F500	FPMV8	FPMV7	FPMV6	FPMV5	FPMV4	FPMV3	FPMV2	FPMV1
	FPMV16	FPMV15	FPMV14	FPMV13	FPMV12	FPMV11	FPMV10	FPMV9
	FPMV24	FPMV23	FPMV22	FPMV21	FPMV20	FPMV19	FPMV18	FPMV17
	FPMV32	FPMV31	FPMV30	FPMV29	FPMV28	FPMV27	FPMV26	FPMV25

 - Direction Move

G501								
F501	FNMV8	FNMV7	FNMV6	FNMV5	FNMV4	FNMV3	FNMV2	FNMV1
	FNMV16	FNMV15	FNMV14	FNMV13	FNMV12	FNMV11	FNMV10	FNMV9
	FNMV24	FNMV23	FNMV22	FNMV21	FNMV20	FNMV19	FNMV18	FNMV17
	FNMV32	FNMV31	FNMV30	FNMV29	FNMV28	FNMV27	FNMV26	FNMV25

 In-Position

G510								
F510	FINP8	FINP7	FINP6	FINP5	FINP4	FINP3	FINP2	FINP1
	FINP16	FINP15	FINP14	FINP13	FINP12	FINP11	FINP10	FINP9
	FINP24	FINP23	FINP22	FINP21	FINP20	FINP19	FINP18	FINP17
	FINP32	FINP31	FINP30	FINP29	FINP28	FINP27	FINP26	FINP25

**Reach Command Speed**

G512								
F512	FCSAGR8	FCSAGR7	FCSAGR6	FCSAGR5	FCSAGR4	FCSAGR3	FCSAGR2	FCSAGR1
	FCSAGR16	FCSAGR15	FCSAGR14	FCSAGR13	FCSAGR12	FCSAGR11	FCSAGR10	FCSAGR9
	FCSAGR24	FCSAGR23	FCSAGR22	FCSAGR21	FCSAGR20	FCSAGR19	FCSAGR18	FCSAGR17
	FCSAGR32	FCSAGR31	FCSAGR30	FCSAGR29	FCSAGR28	FCSAGR27	FCSAGR26	FCSAGR25

**0 (Zero) Speed Reached**

G513								
F513	FZSAGR8	FZSAGR7	FZSAGR6	FZSAGR5	FZSAGR4	FZSAGR3	FZSAGR2	FZSAGR1
	FZSAGR16	FZSAGR15	FZSAGR14	FZSAGR13	FZSAGR12	FZSAGR11	FZSAGR10	FZSAGR9
	FZSAGR24	FZSAGR23	FZSAGR22	FZSAGR21	FZSAGR20	FZSAGR19	FZSAGR18	FZSAGR17
	FZSAGR32	FZSAGR31	FZSAGR30	FZSAGR29	FZSAGR28	FZSAGR27	FZSAGR26	FZSAGR25

**Synchronous Control**

G550	GSYN8	GSYN7	GSYN6	GSYN5	GSYN4	GSYN3	GSYN2	GSYN1
	GSYN16	GSYN15	GSYN14	GSYN13	GSYN12	GSYN11	GSYN10	GSYN9
	GSYN24	GSYN23	GSYN22	GSYN21	GSYN20	GSYN19	GSYN18	GSYN17
	GSYN32	GSYN31	GSYN30	GSYN29	GSYN28	GSYN27	GSYN26	GSYN25
F550	FSYNS8	FSYNS7	FSYNS6	FSYNS5	FSYNS4	FSYNS3	FSYNS2	FSYNS1
	FSYNS16	FSYNS15	FSYNS14	FSYNS13	FSYNS12	FSYNS11	FSYNS10	FSYNS9
	FSYNS24	FSYNS23	FSYNS22	FSYNS21	FSYNS20	FSYNS19	FSYNS18	FSYNS17
	FSYNS32	FSYNS31	FSYNS30	FSYNS29	FSYNS28	FSYNS27	FSYNS26	FSYNS25

**MC External Positioning**

G558	GMCEXP8	GMCEP7	GMCEXP6	GMCEXP5	GMCEXP4	GMCEXP3	GMCEXP2	GMCEXP1
	GMCEXP16	GMCEXP15	GMCEXP14	GMCEXP13	GMCEXP12	GMCEXP11	GMCEXP10	GMCEXP9
	GMCEXP24	GMCEXP23	GMCEXP22	GMCEXP21	GMCEXP20	GMCEXP19	GMCEXP18	GMCEXP17
	GMCEXP32	GMCEXP31	GMCEXP30	GMCEXP29	GMCEXP28	GMCEXP27	GMCEXP26	GMCEXP25
F558	FMCEXPS8	FMCEXPS7	FMCEXPS6	FMCEXPS5	FMCEXPS4	FMCEXPS3	FMCEXPS2	FMCEXPS1
	FMCEXPS16	FMCEXPS15	FMCEXPS14	FMCEXPS13	FMCEXPS12	FMCEXPS11	FMCEXPS10	FMCEXPS9
	FMCEXPS24	FMCEXPS23	FMCEXPS22	FMCEXPS21	FMCEXPS20	FMCEXPS19	FMCEXPS18	FMCEXPS17
	FMCEXPS32	FMCEXPS31	FMCEXPS30	FMCEXPS29	FMCEXPS28	FMCEXPS27	FMCEXPS26	FMCEXPS25

**Position Latch Function**

G561	GPL8	GPL7	GPL6	GPL5	GPL4	GPL3	GPL2	GPL1
	GPL16	GPL15	GPL14	GPL13	GPL12	GPL11	GPL10	GPL9
	GPL24	GPL23	GPL22	GPL21	GPL20	GPL19	GPL18	GPL17
	GPL32	GPL31	GPL30	GPL29	GPL28	GPL27	GPL26	GPL25
F561	GPLS8	GPLS7	GPLS6	GPLS5	GPLS4	GPLS3	GPLS2	GPLS1
	GPLS16	GPLS15	GPLS14	GPLS13	GPLS12	GPLS11	GPLS10	GPLS9
	GPLS24	GPLS23	GPLS22	GPLS21	GPLS20	GPLS19	GPLS18	GPLS17
	GPLS32	GPLS31	GPLS30	GPLS29	GPLS28	GPLS27	GPLS26	GPLS25



Position Compare Function

G565	GPC8	GPC7	GPC6	GPC5	GPC4	GPC3	GPC2	GPC1
	GPC16	GPC15	GPC14	GPC13	GPC12	GPC11	GPC10	GPC9
	GPC24	GPC23	GPC22	GPC21	GPC20	GPC19	GPC18	GPC17
	GPC32	GPC31	GPC30	GPC29	GPC28	GPC27	GPC26	GPC25
F565								

8.8 X/Y Register

8.8.1 Configuration

■ Configuration

Description	Input / Output
Data Type	Bit, Word
Structure	
1. X Register (Input)	① It has 256 registers.
	② Register numbers 0 to 255 are available.
2. Y Register (Output)	① It has 256 registers.
	② Register numbers 0 to 255 are available.

8.9 R Register

8.9.1 Configuration

■ Configuration	
Description	PLC Program Global Variable
Data Type	Bit, Word, Float
Structure	
1. Non-stored register	① 2000
	② Register numbers 0 to 1999 available
2. Stored Registers	① 2000
	② Register numbers from 2000 to 3999 are available

8.10 U Register

8.10.1 Configuration

■ Configuration

Description	Motion Program Global Variable
Data Type	Bit, Word, Float
Structure	
1. Non-stored registers	① Contains 2000 registers.
	② Register numbers 0 to 1999 are available
2. Stored Registers	① It has 2000 registers.
	② Register numbers from 2000 to 3999 are available.

System Diagnostics

This section explains the types and details of errors that occur during operation of the GX-S200/S400, their causes, and countermeasures.

9.1 System Errors

9.1.1 LED Indicator List

9.1.2 System Error List

9.1 System Errors

This section describes the error states displayed on each module of the GX-S200/S400, explains the details of actual errors that occur, and provides countermeasures.

9.1.1 LED Indicator List

■ Power Module Indicator Lights

Indicator Name	Indicator Color	Meaning When Indicator Lights Up	Indicator
Power	Green	Power module is operating	
Error	Red	Power error	

■ CPU Module Indicator

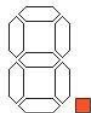
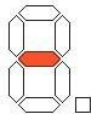
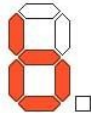
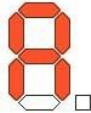
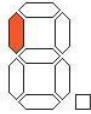
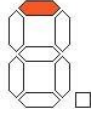
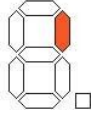
Indicator Name	Indicator Color	Meaning When Indicator Lights Up	Indicator
ERR	Red	Power error	
RUN	Green	CPU Running	
READY	Green	Ready state	
PWR	Red	Power module is operating	

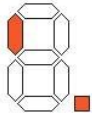
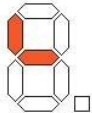
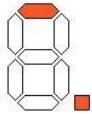
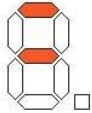
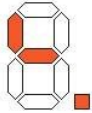
■ Other Module Indicator Lights

Indicator Name	Indicator Color	Meaning when indicator is lit	Indicator light
ERR	Red	Power Error	
RUN	Green	RUN Status	
READY	Green	Ready Status	
PWR	Red	Power module is operating	

■ CPU Module Display Indicator

Composed of 7 segments, it displays the CPU status as follows.

7-Segment Display Content	Purpose	Display Meaning
	Run	CPU isoperating
	Stop	CPU stopped
	Booting (Busy)	CPU booting
	Execute	Processing (7-segment flashing)
	Execute	Processing Complete
	Event (Error)	CPU Event and Error Occurrence
	Diagnosis	RAM Diagnostic Error
	Diagnosis	Register Storage Memory Diagnosis
	Diagnosis	Program Storage Memory Diagnostic Error

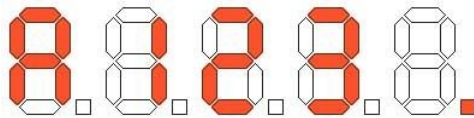
7-Segment Display Content	Purpose	Display Meaning
	Diagnosis	Register Storage Battery Alarm (During CPU Operation)
	Diagnosis	Register Save Battery Alarm (CPU Stopped)
	Diagnosis	Time IC Abnormal (CPU Running)
	Diagnosis	Time IC Abnormal (CPU Stopped)
	Diagnosis	System shutdown due to power off

■ Error Data Display

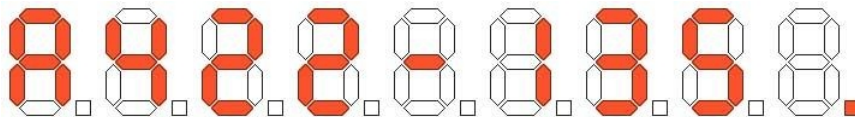
When an error occurs in the system, error data in the following format is displayed sequentially in succession.

- Basic format: [Error symbol] + [Error code] + [End symbol]
- Up to three levels of detailed data can be displayed based on the error code.
- The [-] symbol is used to distinguish between detailed level data.
- When an error occurs, error data from the [error symbol] to the [end symbol] is displayed sequentially and continuously.

◇ Error display for occurrence 345 of error code 123



◇ Occurrence 135 of Error Code 422



9.1.2 System Error List

The following shows the error list generated by each module in the system.

■ System Error List

Category	Error Code	Description
System	108	H/W Module Recognition Error
	113	Momentary Power Outage
	114	Watchdog Reset
	117	Memory Battery Error
	118	Watch IC Error
	127	System Setup Error (Data: Register, Code: Number)
	136	System Start
	137	System Shutdown
	138	System Mode Switch
	143	Event Log Reset
	145	Management History Log Reset
Communication	237	Communication Initialization Error
	240	Communication Data Receive Error
	241	Communication data sending error

Category	Error Code	Description
Motion Program	300	Lex Main Table configuration is incorrect
	301	YACC MAIN TABLE configuration error
	302	YACC SUB TABLE configuration is incorrect
	330	LEX token count exceeds maximum buffer
	331	Undefined characters exist
	332	Number exceeds maximum buffer
	333	One or more decimal points exist
	334	Number of parentheses in expression does not match
	335	YACC token count exceeds maximum buffer
	336	Syntax is incorrect
	337	String composition syntax is incorrect
	338	String exceeds maximum buffer size
	339	Contains invalid characters for the formula
	340	Syntax error in formula
	341	Not a valid macro variable
	342	TANGENT operation error
	343	SQUARE ROOT operation error
	344	Division cannot have a denominator of zero
	345	Exceeded IJK command limit within a single block during macro call
	346	Cannot call modal macros from subprograms
	347	Exceeded the limit for multiple calls to a modal macro
	348	Macro OVERFLOW
	349	User suspension of macro program
	350	Cannot issue simultaneous commands to one block
	351	Commands can only be issued at the beginning of a block
	352	Identical progress blocks exist
	353	Number of door addresses exceeds maximum buffer
	354	Unable to find the next block to proceed to
	355	Subroutine call syntax is incorrect

Error Code	Error Code	Description
Motion Program	356	Exceeded maximum subprogram calls
	357	Programs already called
	358	No duplicate command in subprogram
	359	Repeat command syntax is incorrect
	360	Too many loops
	361	No loop start
	362	Loop connection is incorrect
	363	Exceeded the limit of M commands within a single block
	364	Unused G-code
	365	Unused M code
	366	Cannot issue simultaneous commands in one block
	367	Terminated without a termination command
	368	Could not find the center point of the arc
	369	The center point of the arc is not in the correct position
	370	The specified file does not exist
	371	The file pointer does not exist
	372	The file seek address does not exist
	373	The number of characters per block is limited to 300
	374	';' does not exist
	375	Tool offset number is invalid
	376	Operates only on the main channel
	377	BCD conversion not possible
	378	Data type not usable for SHIFT
	379	Data type not usable with MOVD
	380	Data types that cannot be used with MOV B
	381	Data types not usable with SET D
	382	Data types that cannot be used with SET B
	383	This axis is not permitted in the current CH.
	384	Axis Code Limit Exceeded

Classification	Error Code	Description
Motion Program	385	Beyond Spindle Code
	386	Feed Code Limit Exceeded
	387	Executed only in the Main Program
	388	Incorrect program number instruction
	389	Calculated value is 0 or less
	390	Input axis does not exist
	391	The entered axis does not exist
	392	Axis not permitted on current channel
	393	Value not specified
	394	Arc plane not specified
	395	Radius not specified
	396	Synchronous M code input on unset CH
	397	Input synchronous M codes are different
	398	Input synchronous M code is out of range
	399	SBON SBOF cannot use GD
	400	POVR cannot be commanded independently
	401	Teaching Point command number exceeded
	402	ACC DCC cannot be commanded independently
	403	Teaching Point command number exceeded
	404	Optional Block Skip command number exceeded
	405	CHxx Command is invalid
	406	Exceeded the number of channels (CH) that can issue CHxx commands
	407	Register inaccessible in IPR
	408	Non-buffered M Code and Macro call M Code Cannot be assigned the same number
	409	The same axis cannot be used in each CH
	410	Cannot exceed register range
	411	Cannot input non-integer values in BCD or BIN instructions
	412	Data type not valid for IOW
	413	Only one axis command is possible in EXT mode.

Distinction	Error Code	Description
Motion Control	632	+OT
	633	-OT
	634	+Soft Limit Exceeded
	635	-Soft limit exceeded
	638	Excessive following error during stop
	639	Excessive transfer tracking error
	640	Excessive Synchronization Tracking Error
	643	Servo Alarm
	645	Servo Not Ready
	648	Encoder Disconnection Error
	649	Encoder Phase Error
	651	ABS Encoder Read Error
	655	In position error
	656	Speed Reached Error

Classification	Error Code	Description
PLC	764	Main PLC File Read Error
	765	Sub PLC File (Level 1) Read Error
	766	Sub PLC File (Stage 2) Read Error
	767	Double Call (Calling Itself)
	768	Calling Sub Program at Level 3 or Higher
	769	Subprogram call command optimization failure
	770	Executing a non-existent instruction
	796	First OPERAND value entered incorrectly
	797	Invalid value for the second operand
	798	The value of the third OPERAND is incorrect.
	799	The value of the fourth OPERAND is incorrect.
	800	Incorrect value entered for the fifth OPERAND
	801	Invalid value for sixth OPERAND
	802	Invalid value entered for the seventh OPERAND
	803	Invalid value for the eighth OPERAND
	804	Invalid value for the ninth OPERAND
	805	Invalid value entered for the tenth OPERAND
	806	Operand values are incorrectly set
	807	Overflow occurred during operation
	808	Underflow occurred during operation
	809	Division Error (Division by zero)
	810	Stack overflow occurred
	811	No data stored on stack
	812	Duplicate stack configuration (definition)
	813	Using an uninitialized stack
	814	The data type of the specified register and the data type of the table Type

Distinction	Error Code	Description
External input force (device)	939	Device Abnormal Error
	952	Communication Initialization Error
	953	Communication Error
	956	Slave Module Error